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MAJBURIY MATEMATIKA

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Kitob haqida

Ushbu o'quv qo'llanma majburiy matematikadan tayyorlanayotgan abituriyentlar va o'quvchilar uchun mo'ljallangan. Kitobda mavzular nazariy qism, namunaviy masalalar va mustaqil yechish uchun mashqlar bilan yoritilgan. Har bir mavzu yakunida QR kod orqali qo'shimcha interaktiv testlar ishlash va bilimni mustahkamlash mumkin.

Majburiy matematika telegram kanali

- Imtihon darajasidagi testlar
- Mavzular bo'yicha materiallar
- Interaktiv mashqlar
- Taklif va shikoyatlar uchun

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1.

Natural sonlar va ular ustida amallar

Nazariy qism

Raqam va son

✎ Ta'rif

Raqam — bu sonlarni yozish uchun foydalaniladigan 10 ta belgi: $0,\ 1,\ 2,\ 3,\ 4,\ 5,\ 6,\ 7,\ 8,\ 9$.
Son — raqamlardan tuzilgan miqdor. Masalan: 358 — bitta son, lekin uchta raqamdan iborat.

Natural sonlar

✎ Ta'rif

Natural sonlar — bu sanashda ishlatiladigan sonlardir:

$$\mathbb{N} = \{1,\ 2,\ 3,\ 4,\ 5,\ \ldots\}$$

Eng kichik natural son — **1**. Eng katta natural son mavjud emas (ular cheksiz ko'p). **0** natural son hisoblanmaydi.

✓ Eslatma

1 xonali natural sonlar: 1, 2, 3, 4, 5, 6, 7, 8, 9 — 9 ta
2 xonali natural sonlar: 10, 11, 12, ... , 98, 99 — 90 ta
3 xonali natural sonlar: 100, 101, 102, ... , 998, 999 — 900 ta
4 xonali natural sonlar: 1000, 1001, ... , 9999 — 9000 ta

Arifmetika qonunlari

✠ Asosiy qoida

Qo'shishda:

- O'rin almashtirish: $a + b = b + a$
- Guruhlash: $(a + b) + c = a + (b + c)$

Ko'paytirishda:

- O'rin almashtirish: $a \cdot b = b \cdot a$
- Guruhlash: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
- Taqsimlash: $a \cdot (b + c) = a \cdot b + a \cdot c$

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Amallar tartibi

✠ Asosiy qoida

Murakkab ifodalarda amallarning bajarilish tartibi:

1. Avval **qavslar** ichidagi amallar.
2. So'ngra **ko'paytirish** va **bo'lish** (chapdan o'ngga birinchi kelgani avval bajariladi).
3. Oxirida **qo'shish** va **ayirish** (chapdan o'ngga birinchi kelgani avval bajariladi).

Amaliy qism

1–36. *Hisoblang.*

1 $7 + 3 \cdot 4$

2 $20 - 2 \cdot 5$

3 $4 \cdot 6 + 8$

4 $30 - 5 \cdot 3$

5 $2 \cdot 5 + 3 \cdot 4$

6 $7 \cdot 3 + 6 \cdot 2$

7 $10 \cdot 4 - 5 \cdot 6$

8 $9 \cdot 2 + 8 \cdot 5$

9 $24 : 6 + 5$

10 $18 + 36 : 4$

11 $50 - 40 : 8$

12 $15 : 3 + 12 : 4$

13 $3 \cdot 8 + 5 \cdot 6 - 4 \cdot 7$

14 $50 - 4 \cdot 5 + 18 : 3$

15 $6 \cdot 7 - 24 : 6 + 10$

16 $9 \cdot 4 - 5 \cdot 3 + 7 \cdot 2$

17 $(15 + 25) : 8$

18 $6 \cdot (12 - 7)$

19 $(40 - 16) : 6$

20 $9 \cdot (8 + 5)$

21 $(3 \cdot 4 + 8) : 4$

22 $5 \cdot (20 - 4 \cdot 3)$

23 $(50 - 2 \cdot 15) : 4$

24 $72 : (3 \cdot 4 - 6)$

25 $(24 + 16) : (10 - 2)$

26 $(6 \cdot 5 - 10) : (8 - 4)$

27 $(100 - 28) : (4 + 5)$

28 $(15 + 9) \cdot (12 - 8)$

29 $100 - (50 - (20 - 5))$

30 $84 : (10 + (8 - 6))$

31 $5 \cdot (30 - (12 + 8))$

32 $48 : 6 + 4 \cdot (15 - 8)$

33 $(60 : 5 + 3) \cdot 4 - 25$

34 $90 - 5 \cdot (8 + 6) + 36 : 4$

35 $(144 : 12 + 8) \cdot (25 - 4 \cdot 5)$

36 $7 \cdot (24 : 3 + 5) - 56 : 8$

37–50. *Yig'indini ko'paytmaga almashtirib hisoblang.*

37 $8+8+8+8+8$

38 $100+100+100+100+100+100$

39 $150+150+150+150$

40 $2024+2024+2024+2024+2024$

41 $\underbrace{7+7+7+\ldots+7}_{\text{100 marta}}$

42 $\underbrace{50+50+50+\ldots+50}_{\text{20 marta}}$

43 $\underbrace{125+125+\ldots+125}_{\text{8 marta}}$

44 $\underbrace{8+8+8+\ldots+8}_{\text{1000 marta}}$

45 $(20+30)+(20+30)+(20+30)+(20+30)$

46 $(30-10)+(30-10)+(30-10)$

47 $\underbrace{(12+8)+(12+8)+\ldots+(12+8)}_{\text{20 ta son}}$

48 $\underbrace{(100+50)+(100+50)+\ldots+(100+50)}_{\text{40 ta son}}$

49 $\underbrace{(25-7)+(25-7)+\ldots+(25-7)}_{\text{20 ta son}}$

50 $\underbrace{(100-40)+(100-40)+\ldots+(100-40)}_{\text{30 ta son}}$

51–60. *Qulay usulda hisoblang.*

51 $7+4+3+6$

52 $6+8+4+2$

53 $35+28+65+72$

54 $47+19+53+81$

55 $56+138+144+62$

56 $27+49+173+251$

57 $128+245+172+155$

58 $367+219+133+281$

59 $13+25+38+62+75+87$

60 $24+38+76+62+145+55$

61–70. *Qulay usulda hisoblang .*

61 $12\cdot 7+12\cdot 3$

62 $25\cdot 6+25\cdot 4$

63 $23\cdot 45+23\cdot 55$

64 $84\cdot 73+84\cdot 27$

65 $39\cdot 47-39\cdot 37$

66 $125\cdot 83-125\cdot 73$

67 $18\cdot 83+18\cdot 17+82\cdot 83+17\cdot 82$

68 $32\cdot 57+68\cdot 71+57\cdot 68+32\cdot 71$

69 $9\cdot 73-9\cdot 63+4\cdot 73-4\cdot 63$

70 $18\cdot 39-18\cdot 35+12\cdot 47-12\cdot 43$

71–80. *Matnli masalalar.*

71 Uchta qalamning narxi 4500 so'm bo'lsa, 8 ta qalam necha so'm turadi?

72 Bog'da 25 ta olma daraxti bo'lib, nok daraxtlari olma daraxtlaridan 9 taga ko'p. Bog'da jami nechta daraxt bor?

73 Daftar 5000 so'm, qalam 2000 so'm turadi. 4 ta daftar va 3 ta qalam jami necha so'm turadi?

74 8 ta nonning narxi 24000 so'm bo'lsa, 2 ta nonning narxini toping.

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75 Sentyabrda 40 ta kitob, oktyabrda sentyabrdagidan 15 ta ko'p, noyabrda esa oktyabrdagidan 12 ta kam kitob keltirildi. Shu uch oyda jami nechta kitob keltirilgan?

76 Avtomobil soatiga 60 km tezlik bilan 4 soat yurdi. U qancha masofa bosib o'tgan?

77 Velosipedchi kuniga 25 km yo'l yurib manzilga 6 kunda yetib bordi. Qaytishda 5 kunda qaytmoqchi bo'lsa, u kuniga necha km yo'l yurishi kerak?

78 Piyoda har kuni 18 km yo'l yurib manzilga 10 kunda yetib bordi. Qaytishda 6 kunda yetib kelmoqchi bo'lsa, u kuniga necha km yo'l yurishi kerak?

79 Usta kuniga 8 ta stul, shogirdi 5 ta stul yasaydi. Ular birgalikda 6 kunda nechta stul yasashadi?

80 Bir nasos soatiga 60 litr, ikkinchisi 90 litr suv tortadi. Ikkalasi birga 5 soat ishlasa, jami necha litr suv tortiladi?

81–86. *Berilgan tengsizliklarni qanoatlantiruvchi barcha natural sonlarni yozing.*

81 $1 < n < 10$

82 $4 < n < 9$

83 $7 \leq n \leq 13$

84 $18 \leq n \leq 22$

85 $30 \leq n < 36$

86 $50 < n \leq 54$

87–90. *Quyida berilgan sonli qatorlarda nechta natural son qatnashgan?*

87 $1;\ 2;\ 3;\ \ldots;\ 56;\ 57$

88 $8;\ 9;\ 10;\ \ldots;\ 99;\ 100$

89 $14;\ 15;\ 16;\ \ldots;\ 88;\ 89$

90 $21;\ 22;\ 23;\ \ldots;\ 89;\ 90$

91–94. *Hisoblang.*

91 $7 \cdot 10000 + 4 \cdot 1000 + 6 \cdot 100 + 2 \cdot 10 + 8$

92 $9 \cdot 10000 + 3 \cdot 1000 + 2 \cdot 100 + 4 \cdot 10 + 5$

93 $2 \cdot 10000 + 7 \cdot 1000 + 6 \cdot 10 + 3$

94 $4 \cdot 10000 + 5 \cdot 10$

95–98. *Sonni xona birliklari yig'indisi ko'rinishida yozing.*

95 56243

96 308524

97 472916

98 30506

Mavzu bo'yicha test savollari

1-variant

1. Hisoblang: $25 + 5 \cdot 3$
   A) 90 B) 40 C) 30 D) 75

2. Eng katta to'rt xonali son bilan eng kichik uch xonali sonning ayirmasini toping.
   A) 9000 B) 9899 C) 9989 D) 9999

3. Qulay usulda hisoblang: $35 \cdot 27 + 35 \cdot 73$
   A) 2625 B) 3000 C) 3500 D) 4500

4. Uch xonali natural sonlar nechta?
   A) 899 B) 900 C) 999 D) 1000

5. $5 < n \leq 10$ tengsizlikni qanoatlantiruvchi natural n ning qiymatlari nechta?
   A) 4 B) 5 C) 6 D) 7

6. 56041 sonida o'nliklar xonasidagi raqam qaysi?
   A) 0 B) 4 C) 5 D) 6

7. Hisoblang: $(84 : 4 + 9) \cdot 2$
   A) 30 B) 39 C) 40 D) 60

8. Qulay usulda hisoblang: $18 + 23 + 32 + 27$
   A) 80 B) 90 C) 100 D) 105

9. 3 ta velosipedchining har biri kuniga 25 km dan 4 kun yo'l yursa, ular jami necha km yo'l yuradi?
   A) 100 B) 200 C) 250 D) 300

10. 2 dan 50 gacha bo'lgan (2 va 50 ham kiradi) natural sonlar nechta?
    A) 47 B) 48 C) 49 D) 50

2-variant

1. Hisoblang: $(12 + 8) \cdot 3 - 4$
   A) 56 B) 48 C) 60 D) 32

2. Ikki xonali natural sonlar nechta?
   A) 89 B) 90 C) 99 D) 100

3. Hisoblang: $\underbrace{7 + 7 + 7 + \ldots + 7}_{50 \text{ marta}}$
   A) 49 B) 75 C) 350 D) 700

4. 6, 7, 8 raqamlaridan foydalanib (raqamlar takrorlanmasin) nechta uch xonali son tuzish mumkin?
   A) 3 B) 6 C) 9 D) 27

5. $14;\ 15;\ 16;\ \ldots;\ 50$ qatorda nechta natural son qatnashgan?
   A) 36 B) 37 C) 38 D) 50

6. Hisoblang: $100 - 4 \cdot (12 - 7)$
   A) 480 B) 96 C) 80 D) 75

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7. 12 ta sonning yig'indisi 100 ga teng. Agar har bir qo'shiluvchini 3 ga orttirsak, yig'indi nechaga teng bo'ladi?
   A) 103 B) 112 C) 136 D) 300

8. 2 ta daftarning narxi 16000 so'm bo'lsa, 12 ta daftar uchun necha so'm to'lanadi?
   A) 80 000 B) 96 000 C) 100 000 D) 120 000

9. 1, 5, 7, 9 raqamlaridan iborat (raqamlar takrorlanmasin) eng katta to'rt xonali son qaysi?
   A) 7951 B) 9517 C) 9571 D) 9751

10. 1 dan 30 gacha bo'lgan natural sonlar ketma-ket bitta son sifatida yozildi: 1234 . . . 2930. Hosil bo'lgan son necha xonali?
    A) 30 B) 51 C) 59 D) 60

Javoblar

1. 19
2. 10
3. 32
4. 15
5. 22
6. 33
7. 10
8. 58
9. 9
10. 27
11. 45
12. 8
13. 26
14. 36
15. 48
16. 35
17. 5
18. 30
19. 4
20. 117
21. 5
22. 40
23. 5
24. 12
25. 5
26. 5
27. 8
28. 96
29. 65
30. 7
31. 50
32. 36
33. 35
34. 29
35. 100
36. 84
37. 40
38. 600
39. 600
40. 10 120
41. 700
42. 1000
43. 1000
44. 8000
45. 200
46. 60
47. 200
48. 3000
49. 180
50. 900
51. 20
52. 20
53. 200
54. 200
55. 400
56. 500
57. 700
58. 1000
59. 300
60. 400
61. 120
62. 250
63. 2300
64. 8400
65. 390
66. 1250
67. 10 000
68. 12 800
69. 130
70. 120
71. 12 000 so'm
72. 59 ta daraxt
73. 26 000 so'm
74. 6000 so'm
75. 138 ta kitob
76. 240 km

77. 30 km

78. 30 km

79. 78 ta stul

80. 750 litr

81. 2, 3, 4, 5, 6, 7, 8, 9

82. 5, 6, 7, 8

83. 7, 8, 9, 10, 11, 12, 13

84. 18, 19, 20, 21, 22

85. 30, 31, 32, 33, 34, 35

86. 51, 52, 53, 54

87. 57 ta

88. 93 ta

89. 76 ta

90. 70 ta

91. 74 628

92. 93 245

93. 27 063

94. 40 050

95. $5 \cdot 10\,000 + 6 \cdot 1000 + 2 \cdot 100 + 4 \cdot 10 + 3$

96. $3 \cdot 100\,000 + 8 \cdot 1000 + 5 \cdot 100 + 2 \cdot 10 + 4$

97. $4 \cdot 100\,000 + 7 \cdot 10\,000 + 2 \cdot 1000 + 9 \cdot 100 + 1 \cdot 10 + 6$

98. $3 \cdot 10\,000 + 5 \cdot 100 + 6$

Test javoblari

1-variant: 1) B 2) B 3) C 4) B 5) B 6) B 7) D 8) C 9) D 10) C

2-variant: 1) A 2) B 3) C 4) B 5) B 6) C 7) C 8) B 9) D 10) B

Mavzu bo'yicha interaktiv testlar

QR kodni skanerlab, bilimingizni sinab ko'ring

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2.

Bo'linish belgilari, EKUB va EKUK

Nazariy qism

Bo'luvchi va karrali son

✎ Ta'rif

a soniga qoldiqsiz bo'linadigan sonlar a ning **karralilari** deyiladi.

- 20 ning karralilari: $20, 40, 60, 80, 100, \ldots$

a soni qoldiqsiz bo'linadigan sonlar a ning **bo'luvchilari** deyiladi.

- 20 ning bo'luvchilari: $1, 2, 4, 5, 10, 20$.

✓ Eslatma

- Har qanday a natural sonining karralilari soni **cheksiz**: $a,\ 2a,\ 3a,\ 4a,\ \ldots$
- Har qanday a natural sonining natural bo'luvchilari soni **cheklangan**.

Bo'linish belgilari

✠ Asosiy qoida

- **2 ga bo'linish:** oxirgi raqami juft bo'lsa $(0,\ 2,\ 4,\ 6,\ 8)$.
- **3 ga bo'linish:** raqamlari yig'indisi 3 ga bo'linsa.
- **5 ga bo'linish:** oxirgi raqami 0 yoki 5 bo'lsa.
- **9 ga bo'linish:** raqamlari yig'indisi 9 ga bo'linsa.
- **10 ga bo'linish:** oxirgi raqami 0 bo'lsa.

Tub va murakkab sonlar

✎ Ta'rif

Tub sonlar — faqat ikkita bo'luvchiga (sonning o'zi va 1) ega bo'lgan natural sonlar.
Murakkab sonlar — ikkitadan ortiq bo'luvchiga ega bo'lgan natural sonlar.

1 — **tub son ham emas, murakkab son ham emas**.

✓ **Eslatma**

100 gacha bo'lgan tub sonlar:

$$2,\ 3,\ 5,\ 7,\ 11,\ 13,\ 17,\ 19,\ 23,\ 29,\ 31,\ 37,\ 41,\ 43,\ 47,$$

$$53,\ 59,\ 61,\ 67,\ 71,\ 73,\ 79,\ 83,\ 89,\ 97.$$

Yodda tuting: 2 — yagona juft tub son. Boshqa barcha juft sonlar 2 ga bo'linadi, demak murakkab.

Sonni tub ko'paytuvchilarga ajratish

✓ **Eslatma**

Misol. 360 ni tub ko'paytuvchilarga ajratamiz:

360	2
180	2
90	2
45	3
15	3
5	5
1	

$$360 = 2 \cdot 2 \cdot 2 \cdot 3 \cdot 3 \cdot 5 = 2^3 \cdot 3^2 \cdot 5.$$

EKUB — Eng katta umumiy bo'luvchi

✎ **Ta'rif**

a va b natural sonlarning **eng katta umumiy bo'luvchisi (EKUB)** — bu a ham, b ham bo'linadigan eng katta natural son.
Belgilanishi: EKUB$(a; b)$.

✠ **Asosiy qoida**

EKUBni topish uchun:

1. Har bir son tub ko'paytuvchilarga ajratiladi.
2. Sonlardagi umumiy tub ko'paytuvchilarning eng kichik darajalari olinib, ulardan ko'paytma tuziladi.
3. Tuzilgan ko'paytmaning qiymati hisoblanadi.

Misol. $EKUB(72; 120) = ?$

$$72 = 2^3 \cdot 3^2, \qquad 120 = 2^3 \cdot 3 \cdot 5.$$

Umumiy ko'paytuvchilar: 2 va 3. Eng kichik darajalari: 2^3 va 3^1. Demak,

$$EKUB(72; 120) = 2^3 \cdot 3 = 8 \cdot 3 = 24.$$

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EKUK — Eng kichik umumiy karrali

✎ Ta'rif

a va b natural sonlarning **eng kichik umumiy karralisi (EKUK)** — bu a ga ham, b ga ham bo'linadigan eng kichik natural son.
Belgilanishi: EKUK$(a;b)$.

✠ Asosiy qoida

EKUKni topish uchun:

1. Har bir son tub ko'paytuvchilarga ajratiladi.
2. Sonlardagi umumiy tub ko'paytuvchilarning eng katta darajalari va umumiy bo'lmagan tub ko'paytuvchilardan ko'paytma tuziladi.
3. Tuzilgan ko'paytmaning qiymati hisoblanadi.

Misol. $EKUK(72;120) = ?$

$$72 = 2^3 \cdot 3^2, \qquad 120 = 2^3 \cdot 3 \cdot 5.$$

Umumiy ko'paytuvchilar: 2 va 3. Eng katta darajalari: 2^3 va 3^2. Umumiy bo'lmagan ko'paytuvchi: 5. Demak,

$$EKUK(72;120) = 2^3 \cdot 3^2 \cdot 5 = 8 \cdot 9 \cdot 5 = 360.$$

✠ Asosiy qoida

Har qanday a va b natural sonlari uchun:

$$\boxed{EKUB(a;b) \cdot EKUK(a;b) = a \cdot b}$$

Tekshirish: $EKUB(72;120) \cdot EKUK(72;120) = 24 \cdot 360 = 8640, \quad 72 \cdot 120 = 8640.$ ✓

Amaliy qism

1–8. *Karrali sonlar.*

1 4 ning dastlabki 6 ta karralisini yozing.

2 6 ning dastlabki 5 ta karralisini yozing.

3 9 ning dastlabki 5 ta karralisini yozing.

4 5 ning 50 dan kichik karralilarini yozing.

5 7 ning 50 dan kichik karralilarini yozing.

6 11 ning 100 dan kichik karralilarini nechta?

7 15 ning ikki xonali karralilari nechta?

8 30 ning uch xonali karralilari nechta?

9–16. *Bo'luvchi sonlar.*

9 8 ning barcha natural bo'luvchilarini yozing.

10 15 ning barcha natural bo'luvchilarini yozing.

11 20 ning barcha natural bo'luvchilarini yozing.

12 24 ning barcha natural bo'luvchilarini yozing.

13 36 ning barcha natural bo'luvchilarini yozing.

14 10 ning natural bo'luvchilari yig'indisini toping.

15 18 ning natural bo'luvchilari yig'indisini toping.

16 Nechta ikki xonali son 50 ning bo'luvchisi bo'ladi?

17–22. 2 *ga*, 5 *ga*, 10 *ga bo'linish* (*oxirgi raqamga qarab*).

17 Berilganlardan 2 ga bo'linadiganlarini ajrating: 6, 13, 34, 90, 198, 2981, 74835.

18 Berilganlardan 5 ga bo'linadiganlarini ajrating: 60, 125, 461, 4881, 89975.

19 Berilganlardan 10 ga bo'linadiganlarini ajrating: 65, 180, 770, 4281, 98150.

20 $\overline{72*}$ soni 5 ga bo'linadi. Yulduzcha o'rniga qaysi raqamlarni yozish mumkin?

21 $\overline{4**}$ ko'rinishidagi eng katta uch xonali son 10 ga bo'linadi. Bu sonni toping?

22 5 ga bo'linadigan, lekin 10 ga bo'linmaydigan sonlarni ajrating: 30; 45; 100; 175; 280.

23–28. 3 *ga va* 9 *ga bo'linish* (*raqamlar yig'indisiga qarab*).

23 Berilganlardan 3 ga bo'linadiganlarini ajrating: 42, 96, 108, 1207, 4892, 31980.

24 Berilganlardan 9 ga bo'linadiganlarini ajrating: 72, 135, 404, 6238044, 956011.

25 $\overline{4*3}$ soni 3 ga bo'linadi. Yulduzcha o'rniga qaysi raqamlarni yozish mumkin?

26 $\overline{2*4}$ soni 9 ga bo'linishi uchun $*$ ning o'rniga qanday raqamni qo'yish mumkin?

27 $\overline{5**}$ ko'rinishidagi 9 ga bo'linuvchi eng kichik uch xonali sonni toping.

28 3 ga bo'linadigan, lekin 9 ga bo'linmaydigan sonlarni ajrating: 27, 42, 81, 117, 246.

29–32. *Bir vaqtda bir nechta songa bo'linish.*

29 2 ga ham, 3 ga ham bo'linadigan sonlarni ajrating: 18, 27, 42, 65, 96, 105.

30 3 ga ham, 5 ga ham bo'linadigan sonlarni ajrating: 15, 30, 45, 80, 135, 200.

31 5 ga ham, 9 ga ham bo'linadigan sonlarni ajrating: 45, 90, 150, 180, 225, 360, 305, 405.

32 $\overline{7*5}$ soni 5 ga ham, 3 ga ham bo'linadi. Yulduzcha o'rnida qaysi raqam(lar) bo'lishi mumkin?

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33–38. *Tub sonlar.*

33 Berilgan qatordan tub sonlarni ajrating: $2,\ 7,\ 27,\ 45,\ 57,\ 61,\ 87,\ 91.$

34 Berilgan qatordan tub sonlarni ajrating: $3,\ 9,\ 19,\ 31,\ 43,\ 58,\ 77,\ 81.$

35 1 dan 30 gacha bo'lgan tub sonlar nechta?

36 30 dan 60 gacha bo'lgan tub sonlar nechta?

37 18 ning tub bo'luvchilarini toping va ularning yig'indisini hisoblang.

38 42 ning tub bo'luvchilarini toping va ularning yig'indisini hisoblang.

39–44. *Sonlarni tub ko'paytuvchilarga ajrating.*

39 12

40 36

41 84

42 96

43 120

44 360

45–48. *Umumiy bo'luvchilarni topish.*

45 8 va 12 ning umumiy bo'luvchilarini yozing.

46 12 va 18 ning umumiy bo'luvchilarini yozing.

47 20 va 30 ning umumiy bo'luvchilarini yozing.

48 24 va 36 ning umumiy bo'luvchilarini yozing.

49–60. *EKUBni toping.*

49 $(3;\ 4)$

50 $(12;\ 13)$

51 $(6;\ 8)$

52 $(9;\ 21)$

53 $(20;\ 24)$

54 $(20;\ 60)$

55 $(24;\ 60)$

56 $(72;\ 90)$

57 $(48;\ 72)$

58 $(72;\ 108)$

59 $(18;\ 24;\ 30)$

60 $(36;\ 48;\ 60)$

61–64. *Umumiy karralilarni topish.*

61 4 va 6 ning 50 dan kichik umumiy karralilarini yozing.

62 3 va 5 ning 50 dan kichik umumiy karralilarini yozing.

63 6 va 9 ning 60 dan kichik umumiy karralilarini yozing.

64 8 va 12 ning 100 dan kichik umumiy karralilarini yozing.

65–74. *EKUKni toping.*

65 $(2;\ 14)$

66 $(8;\ 16)$

67 $(9;\ 36)$

68 $(3;\ 5)$

69 $(6;\ 7)$

70 $(24;\ 36)$

71 $(15;\ 25)$

72 $(45;\ 60)$

73 $(6;\ 9;\ 12)$

74 $(8;\ 12;\ 18)$

75–79. *Matnli masalalar.*

75 12 ta qalam va 18 ta daftarni bolalarga teng taqsimlash kerak. Eng ko'pi bilan nechta bolaga taqsimlash mumkin va har biriga nechta qalam, nechta daftar tegadi?

76 Aziz har 4 kunda, Botir har 6 kunda kutubxonaga boradi. Bugun ular kutubxonada uchrashishdi. Keyingi safar necha kundan keyin uchrashishadi?

77 Aziz har 2 kunda, Botir har 3 kunda, Davron har 4 kunda kutubxonaga boradi. Bugun ular kutubxonada uchrashishdi. Keyingi safar necha kundan keyin uchrashishadi?

78 Bekatdan birinchi avtobus har 6 daqiqada, ikkinchisi har 10 daqiqada qatnaydi. Soat 7:00 da ular bekatdan birga jo'nagan bo'lsa, keyingi marta qachon birga jo'naydi?

79 Xonada uchta soat bor: birinchisi har 4 daqiqada, ikkinchisi har 6 daqiqada, uchinchisi har 8 daqiqada bong uradi. Hozir uchalasi birga bong urdi. Keyingi marta necha daqiqadan keyin birga bong uradi?

80–83. *Qo'shimcha masalalar.*

80 1 dan 30 gacha bo'lgan natural sonlar orasida nechtasi 2 ga ham, 3 ga ham bo'linadi?

81 1 dan 20 gacha bo'lgan natural sonlar orasida nechtasi 5 ga bo'linmaydi?

82 Ikki sonning EKUBi 2, EKUKi 12 ga teng. Bu sonlarning ko'paytmasini toping.

83 Ikki sonning ko'paytmasi 36, EKUBi 3 ga teng. Ularning EKUKini toping.

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Mavzu bo'yicha test savollari

1-variant

1. 7 ning 40 dan kichik karralilari nechta?
   A) 4 B) 5 C) 6 D) 7

2. 20 ning natural bo'luvchilari yig'indisini toping.
   A) 36 B) 42 C) 48 D) 56

3. $\overline{5*4}$ soni 3 ga bo'linadi. Yulduzcha o'rniga jami nechta raqam yozish mumkin?
   A) 2 B) 3 C) 4 D) 5

4. Quyidagilardan qaysi biri 2 ga ham, 3 ga ham bo'linadi?
   A) 124 B) 235 C) 342 D) 451

5. 1 dan 30 gacha bo'lgan tub sonlar nechta?
   A) 9 B) 10 C) 11 D) 12

6. 36 va 48 sonlarining umumiy natural bo'luvchilari soni nechta?
   A) 4 B) 5 C) 6 D) 8

7. $EKUK(8;\ 12) = ?$
   A) 20 B) 24 C) 48 D) 96

8. 1 dan 50 gacha bo'lgan natural sonlar orasida nechtasi 6 ga bo'linadi?
   A) 6 B) 7 C) 8 D) 9

9. $\overline{1*5}$ soni 9 ga bo'linadi. Yulduzcha o'rniga qaysi raqam yozilishi kerak?
   A) 0 B) 3 C) 6 D) 9

10. Ikki sonning ko'paytmasi 360, EKUBi 6 ga teng. Ularning EKUKini toping.
    A) 36 B) 60 C) 72 D) 120

2-variant

1. 48 ning barcha natural bo'luvchilari soni nechta?
   A) 8 B) 9 C) 10 D) 12

2. Quyidagi sonlardan qaysi biri 9 ga bo'linadi?
   A) 4258 B) 5418 C) 7531 D) 8912

3. $\overline{82*}$ soni 5 ga bo'linadi. Yulduzcha o'rniga qaysi raqamlarni yozish mumkin?
   A) faqat 0 B) faqat 5 C) 0 yoki 5 D) hech qaysisi

4. Quyidagilardan qaysi biri tub son emas?
   A) 29 B) 37 C) 51 D) 53

5. 84 sonini tub ko'paytuvchilarga ajrating.
   A) $2 \cdot 3 \cdot 14$ B) $2^2 \cdot 3 \cdot 7$ C) $2 \cdot 3^2 \cdot 7$ D) $4 \cdot 3 \cdot 7$

6. $EKUB(36;\ 60) = ?$
   A) 6 B) 9 C) 12 D) 18

7. Ikki sonning EKUBi 6, EKUKi 60 ga teng. Ularning ko'paytmasini toping.
   A) 66 B) 120 C) 360 D) 600

8. Aziz har 8 kunda, Botir har 12 kunda kutubxonaga boradi. Bugun ular birga uchrashishdi. Keyingi safar necha kundan keyin uchrashishadi?
A) 10 B) 12 C) 16 D) 24

9. Quyidagi sonlardan qaysi biri tub son?
A) 21 B) 27 C) 33 D) 41

10. $EKUB(24;\ 30) \cdot EKUK(24;\ 30) = ?$
A) 54 B) 120 C) 360 D) 720

Javoblar

1. $4, 8, 12, 16, 20, 24$

2. $6, 12, 18, 24, 30$

3. $9, 18, 27, 36, 45$

4. $5, 10, 15, 20, 25, 30, 35, 40, 45$

5. $7, 14, 21, 28, 35, 42, 49$

6. 9 ta

7. 6 ta

8. 30 ta

9. $1, 2, 4, 8$

10. $1, 3, 5, 15$

11. $1, 2, 4, 5, 10, 20$

12. $1, 2, 3, 4, 6, 8, 12, 24$

13. $1, 2, 3, 4, 6, 9, 12, 18, 36$

14. 18

15. 39

16. 3 ta $(10, 25, 50)$

17. $6, 34, 90, 198$

18. $60, 125, 89975$

19. $180, 770, 98150$

20. $* = 0, 5$

21. 490

22. $45, 175$

23. $42, 96, 108, 31980$

24. $72, 135, 6238044$

25. $* = 2, 5, 8$

26. $* = 3$

27. 504

28. $42, 246$

29. $18, 42, 96$

30. $15, 30, 45, 135$

31. $45, 90, 180, 225, 360, 405$

32. $* = 0, 3, 6, 9$

33. $2, 7, 61$

34. $3, 19, 31, 43$

35. 10 ta

36. 7 ta

37. Tub bo'luvchilari: $2, 3$. Yig'indi: 5

38. Tub bo'luvchilari: $2, 3, 7$. Yig'indi: 12

39. $2^2 \cdot 3$

40. $2^2 \cdot 3^2$

41. $2^2 \cdot 3 \cdot 7$

42. $2^5 \cdot 3$

43. $2^3 \cdot 3 \cdot 5$

44. $2^3 \cdot 3^2 \cdot 5$

45. $1, 2, 4$

46. $1, 2, 3, 6$

47. $1, 2, 5, 10$

48. $1, 2, 3, 4, 6, 12$

49. 1

50. 1

51. 2

52. 3

53. 4

54. 20

55. 12

56. 18

57. 24

58. 36

59. 6

60. 12

61. $12, 24, 36, 48$

62. $15, 30, 45$

63. $18, 36, 54$

64. $24, 48, 72, 96$

65. 14

66. 16

67. 36

68. 15

69. 42

70. 72

71. 75

72. 180

73. 36

74. 72

75. 6 ta bolaga (2 qalam, 3 daftar)

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76. 12 kundan keyin

77. 12 kundan keyin

78. 7:30 da (30 daqiqadan keyin)

79. 24 daqiqadan keyin

80. 5 ta

81. 16 ta

82. 24

83. 12

Test javoblari

1-variant: 1) B 2) B 3) C 4) C 5) B 6) C 7) B 8) C 9) B 10) B

2-variant: 1) C 2) B 3) C 4) C 5) B 6) C 7) C 8) D 9) D 10) D

Mavzu bo'yicha interaktiv testlar

QR kodni skanerlab, bilimingizni sinab ko'ring

3.

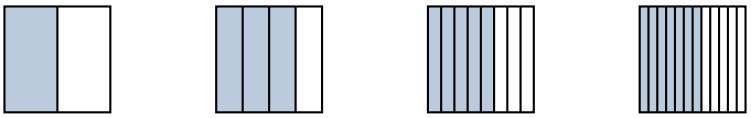
Oddiy kasrlar

Nazariy qism

✎ Ta'rif

$\frac{a}{b}$ ko'rinishidagi yozuvga **oddiy kasr** deyiladi.

- a — **surat** (nechta qism olinganini ko'rsatadi);
- b — **maxraj** (butun nechta teng qismga bo'linganini ko'rsatadi, $b \neq 0$);
- ular orasidagi chiziq — **kasr chizig'i** (bo'lish belgisi vazifasini bajaradi).



$\frac{1}{2}$ $\frac{3}{4}$ $\frac{5}{8}$ $\frac{7}{12}$

✓ Eslatma

- $\frac{a}{b} = a : b$ (kasr chizig'i — bo'lish belgisi).
- Har qanday natural sonni maxraji 1 bo'lgan kasr ko'rinishida yozish mumkin: $7 = \frac{7}{1}$.
- Surati va maxraji teng kasr birga teng: $\frac{5}{5} = 1$.
- Surati nolga teng kasr nolga teng: $\frac{0}{8} = 0$. **Maxraj nolga teng bo'la olmaydi!**

To'g'ri va noto'g'ri kasrlar

✎ Ta'rif

- **To'g'ri kasr** — surati maxrajidan kichik kasr ($a < b$). Bunday kasr birdan kichik: $\frac{a}{b} < 1$.
- **Noto'g'ri kasr** — surati maxrajidan katta yoki teng kasr ($a \geq b$). Bunday kasr birdan katta yoki teng: $\frac{a}{b} \geq 1$.

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✓ Eslatma

Misollar:

To'g'ri kasrlar: $\frac{1}{2}, \frac{3}{7}, \frac{15}{16} \quad (< 1).$

Noto'g'ri kasrlar: $\frac{5}{4}, \frac{9}{9}, \frac{20}{3} \quad (\geq 1).$

Aralash sonlar

✎ Ta'rif

Aralash son — deb noldan farqli butun son bilan to'g'ri kasr yig'indisining qisqa yozuviga aytiladi.

$$2\frac{3}{5} = 2 + \frac{3}{5}.$$

✠ Asosiy qoida

Noto'g'ri kasrni aralash songa aylantirish.

Misol: $\frac{17}{5} = ?$ $\quad 17 : 5 = 3$ qoldiq $2.$

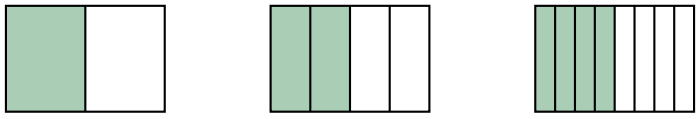
Demak, $\frac{17}{5} = 3\frac{2}{5}.$

Aralash sonni noto'g'ri kasrga aylantirish.

Misol: $4\frac{3}{7} = \frac{4 \cdot 7 + 3}{7} = \frac{31}{7}.$

Teng kasrlar va kasrning asosiy xossasi

Bir butun turlicha bo'linishi mumkin. Quyidagi rasmda — bir xil ulush, lekin uch xil yozuv:



$\frac{1}{2}$ $\qquad$ $\frac{2}{4}$ $\qquad$ $\frac{4}{8}$

Bir xil miqdor — uch xil yozuv: $\frac{1}{2} = \frac{2}{4} = \frac{4}{8}$

✠ Asosiy qoida

Kasrning asosiy xossasi. Surat va maxrajni bir xil noldan farqli songa ko'paytirsak (yoki bo'lsak), kasr qiymati o'zgarmaydi.

$$\frac{a}{b} = \frac{a \cdot k}{b \cdot k} = \frac{a : k}{b : k}, \qquad k \neq 0.$$

Misollar:

$\frac{2}{3} = \frac{2 \cdot 4}{3 \cdot 4} = \frac{8}{12}$ $\qquad$ $\frac{18}{24} = \frac{18 : 6}{24 : 6} = \frac{3}{4}$

Kasrlarni qisqartirish

✠ Asosiy qoida

Qoida. Eng tez qisqartirish uchun surat va maxrajning EKUBini topib, unga bo'lish kerak.

Misol. $\frac{36}{60} = ?$ $\quad EKUB(36;\ 60) = 12.$ $\quad \frac{36}{60} = \frac{36:12}{60:12} = \frac{3}{5}.$

✓ Eslatma

Agar EKUB topish qiyin bo'lsa, bosqichma-bosqich qisqartirsa bo'ladi:

$$\frac{36}{60} \xrightarrow{:2} \frac{18}{30} \xrightarrow{:2} \frac{9}{15} \xrightarrow{:3} \frac{3}{5}.$$

Surati va maxrajining 1 dan boshqa umumiy bo'luvchisi bo'lmagan kasrga qisqarmas kasr deyiladi. Qisqartirish natijasi doimo qisqarmas kasr bo'lishi kerak.

Umumiy maxrajga keltirish

✠ Asosiy qoida

Algoritm.

1. Maxrajlarning EKUKini topamiz — yangi (umumiy) maxraj.
2. Yangi maxrajni har bir eski maxrajga bo'lib, qo'shimcha ko'paytuvchini topamiz.
3. Har bir kasrning surat va maxrajini o'z qo'shimcha ko'paytuvchisiga ko'paytiramiz.

Misol. $\frac{2}{3}$ va $\frac{5}{4}$ ni umumiy maxrajga keltiramiz.

$EKUK(3;\ 4) = 12.$ $\quad 12:3 = 4,$ $\quad 12:4 = 3.$

$$\frac{2}{3} = \frac{2\cdot\mathbf{4}}{3\cdot\mathbf{4}} = \frac{8}{12}, \qquad \frac{5}{4} = \frac{5\cdot\mathbf{3}}{4\cdot\mathbf{3}} = \frac{15}{12}.$$

Kasrlarni taqqoslash

✠ Asosiy qoida

1) Bir xil maxrajli kasrlardan surati katta bo'lgani kattaroq:

$$\frac{3}{8} < \frac{5}{8}.$$

2) Bir xil suratli kasrlardan maxraji kichik bo'lgani kattaroq:

$$\frac{4}{5} > \frac{4}{9}.$$

3) Har xil maxrajli kasrlarni avval umumiy maxrajga keltirib, so'ngra taqqoslaymiz.

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Qo'shish va ayirish

✠ Asosiy qoida

Bir xil maxrajli kasrlar uchun:

$$\frac{a}{c}+\frac{b}{c}=\frac{a+b}{c}, \qquad \frac{a}{c}-\frac{b}{c}=\frac{a-b}{c}.$$

Har xil maxrajli kasrlarni avval umumiy maxrajga keltirib, so'ngra suratlarni qo'shamiz (ayiramiz). Maxraj o'zgarmaydi.

Misol.

$$\frac{2}{3}+\frac{1}{4}=\frac{8}{12}+\frac{3}{12}=\frac{11}{12}.$$

Aralash sonlar ustida amal bajarishda butun va kasr qismlari alohida qo'shiladi (ayiriladi):

$$2\tfrac{1}{4}+1\tfrac{1}{6}=(2+1)+\left(\tfrac{3}{12}+\tfrac{2}{12}\right)=3\tfrac{5}{12}.$$

Ko'paytirish

✠ Asosiy qoida

Kasrlarning surati suratiga, maxraji maxrajiga ko'paytiriladi:

$$\frac{a}{b}\cdot\frac{c}{d}=\frac{a\cdot c}{b\cdot d}.$$

Aralash sonlarni avval noto'g'ri kasrga aylantirish kerak.

✓ Eslatma

Ko'paytirishdan oldin qisqartirilsa, hisob osonlashadi.

Ko'paytirish jarayonida imkon bo'lgan hollarda kasrlarni qisqartirish mumkin.

1-misol.

$$\frac{\cancel{24}^{\,3}}{\cancel{49}_{\,7}}\cdot\frac{\cancel{21}^{\,3}}{\cancel{32}_{\,4}}=\frac{3}{7}\cdot\frac{3}{4}=\frac{3\times 3}{7\times 4}=\frac{9}{28}.$$

Bo'lish

✎ Ta'rif

$\frac{a}{b}$ kasrning **teskarisi** $\frac{b}{a}$.

Misol: $\frac{2}{5}$ ning teskarisi $\frac{5}{2}$; $\quad 7=\frac{7}{1}$ ning teskarisi $\frac{1}{7}$.

Kasrni o'zining teskarisiga ko'paytirsak 1 chiqadi: $\frac{a}{b}\cdot\frac{b}{a}=1$.

✠ Asosiy qoida

Bo'lish — bo'luvchining **teskarisiga** ko'paytirish:

$$\frac{a}{b} : \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c} = \frac{a \cdot d}{b \cdot c}.$$

Misol. $\frac{3}{4} : \frac{9}{8} = \frac{3}{4} \cdot \frac{8}{9} = \frac{1}{1} \cdot \frac{2}{3} = \frac{2}{3}.$

DIQQAT! Keng tarqalgan xatolar

✓ Eslatma

- **Xato:** $\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$ ✗

To'g'ri: avval umumiy maxrajga keltirish kerak: $\frac{1}{2} + \frac{1}{3} = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}.$ ✓

- **Xato:** kasrlarni ko'paytirishda umumiy maxraj kerak. ✗

To'g'ri: kasrlarni ko'paytirishda *umumiy maxraj kerak emas* — to'g'ridan-to'g'ri ko'paytiriladi.

- **Xato:** $5 - 1\frac{2}{3} = 4\frac{2}{3}.$ ✗

To'g'ri: $5 = 4\frac{3}{3}, \quad 4\frac{3}{3} - 1\frac{2}{3} = 3\frac{1}{3}.$ ✓

Amaliy qism

1–6.

1 Pitsa 8 ta teng bo'lakka bo'lingan. 3 ta bo'lak yeyildi. Qancha qismi yeyilgan? Qancha qismi qolgan?

2 Tort 12 ta bo'lakka bo'lingan. Anvar 2 ta, Bobur 3 ta, Davron 1 ta bo'lak yedi. Tortning qancha qismi yeyildi? Qancha qismi qoldi?

3 Kvadrat shaklidagi shokolad 20 ta teng bo'lakka bo'lingan.Umumiydan 7 ta bo'lakni kasr ko'rinishida yozing?

4 1 metr lentadan 35 sm kesib olindi. Qancha qismi kesilgan? Qancha qismi qoldi? (*Maxraji 100 bo'lgan kasr ko'rinishida yozing.*)

5 Bir kun 24 soatdan iborat bo'lsa, 1 soat kunning qanday qismiga teng? 8 soat-chi?

6 Sinfdagi 25 ta o'quvchidan 10 tasi qiz bola, qolgani o'g'il bola. Qiz bolalar jami o'quvchilarning qanday qismini tashkil qiladi? O'g'il bolalar-chi?

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7–18. *Kasrlarni qisqartiring.*

7 $\frac{4}{8}$

8 $\frac{6}{9}$

9 $\frac{12}{16}$

10 $\frac{15}{25}$

11 $\frac{14}{21}$

12 $\frac{18}{24}$

13 $\frac{25}{45}$

14 $\frac{36}{48}$

15 $\frac{32}{56}$

16 $\frac{60}{72}$

17 $\frac{84}{144}$

18 $\frac{120}{180}$

19–24. *? o'rnidagi sonni toping.*

19 $\frac{2}{3} = \frac{?}{12}$

20 $\frac{3}{5} = \frac{?}{20}$

21 $\frac{4}{7} = \frac{?}{28}$

22 $\frac{5}{6} = \frac{25}{?}$

23 $\frac{3}{8} = \frac{15}{?}$

24 $\frac{7}{9} = \frac{42}{?}$

25–28. *To'g'ri va noto'g'ri kasrlar.*

25 Berilgan kasrlardan to'g'ri kasrlarni ajrating: $\frac{2}{5}, \frac{7}{4}, \frac{3}{8}, \frac{9}{9}, \frac{11}{15}, \frac{20}{17}, \frac{6}{13}, \frac{100}{99}$.

26 Berilgan kasrlardan noto'g'ri kasrlarni ajrating: $\frac{5}{4}, \frac{3}{7}, \frac{9}{2}, \frac{12}{12}, \frac{8}{15}, \frac{50}{49}$.

27 $\frac{x}{9}$ to'g'ri kasr bo'ladigan barcha x ning natural qiymatlarini yozing.

28 $\frac{20}{x}$ noto'g'ri kasr bo'ladigan x ning natural qiymatlari nechta?

29–38. *Noto'g'ri kasrni aralash songa aylantiring.*

29 $\frac{7}{3}$

30 $\frac{11}{4}$

31 $\frac{17}{5}$

32 $\frac{23}{6}$

33 $\frac{39}{10}$

34 $\frac{50}{7}$

35 $\frac{71}{8}$

36 $\frac{85}{7}$

37 $\frac{93}{11}$

38 $\frac{100}{9}$

39–48. *Aralash sonni noto'g'ri kasrga aylantiring.*

39 $1\frac{2}{3}$

40 $2\frac{1}{4}$

41 $3\frac{2}{5}$

42 $4\frac{3}{7}$

43 $5\frac{5}{6}$

44 $10\frac{1}{2}$

45 $8\frac{5}{9}$

46 $11\frac{1}{7}$

47 $15\frac{11}{12}$

48 $20\frac{7}{8}$

49–58. *Kasrlarni umumiy maxrajga keltiring.*

49 $\frac{3}{4}$ va $\frac{1}{2}$

50 $\frac{2}{3}$ va $\frac{5}{9}$

51 $\frac{1}{2}$ va $\frac{1}{3}$

52 $\frac{2}{5}$ va $\frac{3}{4}$

53 $\frac{3}{8}$ va $\frac{5}{6}$

54 $\frac{2}{9}$ va $\frac{4}{15}$

55 $\frac{11}{20}$ va $\frac{7}{30}$

56 $\frac{9}{12}$ va $\frac{13}{15}$

57 $\frac{1}{6}$, $\frac{2}{9}$ va $\frac{5}{12}$

58 $\frac{3}{10}$, $\frac{1}{4}$ va $\frac{2}{5}$

59–68. *Kasrlarni taqqoslang* ($>$, $<$, $=$).

59 $\frac{3}{7}$ va $\frac{5}{7}$

60 $\frac{5}{9}$ va $\frac{4}{9}$

61 $\frac{8}{11}$ va $\frac{8}{10}$

62 $\frac{4}{9}$ va $\frac{4}{11}$

63 $\frac{2}{3}$ va $\frac{3}{4}$

64 $\frac{5}{6}$ va $\frac{7}{8}$

65 $\frac{3}{5}$ va $\frac{6}{10}$

66 $\frac{7}{12}$ va $\frac{11}{18}$

67 $\frac{5}{8}$ va $\frac{7}{10}$

68 $\frac{9}{14}$ va $\frac{4}{7}$

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69–86. *Hisoblang (qo'shish va ayirish).*

69 $\frac{1}{5}+\frac{2}{5}$

70 $\frac{5}{8}+\frac{3}{8}$

71 $\frac{7}{9}-\frac{4}{9}$

72 $\frac{1}{2}+\frac{1}{3}$

73 $\frac{3}{4}-\frac{1}{6}$

74 $\frac{5}{6}+\frac{3}{4}$

75 $\frac{7}{8}-\frac{1}{6}$

76 $\frac{2}{3}+\frac{1}{4}+\frac{1}{6}$

77 $\frac{5}{12}-\frac{1}{4}+\frac{1}{3}$

78 $1\frac{2}{5}+2\frac{1}{5}$

79 $3\frac{4}{5}+1\frac{1}{5}$

80 $3\frac{4}{7}-1\frac{2}{7}$

81 $2\frac{1}{3}+1\frac{1}{4}$

82 $5\frac{2}{3}-2\frac{1}{6}$

83 $4\frac{1}{6}-1\frac{2}{3}$

84 $7+3\frac{4}{5}$

85 $7-5\frac{2}{3}$

86 $5-2\frac{3}{7}$

87–96. *Hisoblang (ko'paytirish).*

87 $\frac{1}{2}\cdot\frac{1}{3}$

88 $\frac{2}{3}\cdot\frac{4}{5}$

89 $\frac{3}{4}\cdot\frac{2}{9}$

90 $\frac{5}{6}\cdot\frac{3}{10}$

91 $\frac{7}{8}\cdot 4$

92 $6\cdot\frac{2}{9}$

93 $\frac{4}{15}\cdot\frac{5}{8}$

94 $\frac{9}{14}\cdot\frac{7}{12}$

95 $1\frac{2}{3}\cdot 2\frac{1}{5}$

96 $2\frac{1}{4}\cdot 1\frac{1}{3}$

97–106. *Hisoblang (bo'lish).*

97 $\frac{1}{2}:\frac{1}{4}$

98 $\frac{2}{3}:\frac{4}{9}$

99 $\frac{5}{6}:\frac{10}{3}$

100 $\frac{7}{8}:\frac{14}{16}$

101 $2\frac{1}{4}:1\frac{1}{2}$

102 $5\frac{1}{3}:2\frac{2}{3}$

103 $7\frac{2}{3}:3\frac{2}{7}$

104 $6:\frac{2}{3}$

105 $12:\frac{3}{4}$

106 $\frac{4}{5}:8$

107–136. *Amallar tartibiga oid misollar.*

107 $\frac{1}{2}+\frac{2}{3}\cdot\frac{3}{4}$

108 $\frac{5}{6}-\frac{1}{3}\cdot\frac{1}{2}$

109 $\left(\frac{1}{2}+\frac{1}{3}\right)\cdot\frac{6}{5}$

110 $\left(\frac{3}{4}-\frac{1}{2}\right):\frac{1}{4}$

111 $\frac{2}{3}\cdot\frac{9}{10}+\frac{1}{5}$

112 $\frac{7}{8}:\frac{7}{4}-\frac{1}{4}$

113 $\left(2\frac{1}{2}-1\frac{1}{4}\right)\cdot\frac{4}{5}$

114 $\left(\frac{3}{5}+\frac{1}{2}\right):\left(\frac{3}{5}-\frac{1}{2}\right)$

115 $\frac{1}{3}+\frac{1}{4}\cdot\frac{4}{5}$

116 $\frac{4}{5}\cdot\frac{5}{8}-\frac{1}{4}$

117 $1\frac{1}{2}\cdot\frac{2}{3}+\frac{1}{3}$

118 $\left(\frac{1}{2}+\frac{1}{4}\right):\frac{3}{8}$

119 $5-\left(2\frac{1}{3}+1\frac{1}{2}\right)$

120 $\left(\frac{5}{6}-\frac{1}{3}\right)\cdot 6$

121 $\frac{1}{2}\cdot\frac{3}{4}+\frac{1}{3}\cdot\frac{3}{8}$

122 $2\frac{1}{4}-1\frac{1}{3}\cdot\frac{3}{8}$

123 $\left(\frac{3}{4}-\frac{1}{6}\right)\cdot 12$

124 $1-\frac{2}{3}\cdot\frac{3}{4}$

125 $\frac{1}{3}+2\frac{1}{2}\cdot\frac{1}{5}$

126 $4-1\frac{1}{4}:\frac{3}{4}$

127 $\left(\frac{5}{12}+\frac{1}{6}\right):\frac{7}{24}$

128 $\frac{3}{4}-\frac{1}{2}\cdot\frac{1}{3}$

129 $\left(1+\frac{1}{2}\right)\cdot\left(1+\frac{1}{3}\right)\cdot\left(1+\frac{1}{4}\right)$

130 $\frac{5}{8}\cdot\frac{16}{25}+\frac{1}{5}$

131 $\left(3\frac{1}{3}-1\frac{5}{6}\right):4\frac{1}{2}$

132 $8\cdot\left(\frac{1}{2}+\frac{1}{4}-\frac{1}{8}\right)$

133 $1\frac{1}{2}\cdot\frac{4}{9}+\frac{5}{6}$

134 $2-\left(\frac{1}{2}+\frac{1}{3}+\frac{1}{6}\right)$

135 $\left(\frac{3}{4}-\frac{1}{3}\right)\cdot 12$

136 $\left(1\frac{1}{2}+\frac{2}{3}\right):\left(\frac{5}{6}-\frac{1}{3}\right)$

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Mavzu bo'yicha test savollari

1-variant

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1. $\frac{15}{20}$ kasrni qisqartiring.
   A) $\frac{3}{4}$ B) $\frac{2}{3}$ C) $\frac{5}{6}$ D) $\frac{4}{5}$

2. $4\frac{2}{7}$ aralash sonni noto'g'ri kasrga aylantiring.
   A) $\frac{29}{7}$ B) $\frac{30}{7}$ C) $\frac{14}{7}$ D) $\frac{28}{7}$

3. Hisoblang: $\frac{5}{12} + \frac{1}{4}$
   A) $\frac{6}{16}$ B) $\frac{1}{2}$ C) $\frac{2}{3}$ D) $\frac{1}{3}$

4. Hisoblang: $\frac{2}{5} \cdot \frac{15}{8}$
   A) $\frac{17}{40}$ B) $\frac{3}{4}$ C) $\frac{1}{2}$ D) $\frac{30}{13}$

5. $\frac{2}{3}$ va $\frac{5}{8}$ kasrlardan qaysi biri katta?
   A) $\frac{2}{3}$ B) $\frac{5}{8}$ C) teng D) aniqlash mumkin emas

6. Hisoblang: $1\frac{1}{2} + 2\frac{1}{3}$
   A) $3\frac{2}{5}$ B) $3\frac{5}{6}$ C) $4\frac{1}{6}$ D) $3\frac{1}{6}$

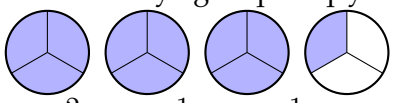
7. $\frac{2}{3}$ kasrga teskari sonni toping.
   A) $\frac{3}{2}$ B) $-\frac{2}{3}$ C) $\frac{2}{3}$ D) $\frac{1}{6}$

8. $\frac{2}{3}$ va $\frac{5}{8}$ kasrlar uchun eng kichik umumiy maxrajni toping.
   A) 11 B) 16 C) 24 D) 48

9. Hisoblang: $\left(\frac{3}{4} - \frac{1}{2}\right) \cdot 8$
   A) 1 B) 2 C) 4 D) 8

10. Hisoblang: $\frac{8}{15} \cdot \frac{25}{16}$
    A) $\frac{5}{6}$ B) $\frac{2}{3}$ C) $\frac{4}{5}$ D) $\frac{8}{15}$

2-variant

1. $\frac{17}{5}$ noto'g'ri kasrni aralash son ko'rinishida yozing.
   A) $3\frac{1}{5}$ B) $3\frac{2}{5}$ C) $4\frac{1}{5}$ D) $2\frac{3}{5}$
2. Quyidagilardan qaysi biri to'g'ri kasr?
   A) $\frac{7}{4}$ B) $\frac{12}{12}$ C) $\frac{5}{8}$ D) $\frac{10}{3}$
3. Hisoblang: $\frac{3}{4} - \frac{1}{6}$
   A) $\frac{2}{2}$ B) $\frac{7}{12}$ C) $\frac{1}{2}$ D) $\frac{5}{12}$
4. Hisoblang: $\frac{4}{9} : \frac{2}{3}$
   A) $\frac{8}{27}$ B) $\frac{2}{3}$ C) $\frac{1}{3}$ D) $\frac{3}{2}$
5. $\frac{18}{x}$ to'g'ri kasr bo'ladigan x ning eng kichik natural qiymatini toping.
   A) 17 B) 18 C) 19 D) 20
6. Rasmda bo'yalgan qism qaysi aralash songa teng?
   
   A) $3\frac{2}{3}$ B) $3\frac{1}{3}$ C) $2\frac{1}{3}$ D) $4\frac{1}{3}$
7. Hisoblang: $2\frac{1}{4} \cdot 1\frac{1}{3}$
   A) $2\frac{1}{12}$ B) 3 C) $2\frac{1}{3}$ D) $2\frac{2}{3}$
8. Hisoblang: $4\frac{1}{2} : 1\frac{1}{2}$
   A) 2 B) 3 C) 4 D) 6
9. Hisoblang: $2\frac{2}{3} + 1\frac{5}{6}$
   A) $3\frac{7}{9}$ B) $4\frac{1}{6}$ C) $4\frac{1}{2}$ D) $4\frac{2}{3}$
10. Hisoblang: $\left(\frac{1}{2} + \frac{1}{3}\right) : \frac{5}{6}$
    A) $\frac{5}{6}$ B) 1 C) $\frac{6}{5}$ D) 2

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Javoblar

1. Yeyilgan: $\frac{3}{8}$, qoldi: $\frac{5}{8}$

2. Yeyilgan: $\frac{6}{12} = \frac{1}{2}$, qoldi: $\frac{1}{2}$

3. $\frac{7}{20}$

4. Kesilgan: $\frac{35}{100} = \frac{7}{20}$, qoldi: $\frac{65}{100} = \frac{13}{20}$

5. 1 soat $= \frac{1}{24}$ qism, 8 soat $= \frac{8}{24} = \frac{1}{3}$ qism

6. Qiz bolalar: $\frac{10}{25} = \frac{2}{5}$, o'g'il bolalar: $\frac{15}{25} = \frac{3}{5}$

7. $\frac{1}{2}$

8. $\frac{2}{3}$

9. $\frac{3}{4}$

10. $\frac{3}{5}$

11. $\frac{2}{3}$

12. $\frac{3}{4}$

13. $\frac{5}{9}$

14. $\frac{3}{4}$

15. $\frac{4}{7}$

16. $\frac{5}{6}$

17. $\frac{7}{12}$

18. $\frac{2}{3}$

19. 8

20. 12

21. 16

22. 30

23. 40

24. 54

25. $\frac{2}{5}, \frac{3}{8}, \frac{11}{15}, \frac{6}{13}$

26. $\frac{5}{4}, \frac{9}{2}, \frac{12}{12}, \frac{50}{49}$

27. $1, 2, 3, 4, 5, 6, 7, 8$

28. 20 ta

29. $2\frac{1}{3}$

30. $2\frac{3}{4}$

31. $3\frac{2}{5}$

32. $3\frac{5}{6}$

33. $3\frac{9}{10}$

34. $7\frac{1}{7}$

35. $8\frac{7}{8}$

36. $12\frac{1}{7}$

37. $8\frac{5}{11}$

38. $11\frac{1}{9}$

39. $\frac{5}{3}$

40. $\frac{9}{4}$

41. $\frac{17}{5}$

42. $\frac{31}{7}$

43. $\frac{35}{6}$

44. $\frac{21}{2}$

45. $\frac{77}{9}$

46. $\frac{78}{7}$

47. $\frac{191}{12}$

48. $\frac{167}{8}$

49. $\frac{3}{4}, \frac{2}{4}$

50. $\frac{6}{9}, \frac{5}{9}$

51. $\frac{3}{6}, \frac{2}{6}$

52. $\frac{8}{20}, \frac{15}{20}$

53. $\frac{9}{24}, \frac{20}{24}$

54. $\frac{10}{45}, \frac{12}{45}$

55. $\frac{33}{60}, \frac{14}{60}$

56. $\frac{45}{60}, \frac{52}{60}$

57. $\frac{6}{36}, \frac{8}{36}, \frac{15}{36}$

58. $\frac{6}{20}, \frac{5}{20}, \frac{8}{20}$

59. $<$

60. $>$

61. $<$

62. $>$

63. $<$

64. $<$

65. $=$

66. $<$

67. $<$

68. $>$

69. $\frac{3}{5}$

70. 1

71. $\frac{1}{3}$

72. $\frac{5}{6}$

73. $\frac{7}{12}$

74. $1\frac{7}{12}$

75. $\frac{17}{24}$

76. $1\frac{1}{12}$

77. $\frac{1}{2}$

78. $3\frac{3}{5}$

79. 5

80. $2\frac{2}{7}$

81. $3\frac{7}{12}$

82. $3\frac{1}{2}$

83. $2\frac{1}{2}$

84. $10\frac{4}{5}$

85. $1\frac{1}{3}$

86. $2\frac{4}{7}$

87. $\frac{1}{6}$

88. $\frac{8}{15}$

89. $\frac{1}{6}$

90. $\frac{1}{4}$

91. $3\frac{1}{2}$

92. $1\frac{1}{3}$

93. $\frac{1}{6}$

94. $\frac{3}{8}$

95. $3\frac{2}{3}$
96. 3
97. 2
98. $1\frac{1}{2}$
99. $\frac{1}{4}$
100. 1
101. $1\frac{1}{2}$
102. 2
103. $2\frac{1}{3}$
104. 9
105. 16
106. $\frac{1}{10}$
107. 1
108. $\frac{2}{3}$
109. 1
110. 1
111. $\frac{4}{5}$
112. $\frac{1}{4}$
113. 1
114. 11
115. $\frac{8}{15}$
116. $\frac{1}{4}$
117. $1\frac{1}{3}$
118. 2
119. $1\frac{1}{6}$
120. 3
121. $\frac{1}{2}$
122. $1\frac{3}{4}$
123. 7
124. $\frac{1}{2}$
125. $\frac{5}{6}$
126. $2\frac{1}{3}$
127. 2
128. $\frac{7}{12}$
129. $2\frac{1}{2}$
130. $\frac{3}{5}$
131. $\frac{1}{3}$
132. 5
133. $1\frac{1}{2}$
134. 1
135. 5
136. $4\frac{1}{3}$

Test javoblari

1-variant: 1) A 2) B 3) C 4) B 5) A 6) B 7) A 8) C 9) B 10) A
2-variant: 1) B 2) C 3) B 4) B 5) C 6) B 7) B 8) B 9) C 10) B

Mavzu bo'yicha interaktiv testlar
QR kodni skanerlab, bilimingizni sinab ko'ring

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4.

O'nli kasrlar

Nazariy qism

O'nli kasr tushunchasi

✎ Ta'rif

Sonning butun qismi kasr qismidan vergul bilan ajratib yozilgan ko'rinishiga **o'nli kasr** deyiladi. Maxraji $10,\ 100,\ 1000,\ \ldots$ bo'lgan har qanday oddiy kasrni o'nli kasr ko'rinishida yozish mumkin.

$$\frac{3}{10} = 0{,}3 \qquad \frac{47}{100} = 0{,}47 \qquad \frac{125}{1000} = 0{,}125 \qquad 5\frac{7}{10} = 5{,}7$$

✓ Eslatma

O'nli kasr oxiriga **nol qo'shilsa, qiymat o'zgarmaydi**:

$$0{,}5 = 0{,}50 = 0{,}500$$

Oddiy kasrni o'nli kasrga aylantirish

✠ Asosiy qoida

Usul 1. Kasrni o'nli kasrga keltirish uchun kasrning maxraji 10 ning darajalaridan $(10,\ 100,\ 1000,\ \ldots)$ iborat bo'lgan teng kasr hosil qilinadi. Hosil bo'lgan o'nli kasr vergul bilan ajratib yoziladi.

$$\frac{1}{2} = \frac{1\cdot 5}{2\cdot 5} = \frac{5}{10} = 0{,}5 \qquad 7\frac{1}{4} = 7\,\frac{1\cdot 25}{4\cdot 25} = 7\frac{25}{100} = 7{,}25$$

Usul 2. Kasrni o'nli kasr ko'rinishiga keltirishda suratni maxrajga bo'lish usuli bilan ham amalga oshirish mumkin.

$$\frac{16}{5} = 16:5 = 3{,}2 \qquad \frac{3}{4} = 3:4 = 0{,}75 \qquad \frac{1}{8} = 1:8 = 0{,}125$$

O'nli kasrni oddiy kasrga aylantirish

✠ Asosiy qoida

Suratga verguldan keyingi raqamlardan tuzilgan son, maxrajga esa 1 va uning ortidan verguldan keyingi xonalar soniga teng miqdorda nol yoziladi. Butun qism o'zgarishsiz qoladi.

$$0{,}9 = \frac{9}{10} \qquad 0{,}25 = \frac{25}{100} = \frac{1}{4} \qquad 0{,}125 = \frac{125}{1000} = \frac{1}{8} \qquad 2{,}04 = 2\frac{4}{100} = 2\frac{1}{25}.$$

Qo'shish va ayirish

✠ Asosiy qoida

Vergullar bir-birining ostiga keltiriladi, kerak bo'lsa nollar qo'shib xonalar tenglashtiriladi.

$$\begin{array}{r} 12{,}45 \\ +\ 7{,}80 \\ \hline 20{,}25 \end{array} \qquad \begin{array}{r} 8{,}30 \\ -\ 2{,}75 \\ \hline 5{,}55 \end{array} \qquad \begin{array}{r} 5{,}00 \\ -\ 2{,}75 \\ \hline 2{,}25 \end{array}$$

10, 100, 1000 ga ko'paytirish va bo'lish

✠ Asosiy qoida

- 10^n ga **ko'paytirganda** vergul **o'ngga** n xonaga suriladi:

$$3{,}45 \cdot 10 = 34{,}5 \qquad 0{,}07 \cdot 100 = 7 \qquad 5{,}1 \cdot 1000 = 5100$$

- 10^n ga **bo'lganda** vergul **chapga** n xonaga suriladi (kerak bo'lsa boshiga nollar qo'shiladi):

$$25{,}8 : 10 = 2{,}58 \qquad 145 : 100 = 1{,}45 \qquad 7 : 1000 = 0{,}007$$

O'nli kasrlarni ko'paytirish

✠ Asosiy qoida

1. Vergulga e'tibor bermay, sonlar ko'paytiriladi.
2. Ko'paytuvchilardagi verguldan keyingi xonalar soni qo'shiladi.
3. Natijaning oxiridan o'shancha xona vergul bilan ajratiladi.

$$0{,}3 \cdot 0{,}4 \Rightarrow\ 3 \cdot 4 = 12 \ \Rightarrow\ 1 + 1 = 2 \ \Rightarrow\ 0{,}12$$

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O'nli kasrlarni bo'lish

✠ Asosiy qoida

Ikkala son bir vaqtda $10,\ 100,\ 1000, \ldots$ ga ko'paytiriladi. Keyin natural songa bo'lgandek bo'linadi.

$$7{,}2 : 0{,}3 = 72 : 3 = 24 \qquad 1{,}5 : 0{,}4 = 15 : 4 = 3{,}75 \qquad 4{,}5 : 0{,}08 = 450 : 8 = 56{,}25$$

Davriy o'nli kasrlar

✎ Ta'rif

Suratni maxrajga bo'lganda cheksiz davom etadigan, lekin ma'lum raqamlar guruhi takrorlanadigan o'nli kasr **davriy o'nli kasr** deyiladi. Takrorlanuvchi qism **davr** bo'lib, qavs ichida yoziladi:

$\frac{1}{3} = 0{,}33333... = 0{,}(3)$ $\qquad$ $\frac{1}{11} = 0{,}090909... = 0{,}(09)$ $\qquad$ $\frac{1}{7} = 0{,}142857142857... = 0{,}(142857)$

Sof va aralash davriy o'nli kasrlar

✎ Ta'rif

- **Sof davriy o'nli kasr** deb — davri verguldan keyin darhol boshlangan kasrga aytiladi. $0{,}(3)$ $\quad 0{,}(27)$ $\quad 0{,}(142)$
- **Aralash davriy o'nli kasr** deb — davrigacha qo'shimcha raqamlar bor bo'lgan kasrga aytiladi. $0{,}1(6)$ $\quad 0{,}18(3)$ $\quad 0{,}2(45)$

Sof davriy o'nli kasrni oddiy kasrga aylantirish

✠ Asosiy qoida

Suratga davrning o'zi yoziladi, maxrajga davrda nechta raqam bo'lsa, shuncha to'qqiz yoziladi:

$$0{,}(\underbrace{a_1a_2\ldots a_n}_{\text{davr}}) = \frac{a_1a_2\ldots a_n}{\underbrace{99\ldots9}_{n \text{ ta}}}$$

Misollar:

$$0{,}(3) = \frac{3}{9} = \frac{1}{3} \qquad 0{,}(27) = \frac{27}{99} = \frac{3}{11} \qquad 0{,}(09) = \frac{9}{99} = \frac{1}{11}$$

Aralash davriy o'nli kasrni oddiy kasrga aylantirish

✠ Asosiy qoida

Davrgacha m ta raqam, davrda n ta raqam bo'lsa:

- **suratga** $\Rightarrow$ (verguldan keyingi barcha raqamlar) $-$ (davrgacha bo'lgan raqamlar);
- **maxrajga** $\Rightarrow$ n ta to'qqiz va keyin m ta nol.

Misollar:

$$0{,}1(6) = \frac{16-1}{90} = \frac{15}{90} = \frac{1}{6} \qquad 0{,}8(3) = \frac{83-8}{90} = \frac{75}{90} = \frac{5}{6}$$

$$0{,}2(45) = \frac{245-2}{990} = \frac{243}{990} = \frac{27}{110}.$$

Amaliy qism

1–20. *Oddiy kasrlarni o'nli kasrlar ko'rinishida yozing.*

1 $\frac{3}{10}$

2 $\frac{7}{10}$

3 $\frac{49}{100}$

4 $\frac{3}{100}$

5 $\frac{125}{1000}$

6 $\frac{1}{2}$

7 $\frac{3}{4}$

8 $\frac{1}{5}$

9 $\frac{17}{25}$

10 $\frac{7}{8}$

11 $\frac{11}{20}$

12 $2\frac{3}{4}$

13 $4\frac{5}{8}$

14 $1\frac{9}{20}$

15 $7\frac{13}{25}$

16 $\frac{17}{40}$

17 $\frac{21}{50}$

18 $\frac{87}{200}$

19 $\frac{47}{250}$

20 $\frac{311}{500}$

21–30. *O'nli kasrlarni oddiy kasrlar ko'rinishida yozing.*

21 0,9

22 0,25

23 0,08

24 0,004

25 0,16

26 0,375

27 1,5

28 3,75

29 0,625

30 2,04

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31–44. *Hisoblang (qo'shish va ayirish).*

31 $1{,}5 + 2{,}3$

32 $4{,}7 + 3{,}25$

33 $5{,}6 - 2{,}3$

34 $8{,}5 - 3{,}27$

35 $12{,}45 + 7{,}8$

36 $20{,}3 - 9{,}57$

37 $5 - 2{,}75$

38 $10 - 0{,}07$

39 $3{,}5 + 0{,}25 + 1{,}75$

40 $5{,}67 + 4{,}33$

41 $15{,}75 - 8{,}9 + 3{,}15$

42 $2{,}3 + 1{,}25 - 0{,}75$

43 $100 - 27{,}68$

44 $50{,}5 - 7{,}25 - 13{,}05$

45–48. *Qulay usulda hisoblang.*

45 $1{,}3 + 2{,}7 + 3{,}3 + 4{,}7$

46 $4{,}28 + 1{,}75 + 0{,}72 + 1{,}25$

47 $25{,}6 - 4{,}7 - 5{,}3$

48 $2{,}5 + 7{,}3 + 1{,}5 + 2{,}7$

49–60. $10,\ 100,\ 1000$ *ga ko'paytiring yoki bo'ling.*

49 $3{,}45 \cdot 10$

50 $0{,}07 \cdot 100$

51 $5{,}1 \cdot 1000$

52 $0{,}008 \cdot 100$

53 $12{,}5 \cdot 10$

54 $0{,}0023 \cdot 1000$

55 $25{,}8 : 10$

56 $145 : 100$

57 $7 : 1000$

58 $0{,}9 : 100$

59 $3500 : 1000$

60 $25 : 10\,000$

61–72. *Hisoblang (ko'paytirish).*

61 $0{,}3 \cdot 0{,}4$

62 $0{,}5 \cdot 0{,}6$

63 $0{,}7 \cdot 0{,}8$

64 $1{,}5 \cdot 0{,}2$

65 $2{,}5 \cdot 0{,}4$

66 $3{,}2 \cdot 1{,}5$

67 $0{,}25 \cdot 0{,}4$

68 $1{,}25 \cdot 0{,}8$

69 $0{,}12 \cdot 0{,}05$

70 $4{,}5 \cdot 2{,}4$

71 $0{,}375 \cdot 1{,}6$

72 $1{,}25 \cdot 1{,}25 \cdot 0{,}8$

73–84. *Hisoblang (bo'lish).*

73 $0{,}8 : 0{,}2$

74 $7{,}2 : 0{,}3$

75 $9{,}6 : 0{,}5$

76 $1{,}5 : 0{,}4$

77 $3{,}6 : 1{,}2$

78 $0{,}48 : 0{,}06$

79 $12{,}5 : 5$

80 $25 : 0{,}4$

81 $4{,}5 : 0{,}8$

82 $0{,}144 : 0{,}12$

83 $7{,}5 : 0{,}25$

84 $5 : 0{,}8$

85–122. *Aralash amallar (amallar tartibiga rioya qiling).*

85 $0{,}3 + 0{,}2 \cdot 0{,}5$

86 $(2{,}5 - 1{,}3) \cdot 4$

87 $1{,}2 \cdot 0{,}5 + 0{,}4$

88 $(0{,}8 + 0{,}2) : 0{,}5$

89 $1 - 0{,}4 \cdot 2$

90 $0{,}7 \cdot 10 - 5$

91 $3{,}6 : 0{,}9 + 0{,}5 \cdot 2$

92 $0{,}5 \cdot 0{,}4 + 0{,}3$

93 $2{,}4 : 0{,}6 - 1$

94 $1{,}2 + 0{,}3 \cdot 0{,}4$

95 $(4{,}5 + 0{,}5) \cdot 0{,}4$

96 $8 : 0{,}4 - 10$

97 $3{,}6 - 0{,}4 \cdot 5$

98 $(5 - 2{,}5) \cdot 0{,}2$

99 $2{,}5 \cdot 0{,}4 + 1{,}5 \cdot 0{,}4$

100 $(1{,}5 + 0{,}5) \cdot (3{,}2 - 1{,}2)$

101 $0{,}5 \cdot 0{,}5 + 0{,}5 \cdot 0{,}5$

102 $(0{,}5 + 1{,}5) : (0{,}5 - 0{,}3)$

103 $0{,}5 \cdot 0{,}2 + 0{,}3 \cdot 0{,}4$

104 $(1{,}2 + 0{,}8) \cdot (0{,}5 - 0{,}1)$

105 $4 \cdot (3{,}2 - 1{,}2) - 0{,}5 \cdot 12$

106 $(0{,}8 - 0{,}3) : 0{,}1 + 5$

107 $(5{,}4 - 0{,}4) \cdot (2{,}5 + 0{,}5)$

108 $1{,}5 \cdot 0{,}8 - 0{,}4 \cdot 1{,}5$

109 $(3{,}6 + 2{,}4) : 1{,}2 - 0{,}5$

110 $0{,}2 \cdot (3{,}5 + 1{,}5) - 0{,}5$

111 $(1{,}2 \cdot 0{,}5 + 0{,}4) : 0{,}2$

112 $(8 - 0{,}4 \cdot 5) \cdot 0{,}1$

113 $2{,}5 \cdot (0{,}5 + 1{,}5) - 1{,}5 \cdot (3{,}2 - 1{,}2)$

114 $(1{,}2 + 0{,}8) \cdot (1{,}5 - 0{,}5) - 0{,}6$

115 $(15 : 0{,}5 - 20) \cdot 0{,}4$

116 $0{,}25 \cdot 1{,}2 + 0{,}75 \cdot 1{,}2 + 1{,}5 \cdot 0{,}8$

117 $(8 - 1{,}25 \cdot 4) : (0{,}5 + 0{,}75)$

118 $1{,}25 \cdot (0{,}8 + 1{,}2) \cdot 0{,}4$

119 $(1{,}5 - 0{,}3) \cdot (1{,}8 - 0{,}8) + 0{,}4$

120 $(15 - 0{,}5 \cdot 18) : (4 - 1{,}5)$

121 $(1{,}25 + 0{,}75) \cdot 0{,}8 - 0{,}4 \cdot 1{,}5$

122 $(4{,}8 - 0{,}8) \cdot 0{,}375 - 0{,}8$

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123–131. *Quyidagi kasrlarni davriy o'nli kasr ko'rinishida yozing va davrini qavs ichida ko'rsating.*

123 $\frac{1}{3}$

124 $\frac{2}{3}$

125 $\frac{1}{9}$

126 $\frac{4}{9}$

127 $\frac{8}{9}$

128 $\frac{1}{6}$

129 $\frac{5}{6}$

130 $\frac{1}{11}$

131 $\frac{2}{11}$

132–133.

132 Berilgan kasrlardan qaysilari davriy o'nli kasr bo'ladi: $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{6}, \frac{1}{8}, \frac{1}{12}, \frac{1}{20}, \frac{1}{25}$.

133 Berilgan kasrlardan qaysilari chekli o'nli kasr bo'ladi: $\frac{3}{40}, \frac{5}{18}, \frac{7}{32}, \frac{11}{30}, \frac{9}{50}, \frac{4}{15}$.

134–141. *Sof davriy o'nli kasrlarni oddiy kasrlarga aylantiring.*

134 $0,(3)$

135 $0,(5)$

136 $0,(8)$

137 $0,(09)$

138 $0,(27)$

139 $0,(36)$

140 $0,(81)$

141 $0,(72)$

142–147. *Aralash davriy o'nli kasrlarni oddiy kasrlarga aylantiring.*

142 $0,1(6)$

143 $0,8(3)$

144 $0,2(7)$

145 $0,5(8)$

146 $0,1(2)$

147 $0,4(1)$

Mavzu bo'yicha test savollari

1-variant

1. $\frac{7}{25}$ ni o'nli kasr ko'rinishida yozing.

A) $0{,}07$ B) $0{,}25$ C) $0{,}28$ D) $0{,}725$

2. Quyidagi sonlardan eng kattasini toping.
A) $0{,}5$ B) $0{,}49$ C) $0{,}505$ D) $0{,}055$

3. Hisoblang: $10 - 3{,}27$
A) $7{,}27$ B) $6{,}73$ C) $7{,}73$ D) $6{,}83$

4. Hisoblang: $58{,}3 : 1000$
A) $5{,}83$ B) $0{,}583$ C) $0{,}0583$ D) 5830

5. Hisoblang: $1{,}25 \cdot 0{,}8$
A) $0{,}1$ B) 1 C) 10 D) $0{,}01$

6. Hisoblang: $4{,}5 : 0{,}8$
A) $5{,}625$ B) $5{,}25$ C) $56{,}25$ D) $5{,}5$

7. $0{,}(27)$ ni oddiy kasr ko'rinishida yozing.

A) $\frac{27}{100}$ B) $\frac{3}{11}$ C) $\frac{27}{999}$ D) $\frac{27}{1000}$

8. Quyidagilardan qaysi biri davriy o'nli kasr bo'ladi?

A) $\frac{3}{8}$ B) $\frac{1}{6}$ C) $\frac{7}{20}$ D) $\frac{3}{25}$

9. Avtomobil 1 soatda $80{,}5$ km yo'l yursa, 3 soatda u necha km yo'l bosadi?
A) $241{,}5$ B) 245 C) $240{,}5$ D) 250

10. Hisoblang: $(8 - 1{,}25 \cdot 4) : (0{,}5 + 0{,}75)$
A) $2{,}4$ B) 3 C) 4 D) $1{,}25$

2-variant

1. $0{,}125$ ni oddiy kasr ko'rinishida yozing.
A) $\frac{1}{8}$ B) $\frac{1}{4}$ C) $\frac{1}{5}$ D) $\frac{12}{100}$

2. Hisoblang: $5{,}6 + 3{,}45$
A) $8{,}51$ B) $9{,}05$ C) $9{,}5$ D) $9{,}11$

3. Hisoblang: $4{,}25 \cdot 100$
A) $42{,}5$ B) 425 C) 4250 D) $0{,}425$

4. Hisoblang: $0{,}3 \cdot 0{,}4$
A) $1{,}2$ B) $0{,}12$ C) $0{,}012$ D) 12

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5. Hisoblang: $1{,}5 : 0{,}4$
A) 3,75 B) 0,375 C) 37,5 D) 4

6. $\frac{1}{6}$ ni davriy o'nli kasr ko'rinishida yozing.

A) $0{,}(16)$ B) $0{,}1(6)$ C) $0{,}(6)$ D) 0,16

7. $\frac{3}{4}$ va 0,7 ni taqqoslang.
A) $\frac{3}{4} > 0{,}7$ B) $\frac{3}{4} < 0{,}7$ C) $\frac{3}{4} = 0{,}7$ D) taqqoslab bo'lmaydi

8. 1 kg yong'oq 35 000 so'm bo'lsa, 2,4 kg yong'oq necha so'mga olinadi?
A) 70 000 B) 80 000 C) 84 000 D) 90 000

9. Hisoblang: $(1{,}5 + 0{,}5) \cdot (3{,}2 - 1{,}2)$
A) 2 B) 4 C) 5 D) 0,4

10. $\frac{1}{3} + \frac{1}{6}$ ni hisoblang va natijani o'nli kasr ko'rinishida yozing.

A) $0{,}(3)$ B) $0{,}1(6)$ C) 0,5 D) $0{,}(5)$

Javoblar

1. 0,3
2. 0,7
3. 0,49
4. 0,03
5. 0,125
6. 0,5
7. 0,75
8. 0,2
9. 0,68
10. 0,875
11. 0,55
12. 2,75
13. 4,625
14. 1,45
15. 7,52
16. 0,425
17. 0,42
18. 0,435
19. 0,188
20. 0,622

21. $\frac{9}{10}$
22. $\frac{1}{4}$
23. $\frac{2}{25}$
24. $\frac{1}{250}$
25. $\frac{4}{25}$
26. $\frac{3}{8}$
27. $1\frac{1}{2}$
28. $3\frac{3}{4}$
29. $\frac{5}{8}$
30. $2\frac{1}{25}$
31. 3,8
32. 7,95
33. 3,3
34. 5,23
35. 20,25

36. 10,73
37. 2,25
38. 9,93
39. 5,5
40. 10
41. 10
42. 2,8
43. 72,32
44. 30,2
45. 12
46. 8
47. 15,6
48. 14
49. 34,5
50. 7
51. 5100
52. 0,8
53. 125
54. 2,3
55. 2,58

56. 1,45
57. 0,007
58. 0,009
59. 3,5
60. 0,0025
61. 0,12
62. 0,3
63. 0,56
64. 0,3
65. 1
66. 4,8
67. 0,1
68. 1
69. 0,006
70. 10,8
71. 0,6
72. 1,25
73. 4
74. 24
75. 19,2

76. 3,75
77. 3
78. 8
79. 2,5
80. 62,5
81. 5,625
82. 1,2
83. 30
84. 6,25
85. 0,4
86. 4,8
87. 1
88. 2
89. 0,2
90. 2
91. 5
92. 0,5
93. 3
94. 1,32
95. 2

96. 10
97. 1,6
98. 0,5
99. 1,6
100. 4
101. 0,5
102. 10
103. 0,22
104. 0,8
105. 2
106. 10
107. 15
108. 0,6

109. 4,5
110. 0,5
111. 5
112. 0,6
113. 2
114. 1,4
115. 4
116. 2,4
117. 2,4
118. 1
119. 1,6
120. 2,4
121. 1

122. 0,7
123. $0,(3)$
124. $0,(6)$
125. $0,(1)$
126. $0,(4)$
127. $0,(8)$
128. $0,1(6)$
129. $0,8(3)$
130. $0,(09)$
131. $0,(18)$
132. Davriy:
$\frac{1}{3}, \frac{1}{6}, \frac{1}{12}$

133. Chekli:
$\frac{3}{40}, \frac{7}{32}, \frac{9}{50}$
134. $\frac{1}{3}$
135. $\frac{5}{9}$
136. $\frac{8}{9}$
137. $\frac{1}{11}$
138. $\frac{3}{11}$
139. $\frac{4}{11}$
140. $\frac{9}{11}$

141. $\frac{8}{11}$
142. $\frac{1}{6}$
143. $\frac{5}{6}$
144. $\frac{5}{18}$
145. $\frac{53}{90}$
146. $\frac{11}{90}$
147. $\frac{37}{90}$

Test javoblari

1-variant: 1) C 2) C 3) B 4) C 5) B 6) A 7) B 8) B 9) A 10) A
2-variant: 1) A 2) B 3) B 4) B 5) A 6) B 7) A 8) C 9) B 10) C

Mavzu bo'yicha interaktiv testlar
QR kodni skanerlab, bilimingizni sinab ko'ring

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5.

Ratsional sonlar

Nazariy qism

Butun sonlar to'plami ($\mathbb{Z}$)

✎ Ta'rif

Butun sonlar deb — natural sonlar $(1, 2, 3, \ldots)$, ularga qarama-qarshi bo'lgan sonlar $(-1, -2, -3, \ldots)$ va **nol** (0) ga aytiladi.

Belgilanishi: $\mathbb{Z}$

$$\mathbb{Z} = \{\ldots,\ -3,\ -2,\ -1,\ 0,\ 1,\ 2,\ 3,\ \ldots\}$$

✓ Eslatma

Butun sonlarning tarkibi:

$$\underbrace{\ldots, -3, -2, -1}_{\text{manfiy butun sonlar}}, \quad \underbrace{0}_{\substack{\text{na musbat,}\\ \text{na manfiy}}}, \quad \underbrace{1, 2, 3, \ldots}_{\substack{\text{musbat butun sonlar}\\ \text{(natural sonlar)}}}$$

- Nol — musbat ham, manfiy ham emas.
- Har bir natural son — butun son. Lekin har bir butun son natural emas (masalan, -7 butun, lekin natural emas).

Ratsional sonlar to'plami ($\mathbb{Q}$)

✎ Ta'rif

Ratsional sonlar bu — $\frac{a}{b}$ kasr ko'rinishida yozish mumkin bo'lgan barcha sonlardir. Bu yerda a — butun son, b — natural son $(a \in \mathbb{Z},\ b \in \mathbb{N})$.

Belgilanishi: $\mathbb{Q}$

Misollar:

$$-\frac{3}{4}, \quad \frac{2}{5}, \quad -7 = \frac{-7}{1}, \quad 0 = \frac{0}{1}, \quad 1{,}25 = \frac{5}{4}, \quad 0{,}(3) = \frac{1}{3}$$

✠ Asosiy qoida

Eng muhim xulosa. Har qanday:

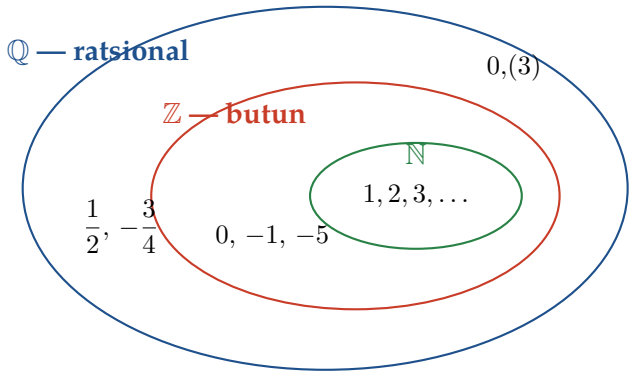
- natural son $(1, 2, 3, \ldots)$,
- butun son $(\ldots, -2, -1, 0, 1, 2, \ldots)$,
- oddiy kasr $(\frac{2}{5},\ -\frac{7}{8}, \ldots)$,
- chekli o'nli kasr $(1{,}25;\ -3{,}047; \ldots)$,
- davriy o'nli kasr $(0{,}(3);\ -1{,}5(27); \ldots)$

— bularning hammasi **ratsional sonlardir**!

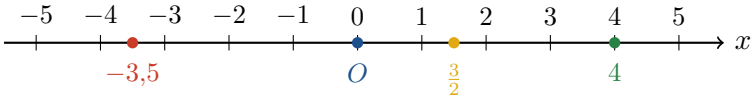
Qisqacha:

$$\boxed{\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q}}$$

Ya'ni: har bir natural son — butun son, har bir butun son — ratsional son.



Koordinata to'g'ri chizig'i



✓ Eslatma

- **Nolning chap**ida joylashgan sonlar — **manfiy**.
- **Nolning o'ng**ida joylashgan sonlar — **musbat**.

Son o'qida masofa, siljitish va kesma o'rtasi

✠ Asosiy qoida

Son o'qida bo'lgan ikki nuqta orasidagi masofani topish uchun ularning kattasidan kichigini ayirish kerak.

Misol. $A(-5)$ va $B(3)$ nuqtalar orasidagi masofa:

$$3 - (-5) = 3 + 5 = 8$$

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✠ Asosiy qoida

Nuqtani siljitish. Son o'qida nuqtani:

- **o'ngga** k birlik siljitsak, koordinatasiga k qo'shamiz;
- **chapga** k birlik siljitsak, koordinatasidan k ayiramiz.

Misol. $M(2)$ nuqta chapga 5 birlik siljitildi: $\quad 2-5=-3 \quad \to N(-3)$.

✠ Asosiy qoida

Kesmaning o'rtasi. Koordinatalari a va b bo'lgan nuqtalarni tutashtiruvchi kesma o'rtasining koordinatasi:

$$\boxed{x_{o'rta}=\frac{a+b}{2}}$$

Misol. $A(-7)$ va $B(3)$ bo'lsa, AB kesma o'rtasining koordinatasi : $\dfrac{-7+3}{2}=\dfrac{-4}{2}=-2$.

Qarama-qarshi sonlar

✠ Asosiy qoida

Qarama-qarshi sonlar deb - yig'indisi **nolga** teng bo'lgan sonlarga aytiladi:

$$a+(-a)=0$$

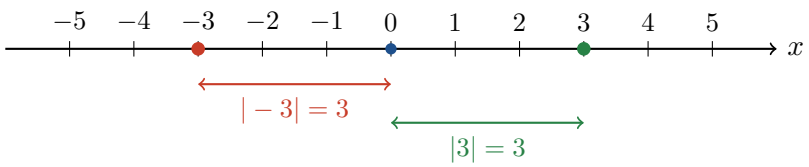
Misol: $7+(-7)=0; \quad -2{,}5+2{,}5=0; \quad \dfrac{3}{8}+\left(-\dfrac{3}{8}\right)=0.$

Sonning moduli

✎ Ta'rif

Sonning **moduli** deb — son o'qida shu songa mos keluvchi nuqtadan sanoq boshi(nol)gacha bo'lgan masofaga aytiladi. Belgilanishi: $|a|$.

$$|a|=\begin{cases} a, & \text{agar } a>0 \quad \text{(musbat bo'lsa)} \\ -a, & \text{agar } a<0 \quad \text{(manfiy bo'lsa)} \\ 0, & \text{agar } a=0 \quad \text{(nol bo'lsa)} \end{cases}$$



Ratsional sonlarni qo'shish

Bir xil ishorali sonlar

✠ Asosiy qoida

Bunda sonlarning modullari qo'shiladi va natijaning oldiga qo'shiluvchilar ishorasi qo'yiladi.

Misollar:

$$(+5)+(+3)=+8 \qquad (-5)+(-3)=-8$$

Har xil ishorali sonlar

✠ Asosiy qoida

Bunda moduli kattasidan moduli kichigi ayriladi va natijaning oldiga moduli katta sonning ishorasi qo'yiladi.

Misollar:

$$(+7)+(-3)=+4 \qquad (-7)+(+3)=-4$$

Ratsional sonlarni ayirish

✠ Asosiy qoida

Ayirishni qo'shishga aylantirib hisoblaymiz, ya'ni ayriluvchini uning **qarama-qarshisi** bilan almashtiramiz.

$$\boxed{a-b=a+(-b)}$$

✓ Eslatma

Hisoblash bosqichlari:

1. Ayirishni qo'shishga aylantiramiz: $-b \rightarrow +(-b)$.
2. Qo'sh ishorani soddalashtiramiz: $+(-) \rightarrow -, \quad -(-) \rightarrow +$.

Misollar:

$$5-8=5+(-8)=-3 \qquad \text{(musbat sonni ayirdik)}$$

$$-7-4=-7+(-4)=-11 \qquad \text{(bir xil ishorali sonlarni qo'shishga aylandi)}$$

$$5-(-3)=5+3=+8 \qquad \text{(manfiy sonni ayirish qo'shishga aylandi)}$$

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Ratsional sonlarni ko'paytirish va bo'lish

✠ Asosiy qoida

Ishoralar qoidasi (ko'paytirish va bo'lishda).

Qisqacha:

- ishoralar **bir xil** bo'lsa — natija **musbat** $(+)$;
- ishoralar **har xil** bo'lsa — natija **manfiy** $(-)$.

Avval sonlarning **modullarini** ko'paytiring(bo'ling), keyin esa qoidaga ko'ra natija oldiga kerakli **ishorani** qo'ying.

Misollar:

$$(+3)\cdot(+4)=+12 \qquad (-3)\cdot(-4)=+12$$
$$(+3)\cdot(-4)=-12 \qquad (-12):(-3)=+4$$

✓ Eslatma

Nol bilan amallar:

- istalgan sonni 0 ga ko'paytirsak — natija 0: $\quad a\cdot 0=0$;
- 0 ni istalgan songa bo'lsak — natija 0: $\quad 0:a=0$ (bunda $a\neq 0$);
- 0 **ga bo'lish mumkin emas!** $\quad a:0$ — aniqlanmagan.

Bir nechta sonni ko'paytirish

✠ Asosiy qoida

Manfiy ko'paytuvchilar sonini sanang:

- agar manfiylar soni **juft** bo'lsa — natija **musbat**;
- agar manfiylar soni **toq** bo'lsa — natija **manfiy**.

Misollar:

$$(-2)\cdot(-3)\cdot(-1)=-6 \qquad (3 \text{ ta manfiy — toq})$$
$$(-1)\cdot(-2)\cdot(-3)\cdot(-4)=+24 \qquad (4 \text{ ta manfiy — juft})$$

Eslatma: ko'paytuvchilardan biri 0 bo'lsa — butun natija 0 ga teng bo'ladi.

Amaliy qism

1–10. *Hisoblang.*

1. $-3+5$
2. $7+(-4)$
3. $-6+(-8)$
4. $-9+9$
5. $-12+5$
6. $-15+(-7)$
7. $-25+13$
8. $-100+250$
9. $-8+(-12)+5$
10. $-15+8+(-3)$

11–20. *Hisoblang.*

11 $-7-5$

12 $-6-(-2)$

13 $-10-(-15)$

14 $5-(-7)$

15 $0-(-9)$

16 $-50-(-25)$

17 $-8+5-3$

18 $10-15-8+6$

19 $-3+7-(-4)-6$

20 $20-35-(-15)+8$

21–32. *Hisoblang.*

21 $-\frac{1}{4}+\frac{3}{4}$

22 $\frac{2}{3}+\left(-\frac{5}{6}\right)$

23 $-\frac{1}{2}-\frac{1}{3}$

24 $-\frac{3}{8}-\left(-\frac{1}{4}\right)$

25 $\frac{3}{4}-\frac{7}{8}$

26 $-2\frac{1}{3}+1\frac{1}{2}$

27 $-2{,}5+1{,}3$

28 $-3{,}7+(-1{,}8)$

29 $-4{,}5-2{,}7$

30 $-1{,}5-(-2{,}5)$

31 $-3{,}25+5{,}1$

32 $-7{,}8-(-3{,}25)$

33–47. *Hisoblang.*

33 $(-3)\cdot 4$

34 $(-5)\cdot(-6)$

35 $7\cdot(-8)$

36 $(-9)\cdot 0$

37 $(-12)\cdot 5$

38 $(-15)\cdot(-4)$

39 $(-100)\cdot(-3)$

40 $(-2)\cdot(-3)\cdot(-5)$

41 $(-1)\cdot(-2)\cdot(-3)\cdot(-4)$

42 $(-1)\cdot 7\cdot(-1)\cdot(-2)$

43 $(-12):3$

44 $(-24):(-6)$

45 $36:(-9)$

46 $0:(-7)$

47 $(-72):(-8)$

48–60. *Hisoblang.*

48 $\left(-\frac{5}{6}\right)\cdot\frac{12}{25}$

49 $\left(-\frac{3}{8}\right)\cdot\left(-\frac{4}{9}\right)$

50 $\frac{2}{7}\cdot\left(-\frac{14}{15}\right)$

51 $1{,}2\cdot(-0{,}5)$

52 $-0{,}25\cdot(-8)$

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53 $-1{,}25 \cdot 0{,}4$

54 $-0{,}5 \cdot (-0{,}6)$

55 $1{,}5 \cdot (-2)$

56 $-\frac{2}{3} : \frac{4}{9}$

57 $\left(-\frac{1}{4}\right) : \left(-\frac{1}{2}\right)$

58 $-2{,}4 : 0{,}3$

59 $-7{,}5 : 2{,}5$

60 $-3{,}6 : (-0{,}9)$

61–78. *Amallar tartibiga rioya qilib hisoblang.*

61 $-5 + 2 \cdot (-3)$

62 $(-2) \cdot 5 - 7$

63 $20 + (-4) \cdot 5$

64 $-3 \cdot 4 + 6 \cdot (-2)$

65 $(-15) : 3 + 7$

66 $-8 + 24 : (-4)$

67 $(-3 + 5) \cdot (-4)$

68 $(-7 - 3) : (-2)$

69 $-6 \cdot (5 - 8)$

70 $(-12) : (1 - 4)$

71 $-3 \cdot (-2) - 5 \cdot (-4)$

72 $\left(-\frac{1}{2}\right) \cdot 6 + \left(-\frac{2}{3}\right) \cdot (-9)$

73 $-3{,}5 + 2 \cdot (-1{,}25)$

74 $-0{,}5 \cdot (-4) - 1{,}5$

75 $-7 - (-3) \cdot (-2) + 4$

76 $\frac{(-12) + 4}{-2}$

77 $\frac{(-3) \cdot 4 + 8}{-2}$

78 $-3 \cdot (-2 + 5) + 4$

79–84. *Modul bilan amallar.*

79 $|-7|$

80 $-|-5|$

81 $|-3| + |-5|$

82 $|3 - 10|$

83 $|-7| \cdot |-2|$

84 $|-|-5||$

85–90. *Berilgan sonning qarama-qarshisini yozing.*

85 9

86 -12

87 $\frac{5}{6}$

88 $-2{,}3$

89 $0{,}45$

90 $1\frac{1}{2}$

91–100. *Matnli masalalar.*

91 Tashkentda kunduzi havo $+18°C$ edi. Kechasi harorat 25 darajaga pasaydi. Kechasi harorat qanday bo'lgan?

92 Termometr ertalab $-7°C$ ni ko'rsatdi. Kun davomida harorat 11 darajaga ko'tarilgan bo'lsa, kunduzi harorat qanday bo'lgan?

93 Yerto'lada harorat $-3°C$, ko'chada $-15°C$. Yerto'la ko'chadan necha darajaga issiq?

94 Muzxonada harorat $-22°C$ dan $-7°C$ gacha ko'tarildi. Harorat necha darajaga o'zgargan?

95 G'orni o'rganuvchi olim yer sathidan 45 m chuqurlikda edi. So'ng tunnel orqali yana 30 m pastga tushdi. U endi qanday chuqurlikda?

96 Bahodirning hisobida $15\,000$ so'm qarz bor edi. Onasi uning hisobiga $50\,000$ so'm pul yubordi. Keyin $20\,000$ so'mga kitob xarid qildi. Shundan so'ng uning hisobida qancha pul qoldi?

97 Tog'chi 4200 m balandlikdan 2750 m balandlikgacha tushdi. U necha metrga tushgan?

98 Muzxonadan $-18°$C li mahsulot olib chiqilgach, har 4 daqiqada $3°$ ga isiydi. 20 daqiqadan keyin uning harorati qancha bo'ladi?

99 O'yinda Aziz 30 ochko yutdi, 45 ochko yutqazdi, yana 20 ochko yutdi. Uning yakuniy ochko soni qancha bo'lgan?

100 Eramizdan avvalgi 530-yilda Pifagor maktabi tashkil etilgan. Eramizning 1980-yilida uning necha yillik yubileyi nishonlangan?

101–110. *Son o'qida masofa, siljitish va kesma o'rtasiga doir masalalar.*

101 Son o'qida $A(-6)$, $B(-1)$, $C(4)$ nuqtalarni belgilangan. AB, BC va AC kesmalarning uzunliklarini toping.

102 Koordinatasi $M(-3)$ bo'lgan nuqtani o'ngga 8 birlik siljitishdan hosil bo'lgan yangi nuqtaning koordinatasini toping.

103 Koordinatasi $K(5)$ bo'lgan nuqtani chapga 12 birlik siljitishdan hosil bo'lgan yangi nuqtaning koordinatasini toping.

104 Son o'qida M nuqtani qaysi tomonga va nechta birlik siljitsak N nuqtaga o'tamiz?

(a) $M(-7)$, $N(-2)$;

(b) $M(4)$, $N(-3)$;

(c) $M(-10)$, $N(-15)$.

105 M nuqta avval o'ngga 9 birlik, so'ng chapga 15 birlik siljitildi va $N(-2)$ nuqtaga keldi. M nuqtaning boshlang'ich koordinatasini toping.

106 Son o'qida A va B nuqtalar $C(-1)$ nuqtaga nisbatan simmetrik joylashgan. Agar A nuqtaning koordinatasi -6 bo'lsa, B nuqtaning koordinatasini toping.

107 Son o'qida $A(-7)$ va $B(11)$ nuqtalar berilgan. AB kesma 3 ta teng bo'lakka bo'lingan. Bo'linish nuqtalarining koordinatalarini toping.

108 Son o'qida K nuqta -5 va 9 koordinatali nuqtalardan teng masofada joylashgan. K nuqtaning koordinatasini va shu masofani toping.

109 Son o'qida $A\left(-\frac{5}{2}\right)$ va $B\left(\frac{7}{2}\right)$ nuqtalar berilgan. AB kesmaning uzunligini va o'rtasining koordinatasini toping.

110 Lift binoning -2-qavatida (yerto'la) edi. So'ng u 8 qavat tepaga, keyin 5 qavat pastga, oxirida yana 4 qavat tepaga ko'tarildi. Lift hozir qaysi qavatda?

Mavzu bo'yicha test savollari

1-variant

1. -7 soni qaysi sonlar to'plamiga tegishli?
   A) faqat $\mathbb{N}$ B) $\mathbb{Z}$ va $\mathbb{Q}$ C) faqat $\mathbb{Q}$ D) $\mathbb{N}$, $\mathbb{Z}$ va $\mathbb{Q}$

2. $|-15|-|-8|$ ning qiymati nechaga teng?
   A) -23 B) 7 C) -7 D) 23

3. $-3{,}5$ sonining qarama-qarshisini toping.
   A) $-3{,}5$ B) $3{,}5$ C) $\dfrac{1}{3{,}5}$ D) 0

4. Hisoblang: $-18+(-7)$
   A) -25 B) -11 C) 25 D) 11

5. Hisoblang: $-6-(-9)$
   A) -15 B) -3 C) 3 D) 15

6. Hisoblang: $(-7)\cdot(-3)$
   A) -21 B) 21 C) -10 D) 10

7. Hisoblang: $-\dfrac{3}{4}:\dfrac{9}{8}$
   A) $-\dfrac{2}{3}$ B) $\dfrac{2}{3}$ C) $-\dfrac{27}{32}$ D) 1

8. Hisoblang: $-5\cdot 3+(-2)\cdot(-4)$
   A) -7 B) 7 C) -23 D) 23

9. Son o'qida $A(-30)$ nuqta o'ng tomonga 15 birlik siljitilsa, B nuqtaga o'tadi. B nuqtaning koordinatasini toping.
   A) -45 B) -15 C) 15 D) 45

10. Son o'qida $A(-3)$ va $B(7)$ nuqtalar berilgan. AB kesma o'rtasining koordinatasini toping.
    A) 5 B) 4 C) 2 D) -2

2-variant

1. Quyidagilardan qaysi biri butun son?
   A) $-\dfrac{5}{3}$ B) $0{,}25$ C) -8 D) $\dfrac{2}{7}$

2. $|-9+4|$ ning qiymati nechaga teng?
   A) -5 B) 5 C) -13 D) 13

3. $\dfrac{2}{7}$ sonining qarama-qarshisini toping.
   A) $\dfrac{2}{7}$ B) $\dfrac{7}{2}$ C) $-\dfrac{2}{7}$ D) $-\dfrac{7}{2}$

4. Hisoblang: $-23+15$
   A) -38 B) -8 C) 8 D) 38

5. Hisoblang: $4-(-6)-3$
   A) -5 B) 1 C) 7 D) 13

6. Hisoblang: $(-12) \cdot 4$
   A) -48 B) 48 C) -16 D) 16

7. Hisoblang: $\frac{5}{8} \cdot \left(-\frac{4}{15}\right)$
   A) $-\frac{1}{6}$ B) $\frac{1}{6}$ C) $-\frac{1}{30}$ D) $-\frac{1}{3}$

8. Hisoblang: $(-4+7) \cdot (-3) - 5$
   A) -14 B) -4 C) 14 D) 4

9. Hisobda $25\,000$ so'm qarz bor edi. So'ng hisobga $40\,000$ so'm tushdi va $10\,000$ so'mga xarid qilindi. Hisobda qancha pul qoldi?
   A) $5\,000$ B) $-5\,000$ C) $15\,000$ D) $25\,000$

10. Son o'qida A nuqta -3 ni, B nuqta 5 ni ifodalaydi. AB kesmaning uzunligini toping.
    A) 2 B) 5 C) 8 D) 15

Javoblar

1. 2
2. 3
3. -14
4. 0
5. -7
6. -22
7. -12
8. 150
9. -15
10. -10
11. -12
12. -4
13. 5
14. 12
15. 9
16. -25
17. -6
18. -7
19. 2
20. 8
21. $\frac{1}{2}$
22. $-\frac{1}{6}$
23. $-\frac{5}{6}$
24. $-\frac{1}{8}$
25. $-\frac{1}{8}$
26. $-\frac{5}{6}$
27. $-1{,}2$
28. $-5{,}5$
29. $-7{,}2$
30. 1
31. $1{,}85$
32. $-4{,}55$
33. -12
34. 30
35. -56
36. 0
37. -60
38. 60
39. 300
40. -30
41. 24
42. -14
43. -4
44. 4
45. -4
46. 0
47. 9
48. $-\frac{2}{5}$
49. $\frac{1}{6}$
50. $-\frac{4}{15}$
51. $-0{,}6$
52. 2
53. $-0{,}5$
54. $0{,}3$
55. -3
56. $-1{,}5$
57. $\frac{1}{2}$
58. -8
59. -3
60. 4
61. -11
62. -17
63. 0
64. -24
65. 2

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66. -14

67. -8

68. 5

69. 18

70. 4

71. 26

72. 3

73. -6

74. $0{,}5$

75. -9

76. 4

77. 2

78. -5

79. 7

80. -5

81. 8

82. 7

83. 14

84. 5

85. -9

86. 12

87. $-\frac{5}{6}$

88. $2{,}3$

89. $-0{,}45$

90. $-1\frac{1}{2}$

91. $-7°\text{C}$

92. $4°\text{C}$

93. $12°$ ga issiq

94. $15°$ ga ko'tarildi

95. 75 m

96. $15\,000$ so'm

97. 1450 m

98. $-3°\text{C}$

99. 5 ochko

100. 2510 yilligi

101. $AB = 5,\ BC = 5,$ $AC = 10$

102. 5

103. -7

104. (a)o'ngga 5; (b) chapga 7; (c) chapga 5

105. $M(4)$

106. $B(4)$

107. -1 va 5

108. $K(2)$; masofa 7

109. uzunlik 6, o'rtasi $\frac{1}{2}$

110. 5-qavat

Test javoblari

1-variant: 1) B 2) B 3) B 4) A 5) C 6) B 7) A 8) A 9) B 10) C

2-variant: 1) C 2) B 3) C 4) B 5) C 6) A 7) A 8) A 9) A 10) C

Mavzu bo'yicha interaktiv testlar

QR kodni skanerlab, bilimingizni sinab ko'ring

6.

Daraja va uning xossalari

Nazariy qism

Daraja tushunchasi

✎ Ta'rif

n natural son va $n \geq 2$ bo'lganda a ning n-darajasi deb har biri a ga teng bo'lgan n ta ko'paytuvchining ko'paytmasiga aytiladi.Bundan tashqari $a^1 = a$ deb qabul qilinadi.

$$a^n = \underbrace{a \cdot a \cdot a \cdot \ldots \cdot a}_{n \text{ ta ko'paytuvchi}}$$

- a — **daraja asosi**;
- n — **daraja ko'rsatkichi**.

$5^1 = 5 \qquad 4^2 = 4 \cdot 4 = 16 \qquad 3^3 = 3 \cdot 3 \cdot 3 = 27 \qquad 2^5 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 32$

O'qilishi:

a^2 — "a ning kvadrati" $\qquad$ a^3 — "a ning kubi"

a^4 — "a ning to'rtinchi darajasi" $\qquad$ a^n — "a ning n-darajasi"

Manfiy sonning darajasi

✠ Asosiy qoida

Manfiy sonning darajasi:

- ko'rsatkich **juft** bo'lsa, darajaning qiymati **musbat**: $(-3)^2 = 9$;
- ko'rsatkich **toq** bo'lsa, darajaning qiymati **manfiy**: $(-3)^3 = -27$.

✓ Eslatma

Bu ikki ifoda bir-biridan farq qiladi:

$$(-2)^4 = (-2)(-2)(-2)(-2) = 16 \quad \text{(musbat)}$$

$$-2^4 = -(2 \cdot 2 \cdot 2 \cdot 2) = -16 \quad \text{(manfiy)}$$

Birinchisida -2 to'rt marta ko'paytiriladi. Ikkinchisida avval $2^4 = 16$ topiladi, keyin minus qo'yiladi.

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Darajaning besh xossasi

1-xossa.

✠ Asosiy qoida

$$a^m \cdot a^n = a^{m+n}$$

Bir xil asosli darajalarni ko'paytirishda asos o'zgarmasdan qoladi, daraja ko'rsatkichlari esa qo'shiladi.

$$a^3 \cdot a^2 = \underbrace{(a \cdot a \cdot a)}_{3\text{ ta}} \cdot \underbrace{(a \cdot a)}_{2\text{ ta}} = \underbrace{a \cdot a \cdot a \cdot a \cdot a}_{5\text{ ta}} = a^5$$

2-xossa.

✠ Asosiy qoida

$$a^m : a^n = a^{m-n}, \qquad a \neq 0$$

Bir xil asosli darajalarni bo'lishda asos o'zgarmasdan qoladi, daraja ko'rsatkichlari esa ayiriladi.

$$\frac{a^7}{a^3} = \frac{a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a}{a \cdot a \cdot a} = a \cdot a \cdot a \cdot a = a^4$$

3-xossa.

✠ Asosiy qoida

$$(a^m)^n = a^{m \cdot n}$$

Darajani darajaga ko'tarishda asos o'zgarmasdan qoladi, daraja ko'rsatkichlari esa o'zaro ko'paytiriladi.

$$(a^3)^4 = \underbrace{a^3 \cdot a^3 \cdot a^3 \cdot a^3}_{4\text{ ta } a^3} = a^{3+3+3+3} = a^{12}$$

4-xossa.

✠ Asosiy qoida

$$(a \cdot b)^n = a^n \cdot b^n$$

Ko'paytmani darajaga ko'tarishda har bir ko'paytuvchi shu darajaga ko'tariladi.

$$(a \cdot b)^5 = \underbrace{ab \cdot ab \cdot ab \cdot ab \cdot ab}_{5\text{ ta } ab} = a^5 \cdot b^5$$

5-xossa.

✠ Asosiy qoida

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}, \qquad b \neq 0$$

Kasrni darajaga ko'tarishda uning surat va maxraji xuddi shu darajaga ko'tariladi.

$$\left(\tfrac{a}{b}\right)^3 = \tfrac{a}{b} \cdot \tfrac{a}{b} \cdot \tfrac{a}{b} = \tfrac{a^3}{b^3}$$

Sonning nolinchi darajasi

✠ Asosiy qoida

$$a^0 = 1, \qquad a \neq 0$$

Noldan farqli har qanday sonning nolinchi darajasi 1 ga teng. *Eslatma:* 0^0 — aniqlanmagan ifoda.

Manfiy butun ko'rsatkichli daraja

✠ Asosiy qoida

$$a^{-n} = \frac{1}{a^n}, \qquad a \neq 0.$$

Misollar:

$$3^{-2} = \frac{1}{3^2} = \frac{1}{9} \qquad 10^{-3} = \frac{1}{1000} = 0{,}001$$

$$\left(\frac{2}{3}\right)^{-1} = \frac{3}{2} \qquad \left(\frac{1}{5}\right)^{-2} = 5^2 = 25$$

Sonlarning standart shakli

✎ Ta'rif

Har bir musbat sonni $a \cdot 10^n$ shaklida yozish mumkin, bunda $1 \leq a < 10$ va n – butun son. Bunday yozuv **sonning standart shakli** deyiladi.

$150\,000\,000 = 1{,}5 \cdot 10^8$ (vergul 8 xona chapga surildi)

$458\,000 = 4{,}58 \cdot 10^5$ (vergul 5 xona chapga surildi)

$0{,}000\,000\,000\,1 = 1 \cdot 10^{-10}$ (vergul 10 xona o'ngga surildi)

$0{,}00027 = 2{,}7 \cdot 10^{-4}$ (vergul 4 xona o'ngga surildi)

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Arifmetik kvadrat ildiz.

✎ Ta'rif

a sonining **kvadrat ildizi** deb — shunday nomanfiy b soniga aytiladiki, uni kvadratga oshirganda a hosil bo'ladi.

$$\sqrt{a} = b \iff b \geq 0 \text{ va } b^2 = a.$$

Kvadrt ildizning yozuvdagi belgilanishi — $\sqrt{a}$.

$$\sqrt{9} = 3, \quad \sqrt{25} = 5, \quad \sqrt{100} = 10, \quad \sqrt{1} = 1, \quad \sqrt{0} = 0.$$

Amaliy qism

1–6. *Ko'paytmani daraja ko'rinishida yozing.*

1 $5 \cdot 5 \cdot 5$

2 $a \cdot a \cdot a \cdot a$

3 $(-2) \cdot (-2) \cdot (-2)$

4 $\frac{1}{3} \cdot \frac{1}{3}$

5 $7 \cdot 7$

6 $b \cdot b \cdot b \cdot b \cdot b$

7–14. *Daraja qiymatini hisoblang.*

7 2^3

8 5^2

9 10^4

10 3^3

11 1^{50}

12 0^7

13 $\left(\frac{1}{2}\right)^3$

14 4^2

15–22. *Manfiy son darajasini hisoblang. Qavslarga e'tibor bering!*

15 $(-2)^2$

16 $(-2)^3$

17 $(-3)^2$

18 $(-3)^3$

19 -2^2

20 -3^2

21 $(-1)^{10}$

22 $(-1)^{11}$

23–38. *Hisoblang.*

23 $(-13)^2 + 14 \cdot (-12)$

24 $(-12)^2 - 11 \cdot 12$

25 $(-10)^2 + 11 \cdot (-9)$

26 $(-1)^{99} + (-1)^{100}$

27 $2^3 + 3^2 - 4^2$

28 $5^2 - (-4)^2$

29 $(-2)^3 + (-3)^2$

30 $3^2 \cdot 2^3$

31 $10^3 - 10^2$

32 $2^4 + (-2)^4$

33 $(-5)^2 - (-3)^2$

34 $2^2 + 3^2 + 4^2$

35 $(-2)^4 - 2^3$

36 $5^3 - 4^3$

37 $100^2 - 99 \cdot 101$

38 $(-1)^{2025} \cdot (-1)^{2024}$

39–44. *Har bir misolda xato bor. Toping va to'g'rilang.*

39 $a^2 \cdot a^3 = a^6$

40 $(a^2)^3 = a^5$

41 $(2a)^3 = 2a^3$

42 $\dfrac{a^7}{a^2} = a^{3,5}$

43 $a^2 + a^3 = a^5$

44 $(-2)^4 = -16$

45–54. ***1-xossa.*** $a^m \cdot a^n = a^{m+n}$.

45 $a^2 \cdot a^3$

46 $x \cdot x^4$

47 $2^3 \cdot 2^4$

48 $b^5 \cdot b^2$

49 $5^2 \cdot 5^3$

50 $y \cdot y \cdot y^2$

51 $a^4 \cdot a^4$

52 $c^3 \cdot c^5$

53 $a^7 \cdot a^3$

54 $10^2 \cdot 10^4$

55–62. ***2-xossa.*** $a^m : a^n = a^{m-n}$.

55 $\dfrac{a^7}{a^3}$

56 $\dfrac{x^5}{x^2}$

57 $\dfrac{2^8}{2^3}$

58 $\dfrac{b^{10}}{b^4}$

59 $\dfrac{c^9}{c^9}$

60 $\dfrac{y^6}{y}$

61 $\dfrac{5^7}{5^4}$

62 $\dfrac{a^8}{a^5}$

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63–70. ***3-xossa.*** $(a^m)^n = a^{mn}$.

63 $(a^2)^3$

64 $(x^4)^2$

65 $(b^3)^3$

66 $(2^2)^4$

67 $(c^5)^2$

68 $(y^7)^2$

69 $(a^5)^3$

70 $(3^2)^2$

71–80. ***4 va 5-xossalar.*** $(ab)^n = a^n b^n$, $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$.

71 $(ab)^3$

72 $(2x)^2$

73 $(3a)^2$

74 $(-2b)^3$

75 $\left(\frac{a}{b}\right)^2$

76 $\left(\frac{x}{3}\right)^2$

77 $\left(\frac{2}{5}\right)^2$

78 $(xyz)^2$

79 $(4a)^2$

80 $\left(-\frac{x}{2}\right)^3$

81–88. *Soddalashtiring (xossalarni birgalikda qo'llang).*

81 $a^3 \cdot a^4 \cdot a$

82 $\frac{x^5 \cdot x^3}{x^4}$

83 $(b^2)^3 \cdot b$

84 $\frac{(c^3)^4}{c^{10}}$

85 $(2a)^2 \cdot a^3$

86 $\frac{(xy)^4}{x^2y^2}$

87 $\frac{a^5 \cdot a^3}{a^4}$

88 $(3b)^2 \cdot b$

89–98. *Nol va manfiy ko'rsatkichli darajani hisoblang.*

89 7^0

90 $(-5)^0$

91 $\left(\frac{2}{3}\right)^0$

92 2^{-1}

93 5^{-1}

94 3^{-2}

95 10^{-2}

96 $\left(\frac{1}{2}\right)^{-1}$

97 $\left(\frac{1}{3}\right)^{-2}$

98 2^{-3}

99–104. *Soddalashtiring (manfiy ko'rsatkich bilan).*

99 $a^{-2} \cdot a^{5}$

100 $a^{-3} \cdot a^{-4}$

101 $\dfrac{x^{4}}{x^{-2}}$

102 $(b^{-2})^{3}$

103 $\dfrac{a^{-5}}{a^{-3}}$

104 $(2a^{-1})^{2}$

105–112. *Sonni standart shaklda yozing* $(a \cdot 10^{n})$.

105 5 000

106 720 000

107 40 000 000

108 0,07

109 0,004

110 0,000 12

111 1 500

112 0,9

113–118. *Standart shaklda berilgan sonlarni oddiy shaklda yozing.*

113 $4 \cdot 10^{3}$

114 $2{,}5 \cdot 10^{5}$

115 $7 \cdot 10^{-2}$

116 $3 \cdot 10^{-4}$

117 $1{,}1 \cdot 10^{6}$

118 $9 \cdot 10^{-1}$

119–124. *Hisoblang.*

119 $(2 \cdot 10^{3}) \cdot (3 \cdot 10^{4})$

120 $(4 \cdot 10^{5}) \cdot (2 \cdot 10^{2})$

121 $\dfrac{6 \cdot 10^{8}}{2 \cdot 10^{3}}$

122 $\dfrac{8 \cdot 10^{10}}{4 \cdot 10^{4}}$

123 $(5 \cdot 10^{-2}) \cdot (2 \cdot 10^{4})$

124 $\dfrac{9 \cdot 10^{7}}{3 \cdot 10^{9}}$

125–128. *Hisoblang.*

125 $2{,}4 \cdot 6\,000\,000 \cdot 10^{-7}$

126 $1{,}5 \cdot 4\,000\,000 \cdot 10^{-5}$

127 $2{,}1 : 3\,000\,000 \cdot 10^{7}$

128 $2{,}4 : 8\,000\,000 \cdot 10^{7}$

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Mavzu bo'yicha test savollari

1-variant

1. Hisoblang: $(-3)^4 - 5 \cdot 4$
   A) 61 B) -61 C) 101 D) -101

2. $a^7 \cdot a^2 \cdot a$ ni soddalashtiring.
   A) a^{14} B) $3a^{10}$ C) a^{10} D) a^9

3. $\dfrac{x^{12}}{x^4}$ ifoda quyidagilardan qaysi biriga teng?
   A) x^3 B) x^8 C) x^{16} D) x^{48}

4. Quyidagi tengliklarning qaysi biri to'g'ri?
   A) $a^3 + a^2 = a^5$ B) $(a^2)^3 = a^5$ C) $(2a)^3 = 6a^3$ D) $a^4 \cdot a^3 = a^7$

5. $(2x)^4$ ifoda quyidagilardan qaysi biriga teng?
   A) $2x^4$ B) $8x^4$ C) $16x^4$ D) $16x$

6. Hisoblang: $5^0 + 5^{-1}$
   A) 0 B) $1{,}2$ C) $5{,}2$ D) $\dfrac{1}{5}$

7. Hisoblang: $(-2)^4 + (-2)^3$
   A) 0 B) 8 C) -8 D) 24

8. Standart shaklda yozing: $0{,}00045$
   A) $4{,}5 \cdot 10^{-4}$ B) $4{,}5 \cdot 10^4$ C) $45 \cdot 10^{-5}$ D) $0{,}45 \cdot 10^{-3}$

9. Standart shaklda berilgan sonni to'liq shaklda yozing: $7{,}2 \cdot 10^5$
   A) $72\,000$ B) $720\,000$ C) $7\,200\,000$ D) 72

10. Hisoblang: $2{,}4 \cdot 5\,000\,000 \cdot 10^{-7}$
    A) $1{,}2$ B) $0{,}12$ C) 12 D) 120

2-variant

1. Hisoblang: $-3^2 + 4^2$
   A) 7 B) -7 C) 25 D) -25

2. Soddalashtiring: $b^5 \cdot b^3$
   A) b^{15} B) b^2 C) b^8 D) $2b^8$

3. Hisoblang: $\dfrac{2^{15}}{2^{12}}$
   A) 4 B) 8 C) 16 D) 32

4. Quyidagi tengliklarning qaysi biri xato?
   A) $a^4 \cdot a^2 = a^6$ B) $(a^3)^2 = a^6$ C) $\dfrac{a^7}{a^3} = a^4$ D) $a^4 + a^2 = a^6$

5. $(b^3)^4 : b^2$ ifoda quyidagilardan qaysi biriga teng?
   A) b^5 B) b^{10} C) b^{14} D) b^{32}

6. Hisoblang: $\left(\frac{1}{3}\right)^{-2}$
   A) $\frac{1}{9}$ B) 9 C) -9 D) $\frac{1}{6}$
7. Hisoblang: $(-1)^{99} - (-1)^{100}$
   A) 0 B) -2 C) 2 D) -1
8. Standart shaklda berilgan sonni to'liq shaklda yozing: $3{,}5 \cdot 10^{-4}$
   A) $0{,}00035$ B) $0{,}0035$ C) $35\,000$ D) $0{,}000035$
9. Standart shaklda yozing: $(3 \cdot 10^4) \cdot (2 \cdot 10^3)$
   A) $5 \cdot 10^7$ B) $6 \cdot 10^{12}$ C) $6 \cdot 10^7$ D) $6 \cdot 10^{34}$
10. Hisoblang: $100^2 - 99 \cdot 101$
    A) 0 B) 1 C) -1 D) 99

Javoblar

1. 5^3
2. a^4
3. $(-2)^3$
4. $\left(\frac{1}{3}\right)^2$
5. 7^2
6. b^5
7. 8
8. 25
9. $10\,000$
10. 27
11. 1
12. 0
13. $\frac{1}{8}$
14. 16
15. 4
16. -8
17. 9
18. -27
19. -4
20. -9
21. 1
22. -1
23. 1
24. 12
25. 1
26. 0
27. 1
28. 9
29. 1
30. 72
31. 900
32. 32
33. 16
34. 29
35. 8
36. 61
37. 1
38. -1
39. $a^2 \cdot a^3 = a^{2+3} = a^5$
40. $(a^2)^3 = a^{2 \cdot 3} = a^6$
41. $(2a)^3 = 2^3 \cdot a^3 = 8a^3$
42. $\frac{a^7}{a^2} = a^{7-2} = a^5$
43. $a^2 + a^3$ ifoda soddalashmaydi.
44. $(-2)^4 = 16$
45. a^5
46. x^5
47. $2^7 = 128$
48. b^7
49. $5^5 = 3125$
50. y^4
51. a^8
52. c^8
53. a^{10}
54. 10^6
55. a^4
56. x^3
57. $2^5 = 32$
58. b^6
59. 1
60. y^5
61. $5^3 = 125$
62. a^3
63. a^6
64. x^8
65. b^9
66. $2^8 = 256$
67. c^{10}

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68. y^{14}

69. a^{15}

70. $3^4 = 81$

71. a^3b^3

72. $4x^2$

73. $9a^2$

74. $-8b^3$

75. $\frac{a^2}{b^2}$

76. $\frac{x^2}{9}$

77. $\frac{4}{25}$

78. $x^2y^2z^2$

79. $16a^2$

80. $-\frac{x^3}{8}$

81. a^8

82. x^4

83. b^7

84. c^2

85. $4a^5$

86. x^2y^2

87. a^4

88. $9b^3$

89. 1

90. 1

91. 1

92. $\frac{1}{2}$

93. $\frac{1}{5}$

94. $\frac{1}{9}$

95. $\frac{1}{100}$

96. 2

97. 9

98. $\frac{1}{8}$

99. a^3

100. $a^{-7} = \frac{1}{a^7}$

101. x^6

102. $b^{-6} = \frac{1}{b^6}$

103. $a^{-2} = \frac{1}{a^2}$

104. $4a^{-2} = \frac{4}{a^2}$

105. $5 \cdot 10^3$

106. $7{,}2 \cdot 10^5$

107. $4 \cdot 10^7$

108. $7 \cdot 10^{-2}$

109. $4 \cdot 10^{-3}$

110. $1{,}2 \cdot 10^{-4}$

111. $1{,}5 \cdot 10^3$

112. $9 \cdot 10^{-1}$

113. 4000

114. $250\,000$

115. $0{,}07$

116. $0{,}0003$

117. $1\,100\,000$

118. $0{,}9$

119. $6 \cdot 10^7$

120. $8 \cdot 10^7$

121. $3 \cdot 10^5$

122. $2 \cdot 10^6$

123. $10 \cdot 10^2 = 10^3 = 1000$

124. $3 \cdot 10^{-2} = 0{,}03$

125. $1{,}44$

126. 60

127. 7

128. 3

Test javoblari

1-variant: 1) A 2) C 3) B 4) D 5) C 6) B 7) B 8) A 9) B 10) A

2-variant: 1) A 2) C 3) B 4) D 5) B 6) B 7) B 8) A 9) C 10) B

Mavzu bo'yicha interaktiv testlar

QR kodni skanerlab, bilimingizni sinab ko'ring

7.

Qism va foiz

Nazariy qism

Sonning qismini topish

✠ Asosiy qoida

Sonning berilgan qismini topish uchun shu sonni uning qismini ifodalovchi kasrga ko'paytirish kerak.

$$a \cdot \frac{m}{n}$$

✓ Eslatma

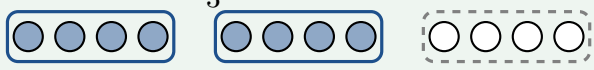
Misol 1. 12 ning $\frac{2}{3}$ qismini toping.

1-usul:

1-qadam: $12 : 3 = 4$ (1 bo'lakda 4 ta).

2-qadam: $4 \cdot 2 = 8$ (2 ta bo'lak).

Demak, 12 ning $\frac{2}{3}$ qismi 8 ga teng.



1-bo'lak (4 ta) 2-bo'lak (4 ta) 3-bo'lak (qoldi)

2-usul:

$12 \cdot \frac{2}{3} = 8$

Misol 2. 200 ning $\frac{3}{8}$ qismini toping.

$200 \cdot \frac{3}{8} = 75$

Misol 3. Sinfda 32 ta o'quvchi bor. Ularning $\frac{5}{8}$ qismini o'g'il bolalar tashkil qilsa, sinfda nechta o'g'il bola bor?

$32 \cdot \frac{5}{8} = 20$. Demak, sinfda 20 ta o'g'il bola bor.

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Qismiga ko'ra sonni topish

✠ Asosiy qoida

Qismiga ko'ra sonni topish uchun berilgan sonni qismni ifodalovchi kasrga bo'lish kerak.

$$b : \frac{m}{n}$$

✓ Eslatma

Misol 1. Sonning $\frac{3}{4}$ qismi 24 ga teng. Sonni toping.

$$24 : \frac{3}{4} = 24 \cdot \frac{4}{3} = \frac{96}{3} = 32.$$

Tekshirish: $32 \cdot \frac{3}{4} = 24$ ✓

Misol 2. Sonning $\frac{2}{5}$ qismi 18 ga teng. Sonni toping.

$$18 : \frac{2}{5} = 18 \cdot \frac{5}{2} = \frac{90}{2} = 45.$$

Foiz tushunchasi

✎ Ta'rif

Miqdorning (sonning) yuzdan bir qismi uning **bir foizi** deb ataladi. Belgisi: **%**.

$$1\% = \frac{1}{100} = 0{,}01$$

Foizni kasrga va kasrni foizga aylantirish

✠ Asosiy qoida

Foizni kasrda ifodalash uchun berilgan foizni 100 ga bo'lish kifoya:

$$p\% = \frac{p}{100}$$

Misollar:

$$25\% = \frac{25}{100} = 0{,}25 \qquad 7\% = 0{,}07 \qquad 150\% = 1{,}5 \qquad 0{,}5\% = 0{,}005$$

✠ Asosiy qoida

Kasrni foiz ko'rinishida ifodalash uchun shu kasrni 100 ga ko'paytirish va natijaga % belgisini qo'yish kifoya:

$$0{,}3 = 30\% \qquad 0{,}07 = 7\% \qquad 1{,}25 = 125\% \qquad 0{,}004 = 0{,}4\%$$

✓ Eslatma

Eng ko'p uchraydigan foizlarni yodlab oling:

Foiz	O'nli kasr	Oddiy kasr	Foiz	O'nli kasr	Oddiy kasr
1%	0,01	$\frac{1}{100}$	50%	0,5	$\frac{1}{2}$
5%	0,05	$\frac{1}{20}$	60%	0,6	$\frac{3}{5}$
10%	0,1	$\frac{1}{10}$	75%	0,75	$\frac{3}{4}$
20%	0,2	$\frac{1}{5}$	80%	0,8	$\frac{4}{5}$
25%	0,25	$\frac{1}{4}$	100%	1	1

Sonning foizini topish

✠ Asosiy qoida

a sonning p foizini topish:

$$a \cdot \frac{p\%}{100\%}$$

✓ Eslatma

Misol 1. 400 ning 25% ini toping.

$$400 \cdot \frac{25}{100} = 400 \cdot 0{,}25 = 100.$$

Misol 2. 80 ning 15% ini toping.

$$80 \cdot 0{,}15 = 12.$$

Misol 3. Telefonning narxi 2 500 000 so'm edi. 20% chegirma e'lon qilinsa, chegirmali narx qanchani tashkil etadi?

$$2\,500\,000 \cdot 0{,}2 = 500\,000 \text{ so'm.}$$

Telefonning chegirmali narxi: $2\,500\,000 - 500\,000 = 2\,000\,000$ so'm.

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Foizi bo'yicha sonni topish

✠ Asosiy qoida

p foizi b ga teng bo'lgan sonni topish:

$$\frac{b}{p\%} \cdot 100\%$$

✓ Eslatma

Misol 1. Sonning 30% i 90 ga teng. Sonni toping.

$$\frac{90}{30} \cdot 100 = 3 \cdot 100 = 300.$$

Misol 2. Maktab o'quvchilarining 40% i sport bilan shug'ullanadi va ularning soni 200 ta. Maktabda jami nechta o'quvchi bor?

$$\frac{200}{40} \cdot 100 = 5 \cdot 100 = 500 \text{ ta o'quvchi.}$$

Ikki sonning foizli nisbati

✠ Asosiy qoida

b son a sonning necha foizini tashkil qilishini topish:

$$\boxed{\frac{b}{a} \cdot 100\%}$$

✓ Eslatma

Misol 1. 30 soni 120 ning necha foizi?

$$\frac{30}{120} \cdot 100\% = 0{,}25 \cdot 100\% = 25\%.$$

Misol 2. Sinfda 25 ta o'quvchi bor, shulardan 20 tasi mavzuni o'zlashtirgan. Necha foiz o'quvchi mavzuni o'zlashtirgan?

$$\frac{20}{25} \cdot 100\% = 0{,}8 \cdot 100\% = 80\%.$$

Foizga oshirish va kamaytirish

✠ Asosiy qoida

Sonni $p\%$ ga oshirish:

$$a \cdot \left(1 + \frac{p}{100}\right)$$

Sonni $p\%$ ga kamaytirish:

$$a \cdot \left(1 - \frac{p}{100}\right)$$

Misollar:

- 500 ni 20% ga oshiring: $500 \cdot 1{,}2 = 600$.
- 500 ni 20% ga kamaytiring: $500 \cdot 0{,}8 = 400$.

Masshtab

✎ Ta'rif

Masshtab — chizmadagi o'lchamning haqiqiy o'lchamga nisbati.

✓ Eslatma

Uzunlik birliklarini eslab oling:

$$1\ \text{m} = 100\ \text{cm} \qquad 1\ \text{km} = 1000\ \text{m} = 100\,000\ \text{cm}$$

$1 : 50\,000$ masshtabda xaritadagi 1 cm haqiqatda $50\,000$ cm $= 500$ m bo'ladi.

Misol 1. $1 : 1\,000\,000$ masshtabli xaritada 5 cm masofa haqiqiy o'lchamda necha km?

$$5 \times 1\,000\,000 = 5\,000\,000\ \text{cm} = 50\,000\ \text{m} = \boxed{50\ \text{km}}.$$

Misol 2. Bog'ning haqiqiy uzunligi 80 m bo'lsa, $1 : 200$ masshtabli xaritada bog' necha cm bo'ladi?
Avval 80 m ni cm ga aylantiramiz: $80\ \text{m} = 8000\ \text{cm}$.

$$8000 \div 200 = \boxed{40\ \text{cm}}.$$

Misol 3 (masshtabni topish). Xaritada 5 cm masofaning haqiqiy o'lchami 10 km bo'lsa, masshtabni toping.
$10\ \text{km} = 1\,000\,000\ \text{cm}$.
Xaritadagi 1 cm masofaning haqiqiy o'lchami $1\,000\,000 : 5 = 200\,000$ cm.

$$\boxed{\text{Masshtab} = 1 : 200\,000}.$$

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Amaliy qism

1 40 ning $\frac{1}{4}$ qismini toping.

2 90 ning $\frac{2}{3}$ qismini toping.

3 120 ning $\frac{3}{4}$ qismini toping.

4 200 ning $\frac{2}{5}$ qismini toping.

5 84 ning $\frac{5}{6}$ qismini toping.

6 96 ning $\frac{3}{8}$ qismini toping.

7 250 ning $\frac{7}{10}$ qismini toping.

8 480 ning $\frac{7}{16}$ qismini toping.

9 Sonning $\frac{1}{2}$ qismi 18 ga teng. Sonni toping.

10 Sonning $\frac{2}{3}$ qismi 40 ga teng. Sonni toping.

11 Sonning $\frac{3}{4}$ qismi 60 ga teng. Sonni toping.

12 Sonning $\frac{4}{5}$ qismi 80 ga teng. Sonni toping.

13 Sonning $\frac{3}{8}$ qismi 48 ga teng. Sonni toping.

14 Sonning $\frac{5}{12}$ qismi 75 ga teng. Sonni toping.

15 Sonning $\frac{9}{25}$ qismi 108 ga teng. Sonni toping.

16–25. *Foizlarni kasrlarda ifodalang.*

16 50%

17 20%

18 7%

19 36%

20 40%

21 4%

22 125%

23 48%

24 0,5%

25 12,5%

26–35. *Kasrlarni foizlarda ifodalang.*

26 $0{,}3$

27 $\frac{1}{4}$

28 $0{,}07$

29 $\frac{3}{4}$

30 $1{,}5$

31 $\frac{2}{5}$

32 $0{,}625$

33 $\frac{3}{20}$

34 $0{,}05$

35 $\frac{7}{25}$

36 400 ning 25% ini toping.

37 Sonning 20% i 40 ga teng. Sonni toping.

38 250 ning 20% ini toping.

39 Sonning 25% i 90 ga teng. Sonni toping.

40 1200 ning 5% ini toping.

41 Sonning 50% i 135 ga teng. Sonni toping.

42 640 ning 15% ini toping.

43 Sonning 10% i 7 ga teng. Sonni toping.

44 750 ning 12% ini toping.

45 Sonning 40% i 120 ga teng. Sonni toping.

46 1500 ning 30% ini toping.

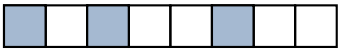
47 Sonning 15% i 63 ga teng. Sonni toping.

48 2400 ning 7% ini toping.

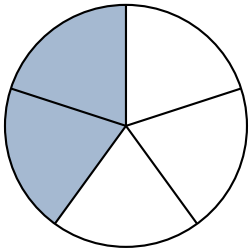
49 Sonning 4% i 80 ga teng. Sonni toping.

50 80 ning 125% ini toping.

51 Quyidagi shaklning bo'yalgan qismini oddiy kasr va foiz ko'rinishida yozing.

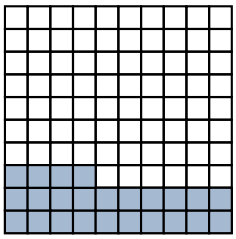


52 Quyidagi doira teng besh sektorga bo'lingan. Doiraning necha foizi bo'yalgan?



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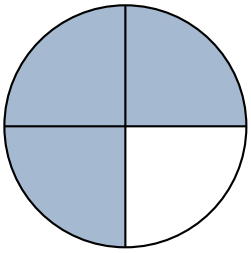
53 Quyidagi 10×10 to'rda nechta katak bo'yalgan? Bo'yalgan qismni qisqartirilgan oddiy kasr va foiz ko'rinishida yozing.



54 Quyidagi shaklning bo'yalgan qismi sinfdagi qiz bolalar ulushini ko'rsatadi. Agar sinfda 12 ta qiz bola bo'lsa, sinfda jami nechta o'quvchi bor?



55 Quyidagi doiraviy diagramma bog'dagi olma daraxtlari ulushini ko'rsatadi. Bog'da jami 80 ta daraxt bo'lsa, olma daraxtlari soni nechta?



56 Kitobxonlik tanlovida 50 ta kitobdan Sardor 30 tasini o'qib chiqdi. U kitoblarning necha foizini o'qigan?

57 Maktabda 200 ta o'quvchidan 80 tasi sport bilan shug'ullanadi. Sport bilan shug'ullanuvchilar maktab o'quvchilarining necha foizini tashkil etadi?

58 Sinfda 30 ta o'quvchi bor. Ularning $\frac{2}{5}$ qismi qiz bolalar. Sinfda nechta qiz bola bor?

59 Telefonning narxi $3\,000\,000$ so'm edi. Unga 15% chegirma e'lon qilindi. Telefonning yangi narxini toping.

60 Do'kondagi ko'ylakning narxi $250\,000$ so'm edi. U 20% ga arzonlashdi. Hozir ko'ylakning narxi qancha?

61 Fermer $5\,000\,000$ so'mga ko'chat ekib, hosildan 14% sof foyda ko'rdi. U boshlang'ich kapitali bilan birga jami qancha pulga ega bo'ldi?

62 Maktabda 600 ta o'quvchi bor. Ulardan 45% i sport to'garagiga qatnashadi. Sport to'garagida nechta o'quvchi bor?

63 Anordan olingan sharbat anor og'irligining $\frac{3}{4}$ qismini tashkil qilsa, 20 kg anordan necha kg sharbat olinadi?

64 Imtihonda 80 ta savol berildi. O'quvchi 68 tasiga to'g'ri javob berdi. U imtihonni necha foizga topshirgan?

65 Tovarning narxi avval 20% ga oshdi, so'ng yangi narxidan 20% ga kamaytirildi. Tovarning narxi avvalgi narxiga nisbatan jami necha foizga o'zgargan?

66 Sinfdagi 12 ta o'quvchi shaxmat o'ynaydi va ular sinfdagi jami o'quvchilarning 40% ini tashkil qiladi. Sinfda jami nechta o'quvchi bor?

67 Kompyuterning narxi $4\,000\,000$ so'm edi. Yil oxirida narxi $5\,200\,000$ so'mga ko'tarildi. Narx necha foizga oshgan?

68 Bug'doyni un qilishda $2{,}5\%$ chiqindi chiqsa, 400 kg bug'doydan necha kg un olinadi?

69 Tovarning narxi avval 25% ga oshdi, keyin yangi narxi 20% ga kamaytirildi. Boshlang'ich narxi $200\,000$ so'm bo'lsa, hozirgi narxi qancha?

70 Tovarning narxi 3 marta 10% ga oshirildi. Yakuniy narxi avvalgisidan necha foizga oshdi?

71–80. *Masshtabga doir masalalar.*

71 Xaritaning masshtabi $1:1000$ bo'lsa, xaritadagi 5 cm joyning haqiqiy masofasi necha metrga to'g'ri keladi?

72 Xaritaning masshtabi $1:50\,000$ bo'lsa, xaritadagi 4 cm joyning haqiqiy masofasi necha kilometrga to'g'ri keladi?

73 Xona rejasining masshtabi $1:100$. Rejada divanning uzunligi 2 cm bo'lsa, divanning haqiqiy uzunligini toping.

74 Toshkent va Samarqand orasidagi masofa 300 km bo'lsa, $1:3\,000\,000$ masshtabli xaritada bu masofa necha cm bo'lib ko'rsatiladi?

75 Maktab maydonining uzunligi 80 m bo'lsa, $1:400$ masshtabli rejada uning uzunligi necha cm bo'ladi?

76 Avtomobilning modeli $1:25$ masshtabda yasalgan. Modelning uzunligi 18 cm bo'lsa, haqiqiy avtomobilning uzunligi qancha?

77 $1:200\,000$ masshtabli xaritada A va B shaharlar orasidagi masofa $7{,}5$ cm bo'lsa, shaharlar orasidagi haqiqiy masofani km da toping.

78 Bog'cha hovlisining haqiqiy uzunligi 60 m, eni 40 m bo'lsa, $1:500$ masshtabli rejada uning o'lchamlari qanday bo'ladi?

79 Xaritada ikki qishloq orasidagi masofa 12 cm, haqiqiy masofa 24 km bo'lsa, xaritaning masshtabini toping.

80 Globusning masshtabi $1:50\,000\,000$. Yer ekvatorining uzunligi taxminan $40\,000$ km. Globusdagi ekvatorning uzunligini toping (cm da).

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Mavzu bo'yicha test savollari

1-variant

1. 80 ning $\frac{3}{4}$ qismini toping.
   A) 20 B) 40 C) 60 D) 75

2. Sonning $\frac{2}{5}$ qismi 30 ga teng. Sonni toping.
   A) 12 B) 50 C) 60 D) 75

3. $0{,}07$ ni foiz ko'rinishida yozing.
   A) $0{,}07\%$ B) $0{,}7\%$ C) 7% D) 70%

4. 25% ni qisqartirilgan oddiy kasrga aylantiring.
   A) $\frac{1}{4}$ B) $\frac{1}{5}$ C) $\frac{1}{25}$ D) $\frac{25}{100}$

5. 300 ning 20% ini toping.
   A) 30 B) 60 C) 80 D) 150

6. Sonning 15% i 90 ga teng. Sonni toping.
   A) 135 B) 300 C) 600 D) 1350

7. Telefon $4\,000\,000$ so'm edi. Unga 15% chegirma berildi. Yangi narxini toping.
   A) $3\,400\,000$ B) $3\,600\,000$ C) $3\,850\,000$ D) $600\,000$

8. Maktabda 800 ta o'quvchidan 35% i ingliz tilini chuqur o'rganadi. Ularning soni nechta?
   A) 240 B) 260 C) 280 D) 320

9. Sinfda 24 ta o'quvchidan $\frac{5}{8}$ qismi sport to'garagiga qatnashadi. Sport to'garagiga qatnashmaydiganlar soni nechta?
   A) 9 B) 12 C) 15 D) 18

10. Xaritaning masshtabi $1:50\,000$ bo'lsa, xaritadagi 4 cm masofa haqiqatda necha kilometrga to'g'ri keladi?
    A) $0{,}2$ km B) 2 km C) 20 km D) 200 km

2-variant

1. $\frac{3}{20}$ ni foiz ko'rinishida yozing.
   A) 3% B) 6% C) 15% D) 20%

2. 250 ning $\frac{2}{5}$ qismini toping.
   A) 50 B) 100 C) 125 D) 150

3. 500 ning 8% ini toping.
   A) 40 B) 50 C) 60 D) 80

4. Sonning 25% i 60 ga teng. Sonni toping.
   A) 80 B) 150 C) 240 D) 300

5. 18 soni 72 ning necha foizini tashkil qiladi?
   A) 4% B) 18% C) 25% D) 40%

6. Tovarning narxi 200 000 so'm edi. Narxi avval 20% ga oshdi, keyin yangi narxidan 20% ga kamaydi. Hozirgi narxi qancha?
   A) 192 000 B) 200 000 C) 208 000 D) 240 000

7. Tadbirkor 2 500 000 so'mga mol sotib olib, ustiga 12% ustama qo'yib sotdi. Tadbirkor savdodan jami qancha pul oldi?
   A) 2 650 000 B) 2 700 000 C) 2 800 000 D) 2 950 000

8. Sinfdagi 16 ta o'quvchi sport bilan shug'ullanadi va ular jami o'quvchilarning 40% ini tashkil qiladi. Sinfdagi jami o'quvchilar sonini toping.
   A) 32 B) 40 C) 48 D) 50

9. Birinchi kuni 100 betlik kitobning $\frac{1}{4}$ qismi, ikkinchi kuni qolganining $\frac{1}{3}$ qismi o'qildi. Kitobning qancha beti o'qilmay qoldi?
   A) 25 B) 40 C) 50 D) 60

10. Maktab maydonining haqiqiy uzunligi 100 m bo'lsa, $1:500$ masshtabli rejadada uning uzunligi necha cm bo'ladi?
    A) 5 B) 10 C) 20 D) 50

Javoblar

1. 10
2. 60
3. 90
4. 80
5. 70
6. 36
7. 175
8. 210
9. 36
10. 60
11. 80
12. 100
13. 128
14. 180
15. 300
16. 0,5 $\left(\frac{1}{2}\right)$
17. 0,2 $\left(\frac{1}{5}\right)$
18. 0,07 $\left(\frac{7}{100}\right)$
19. 0,36 $\left(\frac{9}{25}\right)$
20. 0,4 $\left(\frac{2}{5}\right)$
21. 0,04 $\left(\frac{1}{25}\right)$
22. 1,25 $\left(\frac{5}{4}\right)$
23. 0,48 $\left(\frac{12}{25}\right)$
24. 0,005 $\left(\frac{1}{200}\right)$
25. 0,125 $\left(\frac{1}{8}\right)$
26. 30%
27. 25%
28. 7%
29. 75%
30. 150%
31. 40%
32. 62,5%
33. 15%
34. 5%
35. 28%
36. 100
37. 200
38. 50
39. 360
40. 60
41. 270
42. 96
43. 70
44. 90
45. 300
46. 450
47. 420
48. 168
49. 2000

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50. 100

51. $\frac{3}{8} = 37{,}5\%$

52. 40%

53. $\frac{6}{25} = 24\%$ (24 ta katak)

54. 30 ta o'quvchi

55. 60 ta olma daraxti

56. 60%

57. 40%

58. 12 ta qiz

59. 2 550 000 so'm

60. 200 000 so'm

61. 5 700 000 so'm

62. 270 ta o'quvchi

63. 15 kg sharbat

64. 85%

65. 4% ga kamaygan

66. 30 ta o'quvchi

67. 30%

68. 390 kg

69. 200 000 so'm

70. 33,1%

71. 50 m

72. 2 km

73. 2 m

74. 10 cm

75. 20 cm

76. 4,5 m

77. 15 km

78. 12 cm $\times$ 8 cm

79. 1 : 200 000

80. 80 cm

Test javoblari

1-variant: 1) C 2) D 3) C 4) A 5) B 6) C 7) A 8) C 9) A 10) B

2-variant: 1) C 2) B 3) A 4) C 5) C 6) A 7) C 8) B 9) C 10) C

Mavzu bo'yicha interaktiv testlar

QR kodni skanerlab, bilimingizni sinab ko'ring

8.

Geometrik shakllarning perimetri va yuzi

Nazariy qism

Asosiy tushunchalar

✎ Ta'rif

Perimetr — ko'pburchak tomonlari uzunliklarining yig'indisi. P harfi bilan belgilanadi. O'lchov birliklari: cm, m, km, ...

Yuza — shakl tekislikning qancha joyini egallab turganini ko'rsatadi. S harfi bilan belgilanadi. O'lchov birliklari: cm^2, m^2, km^2, ...

✓ Eslatma

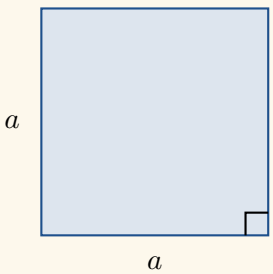
Yer maydoni o'lchov birliklari:

- 1 ar $= 100\ \text{m}^2$ (1 sotix)
- 1 ga (gektar) $= 100$ ar $= 10\,000\ \text{m}^2$
- $1\ \text{km}^2 = 100$ ga $= 1\,000\,000\ \text{m}^2$

Kvadrat

✠ Asosiy qoida

Tomoni a ga teng kvadrat uchun:



$$P = 4a \qquad S = a^2$$

1-misol. Tomoni 5 cm bo'lgan kvadratning perimetri va yuzini toping.

$P = 4 \cdot 5 = 20$ cm, $\quad S = 5^2 = 25\ \text{cm}^2$.

2-misol. Yuzi $49\ \text{cm}^2$ bo'lgan kvadratning perimetrini toping.

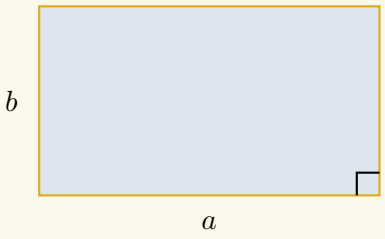
$a^2 = 49 \Rightarrow a = 7$ cm, $\quad P = 4 \cdot 7 = 28$ cm.

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To'g'ri to'rtburchak

✠ Asosiy qoida

Tomonlari a va b bo'lgan to'g'ri to'rtburchak uchun:



$$P = 2(a+b) \qquad S = a \cdot b$$

Misol. Bo'yi 7 cm, eni 4 cm bo'lgan to'g'ri to'rtburchakning perimetri va yuzini toping.

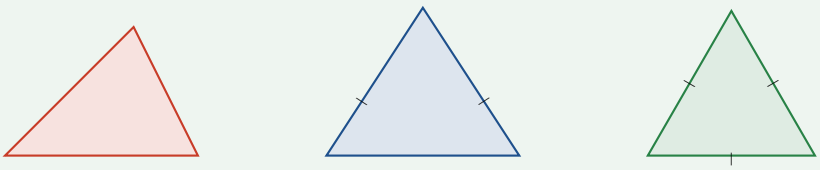
$$P = 2 \cdot (7+4) = 22 \text{ cm}, \qquad S = 7 \cdot 4 = 28 \text{ cm}^2.$$

Uchburchak

✓ Eslatma

Tomonlariga ko'ra uchburchak turlari:

- **Turli tomonli** — barcha tomonlari har xil uzunlikda.
- **Teng yonli** — ikkita tomoni o'zaro teng (*yon tomonlar*); uchinchi tomoni — *asos*.
- **Teng tomonli** (muntazam) — uchala tomoni o'zaro teng.



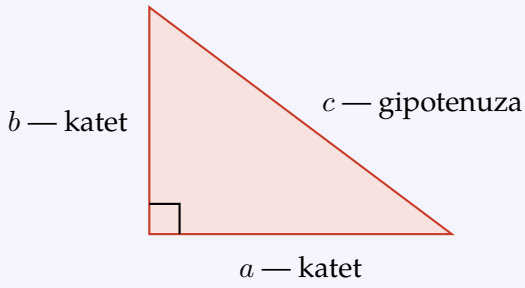
Burchaklariga ko'ra uchburchak turlari:

- **O'tkir burchakli** — barcha burchaklari o'tkir burchak.
- **To'g'ri burchakli** — burchaklaridan biri to'g'ri burchak.
- **O'tmas burchakli** — burchaklaridan biri o'tmas burchak.

✎ Ta'rif

To'g'ri burchakli uchburchak — bitta burchagi 90° ga teng bo'lgan uchburchak.

- **Katetlar** — to'g'ri burchakni hosil qiluvchi ikkita tomonga aytiladi.
- **Gipotenuza** — 90° li burchak qarshisida yotuvchi tomon. U uchburchakning eng uzun tomoni hisoblanadi.



✠ Asosiy qoida

Pifagor teoremasi. To'g'ri burchakli uchburchakda katetlar uzunliklari kvadratlarining yig'indisi gipotenuza uzunligining kvadratiga teng:

$$a^2 + b^2 = c^2$$

Misol. Katetlari 3 cm va 4 cm bo'lgan to'g'ri burchakli uchburchakning gipotenuzasini toping.

$$3^2 + 4^2 = c^2 \Rightarrow c^2 = 25 \Rightarrow c = 5 \text{ cm}.$$

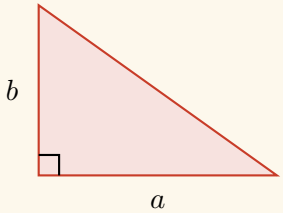
✠ Asosiy qoida

Uchburchakning perimetri uchburchakning tomonlari uzunliklari yig'indisiga teng.

$$P = a + b + c$$

✠ Asosiy qoida

To'g'ri burchakli uchburchakning yuzi katetlari ko'paytmasining yarmiga teng.

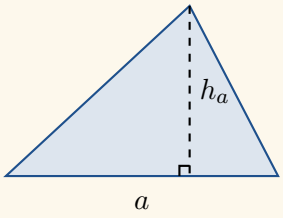


$$S = \frac{a \cdot b}{2}$$

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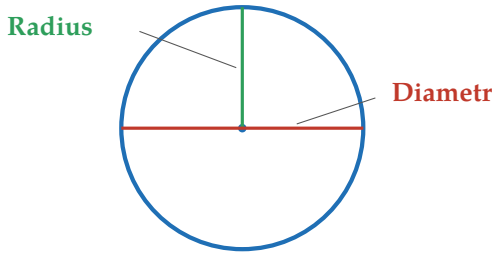
✠ Asosiy qoida

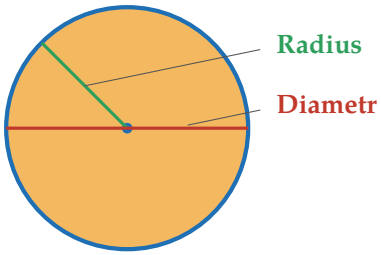
Uchburchakning bir tomoni va shu tomonga tushirilgan perpendikulyar orqali uchburchak yuzini topish mumkin:



$$S = \frac{a \cdot h_a}{2}$$

Aylana va doira





✎ Ta'rif

Aylana — tekislikda berilgan nuqtadan bir xil masofada joylashgan nuqtalar to'plami.
Doira — tekislikning aylana bilan chegaralangan sohasi .

- **Radius** (r) — aylana markazidan aylanagacha masofa.
- **Diametr** (d) — aylana markazidan o'tuvchi vatar. $\Rightarrow \boxed{d = 2r}$.

✠ Asosiy qoida

Aylana uzunligi:

$$C = 2\pi r = \pi d$$

Doira yuzi:

$$S = \pi r^2$$

$\pi \approx 3{,}14$ (*irratsional son*). Masalalarda ko'pincha javob π orqali yoziladi.

Misol. Radiusi 5 cm bo'lgan aylana uzunligi va doira yuzini toping.

$$C = 2\pi \cdot 5 = 10\pi \text{ cm}, \qquad S = \pi \cdot 5^2 = 25\pi \text{ cm}^2.$$

Amaliy qism

1–12. *Kvadrat.*

1 Tomoni 8 m ga teng bo'lgan kvadratning yuzini toping.

2 Tomoni 12 cm ga teng bo'lgan kvadratning perimetrini toping.

3 Tomoni 9 dm ga teng kvadratning perimetrini toping.

4 Perimetri 16 dm ga teng kvadratning yuzini toping.

5 Perimetri 24 m ga teng kvadratning yuzini toping.

6 Yuzi 121 cm^2 ga teng kvadratning perimetrini toping.

7 Yuzi 64 m^2 ga teng kvadratning tomonini toping.

8 Tomoni $1{,}5$ m bo'lgan kvadratning yuzini toping.

9 Yuzi 144 ar bo'lgan kvadrat shaklidagi yer maydonining tomoni necha metr?

10 Tomoni 24 cm bo'lgan kvadratni yuzi 144 cm^2 bo'lgan nechta kvadratchaga bo'lish mumkin?

11 Tomoni 2 m ga teng bo'lgan kvadrat shaklidagi qog'ozni tomoni 1 dm bo'lgan kvadratchalarga bo'lishdi. Nechta bo'lakcha hosil bo'ldi?

12 Yuzi 169 gektar bo'lgan kvadrat shaklidagi yerning bir tomoni va perimetrining uzunligini toping.

13–24. *To'g'ri to'rtburchak.*

13 Tomonlari 4 cm va 6 cm ga teng to'g'ri to'rtburchakning perimetrini toping.

14 Tomonlari 12 cm va 15 cm ga teng to'g'ri to'rtburchakning perimetrini toping.

15 Perimetri 24 m ga teng to'g'ri to'rtburchakning eni 5 m bo'lsa, to'g'ri to'rtburchakning bo'yini toping.

16 Perimetri 28 cm ga teng to'g'ri to'rtburchakning bo'yi enidan 3 cm ga uzun. To'g'ri to'rtburchakning tomonlarini toping.

17 Yuzi 28 cm^2 ga teng to'g'ri to'rtburchakning eni 4 cm. To'g'ri to'rtburchakning perimetrini toping.

18 Yuzi 51 m^2 ga teng to'g'ri to'rtburchakning eni 3 m. Perimetrini toping.

19 To'g'ri to'rtburchak shaklidagi yer maydonining eni 65 m, bo'yi 80 m. To'g'ri to'rtburchakning yuzini toping va arda ifodalang.

20 To'g'ri to'rtburchakning bo'yi 30 dm, eni bo'yidan 5 dm qisqa bo'lsa, yuzini m^2 da toping.

21 To'g'ri to'rtburchak shaklidagi yer maydonining o'lchamlari 700 m va 400 m. Yer maydonining yuzini ga da ifodalang.

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22 Bog' maydoni 70 m^2. Bo'yi 7 m bo'lsa, perimetrini toping.

23 To'g'ri to'rtburchakning bo'yi 4 dm, eni 360 cm. To'g'ri to'rtburchakning yuzini m^2 da ifodalang.

24 Tomonlari 56 cm va 44 cm bo'lgan to'g'ri to'rtburchak perimetriga perimetri teng kvadratning tomonini toping.

25–36. *Uchburchak.*

25 Tomonlari 3 cm, 4 cm va 5 cm bo'lgan uchburchakning perimetrini toping.

26 Tomoni uzunligi 6 m bo'lgan teng tomonli uchburchakning perimetrini toping.

27 Yon tomonlari 5 cm, asosi 4 cm bo'lgan teng yonli uchburchakning perimetrini toping.

28 Uchburchakning eng katta tomoni 12 cm. Qolgan tomonlari katta tomonidan 2 cm va 3 cm ga qisqa bo'lsa, uchburchakning perimetrini toping.

29 To'g'ri burchakli uchburchakning katetlari 5 cm va 4 cm. To'g'ri burchakli uchburchakning yuzini toping.

30 To'g'ri burchakli uchburchakning katetlari 12 m va 6 m. To'g'ri burchakli uchburchakning yuzini toping.

31 To'g'ri burchakli uchburchakning gipotenuzasi 10 m, gipotenuzaga tushirilgan balandligi 4 m. To'g'ri burchakli uchburchakning yuzini toping.

32 To'g'ri burchakli uchburchakning yuzi 24 m^2. Bir kateti 6 m bo'lsa, ikkinchi katetini toping.

33 To'g'ri burchakli uchburchak yuzi 16 cm^2. Gipotenuzaga tushirilgan balandligi 4 cm. Gipotenuza uzunligini toping.

34 Bitta kateti 6 va gipotenuzasi 10 ga teng to'g'ri burchakli uchburchakning yuzini toping.

35 Katetlari 9 va 12 ga teng to'g'ri burchakli uchburchakning gipotenuzasini va perimetrini toping.

36 Bir tomoni 24 ga teng uchburchakning qolgan ikki tomoni $4:5$ nisbatda. Agar uchburchakning perimetri 60 ga teng bo'lsa, uning kichik tomonni toping.

37–46. *Aylana va doira.*

37 Radiusi 5 cm bo'lgan aylana uzunligini toping.

38 Radiusi 2 cm bo'lgan aylana uzunligini toping.

39 Diametri 12 cm bo'lgan aylana uzunligini toping.

40 Uzunligi 16π m bo'lgan aylana radiusini toping.

41 Uzunligi 12π cm bo'lgan aylana diametrini toping.

42 Radiusi 6 cm bo'lgan doira yuzini toping.

43 Radiusi 12 cm bo'lgan doira yuzini toping.

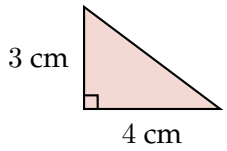
44 Diametri 14 cm bo'lgan doira yuzini toping.

45 Yuzi 121π cm^2 bo'lgan doira radiusini toping.

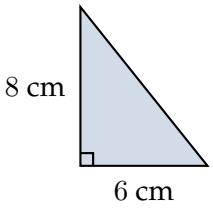
46 Yuzi 144π cm^2 bo'lgan doira diametrini toping.

47–70. *Chizmali masalalar.*

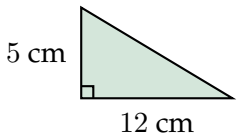
47 To'g'ri burchakli uchburchakning yuzini toping.



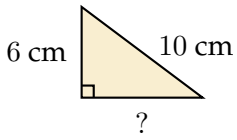
48 To'g'ri burchakli uchburchakning yuzini toping.



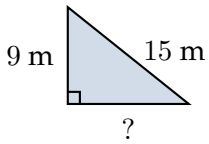
49 To'g'ri burchakli uchburchakning yuzini toping.



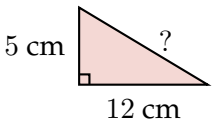
50 Kateti 6 cm va gipotenuzasi 10 cm bo'lgan to'g'ri burchakli uchburchakning yuzini va perimetrini toping.



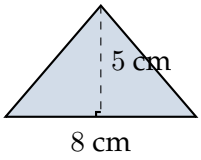
51 Kateti 9 m va gipotenuzasi 15 m bo'lgan to'g'ri burchakli uchburchakning yuzini toping.



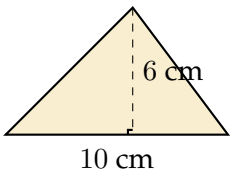
52 To'g'ri burchakli uchburchakning katetlari 5 cm va 12 cm. Gipotenuzasini toping.



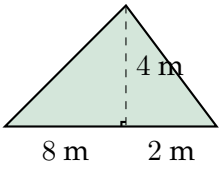
53 Berilgan uchburchakning yuzini toping.



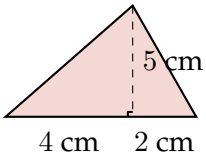
54 Berilgan uchburchakning yuzini toping.



55 Uchburchakning yuzini toping.

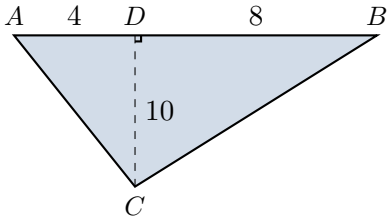


56 Berilgan uchburchakning yuzini toping.

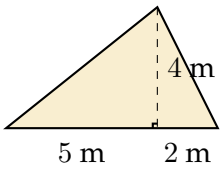


57 ABC uchburchakda $AD = 4$ cm, $DB = 8$ cm, $CD = 10$ cm va $CD \perp AB$. Uchburchakning yuzini toping.

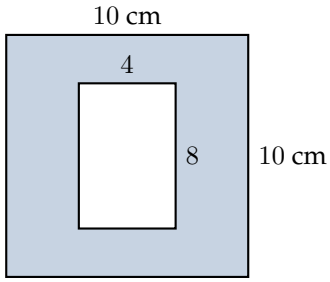
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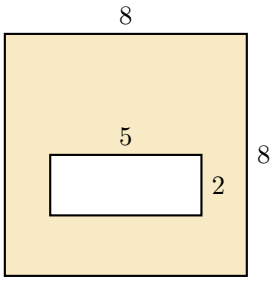
58 Uchburchak yuzini toping.



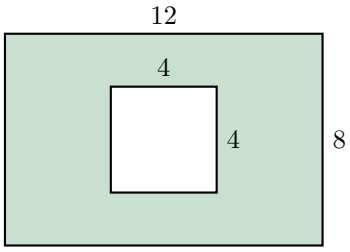
59 Tomoni 10 cm bo'lgan kvadrat ichida tomonlari 4 cm va 8 cm bo'lgan to'g'ri to'rtburchak shaklidagi rasm joylashgan. Bo'yalgan soha yuzini toping.



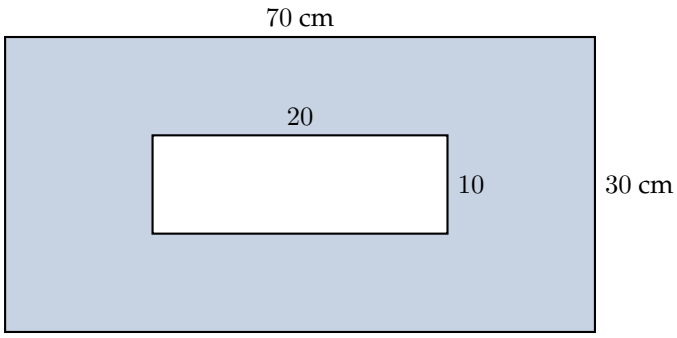
60 Tomoni 8 cm bo'lgan kvadrat ichida tomonlari 5 cm va 2 cm bo'lgan to'g'ri to'rtburchak joylashgan. Bo'yalgan soha yuzini toping.



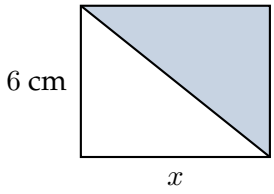
61 Tomonlari 12 cm va 8 cm bo'lgan to'g'ri to'rtburchak ichida tomoni 4 cm bo'lgan kvadrat shaklidagi oyna kesib olingan. Bo'yalgan soha yuzini toping.



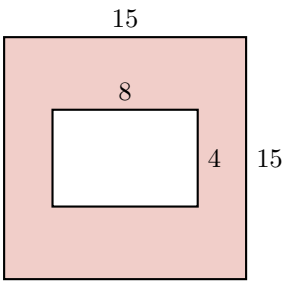
62 Tomonlari 70 cm va 30 cm bo'lgan to'g'ri to'rtburchak shaklidagi taxta o'rtasidan 20 cm $\times$ 10 cm o'lchamli teshik o'yib olindi. Bo'yalgan (qolgan) soha yuzini toping.



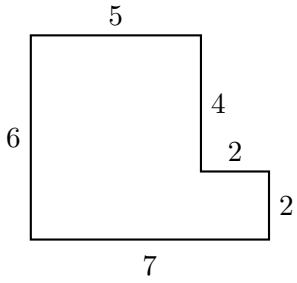
63 Berilgan shaklda bo'yalgan soha yuzi 30 cm^2 ga teng. x uzunligini (cm) toping.



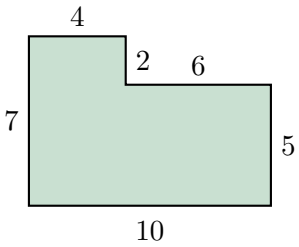
64 Tomoni 15 ga teng bo'lgan $ABCD$ kvadrat ichida tomonlari 8 va 4 ga teng to'g'ri to'rtburchak joylashtirilgan. Bo'yalgan soha yuzini toping.



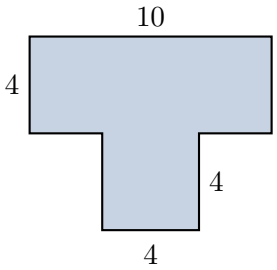
65 Quyidagi shakl perimetrini toping.



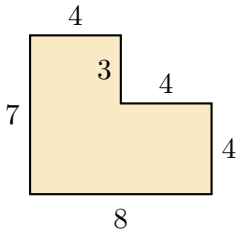
66 L-shaklli figura yuzini toping .



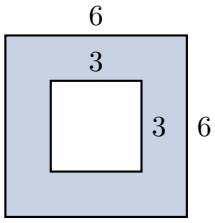
67 Quyidagi T-shaklli figura yuzini toping .



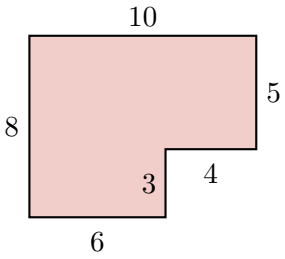
68 Quyidagi shakl yuzini toping.



69 Tomoni 6 cm bo'lgan kvadrat ichida tomoni 3 cm bo'lgan kichik kvadrat joylashgan. Bo'yalgan soha yuzini toping.



70 Quyidagi murakkab shakl yuzini toping .



71–80. *Matnli masalalar.*

71 Futbol maydonining bo'yi 115 m, eni 75 m. Maydonning yuzini hisoblang.

72 14 gektar dalaning 80 % iga paxta ekilgan. Dalaning qancha yeriga paxta ekilgan?

73 12 gektar yerning 80 % iga paxta ekildi. Qolgan yer necha gektarni tashkil qiladi?

74 15 gektar yer yosh oilalarga tomorqa uchun taqsimlab berildi. Agar bitta tomorqa maydoni 3 sotix bo'lsa, jami nechta oila tomorqa olgan?

75 40 700 m^2 ni gektar va arda ifodalang.

76 Tomonlari 5 m va 3 m bo'lgan to'g'ri to'rtburchak shaklidagi xona poliga tomoni 50 cm bo'lgan kvadrat plitkalar yotqizilmoqchi. Polni to'liq qoplash uchun necha dona plitka kerak?

77 Bog' uchun ajratilgan to'g'ri to'rtburchak shaklidagi yerning bo'yi 40 m, eni 25 m. Bog' atrofini sim bilan o'rash uchun necha metr sim kerak bo'ladi?

78 Devor yuzasining 15 % qismi bo'yalgan bo'lsa, uning qancha qismi bo'yalmagan? (Javobni oddiy kasr ko'rinishida yozing.)

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79 Doira shaklidagi havzaning radiusi 7 m. Havza atrofini aylantirib o'rab chiqishga necha metr sim kerak? ($\pi = \frac{22}{7}$ deb oling.)

80 Tomoni 20 m bo'lgan kvadrat shaklidagi yer markazida tomoni 5 m bo'lgan kvadrat hovuz qurildi. Hovuzdan tashqaridagi yer maydonining yuzini toping.

Mavzu bo'yicha test savollari

1-variant

1. Tomoni 6 cm bo'lgan kvadratning yuzini toping.
   A) 12 cm^2 B) 24 cm^2 C) 36 cm^2 D) 64 cm^2
2. Yuzi 100 m^2 bo'lgan kvadratning tomonini toping.
   A) 10 m B) 20 m C) 25 m D) 50 m
3. Tomonlari 5 cm va 8 cm bo'lgan to'g'ri to'rtburchakning perimetrini toping.
   A) 13 cm B) 26 cm C) 40 cm D) 20 cm
4. Tomonlari 7 cm va 4 cm bo'lgan to'g'ri to'rtburchakning yuzini toping.
   A) 11 cm^2 B) 22 cm^2 C) 28 cm^2 D) 32 cm^2
5. Tomoni 5 m bo'lgan teng tomonli uchburchakning perimetrini toping.
   A) 10 m B) 15 m C) 20 m D) 25 m
6. Katetlari 3 cm va 4 cm bo'lgan to'g'ri burchakli uchburchakning gipotenuzasini toping.
   A) 5 cm B) 7 cm C) 12 cm D) 25 cm
7. Katetlari 8 cm va 6 cm bo'lgan to'g'ri burchakli uchburchakning yuzini toping.
   A) 14 cm^2 B) 24 cm^2 C) 48 cm^2 D) 30 cm^2
8. Radiusi 4 cm bo'lgan aylananing uzunligini toping.
   A) 4π cm B) 8π cm C) 16π cm D) 2π cm
9. Radiusi 3 cm bo'lgan doiraning yuzini toping.
   A) 6π cm^2 B) 9π cm^2 C) 3π cm^2 D) 12π cm^2
10. 1 ga necha m^2 ga teng?
    A) 100 B) 1000 C) 10 000 D) 100 000

2-variant

1. Perimetri 20 cm bo'lgan kvadratning yuzini toping.
   A) 16 cm^2 B) 25 cm^2 C) 36 cm^2 D) 100 cm^2
2. Yuzi 144 m^2 bo'lgan kvadratning perimetrini toping.
   A) 24 m B) 36 m C) 48 m D) 72 m
3. To'g'ri to'rtburchakning perimetri 32 cm. Bo'yi 9 cm bo'lsa, enini toping.
   A) 5 cm B) 7 cm C) 14 cm D) 23 cm
4. To'g'ri to'rtburchakning yuzi 60 cm^2 ga teng. Eni 5 cm bo'lsa, perimetrini toping.
   A) 17 cm B) 26 cm C) 34 cm D) 20 cm

5. To'g'ri burchakli uchburchakning kateti 9 m, gipotenuzasi 15 m. Yuzasini toping.
   A) 36 m^2 B) 54 m^2 C) 60 m^2 D) 67,5 m^2

6. Katetlari 8 va 15 bo'lgan to'g'ri burchakli uchburchakning gipotenuzasini toping.
   A) 13 B) 15 C) 17 D) 21

7. Uzunligi 20π cm bo'lgan aylananing radiusini toping.
   A) 5 cm B) 10 cm C) 20 cm D) 40 cm

8. Yuzi 100π cm^2 bo'lgan doiraning diametrini toping.
   A) 5 cm B) 10 cm C) 20 cm D) 50 cm

9. Tomoni 10 cm bo'lgan kvadrat ichida tomoni 6 cm bo'lgan kvadrat shaklidagi teshik kesib olingan. Qolgan sohaning yuzini toping.
   A) 16 cm^2 B) 36 cm^2 C) 64 cm^2 D) 100 cm^2

10. Tomonlari 30 m va 50 m bo'lgan to'g'ri to'rtburchak shaklidagi yer maydoni necha ar?
    A) 15 ar B) 150 ar C) 1,5 ar D) 1500 ar

Javoblar

1. 64 m^2

2. 48 cm

3. 36 dm

4. 16 dm^2

5. 36 m^2

6. 44 cm

7. 8 m

8. 2,25 m^2

9. 120 m

10. 4 ta

11. 400 ta

12. 1300 m; 5200 m

13. 20 cm

14. 54 cm

15. 7 m

16. 8,5 cm va 5,5 cm

17. 22 cm

18. 40 m

19. 5200 $m^2 = 52$ ar

20. $30 \cdot 25 = 750$ $dm^2 = 7{,}5$ m^2

21. 280 000 $m^2 = 28$ ga

22. 34 m

23. 1,44 m^2

24. 50 cm

25. 12 cm

26. 18 m

27. 14 cm

28. 31 cm

29. 10 cm^2

30. 36 m^2

31. 20 m^2

32. 8 m

33. 8 cm

34. 24

35. $c = 15$, $P = 36$

36. 16

37. 10π cm

38. 4π cm

39. 12π cm

40. 8 m

41. 12 cm

42. 36π cm^2

43. 144π cm^2

44. 49π cm^2

45. 11 cm

46. 24 cm

47. 6 cm^2

48. 24 cm^2

49. 30 cm^2

50. Ikkinchi kateti 8, $S = 24$ cm^2, $P = 24$ cm

51. Ikkinchi kateti 12, $S = 54$ m^2

52. $c = 13$ cm

53. 20 cm^2

54. 30 cm^2

55. 20 m^2

56. 15 cm^2

57. 60 cm^2

58. 14 m^2

59. 68

60. 54

61. 80 cm^2

62. 1900 cm^2

63. $x = 10$

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64. 193

65. 26 cm

66. 58 cm^2

67. 56 cm^2

68. 44 cm^2

69. 27 cm^2

70. 68 cm^2

71. 8625 m^2

72. 11,2 ga

73. 2,4 ga

74. 500 ta oila

75. 4 ga 7 ar

76. 60 ta

77. 130 m

78. $\frac{17}{20}$

79. 44 m

80. 375 m^2

Test javoblari

1-variant: 1) C 2) A 3) B 4) C 5) B 6) A 7) B 8) B 9) B 10) C

2-variant: 1) B 2) C 3) B 4) C 5) B 6) C 7) B 8) C 9) C 10) A

Mavzu bo'yicha interaktiv testlar

QR kodni skanerlab, bilimingizni sinab ko'ring

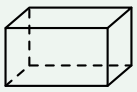
9.

Fazoviy jismlar: kub, parallelepiped, prizma va piramida

Nazariy qism

To'g'ri burchakli parallelepiped

✓ Eslatma



Parallelepiped

- Uchlari soni — **8** ta
- Qirralari soni — **12**
- Yoqlari soni— **6** ta

✠ Asosiy qoida

To'g'ri burchakli parallelepiped formulalari (o'lchamlari a, b, c):

- **Qirralar uzunliklari yig'indisi:**

$$L = 4(a + b + c)$$

- **Yon sirt yuzi**:

$$S_{\text{yon}} = 2(a + b) \cdot c = P_{\text{asos}} \cdot H$$

- **To'la sirt yuzi**:

$$S_{\text{to'la}} = 2(ab + bc + ac)$$

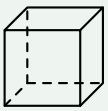
- **Hajmi:**

$$V = a \cdot b \cdot c \qquad \text{yoki} \qquad V = S_{\text{asos}} \cdot H$$

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Kub

✓ Eslatma



Kub

- Uchlari soni — **8** ta
- Qirralari soni — **12** ta
- Yoqlari soni— **6** ta

✠ Asosiy qoida

Kub formulalari (qirrasi a):

- **Qirralar uzunliklari yig'indisi:** $L = 12a$
- **Bir yog'ining yuzi:** $S_1 = a^2$
- **Yon sirt yuzi:** $S_{\text{yon}} = 4a^2$
- **To'la sirt yuzi:** $S_{\text{to'la}} = 6a^2$
- **Hajmi:** $V = a^3$

n-burchakli prizma

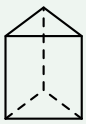
✠ Asosiy qoida

n-burchakli prizmaning:

- Uchlari soni: **2n**
- Qirralari soni: **3n**
- Yoqlari soni: **n + 2**

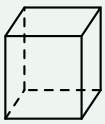
✓ Eslatma

Xususiy hollar:



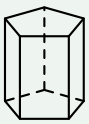
3-burchakli prizma

- Uchlari soni: 6 ta
- Qirralari soni: 9 ta
- Yoqlari soni: 5 ta



4-burchakli prizma

- Uchlari soni: 8 ta
- Qirralari soni: 12 ta
- Yoqlari soni: 6 ta



5-burchakli prizma

- Uchlari soni: 10 ta
- Qirralari soni: 15 ta
- Yoqlari soni: 7 ta

n-burchakli piramida

✠ Asosiy qoida

n-burchakli piramidaning:

- Uchlari soni: $\mathbf{n+1}$
- Qirralari soni: $\mathbf{2n}$
- Yoqlari soni: $\mathbf{n+1}$

✓ Eslatma

Xususiy hollar:



3-burchakli piramida

- Uchlari soni: 4 ta
- Qirralari soni: 6 ta
- Yoqlari soni: 4 ta



4-burchakli piramida

- Uchlari soni: 5 ta
- Qirralari soni: 8 ta
- Yoqlari soni: 5 ta



5-burchakli piramida

- Uchlari soni: 6 ta
- Qirralari soni: 10 ta
- Yoqlari soni: 6 ta

✓ Eslatma

$$1 \text{ litr} = 1 \text{ dm}^3, \quad 1 \text{ m}^3 = 1000 \text{ dm}^3 = 1\,000\,000 \text{ cm}^3$$

Amaliy qism

1–12. *Uchlar, qirralar va yoqlar soni.*

1 Kubning nechta uchi bor?

2 Parallelepipedning nechta qirrasi bor?

3 Kubning nechta yon yog'i bor?

4 Parallelepipedning nechta yog'i bor?

5 Uchburchakli prizmaning uchlari, qirralari va yoqlari sonini toping.

6 5-burchakli prizmaning uchlari, qirralari va yoqlari sonini toping.

7 6-burchakli prizmaning uchlari, qirralari va yoqlari sonini toping.

8 4-burchakli piramidaning uchlari, qirralari va yoqlari sonini toping.

9 Uchburchakli piramidaning uchlari, qirralari va yoqlari sonini toping.

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10 7-burchakli piramidaning uchlari, qirralari va yoqlari sonini toping.

11 Prizmaning yoqlari soni 9 ta. Bu necha burchakli prizma? Uchlari va qirralari sonini toping.

12 Piramidaning qirralari soni 14 ta. Bu necha burchakli piramida? Uchlari va yoqlari sonini toping.

13–32. *Kub.*

13 Qirrasi 3 cm bo'lgan kubning qirralari uzunliklari yig'indisini toping.

14 Qirrasi 7 cm bo'lgan kubning to'la sirtining yuzini toping.

15 Qirrasi 4 cm bo'lgan kubning hajmini toping.

16 Qirrasi 9 cm bo'lgan kubning yon sirtining yuzini toping.

17 Qirrasi 5 cm bo'lgan kubning to'la sirtining yuzini va hajmini toping.

18 Qirrasi 10 cm bo'lgan kubning qirralari uzunliklari yig'indisini va hajmini toping.

19 Qirrasi 2 dm bo'lgan kubning to'la sirtining yuzini toping.

20 Qirrasi 6 cm bo'lgan kubning yon sirtining yuzini toping.

21 Qirrasi 1 m bo'lgan kubning hajmini cm^3 da toping.

22 Qirrasi 8 cm bo'lgan kubning to'la sirt yuzi va yon sirt yuzi farqini toping.

23 Kubning hajmi 8 cm^3 ga teng. Qirrasining uzunligini toping.

24 Kubning hajmi 216 cm^3 ga teng. Qirrasining uzunligini toping.

25 Kubning to'la sirtining yuzi 54 cm^2 ga teng. Qirrasining uzunligini toping.

26 Kubning to'la sirtining yuzi 96 cm^2 ga teng. Kubning hajmini toping.

27 Kubning yon sirtining yuzi 36 cm^2 ga teng. Qirrasining uzunligini toping.

28 Kubning yon sirt yuzi 100 cm^2 ga teng. Kubning hajmini toping.

29 Kubning qirralari uzunliklari yig'indisi 48 cm. Kubning qirrasini va hajmini toping.

30 Kubning qirralari uzunliklari yig'indisi 84 cm. To'la sirtining yuzini toping.

31 Kubning hajmi 125 cm^3 ga teng. To'la sirtining yuzini toping.

32 Bir yog'ining yuzi 49 cm^2 bo'lgan kubning hajmini toping.

33–54. *Parallelepiped.*

33 O'lchamlari 3 cm, 4 cm, 5 cm bo'lgan parallelepipedning hajmini toping.

34 O'lchamlari 2 cm, 3 cm, 4 cm bo'lgan parallelepipedning to'la sirt yuzini toping.

35 O'lchamlari 5 cm, 6 cm, 7 cm bo'lgan parallelepipedning qirralari uzunliklari yig'indisini toping.

36 O'lchamlari 4 cm, 5 cm, 6 cm bo'lgan parallelepipedning to'la sirt yuzini toping.

37 O'lchamlari 8 cm, 5 cm, 3 cm bo'lgan parallelepipedning hajmini toping.

38 O'lchamlari 10 cm, 6 cm, 4 cm bo'lgan parallelepipedning qirralari uzunliklari yig'indisini toping.

39 O'lchamlari 9 cm, 4 cm, 5 cm bo'lgan parallelepipedning yon sirt yuzini toping (balandligi 5 cm).

40 Eni 4 cm, bo'yi 6 cm, balandligi 8 cm bo'lgan parallelepipedning hajmini toping.

41 Eni 3 cm, bo'yi 7 cm, balandligi 5 cm bo'lgan parallelepipedning to'la sirt yuzini toping.

42 Asosing yuzi 24 cm^2, balandligi 5 cm bo'lgan parallelepipedning hajmini toping.

43 O'lchamlari 7 cm, 4 cm, 2 cm bo'lgan parallelepipedning yon sirt yuzini toping (balandligi 2 cm).

44 Asosining perimetri 20 cm, balandligi 7 cm bo'lgan parallelepipedning yon sirtining yuzini toping.

45 Hajmi 240 cm^3, eni 4 cm, bo'yi 6 cm bo'lgan parallelepipedning balandligini toping.

46 Hajmi 180 cm^3, balandligi 5 cm bo'lgan parallelepiped asosining yuzini toping.

47 Asosining yuzi 36 cm^2, hajmi 144 cm^3 bo'lgan parallelepipedning balandligini toping.

48 Qirralari uzunliklari yig'indisi 80 cm, eni 4 cm, bo'yi 6 cm bo'lgan parallelepipedning balandligini toping.

49 Qirralari uzunliklari yig'indisi 56 cm bo'lgan parallelepipedning eni, bo'yi va balandligi $1:2:4$ nisbatda. O'lchamlarini toping.

50 Eni va bo'yining uzunliklari 5 va 6 ga teng bo'lgan parallelepipedning qirralari uzunliklari yig'indisi 60 ga teng. Balandligini toping.

51 Asosining perimetri 14 ga teng, balandligi 6 ga teng parallelepipedning yon sirt yuzini toping.

52 Hajmi 120 ga, asosining yuzi 24 ga teng parallelepipedning balandligini toping.

53 Eni 3 cm, bo'yi 4 cm bo'lgan parallelepipedning to'la sirt yuzi 108 cm^2. Balandligini toping.

54 Eni 5 cm, balandligi 6 cm bo'lgan parallelepipedning hajmi 240 cm^3. Bo'yini toping.

55–62. *Prizma va piramida.*

55 n-burchakli prizmaning uchlari 16 ta. n ni va prizma qirralari sonini toping.

56 n-burchakli piramidaning yoqlari 9 ta. Uchlari va qirralari sonini toping.

57 Prizmaning qirralari soni 24 ta. Bu prizmaning uchlari va yoqlari sonini toping.

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58 Piramidaning uchlari soni 11 ta. Uning qirralari va yoqlari sonini toping.

59 Bir prizmaning uchlari 20 ta. Bu necha burchakli prizma? Yon yoqlari sonini toping.

60 Muntazam 5-burchakli prizmaning umumiy elementlari (uch + qirra + yoq) soni nechta?

61 Muntazam 6-burchakli piramidaning umumiy elementlari (uch + qirra + yoq) soni nechta?

62 Bir piramidaning qirralari soni 20. Bu piramida necha burchakli? Uchlari va yoqlari sonini toping.

63–70. *Aralash masalalar.*

63 Xonaning bo'yi 5 m, eni 4 m, balandligi 3 m. Xonada necha kub metr havo bor?

64 Akvariumning eni 4 dm, bo'yi 5 dm, balandligi 3 dm. Akvariumga necha litr suv sig'adi?

65 Suv bakining o'lchamlari 6 dm, 5 dm va 4 dm. Suv bakining hajmini toping.

66 Qirrasi 6 cm bo'lgan kubni qirrasi 2 cm bo'lgan kichik kublarga ajratish mumkin. Nechta kichik kub hosil bo'ladi?

67 Qirrasi 15 cm bo'lgan kubni qirrasi 3 cm bo'lgan kichik kublarga ajratish mumkin. Nechta kichik kub hosil bo'ladi?

68 Atirsovunning o'lchamlari 8 cm, 4 cm, 2 cm. Sovundan har kuni 4 cm^3 ishlatilsa, sovundan necha kun foydalanish mumkin?

69 Temirdan qirrasi 5 cm bo'lgan kub shaklidagi detal yasaldi. 5 cm^3 temirning massasi 40 g bo'lsa, detalning massasini toping.

70 1 cm^2 yuzani bo'yash uchun 0,5 g bo'yoq sarflanadi. Qirrasi 4 cm bo'lgan kub shaklidagi qutini bo'yash uchun necha gramm bo'yoq kerak?

Mavzu bo'yicha test savollari

1-variant

---

1. Qirrasi 5 cm bo'lgan kubning hajmini toping.
   A) 25 cm^3 B) 75 cm^3 C) 125 cm^3 D) 150 cm^3

2. Parallelepiped o'lchamlari 4 cm, 5 cm, 6 cm. Hajmini toping.
   A) 90 cm^3 B) 120 cm^3 C) 148 cm^3 D) 180 cm^3

3. Qirrasi 7 cm bo'lgan kubning to'la sirt yuzini toping.
   A) 196 cm^2 B) 245 cm^2 C) 294 cm^2 D) 343 cm^2

4. Kubning hajmi 64 cm^3 ga teng. Qirrasini toping.
   A) 2 cm B) 4 cm C) 6 cm D) 8 cm

5. 5-burchakli prizmaning qirralari soni nechta?
   A) 10 B) 12 C) 15 D) 20

6. Parallelepiped o'lchamlari 3 cm, 4 cm, 5 cm bo'lsa, qirralari uzunliklari yig'indisi nechaga teng?
   A) 36 cm B) 48 cm C) 60 cm D) 72 cm

7. 6-burchakli piramidaning yoqlari soni nechta?
   A) 6 B) 7 C) 8 D) 12

8. Kubning to'la sirt yuzi 24 cm^2. Hajmini toping.
   A) 4 cm^3 B) 8 cm^3 C) 16 cm^3 D) 27 cm^3

9. Akvarium o'lchamlari 5 dm, 4 dm, 3 dm. Necha litr suv sig'adi?
   A) 30 B) 45 C) 60 D) 90

10. Hajmi 343 cm^3 bo'lgan kubning qirralari uzunliklari yig'indisi nechaga teng?
    A) 42 cm B) 56 cm C) 72 cm D) 84 cm

2-variant

---

1. Qirrasi 6 cm bo'lgan kubning to'la sirt yuzini toping.
   A) 144 cm^2 B) 180 cm^2 C) 216 cm^2 D) 256 cm^2

2. Parallelepiped o'lchamlari 2 cm, 3 cm, 5 cm. To'la sirt yuzini toping.
   A) 30 cm^2 B) 46 cm^2 C) 54 cm^2 D) 62 cm^2

3. Kubning qirralari uzunliklari yig'indisi 36 cm. Qirrasini toping.
   A) 2 cm B) 3 cm C) 4 cm D) 6 cm

4. Uchburchakli prizmaning uchlari soni nechta?
   A) 4 B) 5 C) 6 D) 9

5. Parallelepipedning hajmi 48 cm^3, eni 2 cm, bo'yi 4 cm. Balandligini toping.
   A) 4 cm B) 6 cm C) 8 cm D) 12 cm

6. Qirrasi a bo'lgan kubning to'la sirt yuzi formulasi qaysi?
   A) $4a^2$ B) $6a^2$ C) a^3 D) $12a$

7. 4-burchakli piramidaning qirralari soni nechta?
   A) 4 B) 5 C) 6 D) 8

8. Xonaning bo'yi 6 m, eni 5 m, balandligi 4 m. Necha kub metr havo bor?
   A) 60 B) 90 C) 120 D) 150

9. Kubning yon sirt yuzi 64 cm^2. Qirrasini toping.
   A) 2 cm B) 4 cm C) 6 cm D) 8 cm

10. Bir prizmaning yoqlari soni 7 ga teng. Bu prizmaning uchlari soni nechaga teng?
    A) 5 B) 6 C) 10 D) 12

Javoblar

1. 8 ta

2. 12 ta

3. 4 ta

4. 6 ta

5. 6 uchi, 9 qirrasi, 5 yog'i

6. 10 uchi, 15 qirrasi, 7 yog'i

7. 12 uchi, 18 qirrasi, 8 yog'i

8. 5 uchi, 8 qirrasi, 5 yog'i

9. 4 uchi, 6 qirrasi, 4 yog'i

10. 8 uchi, 14 qirrasi, 8 yog'i

11. 7-burchakli; 14 uchi, 21 qirrasi

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12. 7-burchakli; 8 uchi, 8 yog'i

13. 36 cm

14. 294 cm^2

15. 64 cm^3

16. 324 cm^2

17. $S = 150\ \text{cm}^2$, $V = 125\ \text{cm}^3$

18. $L = 120$ cm, $V = 1000\ \text{cm}^3$

19. 24 dm^2

20. 144 cm^2

21. 1 000 000 cm^3

22. 128 cm^2

23. 2 cm

24. 6 cm

25. 3 cm

26. 64 cm^3

27. 3 cm

28. 125 cm^3

29. $a = 4$ cm, $V = 64\ \text{cm}^3$

30. 294 cm^2

31. 150 cm^2

32. 343 cm^3

33. 60 cm^3

34. 52 cm^2

35. 72 cm

36. 148 cm^2

37. 120 cm^3

38. 80 cm

39. 130 cm^2

40. 192 cm^3

41. 142 cm^2

42. 120 cm^3

43. 44 cm^2

44. 140 cm^2

45. 10 cm

46. 36 cm^2

47. 4 cm

48. 10 cm

49. $a = 2,\ b = 4,\ c = 8$

50. 4

51. 84

52. 5

53. 6 cm

54. 8 cm

55. $n = 8$, qirralari 24 ta

56. $n = 8$, 9 uchi, 16 qirrasi

57. 8-burchakli; 16 uchi, 10 yog'i

58. 10-burchakli; 20 qirrasi, 11 yog'i

59. 10-burchakli; 10 yon yog'i

60. 32

61. 26

62. 10-burchakli; 11 uchi, 11 yog'i

63. 60 m^3

64. 60 litr

65. 120 dm^3

66. 27 ta

67. 125 ta

68. 16 kun

69. 1000 g = 1 kg

70. 48 g

Test javoblari

1-variant: 1) C 2) B 3) C 4) B 5) C 6) B 7) B 8) B 9) C 10) D

2-variant: 1) C 2) D 3) B 4) C 5) B 6) B 7) D 8) C 9) B 10) C

Mavzu bo'yicha interaktiv testlar

QR kodni skanerlab, bilimingizni sinab ko'ring

10.

O'lchov birliklari

Nazariy qism

1. Uzunlik o'lchov birliklari

✠ Asosiy qoida

$$1\text{ km} = 1000\text{ m}, \qquad 1\text{ m} = 10\text{ dm} = 100\text{ cm} = 1000\text{ mm}.$$
$$1\text{ dm} = 10\text{ cm} = 100\text{ mm}, \qquad 1\text{ cm} = 10\text{ mm}.$$

km	m	dm	cm	mm
$\xrightarrow{\times 1000}$	$\xrightarrow{\times 10}$	$\xrightarrow{\times 10}$	$\xrightarrow{\times 10}$	—

Misol 1. 3 km 40 m necha metrga teng?
Yechish: $3\text{ km} = 3 \cdot 1000 = 3000\text{ m}$. Demak, $3\text{ km }40\text{ m} = 3000 + 40 = 3040\text{ m}$.

Misol 2. 5000 mm necha metr?
Yechish: $1\text{ m} = 1000\text{ mm}$, demak $5000 : 1000 = 5\text{ m}$.

2. Yuza o'lchov birliklari

✠ Asosiy qoida

$$1\text{ m}^2 = 100\text{ dm}^2 = 10\,000\text{ cm}^2, \qquad 1\text{ dm}^2 = 100\text{ cm}^2 = 10\,000\text{ mm}^2.$$
$$1\text{ cm}^2 = 100\text{ mm}^2, \qquad 1\text{ km}^2 = 1\,000\,000\text{ m}^2.$$

Yer maydoni uchun:

$$1\text{ ar} = 100\text{ m}^2, \qquad 1\text{ ga} = 100\text{ ar} = 10\,000\text{ m}^2, \qquad 1\text{ km}^2 = 100\text{ ga}.$$

km²	m²	dm²	cm²	mm²
$\xrightarrow{\times 10^6}$	$\xrightarrow{\times 100}$	$\xrightarrow{\times 100}$	$\xrightarrow{\times 100}$	—

Misol 1. $8\text{ m}^2\ 23\text{ dm}^2$ necha dm^2?
Yechish: $8 \cdot 100 + 23 = 800 + 23 = 823\text{ dm}^2$.

Misol 2. $50\,000\text{ m}^2$ necha gektar?
Yechish: $50\,000 : 10\,000 = 5\text{ ga}$.

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3. Hajm o'lchov birliklari

✠ Asosiy qoida

$$1\ \text{m}^3 = 1000\ \text{dm}^3 = 1\,000\,000\ \text{cm}^3.$$
$$1\ \text{dm}^3 = 1000\ \text{cm}^3 = 1\,000\,000\ \text{mm}^3.$$
$$1\ \text{cm}^3 = 1000\ \text{mm}^3.$$

Suyuqlik uchun — litr (l):

$\boxed{1\ \text{dm}^3 = 1\ \text{litr}}$ $\quad 1\ \text{cm}^3 = 1\ \text{ml}$ $\quad 1\ \text{m}^3 = 1000\ \text{litr}.$

m^3	dm^3 (litr)	cm^3 (ml)	mm^3
$\xrightarrow{\times 1000}$	$\xrightarrow{\times 1000}$	$\xrightarrow{\times 1000}$	—

Misol 1. $4\ \text{dm}^3$ necha cm^3 va necha litr?
Yechish: $4 \cdot 1000 = 4000\ \text{cm}^3$, $4\ \text{dm}^3 = 4$ litr.

Misol 2. $60\,000\ \text{cm}^3$ necha litr?
Yechish: 1 litr $= 1000\ \text{cm}^3$,
demak $60\,000 : 1000 = 60$ litr.

4. Vazn (og'irlik) o'lchov birliklari

✠ Asosiy qoida

$1\ \text{t} = 10\ \text{sr} = 1000\ \text{kg}, \quad 1\ \text{sr} = 100\ \text{kg}, \quad 1\ \text{kg} = 1000\ \text{g}, \quad 1\ \text{g} = 1000\ \text{mg}.$

(**sr** — sentner; **t** — tonna; **g** — gramm; **mg** — milligramm.)

t	sr	kg	g	mg
$\xrightarrow{\times 10}$	$\xrightarrow{\times 100}$	$\xrightarrow{\times 1000}$	$\xrightarrow{\times 1000}$	—

Misol 1. 5 t 5 sr 5 kg necha kg?
Yechish: $5 \cdot 1000 + 5 \cdot 100 + 5 = 5000 + 500 + 5 = 5505$ kg.

5. Vaqt o'lchov birliklari

✠ Asosiy qoida

1 minut $= 60$ sek, $\quad 1$ soat $= 60$ min $= 3600$ sek.

1 sutka $= 24$ soat, $\quad 1$ hafta $= 7$ sutka, $\quad 1$ yil $= 12$ oy.

sutka	soat	min	sek
$\xrightarrow{\times 24}$	$\xrightarrow{\times 60}$	$\xrightarrow{\times 60}$	—

Misol 1. 2 soat 15 minut necha minut?
Yechish: $2 \cdot 60 + 15 = 120 + 15 = 135$ min.

✓ **Eslatma**

Asosiy qoida: *kattadan kichikka* o'tganda **ko'paytirish**, *kichikdan kattaga* o'tganda — **bo'lish** bajariladi.

Murakkab ifodalarda har bir qism alohida hisoblanadi, so'ngra qo'shiladi:

$$3\ \text{m}\ 4\ \text{dm}\ 5\ \text{cm} = 3 \cdot 100 + 4 \cdot 10 + 5 = 345\ \text{cm}.$$

Amaliy qism

1–10. *Uzunlik birliklarini ifodalang.*

1. $4\ \text{km} = \ldots\ \text{m}$
2. $6\ \text{m} = \ldots\ \text{cm}$
3. $8\ \text{dm} = \ldots\ \text{mm}$
4. $5\ \text{km}\ 320\ \text{m} = \ldots\ \text{m}$
5. $3\ \text{m}\ 4\ \text{dm}\ 2\ \text{cm} = \ldots\ \text{cm}$
6. $2\ \text{km}\ 50\ \text{m} = \ldots\ \text{cm}$
7. $9000\ \text{m} = \ldots\ \text{km}$
8. $8000\ \text{mm} = \ldots\ \text{m}$
9. $7000\ \text{cm} = \ldots\ \text{m}$
10. $50\,000\ \text{mm} = \ldots\ \text{dm}$

11–20. *Yuza birliklarini ifodalang.*

11. $5\ \text{m}^2 = \ldots\ \text{dm}^2$
12. $7\ \text{km}^2 = \ldots\ \text{m}^2$
13. $4\ \text{dm}^2 = \ldots\ \text{cm}^2$
14. $3\ \text{m}^2\ 25\ \text{dm}^2 = \ldots\ \text{dm}^2$
15. $8\ \text{ga} = \ldots\ \text{m}^2$
16. $6\ \text{ar}\ 50\ \text{m}^2 = \ldots\ \text{m}^2$
17. $80\,000\ \text{m}^2 = \ldots\ \text{ga}$
18. $9000\ \text{dm}^2 = \ldots\ \text{m}^2$
19. $600\ \text{ar} = \ldots\ \text{ga}$
20. $5\,000\,000\ \text{m}^2 = \ldots\ \text{km}^2$

21–30. *Hajm birliklarini ifodalang.*

21. $7\ \text{m}^3 = \ldots\ \text{dm}^3$
22. $4\ \text{dm}^3 = \ldots\ \text{cm}^3$
23. $6\ \text{litr} = \ldots\ \text{cm}^3$
24. $3\ \text{m}^3\ 250\ \text{dm}^3 = \ldots\ \text{dm}^3$
25. $5\ \text{dm}^3 = \ldots\ \text{litr}$
26. $2\ \text{m}^3\ 40\ \text{dm}^3 = \ldots\ \text{litr}$
27. $8000\ \text{dm}^3 = \ldots\ \text{m}^3$
28. $9000\ \text{cm}^3 = \ldots\ \text{dm}^3$
29. $12\,000\ \text{ml} = \ldots\ \text{litr}$
30. $60\,000\ \text{cm}^3 = \ldots\ \text{dm}^3$

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31–40. *Massa birliklarini ifodalang.*

31 7 kg = ... g

32 5 t = ... kg

33 8 sr = ... kg

34 4 kg 250 g = ... g

35 3 t 5 sr = ... kg

36 6 t 4 sr 50 kg = ... kg

37 9000 g = ... kg

38 8000 kg = ... t

39 7000 kg = ... sr

40 60 000 g = ... kg

41–50. *Vaqt birliklarini ifodalang.*

41 5 soat = ... min

42 4 min = ... sek

43 2 sutka = ... soat

44 3 soat 20 min = ... min

45 1 soat 30 min = ... sek

46 1 sutka 5 soat = ... min

47 480 min = ... soat

48 7200 sek = ... soat

49 96 soat = ... sutka

50 540 sek = ... min

51–60. *Hisoblang.*

51 1 km − 250 m = ... m

52 2 m + 70 cm = ... cm

53 1 soat − 15 min = ... min

54 1 kg − 300 g = ... g

55 $1\ m^2 - 25\ dm^2 = ...\ dm^2$

56 1 sutka − 8 soat = ... soat

57 3 m 25 cm + 1 m 40 cm = ... cm

58 5 kg 200 g − 2 kg 800 g = ... g

59 1 t − 250 kg = ... kg

60 1 litr − 350 ml = ... ml

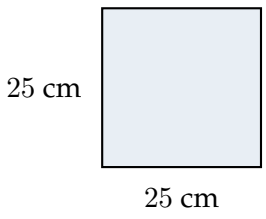
61–80. *Matnli va chizmali masalalar.*

61 Akvariumning ichki o'lchamlari 50 cm × 40 cm × 30 cm. U to'la suv bilan to'ldirilsa, necha litr suv sig'adi?

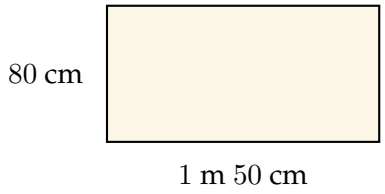
62 To'g'ri to'rtburchak shaklidagi yer maydonining bo'yi 40 m, eni 25 m. Maydonning yuzi necha arga teng?

63 Qirrasi 2 dm bo'lgan kubning hajmi necha cm^3?

64 Quyidagi rasmda kvadratning tomoni 25 cm ga teng. Uning perimetri necha metr bo'ladi?



65 Rasmdagi to'g'ri to'rtburchakning perimetri necha santimetr?

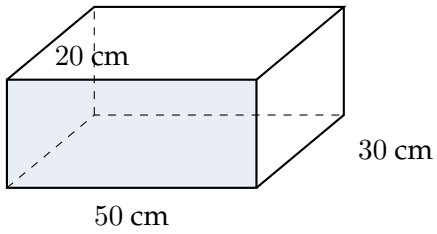


66 Aziz 4 km masofani bosib o'tdi, so'ngra yana 750 m yo'l yurdi. Aziz jami necha metr yo'l bosib o'tgan?

67 Poyezd manzilga soat 9:15 da jo'nab, 12:40 da yetib bordi. Poyezd yo'lda necha minut bo'lgan?

68 5 kg un solingan qopdan 1200 g un olindi. Qopda yana necha gramm un qoldi?

69 Rasmdagi to'g'ri burchakli parallelepiped shaklidagi idishning hajmi necha litr?

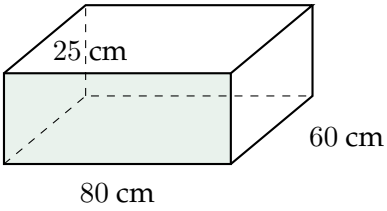


70 4 ta yuk mashinasi har birida 3 t 500 kg dan yuk olib keldi. Jami necha kg yuk olib kelingan?

71 Maktabda dars davomiyligi 45 min, tanaffuslar 10 mindan. 6 dars va 5 tanaffus jami necha soat va necha minutni tashkil qiladi?

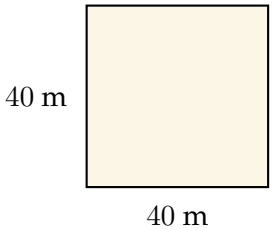
72 Bir arqonni 8 ta teng bo'lakka bo'lishdi. Har bir bo'lak 75 cm dan bo'ldi. Arqonning to'liq uzunligi necha metr edi?

73 Rasmda to'g'ri burchakli parallelepiped shaklidagi quti tasvirlangan. Uning hajmi necha dm^3?



74 Kino soat 18:45 da boshlanib, 1 soat 50 min davom etdi. Kino soat nechada tugagan?

75 Rasmda kvadrat shaklidagi yer maydoni tasvirlangan. Maydonning yuzi necha arga teng?



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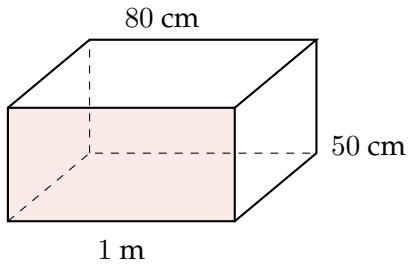
76 To'g'ri burchakli parallelepiped shaklidagi akvariumning bo'yi 60 cm, eni 40 cm, balandligi 50 cm. Akvariumga suv quyildi, lekin yuqori qismida 10 cm joy bo'sh qoldi. Akvariumdagi suvning hajmi necha litr?

77 Avtomobil soatiga 80 km tezlikda yurmoqda. U 2 soat 30 min davomida qancha masofa bosib o'tadi?

78 1 gektar bug'doy maydonidan o'rtacha 25 sentner bug'doy olinadi. 6 gektar maydondan jami necha tonna bug'doy yig'iladi?

79 Bir suv havzasi 6 ta bir xil nasos yordamida 5 soatda to'ldiriladi. Agar 3 ta shunday nasos ishlatilsa, havza necha soatda to'ladi?

80 Quyidagi rasmda parallelepiped shaklidagi quti tasvirlangan. Qutining hajmi necha litr?



Mavzu bo'yicha test savollari

1-variant

1. 5 t 5 sr 5 kg $=\boxed{?}$ kg
   A) 5055 B) 5505 C) 5550 D) 5555

2. 6 dm 6 mm $=\boxed{?}$ mm
   A) 66 B) 606 C) 660 D) 6006

3. 7 m^2 25 dm^2 $=\boxed{?}$ dm^2
   A) 725 B) 7025 C) 70 025 D) 7250

4. 1 sutka 2 soat 30 minut necha minut?
   A) 1500 B) 1530 C) 1560 D) 1590

5. Qirrasi 3 dm bo'lgan kubning hajmi necha cm^3?
   A) 9 B) 27 C) 2700 D) 27 000

6. 9 km $=\boxed{?}$ cm
   A) 9000 B) 90 000 C) 900 000 D) 9 000 000

7. Akvariumning bo'yi 40 cm, eni 30 cm, balandligi 25 cm. U to'la suv bilan to'ldirilsa, necha litr suv sig'adi?
   A) 3 B) 30 C) 95 D) 300

8. 4 ga 25 ar $=\boxed{?}$ m^2
   A) 4250 B) 40 250 C) 42 500 D) 425 000

9. Avtomobil soatiga 90 km tezlikda 4 soat 20 min yurdi. U qancha masofa bosib o'tgan?
   A) 300 km B) 360 km C) 380 km D) 390 km

10. 1 sentner 1 tonnaning qanday qismini tashkil etadi?
    A) $\frac{1}{100}$ B) $\frac{1}{10}$ C) $\frac{1}{1000}$ D) $\frac{1}{4}$

2-variant

---

1. 8 t 8 sr 8 kg = $\boxed{?}$ kg
   A) 8088 B) 8808 C) 8880 D) 8888

2. 9 m 9 cm = $\boxed{?}$ cm
   A) 99 B) 909 C) 990 D) 9009

3. 6 m^3 40 dm^3 = $\boxed{?}$ litr
   A) 604 B) 640 C) 6040 D) 6400

4. 2 soat 15 minut necha sekund?
   A) 135 B) 8100 C) 7500 D) 7350

5. 80 000 cm^2 necha m^2?
   A) 0,8 B) 8 C) 80 D) 800

6. 1 t 250 kg + 750 kg = $\boxed{?}$ t
   A) 1 B) 1,5 C) 2 D) 2,5

7. Yer maydoni o'lchamlari 200 m × 250 m ga teng bo'lsa, uning yuzi necha gektarga teng?
   A) 5 B) 50 C) 500 D) 5000

8. Bir bochkaga 200 litr suv sig'sa, unga necha cm^3 suv sig'adi?
   A) 200 B) 2000 C) 20 000 D) 200 000

9. Treningda 3 ta dars 40 minutdan, 2 ta dars 30 minutdan o'tkazildi. Trening jami necha soat davom etgan?
   A) 2 soat 30 min B) 2 soat 45 min C) 3 soat D) 3 soat 30 min

10. Yuk mashinasi 2 sentner yuk olib ketmoqda. Mashina yuk vaznini 1 tonnaga yetkazish uchun yana qancha yuk ortishi kerak?
    A) 200 kg B) 500 kg C) 800 kg D) 980 kg

Javoblar

1. 4000
2. 600
3. 800
4. 5320
5. 342
6. 205 000
7. 9
8. 8
9. 70
10. 500
11. 500
12. 7 000 000
13. 400
14. 325
15. 80 000
16. 650
17. 8
18. 90

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19. 6
20. 5
21. 7000
22. 4000
23. 6000
24. 3250
25. 5
26. 2040
27. 8
28. 9
29. 12
30. 60
31. 7000
32. 5000
33. 800
34. 4250
35. 3500
36. 6450
37. 9
38. 8
39. 70
40. 60
41. 300
42. 240
43. 48
44. 200
45. 5400
46. 1740
47. 8
48. 2
49. 4
50. 9
51. 750
52. 270
53. 45
54. 700
55. 75
56. 16
57. 465
58. 2400
59. 750
60. 650
61. 60 litr
62. 10 ar
63. 8000 cm^3
64. 1 m
65. 460 cm
66. 4750 m
67. 205 min
68. 3800 g
69. 30 litr
70. 14 000 kg
71. 5 soat 20 min
72. 6 m
73. 120 dm^3
74. 20:35
75. 16 ar
76. 96 litr
77. 200 km
78. 15 t
79. 10 soat
80. 400 litr

Test javoblari

1-variant: 1) B 2) B 3) A 4) D 5) D 6) C 7) B 8) C 9) D 10) B
2-variant: 1) B 2) B 3) C 4) B 5) B 6) C 7) A 8) D 9) C 10) C

Mavzu bo'yicha interaktiv testlar
QR kodni skanerlab, bilimingizni sinab ko'ring

11.

Birhadlar

Nazariy qism

Birhad tushunchasi

✎ Ta'rif

Birhad deb — sonli va harfiy ko'paytuvchilar ko'paytmasidan iborat algebraik ifodaga aytiladi.

Misollar: $5a,\quad -3xy,\quad \frac{1}{2}ab^2,\quad -7,\quad x,\quad 4a^2b^3c.$

Birhadning standart shakli

✠ Asosiy qoida

Birhadni standart shaklga keltirish uchun:

1. Barcha sonli ko'paytuvchilarning ko'paytmasi topiladi.
2. Barcha harfiy ko'paytuvchilarning alohida ko'paytmasi topiladi.
3. Topilgan sonli ko'paytuvchi harflar oldiga yoziladi.

Misollar:

$$6a \cdot 2b = (6 \cdot 2) \cdot (a \cdot b) = 12ab.$$

$$\tfrac{1}{2}a \cdot \left(-\tfrac{4}{5}b\right) \cdot 3ac = \left(\tfrac{1}{2} \cdot \left(-\tfrac{4}{5}\right) \cdot 3\right) \cdot a \cdot a \cdot b \cdot c = -\tfrac{6}{5}a^2bc.$$

Koeffitsiyent va daraja

✎ Ta'rif

Standart shakldagi birhadning sonli ko'paytuvchisi — uning **koeffitsiyenti** deyiladi.
Birhadda qatnashgan barcha harflar darajalarining yig'indisi — uning **darajasi** deyiladi.

Misol: $-5a^3b^2c$ birhadi uchun:

- Koeffitsiyent: (-5).
- Daraja: $3 + 2 + 1 = 6$.

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✓ **Eslatma**

Agar birhad oldida koeffitsiyent yozilmagan bo'lsa (masalan: abc), uning koeffitsiyenti 1 ga teng bo'ladi. Agar oldida faqat minus tursa (masalan: $-x^2y$), koeffitsiyent (-1) ga teng bo'ladi.

Qavslarni ochish qoidasi

✠ **Asosiy qoida**

Agar qavs oldida $+$ ishorasi turgan bo'lsa, qavslarni ochishda qavs ichidagi qo'shiluvchilarning ishoralarini o'zgartirmay qavsni ochish kerak:

$$a + (b - c + d) = a + b - c + d.$$

Agar qavs oldida $-$ ishorasi turgan bo'lsa, qavslarni ochishda qavs ichidagi qo'shiluvchilarning ishoralarini qarama-qarshisiga almashtirib, qavsni ochish kerak:

$$a - (b - c + d) = a - b + c - d.$$

Ichma-ich qavslarni ochishda eng ichkaridan boshlab ochish qulay:

$$a - \big(b - (c + d)\big) = a - (b - c - d) = a - b + c + d.$$

Birhadlarni ko'paytirish

✠ **Asosiy qoida**

Birhadlar ko'paytirilganda:

1. Koeffitsiyentlar ko'paytiriladi.
2. Bir xil asosli darajalar uchun ko'rsatkichlar qo'shiladi:

$$a^m \cdot a^n = a^{m+n}.$$

Misol:

$$3a^2b \cdot (-4ab^3) = (3 \cdot (-4)) \cdot a^{2+1} \cdot b^{1+3} = -12a^3b^4.$$

✓ **Eslatma**

Daraja qavs ichidagi ifodaga qo'llanilsa, qavs ichidagi har bir ko'paytuvchi darajaga ko'tariladi:

$$(ab)^n = a^n \cdot b^n, \qquad (a^m)^n = a^{m \cdot n}.$$

Misol:

$$(2x^2y)^3 \cdot xy^2 = 2^3x^6y^3 \cdot xy^2 = 8x^{6+1}y^{3+2} = 8x^7y^5.$$

Birhadlarni bo'lish

✠ Asosiy qoida

Birhadni birhadga bo'lish uchun:

1. Koeffitsiyentlar bo'linadi.
2. Bir xil asosli darajalar uchun ko'rsatkichlar ayriladi:

$$a^m : a^n = a^{m-n}.$$

Misol:

$$12a^5b^3 : (4a^2b) = (12 : 4) \cdot a^{5-2} \cdot b^{3-1} = 3a^3b^2.$$

Amaliy qism

1–10. *Qavslarni oching.*

1 $a + (b - c + d)$

2 $5 + (2a - 3b + c)$

3 $x - (y + z - t)$

4 $4 - (2x - 3y + z)$

5 $7a - (-2b + 3c - d)$

6 $a - \big(b - (c + d)\big)$

7 $10 + \big(3a - (2b - c)\big)$

8 $x - \big(2y + (3z - t)\big)$

9 $a + \Big(b - \big(c + (d - e)\big)\Big)$

10 $5a - \Big(2b - \big(3c + (d - e)\big)\Big)$

11–20. *Birhadning koeffitsiyenti va darajasini toping.*

11 $-6a^3b$

12 $4a^2b^3$

13 $\frac{1}{3}abc$

14 $-7x^4y$

15 $\frac{2}{5}x^2y^3z$

16 $-a^5b^2$

17 $8m^2n^4p$

18 $-\frac{3}{4}a^2b^2c^2$

19 $9x^3y$

20 $-12a^4bc^3$

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21–35. *Ko'paytmani bajaring.*

21 $3a \cdot 4b$

22 $-5x \cdot 2y$

23 $\frac{1}{2}a \cdot 6b$

24 $2ab^2 \cdot 3a^3b$

25 $-4x^2y \cdot 5xy^3$

26 $\frac{2}{3}a^2b \cdot \frac{3}{4}ab^4$

27 $-7m^3n^2 \cdot (-2mn^5)$

28 $0{,}5x^4y \cdot 8x^2y^3$

29 $(2a^2)^3 \cdot 3ab$

30 $(-3xy^2)^2 \cdot 2x^3y$

31 $(2a^2b)^3 \cdot (-ab^2)$

32 $(-x^3)^2 \cdot (2x^2y)^2$

33 $(3a^2b)^2 \cdot (-2ab^3)^2$

34 $\left(\frac{1}{2}xy^2\right)^3 \cdot 8x^2y$

35 $(-2a^2b)^3 \cdot (3ab^2)^2$

36–50. *Birhadni birhadga bo'ling.*

36 $12a^5 : (4a^2)$

37 $-18b^6 : (6b^3)$

38 $20x^3y^4 : (5xy^2)$

39 $-15a^4b^5 : (-3a^2b)$

40 $\frac{2}{3}x^5y^3 : \left(\frac{1}{6}x^2y\right)$

41 $36m^7n^4 : (9m^3n^2)$

42 $-24a^3b^4c : (8abc)$

43 $48x^5y^3z^2 : (-6x^2yz)$

44 $\frac{3}{5}p^4q^6 : \left(\frac{3}{10}p^2q^3\right)$

45 $-100a^8b^5 : (-25a^4b^2)$

46 $54a^6b^5c^4 : (6a^2b^3c)$

47 $(2a^3b)^2 : (4a^2b)$

48 $(-3xy^2)^3 : (9x^2y^4)$

49 $(4a^2b^3)^2 : (2ab)^3$

50 $(-2x^2y)^4 : (2xy)^2$

51–57. *Birhadning qiymatini toping.*

51 $2a^2b$, agar $a = 2$; $b = 3$ bo'lsa

52 $-3x^3y$, agar $x = -1$; $y = 4$ bo'lsa

53 $\frac{1}{5}a^3b^2$, agar $a = -2$; $b = 5$ bo'lsa

54 $4m^2n$, agar $m = 3$; $n = -2$ bo'lsa

55 $8a^2b$, agar $a = 0{,}5$; $b = 4$ bo'lsa

56 $-2x^2y^3$, agar $x = -3$; $y = 2$ bo'lsa

57 $-\frac{3}{4}a^2b^5$, agar $a = 4$; $b = -1$ bo'lsa

58–65. *Standart shaklga keltiring va son qiymatini toping.*

58 $\frac{1}{2}a \cdot 4ab$, bunda $a = 2, b = 3$

59 $(-2x)^2 \cdot xy$, bunda $x = -1, y = 3$

60 $3m^2n \cdot (-2mn)$, bunda $m = 2, n = -3$

61 $(-a^2b)^2 \cdot 2b$, bunda $a = -1, b = 4$

62 $6a^3b^4 : (2ab^2)$, bunda $a = 3, b = 2$

63 $(2xy)^3 : (4y)$, bunda $x = -2, y = 5$

64 $\frac{1}{3}abc \cdot 6a^2b^2$, bunda $a = 2, b = -1, c = 3$

65 $(-x^2y)^3 \cdot 4yz$, bunda $x = 1, y = -2, z = 3$

66–70. *Berilgan shaklning yuzini yoki hajmini birhad ko'rinishida yozing.*

66 Tomonlari $3a$ va $4b$ bo'lgan to'g'ri to'rtburchakning yuzini toping.

67 Tomoni $2x$ bo'lgan kvadratning yuzini toping.

68 Asosi $6m$, balandligi $4n$ bo'lgan uchburchakning yuzini toping.

69 O'lchamlari $2a$, $3b$, $5c$ bo'lgan to'g'ri burchakli parallelepipedning hajmini toping.

70 Qirrasi $3x$ bo'lgan kubning hajmini toping.

Mavzu bo'yicha test savollari

1-variant

1. Qavsni oching: $8 + (a - b)$
   A) $8 - a + b$ B) $8 + a - b$ C) $8 - a - b$ D) $8 + a + b$

2. Soddalashtiring: $5a - (3a - 2b)$
   A) $2a - 2b$ B) $8a - 2b$ C) $2a + 2b$ D) $8a + 2b$

3. $-4a^3b^2c$ birhadining darajasini toping.
   A) 4 B) 5 C) 6 D) -4

4. Ko'paytmani toping: $3a^2b \cdot (-2ab^3)$
   A) $6a^3b^4$ B) $-6a^3b^4$ C) $-6a^2b^3$ D) $-5a^3b^4$

5. $(2x^2y)^3$ ni soddalashtiring.
   A) $6x^6y^3$ B) $8x^5y^3$ C) $8x^6y^3$ D) $8x^6y$

6. Birhadni birhadga bo'ling: $15a^5b^3 : (5a^2b)$
   A) $3a^3b^2$ B) $3a^7b^4$ C) $10a^3b^2$ D) $3a^3b$

7. $-\frac{1}{2}a^2b$ ning qiymatini toping, bunda $a = -2, b = 3$.
   A) 6 B) -6 C) 3 D) -3

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8. Tomonlari $2a$ va $5b$ bo'lgan to'g'ri to'rtburchakning yuzini toping.
   A) $7ab$ B) $10ab$ C) $10a^2b^2$ D) $25ab$

9. $-\frac{2}{3}a^4b^2$ birhadining koeffitsiyentini toping.
   A) $-\frac{2}{3}$ B) $\frac{2}{3}$ C) $-\frac{3}{2}$ D) 4

10. Qavslarni oching: $a-\big(b-(c-d)\big)$
    A) $a-b+c-d$ B) $a+b-c+d$ C) $a-b-c+d$ D) $a-b+c+d$

2-variant

---

1. Qavsni oching: $10-(3x-2y+z)$
   A) $10-3x+2y-z$ B) $10-3x-2y-z$ C) $10-3x-2y+z$ D) $10-3x+2y+z$

2. Soddalashtiring: $7+b+a-(a+b)$
   A) $7-2a$ B) $7-2b$ C) 7 D) $7-a-b$

3. $5a^2b^4c$ birhadining darajasini toping.
   A) 5 B) 6 C) 7 D) 11

4. Ko'paytmani toping: $-4xy^2\cdot 3x^3y$
   A) $-12x^4y^3$ B) $12x^4y^3$ C) $-12x^3y^2$ D) $-7x^4y^3$

5. $(-3a^2b)^2$ ni soddalashtiring.
   A) $-9a^4b^2$ B) $9a^4b^2$ C) $9a^2b^2$ D) $6a^4b^2$

6. Birhadni birhadga bo'ling: $-24m^6n^4:(6m^3n^2)$
   A) $4m^3n^2$ B) $-4m^2n^2$ C) $-4m^3n^2$ D) $-4m^9n^6$

7. $2x^2y^3$ ning qiymatini toping, bunda $x=3$, $y=-1$.
   A) 18 B) -18 C) 6 D) -6

8. Qirrasi $4a$ bo'lgan kubning hajmini toping.
   A) $12a^3$ B) $16a^3$ C) $64a^3$ D) $4a^3$

9. Quyidagilardan qaysi biri birhad emas?
   A) $-7x^3y$ B) $\frac{1}{2}abc$ C) 5 D) $\frac{2a}{b}$

10. $(2a^2b)^3\cdot ab^2$ ko'paytmaning standart shaklini toping.
    A) $6a^7b^5$ B) $8a^6b^4$ C) $8a^7b^5$ D) $2a^7b^5$

Javoblar

1. $a+b-c+d$

2. $5+2a-3b+c$

3. $x-y-z+t$

4. $4-2x+3y-z$

5. $7a+2b-3c+d$

6. $a-b+c+d$

7. $10+3a-2b+c$

8. $x-2y-3z+t$

9. $a+b-c-d+e$

10. $5a-2b+3c+d-e$

11. Koeffitsiyent: -6; daraja: 4

12. Koeffitsiyent: 4; daraja: 5

13. Koeffitsiyent: $\frac{1}{3}$; daraja: 3

14. Koeffitsiyent: -7; daraja: 5

15. Koeffitsiyent: $\frac{2}{5}$; daraja: 6

16. Koeffitsiyent: -1; daraja: 7

17. Koeffitsiyent: 8; daraja: 7

18. Koeffitsiyent: $-\frac{3}{4}$; daraja: 6

19. Koeffitsiyent: 9; daraja: 4

20. Koeffitsiyent: -12; daraja: 8

21. $12ab$

22. $-10xy$

23. $3ab$

24. $6a^4b^3$

25. $-20x^3y^4$

26. $\frac{1}{2}a^3b^5$

27. $14m^4n^7$

28. $4x^6y^4$

29. $24a^7b$

30. $18x^5y^5$

31. $-8a^7b^5$

32. $4x^{10}y^2$

33. $36a^6b^8$

34. x^5y^7

35. $-72a^8b^7$

36. $3a^3$

37. $-3b^3$

38. $4x^2y^2$

39. $5a^2b^4$

40. $4x^3y^2$

41. $4m^4n^2$

42. $-3a^2b^3$

43. $-8x^3y^2z$

44. $2p^2q^3$

45. $4a^4b^3$

46. $9a^4b^2c^3$

47. a^4b

48. $-3xy^2$

49. $2ab^3$

50. $4x^6y^2$

51. 24

52. 12

53. -40

54. -72

55. 8

56. -144

57. 12

58. 24

59. -12

60. -432

61. 128

62. 108

63. -400

64. -48

65. -192

66. $12ab$

67. $4x^2$

68. $12mn$

69. $30abc$

70. $27x^3$

Test javoblari

1-variant: 1) B 2) C 3) C 4) B 5) C 6) A 7) B 8) B 9) A 10) A

2-variant: 1) A 2) C 3) C 4) A 5) B 6) C 7) B 8) C 9) D 10) C

Mavzu bo'yicha interaktiv testlar

QR kodni skanerlab, bilimingizni sinab ko'ring

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12.

Ko'phadlar

Nazariy qism

Ko'phadlar

✎ Ta'rif

Bir nechta birhadning algebraik yig'indisi **ko'phad** deyiladi.

Ko'phadni tashkil qiluvchi birhadlar shu **ko'phadning hadlari** deyiladi.

O'xshash hadlar

✎ Ta'rif

O'xshash hadlar deb harfli qismi (harflar va ularning darajalari) bir xil bo'lgan hadlarga aytiladi.

- $5a$ va $7a$ — o'xshash hadlar;
- $3x^2$ va $-8x^2$ — o'xshash hadlar;
- $4ab$ va $4a^2b$ — o'xshash emas (darajalari farqli).

✠ Asosiy qoida

Ifodani soddalashtirish uchun faqat o'xshash hadlar qo'shiladi yoki ayiriladi. Bunda harflar o'zgarmaydi, faqat koeffitsiyentlar hisoblanadi.

Misol: $5a + 8a - 3a = (5 + 8 - 3)a = 10a.$

Birhadni ko'phadga ko'paytirish

✠ Asosiy qoida

Birhadni ko'phadga ko'paytirish uchun birhadni ko'phadning har bir hadiga ko'paytirish va hosil bo'lgan ko'paytmalarni qo'shish kerak.

$$k(a + b - c) = ka + kb - kc.$$

Misollar:

$$3(2x - 5) = 6x - 15; \qquad -2a(3a + 4b) = -6a^2 - 8ab.$$

Ko'phadni ko'phadga ko'paytirish

✠ Asosiy qoida

Ko'phadni ko'phadga ko'paytirish uchun birinchi ko'phadning har bir hadini ikkinchi ko'phadning har bir hadiga ko'paytirish va hosil bo'lgan ko'paytmalarni qo'shish kerak.

$$(a+b)(c+d) = ac + ad + bc + bd.$$

Misol:

$$\begin{aligned}(x+2)(x-3) &= x \cdot x + x \cdot (-3) + 2 \cdot x + 2 \cdot (-3) \\ &= x^2 - 3x + 2x - 6 = x^2 - x - 6.\end{aligned}$$

Ko'phadni birhadga bo'lish

✠ Asosiy qoida

Ko'phadni birhadga bo'lish uchun ko'phadning har bir hadini berilgan birhadga bo'lish va natijalarni qo'shish kerak.

$$(a+b) : k = a : k + b : k, \qquad \frac{a+b}{k} = \frac{a}{k} + \frac{b}{k}.$$

Misollar:

$$(6a+8) : 2 = 3a + 4, \qquad \frac{18a^2 + 12a}{3a} = 6a + 4.$$

Ko'paytuvchilarga ajratish (umumiy ko'paytuvchini qavs tashqarisiga chiqarish)

✠ Asosiy qoida

Agar ko'phadning har bir hadida bir xil ko'paytuvchi bo'lsa, uni qavs tashqarisiga chiqarish mumkin:

$$ab + ac = a(b+c), \qquad ab - ac = a(b-c).$$

Misollar: $12a^2 + 9a = 3a(4a+3)$ $\qquad 6ab^2c - 18abc = 6abc(b-3)$

Agar $xy = 2$ va $x + y = 3$ bo'lsa, $x^2y + xy^2$ ning qiymatini toping.

$x^2y + xy^2 = xy(x+y) = 2 \cdot 3 = 6.$

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Qisqa ko'paytirish formulalari

✓ Eslatma

Yig'indining kvadrati:

$$(a+b)^2 = a^2 + 2ab + b^2$$

Ayirmaning kvadrati:

$$(a-b)^2 = a^2 - 2ab + b^2$$

Yig'indining kubi:

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

Ayirmaning kubi:

$$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

Kvadratlar ayirmasi:

$$a^2 - b^2 = (a-b)(a+b)$$

Kublar ayirmasi:

$$a^3 - b^3 = (a-b)(a^2+ab+b^2)$$

Kublar yig'indisi:

$$a^3 + b^3 = (a+b)(a^2-ab+b^2)$$

Amaliy qism

1–55. *Ifodani soddalashtiring.*

1 $\underbrace{2a+2a+\ldots+2a}_{11\text{ ta}} - 21a$

2 $\underbrace{3a+3a+\ldots+3a}_{15\text{ ta}} - 44a$

3 $2a+6b+5a-8b$

4 $3x+5y-7x+2y$

5 $4a+7b-9a-2b+5a$

6 $-6m+8n+11m-15n+3m$

7 $(2a+5b)-(7a-3b)$

8 $(3x-4y)+(5x-7y)$

9 $4a-(5a-9b)+(8a-6b)$

10 $-(2x+3y)-(5x-4y)$

11 $3a+8b+(2a+5b-2c)-(-4a+9b-3c)$

12 $-(7m-3n)+(5m+8n)-(2m-n)$

13 $(6x-2y+z)-(3x+5y-2z)$

14 $5a-(3a+2b)-(4b-6a)$

15 $2x+(3x-y)-(5x-7y)+(y-x)$

16 $-(a-b+c)+(b-c+a)-(c-a+b)$

17 $4(a+2)$

18 $3(2a-5)$

19 $-2(3x-7)$

20 $5(2y+4)$

21 $2(3a+2)-3(a+1)$

22 $2(a-3b)+3(2a+b)$

23 $-4(3a-2b)+3(5a-4b)$

24 $5(2x-3y)-2(4x+y)$

25 $3(a-2b)-(2a-5b)$

26 $4(x+3)-3(2x-1)+2(x-4)$

27 $2(a+b-c)+3(2a-b+c)$

28 $5(x-y+2z)-3(2x+y-z)$

29 $-2(3a-b+2c)+4(a+2b-c)$

30 $3(2x-y+z)-2(x+y-2z)+(x-3y+z)$

31 $2x(3x-2)$

32 $3x(2x+5)$

33 $-4y(3y-1)$

34 $5a(2a+7)$

35 $4y(5y+2)-2y(8y+3)$

36 $5a(a+3)-2a(3a-3)$

37 $2a(3a+2b)-4a(a+2b)$

38 $3x(2x-y)+2x(x+3y)$

39 $-2m(m+5)+3m(2m-4)$

40 $(x+2)(x+3)$

41 $(x-3)(x+5)$

42 $(2x+3)(x-2)$

43 $(3x-1)(2x+4)$

44 $(x+2)(x^2-2x+4)$

45 $(x-3)(x^2+3x+9)$

46 $(6a+8):2$

47 $(8a-4):(-4)$

48 $(10x+15):5$

49 $(12y-18):(-6)$

50 $(20a-25b):5$

51 $(-18x+27y):(-9)$

52 $(18a^2+12a):(3a)$

53 $(24x^2-16x):(8x)$

54 $(15y^3+10y^2):(5y^2)$

55 $(28a^2b-14ab^2):(7ab)$

56–70. *Ifodani ko'paytuvchilarga ajrating.*

56 $8a+6$

57 $16a-20$

58 $15x+25$

59 $21y-14$

60 $12a+18b$

61 $24x-32y$

62 $12a^2+9a$

63 $18x^2-24x$

64 $25y^3+15y$

65 $32a^3-16a^2$

66 a^2b+ab^2

67 a^2b-ab^2

68 $a^3b^2+a^2b^3$

69 $a^3b^2-a^2b^3$

70 $6ab^2c-18abc$

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71–80. *Berilgan ifodaning qiymatini toping.*

71 Agar $a = 5$ va $b = 3$ bo'lsa, $0{,}4a^2 + ab$ ning qiymatini toping.

72 Agar $a = 5$ va $b = 2$ bo'lsa, $0{,}3a^2 + ab$ ning qiymatini toping.

73 Agar $x = 4$ va $y = -2$ bo'lsa, $2x^2 - 3xy + y^2$ ning qiymatini toping.

74 Agar $x = 3$ va $y = 2$ bo'lsa, $x^2 + xy - y^2$ ning qiymatini toping.

75 Agar $xy = 2$ va $x + y = 3$ bo'lsa, $x^2y + xy^2$ ning qiymatini toping.

76 Agar $xy = 4$ va $x + y = 5$ bo'lsa, $x^2y + xy^2$ ning qiymatini toping.

77 Agar $a = 2$ va $b = 4$ bo'lsa, $a^2 + 2ab + b^2$ ning qiymatini toping.

78 Agar $a = 3$ va $b = -1$ bo'lsa, $a^2 - 2ab + b^2$ ning qiymatini toping.

79 Agar $a + b = 3$ bo'lsa, $a^3 + 3a^2b + 3ab^2 + b^3$ ning qiymatini toping.

80 Agar $a - b = 5$ bo'lsa, $a^3 - 3a^2b + 3ab^2 - b^3$ ning qiymatini toping.

81–88. *Matnli masalalar.*

81 To'g'ri to'rtburchak shaklidagi bog'ning eni a m, bo'yi esa enidan 5 m uzunroq. Bog'ning perimetri uchun ifoda tuzing va $a = 12$ m bo'lganda perimetrini toping.

82 Sinfda x ta o'g'il bola va y ta qiz bola bor. Har bir o'g'il bolada 4 tadan, har bir qiz bolada 6 tadan daftar bor. Jami daftarlar soni uchun ifoda tuzing va $x = 10$, $y = 15$ bo'lgandagi qiymatini toping.

83 Avtomobil 3 soat davomida soatiga v km, keyin yana 2 soat davomida soatiga $(v+10)$ km tezlikda yurdi. Bosib o'tilgan umumiy masofa uchun ifoda tuzing va $v = 50$ km/soat bo'lgandagi qiymatini toping.

84 Do'konda x ta daftar va y ta ruchka sotildi. Bitta daftar 5000 so'm, bitta ruchka 3000 so'm turadi. Tushum uchun ifoda tuzing va $x = 8$, $y = 12$ bo'lgandagi tushumni hisoblang.

85 To'g'ri to'rtburchakning tomonlari $(2x+1)$ dm va $(x+3)$ dm ga teng. Uning yuzi uchun ifoda tuzing va $x = 4$ bo'lgandagi yuzasini toping.

86 Brigada birinchi kuni $2x$ ta detal, ikkinchi kuni birinchi kundagidan 5 ta ko'p detal, uchinchi kuni esa $(3x - 4)$ ta detal yasadi. Uch kunda jami yasalgan detallar soni uchun ifoda tuzing va ifodaning $x = 10$ bo'lgandagi son qiymatini toping.

87 Bog' to'g'ri to'rtburchak shaklida bo'lib, eni a, bo'yi b metr. Eni 3 metrga uzaytirildi, bo'yi esa 2 metrga qisqartirildi. Yangi bog'ning yuzi uchun ifoda tuzing va $a = 10$, $b = 8$ bo'lgandagi qiymatini toping.

88 To'g'ri to'rtburchak shaklidagi yer maydonining yuzi $a(a + 3)$ m^2 ga teng. Bu maydonda har birining yuzi a m^2 bo'lgan 4 ta gulzor barpo qilindi. Gulzorlardan tashqari qolgan maydon yuzi uchun ifoda tuzing va $a = 5$ bo'lgandagi qiymatini toping.

89–90.

89 Agar $P = 2a + 3b$ va $Q = 3a + 4b$ bo'lsa, $P + Q$ ni toping.

90 Agar $P = 7x - 4y$ va $Q = 3x - 5y$ bo'lsa, $P - Q$ ni toping.

Mavzu bo'yicha test savollari

1-variant

1. Soddalashtiring: $\underbrace{3a + 3a + \ldots + 3a}_{10\text{ ta}} - 29a$
   A) $-a$ B) 0 C) a D) $30a$
2. Soddalashtiring: $(4x - 3y) - (2x + 5y)$
   A) $6x + 2y$ B) $2x - 8y$ C) $2x + 2y$ D) $6x - 8y$
3. Soddalashtiring: $3(2a - 5) - 2(a - 7)$
   A) $4a - 29$ B) $8a + 1$ C) $4a + 1$ D) $4a - 1$
4. Soddalashtiring: $3x(2x - 5) - 2x(x - 8)$
   A) $4x^2 - x$ B) $4x^2 + x$ C) $8x^2 + x$ D) $4x^2 - 31x$
5. Agar $P = 3a - 5b$ va $Q = 5b - 3a$ bo'lsa, $P + Q$ ni toping.
   A) $6a - 10b$ B) $10b$ C) $6a$ D) 0
6. $a^2b + ab^2$ ni ko'paytuvchilarga ajrating.
   A) $ab(a + b)$ B) $a(b + a^2b)$ C) $ab(a^2 + b^2)$ D) $(ab)^2$
7. Agar $a = 4,\ b = -1$ bo'lsa, $2a^2 - ab$ ning qiymati nechaga teng?
   A) 30 B) 32 C) 36 D) 28
8. Agar $x = 2$ va $y = 3$ bo'lsa, $x^2 + 2xy + y^2$ ning qiymatini toping.
   A) 28 B) 22 C) 31 D) 25
9. Soddalashtiring: $(12a^2 - 18a) : (6a)$
   A) $2a - 18$ B) $2a - 3$ C) $6a - 12$ D) $2a^2 - 3$
10. Soddalashtiring: $(x + 2)(x^2 - 2x + 4) - 4$
    A) $x^3 - 4$ B) $x^3 - 8$ C) $x^3 + 8$ D) $x^3 + 4$

2-variant

1. Soddalashtiring: $\underbrace{4a + 4a + \ldots + 4a}_{12\text{ ta}} - 47a$
   A) $48a$ B) a C) $-a$ D) 0
2. Soddalashtiring: $4a - (3a - 2b) + (a - 5b)$
   A) $2a - 7b$ B) $8a - 3b$ C) $2a - 3b$ D) $2a + 3b$
3. Soddalashtiring: $-2(3a - 4) + 5(a - 2)$
   A) $-a + 2$ B) $a - 2$ C) $11a - 18$ D) $-a - 2$
4. Soddalashtiring: $4y(2y + 3) - 3y(y - 5)$
   A) $11y^2 + 27y$ B) $y^2 + 27y$ C) $5y^2 - 3y$ D) $5y^2 + 27y$

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5. Agar $P = 5a - 2b$ va $Q = 5b - 2a$ bo'lsa, $P + Q$ ni toping.
   A) $3(a + b)$ B) $3(a - b)$ C) $7(a - b)$ D) $7(a + b)$

6. $a^3b^2 - a^2b^3$ ni ko'paytuvchilarga ajrating.
   A) $ab(a^2 - b^2)$ B) $a^3b^3(a - b)$ C) $a^2b^2(b - a)$ D) $a^2b^2(a - b)$

7. Agar $a = 3,\ b = 2$ bo'lsa, $a^2 + 2ab$ ning qiymati nechaga teng?
   A) 13 B) 17 C) 15 D) 21

8. Agar $x = 4$ va $y = 5$ bo'lsa, $x^2 - 2xy + y^2$ ning qiymatini toping.
   A) 9 B) -1 C) 1 D) 4

9. Soddalashtiring: $(20a^2b - 15ab) : (5ab)$
   A) $4ab - 3$ B) $4a - 3$ C) $4a^2 - 3$ D) $5a - 3$

10. Soddalashtiring: $(x - 2)(x^2 + 2x + 4) + 4$
    A) $x^3 + 8$ B) $x^3 + 4$ C) $x^3 - 4$ D) $x^3 - 8$

Javoblar

1. a
2. a
3. $7a - 2b$
4. $-4x + 7y$
5. $5b$
6. $8m - 7n$
7. $-5a + 8b$
8. $8x - 11y$
9. $7a + 3b$
10. $-7x + y$
11. $9a + 4b + c$
12. $-4m + 12n$
13. $3x - 7y + 3z$
14. $8a - 6b$
15. $-x + 7y$
16. $a + b - 3c$
17. $4a + 8$
18. $6a - 15$
19. $-6x + 14$
20. $10y + 20$
21. $3a + 1$
22. $8a - 3b$
23. $3a - 4b$
24. $2x - 17y$
25. $a - b$
26. 7
27. $8a - b + c$
28. $-x - 8y + 13z$
29. $-2a + 10b - 8c$
30. $5x - 8y + 8z$
31. $6x^2 - 4x$
32. $6x^2 + 15x$
33. $-12y^2 + 4y$
34. $10a^2 + 35a$
35. $4y^2 + 2y$
36. $-a^2 + 21a$
37. $2a^2 - 4ab$
38. $8x^2 + 3xy$
39. $4m^2 - 22m$
40. $x^2 + 5x + 6$
41. $x^2 + 2x - 15$
42. $2x^2 - x - 6$
43. $6x^2 + 10x - 4$
44. $x^3 + 8$
45. $x^3 - 27$
46. $3a + 4$
47. $-2a + 1$
48. $2x + 3$
49. $-2y + 3$
50. $4a - 5b$
51. $2x - 3y$
52. $6a + 4$
53. $3x - 2$
54. $3y + 2$
55. $4a - 2b$
56. $2(4a + 3)$
57. $4(4a - 5)$
58. $5(3x + 5)$
59. $7(3y - 2)$
60. $6(2a + 3b)$
61. $8(3x - 4y)$
62. $3a(4a + 3)$
63. $6x(3x - 4)$
64. $5y(5y^2 + 3)$
65. $16a^2(2a - 1)$
66. $ab(a + b)$

67. $ab(a-b)$

68. $a^2b^2(a+b)$

69. $a^2b^2(a-b)$

70. $6abc(b-3)$

71. 25

72. 17,5

73. 60

74. 11

75. 6

76. 20

77. 36

78. 16

79. 27

80. 125

81. $P = 4a + 10$; 58 m

82. $4x + 6y$; 130 ta

83. $5v + 20$; 270 km

84. $5000x + 3000y$; 76 000 so'm

85. $S = 2x^2 + 7x + 3$; 63 dm^2

86. $7x + 1$; 71 ta

87. $S = ab + 3b - 2a - 6$; 78 m^2

88. $S = a^2 - a = a(a-1)$; 20 m^2

89. $5a + 7b$

90. $4x + y$

Test javoblari

1-variant: 1) C 2) B 3) D 4) B 5) D 6) A 7) C 8) D 9) B 10) D

2-variant: 1) B 2) C 3) D 4) D 5) A 6) D 7) D 8) C 9) B 10) C

Mavzu bo'yicha interaktiv testlar

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13.

Chiziqli tenglamalar

Nazariy qism

Tenglama nima?

✎ Ta'rif

Harf bilan belgilangan noma'lum sonni o'z ichiga olgan tenglik **tenglama** deyiladi.

Noma'lum sonning berilgan tenglamani to'g'ri tenglikka aylantiruvchi son qiymati tenglamaning **ildizi** (yoki **yechimi**) deyiladi.

Tenglamani yechish deb uning barcha ildizlarini topish yoki ildizi yo'qligini ko'rsatishga aytiladi.

✓ Eslatma

$x + 5 = 12$ tenglamani qaraylik. Bu yerda x ning o'rniga 7 ni qo'ysak: $7 + 5 = 12$ — to'g'ri tenglik hosil bo'ladi. Demak, $x = 7$ shu tenglamaning ildizi. Boshqa biror son qo'yilganda tenglik bajarilmaydi.

Chiziqli tenglama

✎ Ta'rif

$ax = b$ ko'rinishidagi tenglama **bir noma'lumli chiziqli tenglama** deyiladi. Bu yerda a va b — berilgan sonlar, x — noma'lum.

✠ Asosiy qoida

Chiziqli tenglamaning yechimlari soni a va b ning qiymatiga bog'liq ravishda quyidagicha bo'ladi:

Holat	Yechim	Misol
$a \neq 0$	yagona ildiz: $x = \frac{b}{a}$	$3x = 12 \Rightarrow x = 4$
$a = 0,\ b = 0$	cheksiz ko'p ildiz	$0 \cdot x = 0$
$a = 0,\ b \neq 0$	ildiz mavjud emas	$0 \cdot x = 5$

To'g'ri tenglikning xossalari

✠ Asosiy qoida

1-xossa. Agar to'g'ri tenglikning ikkala qismiga bir xil son qo'shilsa yoki ayirilsa, yana to'g'ri tenglik hosil bo'ladi.

$$A = B \iff A + c = B + c \iff A - c = B - c.$$

2-xossa. Agar to'g'ri tenglikning ikkala qismi noldan farqli ayni bir songa ko'paytirilsa yoki bo'linsa, u holda yana to'g'ri tenglik hosil bo'ladi.

$$A = B \iff A \cdot c = B \cdot c \iff A : c = B : c \quad (c \neq 0).$$

Proporsiya

✎ Ta'rif

Ikki nisbatning tengligi **proporsiya** deyiladi:

$$\frac{a}{b} = \frac{c}{d} \qquad \text{yoki} \qquad a : b = c : d.$$

Bu yerda a va d — proporsiyaning **chetki** hadlari, b va c — **o'rta** hadlari.

✠ Asosiy qoida

Proporsiyaning asosiy xossasi:

$$\frac{a}{b} = \frac{c}{d} \iff a \cdot d = b \cdot c.$$

Ya'ni chetki hadlar ko'paytmasi o'rta hadlar ko'paytmasiga teng.

Misol. $\frac{x}{6} = \frac{4}{8} \Rightarrow 8x = 24 \Rightarrow x = 3.$

Namunaviy misollar

✓ Eslatma

1-namuna. $4x - 7 = 13$ tenglamani yeching.
Yechish:

$$\begin{aligned} 4x - 7 &= 13 \\ 4x &= 13 + 7 \qquad (-7 \text{ ni o'ng tomonga } +7 \text{ qilib o'tkazamiz}) \\ 4x &= 20 \\ x &= 20 : 4 = 5. \end{aligned}$$

Tekshirish: $4 \cdot 5 - 7 = 20 - 7 = 13$ ✓. **Javob:** $x = 5$.

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✓ **Eslatma**

2-namuna. $3(2x-1)-2(x+4)=7$ tenglamani yeching.
Yechish:

$$\begin{aligned} 3(2x-1)-2(x+4)&=7 \\ 6x-3-2x-8&=7 && \text{(qavslar ochildi)} \\ 4x-11&=7 && \text{(o'xshash hadlar ixchamlandi)} \\ 4x&=7+11 \\ 4x&=18 \\ x&=\tfrac{18}{4}=4{,}5. \end{aligned}$$

Javob: $x=4{,}5$.

✓ **Eslatma**

3-namuna. $\dfrac{2x+1}{3}-\dfrac{x-1}{2}=2$ tenglamani yeching.
Yechish: Maxrajlar 3 va 2 bo'lgani uchun ularning eng kichik umumiy maxraji (ya'ni EKUKi) 6 ga teng. Kasrlarni umumiy maxrajga keltiramiz:

$$\begin{aligned} \frac{2(2x+1)}{6}-\frac{3(x-1)}{6}&=\frac{6\cdot 2}{6} \\ 2(2x+1)-3(x-1)&=12 \\ 4x+2-3x+3&=12 \\ x+5&=12 \\ x&=7. \end{aligned}$$

Javob: $x=7$.

Yechimi yo'q va cheksiz ko'p yechimli tenglamalar

✓ **Eslatma**

4-namuna. $2x+5=2x+9$ tenglamani yeching.
Yechish:

$$\begin{aligned} 2x+5&=2x+9 \\ 2x-2x&=9-5 && \text{(x li hadlarni chapga o'tkazamiz)} \\ 0\cdot x&=4 \\ 0&=4 \quad \text{— noto'g'ri tenglik!} \end{aligned}$$

x ning hech qanday qiymatida $0=4$ bo'la olmaydi. **Javob:** yechim yo'q, $\varnothing$.

✓ **Eslatma**

5-namuna. $3(x+2)=3x+6$ tenglamani yeching.
Yechish:

$$\begin{aligned} 3(x+2) &= 3x+6 \\ 3x+6 &= 3x+6 \qquad \text{(qavs ochildi)} \\ 3x-3x &= 6-6 \\ 0 &= 0 \quad \text{— to'g'ri tenglik!} \end{aligned}$$

Bu tenglik x ning **istalgan** qiymatida bajariladi. **Javob:** cheksiz ko'p yechim (x — istalgan son).

Amaliy qism

1–10. *Tenglamalarni yeching.*

1 $x+6=13$

2 $x-7=15$

3 $9-x=4$

4 $18-x=11$

5 $4x=28$

6 $7x=-42$

7 $-5x=35$

8 $\dfrac{x}{6}=4$

9 $\dfrac{x}{-3}=5$

10 $-\dfrac{x}{7}=-2$

11–25. *Tenglamalarni yeching.*

11 $3x+5=17$

12 $4x-11=13$

13 $6x+8=-16$

14 $-2x+5=-13$

15 $-7x-3=18$

16 $9-2x=1$

17 $12-5x=27$

18 $8x-11=-3$

19 $15+4x=7$

20 $-3x+14=2$

21 $7-6x=-11$

22 $10x-13=27$

23 $-8+5x=17$

24 $25-7x=4$

25 $-4x-9=7$

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26–40. *Tenglamalarni yeching.*

26 $3(2x-1)=9$

27 $2(x+5)=14$

28 $-4(3x-2)=-16$

29 $5(2x+3)=25$

30 $7(x-4)=-14$

31 $-2(5x+1)=8$

32 $6(2x-5)=18$

33 $4(3x+1)=-20$

34 $-3(4x-7)=9$

35 $9(x+2)=27$

36 $-5(2x+3)=-25$

37 $8(x-3)=16$

38 $2(7x-1)=12$

39 $3(2-5x)=21$

40 $-6(3-x)=12$

41–50. *Tenglamalarni yeching.*

41 $3(x+2)-2(x-1)=10$

42 $4(2x-1)+3(x+5)=11$

43 $5(x-3)-2(2x+1)=-8$

44 $2(3x+1)-4(x-2)=6$

45 $7(x+2)-3(2x-1)=16$

46 $4(x+2)-3(x-1)=2(x+3)$

47 $5(2x-1)-3(x+4)=4(x-5)$

48 $2(3x+1)+4(x-3)=5(x+2)$

49 $6(x-1)-2(x-4)=3(x+1)$

50 $4(2x+3)-5(x-1)=2(x+7)$

51–60. *Tenglamalarni yeching.*

51 $\frac{x}{3}+5=11$

52 $\frac{x}{4}-2=7$

53 $\frac{x+3}{5}=4$

54 $\frac{2x-1}{3}=5$

55 $\frac{x}{-2}+7=4$

56 $\frac{3x+6}{4}=9$

57 $\frac{x}{6}-3=-1$

58 $\frac{5-x}{2}=6$

59 $\frac{2x+7}{3}=-1$

60 $\frac{4-3x}{5}=2$

61–70. *Tenglamalarni yeching.*

61 $\frac{x}{2}+\frac{x}{3}=5$

62 $\frac{x}{3}-\frac{x}{5}=4$

63 $\frac{x+1}{2}+\frac{x}{3}=8$

64 $\frac{x}{4}+\frac{x-2}{6}=3$

65 $\frac{2x+1}{3}-\frac{x-1}{2}=1$

66 $\frac{3x-2}{4}+\frac{x+1}{2}=5$

67 $\frac{x}{5}-\frac{x+3}{2}=-3$

68 $\frac{x-2}{3}+\frac{x+1}{2}=4$

69 $\frac{2x+3}{5}-\frac{x-1}{4}=1$

70 $\frac{2x+5}{3}-\frac{x-2}{6}=4$

71–80. *Tenglamalarni yeching.*

71 $x:3=8:6$

72 $4:x=12:9$

73 $\frac{x+2}{5}=\frac{6}{10}$

74 $\frac{7}{x}=\frac{21}{15}$

75 $(x-3):4=5:2$

76 $\frac{3}{x+1}=\frac{6}{8}$

77 $\frac{2x}{9}=\frac{4}{6}$

78 $\frac{5}{x-2}=\frac{10}{8}$

79 $\frac{x+5}{3}=\frac{x-1}{2}$

80 $\frac{2x-1}{3}=\frac{x+3}{2}$

81–90. *Tenglamalarni yeching.*

81 $3(2x+1)=6x+5$

82 $5(x-2)+3=5x-7$

83 $2(3x-4)+5=6x-2$

84 $7x+2(1-x)=5x+2$

85 $\frac{6x+4}{2}=3x+1$

86 $8x-3(x+1)=5x+4$

87 $4(x-1)-2(x-3)=2x+2$

88 $4x-7=2(2x-3)-1$

89 $\frac{10x-5}{5}=2x-1$

90 $3(x+2)-(x-4)=2x+10$

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Mavzu bo'yicha test savollari

1-variant

1. $5x - 3 = 17$ tenglamani yeching.
   A) 2 B) 3 C) 4 D) 5

2. $12 - 4x = -8$ tenglamani yeching.
   A) 4 B) 5 C) -5 D) -1

3. $2(3x - 1) = 16$ tenglamani yeching.
   A) 2 B) 3 C) 4 D) 5

4. $4(x + 1) - 3(x - 2) = 14$ tenglamani yeching.
   A) 2 B) 3 C) 4 D) 5

5. $\dfrac{2x - 3}{5} = 1$ tenglamani yeching.
   A) 4 B) 5 C) 2 D) 6

6. $\dfrac{x}{2} + \dfrac{x}{4} = 6$ tenglamani yeching.
   A) 6 B) 8 C) 12 D) 4

7. $x : 6 = 6 : 12$ tenglamani yeching.
   A) 2 B) 3 C) 4 D) 8

8. Ketma-ket kelgan ikkita natural sonning yig'indisi 35. Shu sonlarning kichigini toping.
   A) 16 B) 17 C) 18 D) 19

9. Otaning yoshi o'g'lining yoshidan 4 marta katta. 12 yildan keyin otaning yoshi o'g'lining yoshidan 2 marta katta bo'ladi. Hozir o'g'il necha yoshda?
   A) 4 B) 5 C) 6 D) 8

10. To'g'ri to'rtburchakning bo'yi enidan 4 cm ortiq, perimetri 28 cm. To'g'ri to'rtburchakning bo'yini toping.
    A) 7 cm B) 8 cm C) 9 cm D) 10 cm

2-variant

1. $3x + 7 = 22$ tenglamani yeching.
   A) 3 B) 4 C) 5 D) 6

2. $8 - 3x = -7$ tenglamani yeching.
   A) 4 B) 5 C) -5 D) 3

3. $5(2x + 1) = 35$ tenglamani yeching.
   A) 2 B) 3 C) 4 D) 5

4. $3(2x - 1) - 2(x + 2) = 9$ tenglamani yeching.
   A) 3 B) 4 C) 5 D) 2

5. $\dfrac{3x + 2}{4} = 2$ tenglamani yeching.
   A) 2 B) 3 C) 1 D) 4

6. $\frac{x}{3} + \frac{x}{6} = 3$ tenglamani yeching.
   A) 4 B) 5 C) 6 D) 9

7. $4 : x = 12 : 15$ tenglamani yeching.
   A) 4 B) 5 C) 6 D) 8

8. Ikkita sonning ayirmasi 8, yig'indisi 30 ga teng. Sonlarning kattasini toping.
   A) 11 B) 17 C) 19 D) 22

9. Kitobning narxi daftar narxidan 5000 so'm ortiq. 2 ta kitob va 3 ta daftar uchun jami 30 000 so'm to'langan. Bitta kitob necha so'm?
   A) 4000 B) 7000 C) 8000 D) 9000

10. Kvadrat shaklidagi yer maydonining tomoni 2 m ga oshirilsa, uning perimetri 40 m bo'ladi. Kvadrat tomonining dastlabki uzunligini (m) toping.
    A) 6 B) 7 C) 8 D) 10

Javoblar

1. $x = 7$
2. $x = 22$
3. $x = 5$
4. $x = 7$
5. $x = 7$
6. $x = -6$
7. $x = -7$
8. $x = 24$
9. $x = -15$
10. $x = 14$
11. $x = 4$
12. $x = 6$
13. $x = -4$
14. $x = 9$
15. $x = -3$
16. $x = 4$
17. $x = -3$
18. $x = 1$
19. $x = -2$
20. $x = 4$
21. $x = 3$
22. $x = 4$
23. $x = 5$
24. $x = 3$
25. $x = -4$
26. $x = 2$
27. $x = 2$
28. $x = 2$
29. $x = 1$
30. $x = 2$
31. $x = -1$
32. $x = 4$
33. $x = -2$
34. $x = 1$
35. $x = 1$
36. $x = 1$
37. $x = 5$
38. $x = 1$
39. $x = -1$
40. $x = 5$
41. $x = 2$
42. $x = 0$
43. $x = 9$
44. $x = -2$
45. $x = -1$
46. $x = 5$
47. $x = -1$
48. $x = 4$
49. $x = 1$
50. $x = -3$
51. $x = 18$
52. $x = 36$
53. $x = 17$
54. $x = 8$
55. $x = 6$
56. $x = 10$
57. $x = 12$
58. $x = -7$
59. $x = -5$
60. $x = -2$
61. $x = 6$
62. $x = 30$
63. $x = 9$
64. $x = 8$
65. $x = 1$
66. $x = 4$
67. $x = 5$
68. $x = 5$
69. $x = 1$

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70. $x = 4$

71. $x = 4$

72. $x = 3$

73. $x = 1$

74. $x = 5$

75. $x = 13$

76. $x = 3$

77. $x = 3$

78. $x = 6$

79. $x = 13$

80. $x = 11$

81. Yechim yo'q: $\varnothing$

82. Cheksiz ko'p yechim: x — istalgan son

83. Yechim yo'q: $\varnothing$

84. Cheksiz ko'p yechim: x — istalgan son

85. Yechim yo'q: $\varnothing$

86. Yechim yo'q: $\varnothing$

87. Cheksiz ko'p yechim: x — istalgan son

88. Cheksiz ko'p yechim: x — istalgan son

89. Cheksiz ko'p yechim: x — istalgan son

90. Cheksiz ko'p yechim: x — istalgan son

Test javoblari

1-variant: 1) C 2) B 3) B 4) C 5) A 6) B 7) B 8) B 9) C 10) C

2-variant: 1) C 2) B 3) B 4) B 5) A 6) C 7) B 8) C 9) D 10) C

Mavzu bo'yicha interaktiv testlar

QR kodni skanerlab, bilimingizni sinab ko'ring

14.

Ikki noma'lumli chiziqli tenglamalar sistemasi

Nazariy qism

Tenglamalar sistemasi nima?

✓ Eslatma

Ikki do'st birga 1000 so'm pul yig'ishdi. Birinchisi ikkinchisidan 200 so'm ko'p pul yig'gan bo'lsa, har biri qanchadan yig'gan?

Bu masalada **ikkita noma'lum** bor bo'lib (Anvar va Bahodirning pullari) va **ikkita shart** berilgan:

$$x + y = 1000, \qquad x - y = 200.$$

Birgalikda qaraladigan bunday tenglamalar tenglamalar sistemasini tashkil qiladi.

✎ Ta'rif

Ikki noma'lumli ikkita chiziqli tenglamadan iborat sistema — chiziqli tenglamalar sistemasi deyiladi.

$$\begin{cases} x + y = 10 \\ x - y = 4 \end{cases}$$

✎ Ta'rif

Sistemaning **yechimi** deb ikkala tenglamani ham bir paytda to'g'ri tenglikka aylantiruvchi $(x;\ y)$ **sonlar juftligi**ga aytiladi.

Misol: Yuqoridagi sistemaning yechimi $(x;\ y) = (7;\ 3)$, chunki:

$$7 + 3 = 10\ \checkmark \qquad 7 - 3 = 4\ \checkmark$$

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Tenglashtirish usuli

✓ **Eslatma**

1-misol. Tenglamalar sistemasini yeching:

$$\begin{cases} y = 2x + 1 \\ y = 3x - 2 \end{cases}$$

Ikkala tenglamaning ham chap tomoni y ga teng. Demak, ularning o'ng tomonlari o'zaro teng:

$$2x + 1 = 3x - 2 \implies x = 3$$

$y = 2 \cdot 3 + 1 = 7$ Yechim: $(x;\, y) = (3;\, 7)$

O'rniga qo'yish usuli

✓ **Eslatma**

2-misol. Tenglamalar sistemasini yeching:

$$\begin{cases} y = 2x \\ x + y = 6 \end{cases}$$

$y = 2x$ ni ikkinchi tenglamadagi y ning o'rniga qo'yamiz:

$$x + 2x = 6 \implies 3x = 6 \implies x = 2$$

$y = 2 \cdot 2 = 4$ Yechim: $(2;\, 4)$

3-misol. Tenglamalar sistemasini yeching:

$$\begin{cases} x = 3y \\ 2x - y = 10 \end{cases}$$

$x = 3y$ ni ikkinchi tenglamadagi x ning o'rniga qo'yamiz:

$2 \cdot 3y - y = 10 \Rightarrow 5y = 10 \Rightarrow y = 2$

$x = 3 \cdot 2 = 6$ Yechim: $(6;\, 2)$

✓ **Eslatma**

Agar tenglamalarning birida $\frac{y}{x} = k$ ko'rinishidagi nisbat berilgan bo'lsa, uni $y = kx$ shaklga keltirib, so'ngra o'rniga qo'yish usulini qo'llash kerak.

$$\frac{y}{x} = 3 \iff y = 3x \qquad \frac{x}{y} = 2 \iff x = 2y$$

Qo'shish usuli

✓ **Eslatma**

4-misol. Tenglamalar sistemasini yeching:

$$\begin{cases} x + y = 13 \\ x - y = 5 \end{cases}$$

Tenglamalarni qo'shamiz: $2x = 18 \Rightarrow x = 9$

$9 + y = 13 \Rightarrow y = 4$ Yechim: $(9;\ 4)$

5-misol. Tenglamalar sistemasini yeching:

$$\begin{cases} 2x + y = 7 \\ -2x + 3y = 5 \end{cases}$$

Qo'shsak, x yo'qoladi: $4y = 12 \Rightarrow y = 3$

$2x + 3 = 7 \Rightarrow x = 2$ Yechim: $(2;\ 3)$

✓ **Eslatma**

Sistemalarni yechishda yechimni **tekshirib ko'rish** muhim. Topilgan $(x;\ y)$ juftlik ikkala tenglamaga ham qo'yiladi. Agar ikkala tenglik ham to'g'ri bo'lsa, yechim to'g'ri topilgan.

Matnli masalalarni sistemalar yordamida yechish

✠ **Asosiy qoida**

1. Masaladagi noma'lumlar x va y orqali belgilanadi.
2. Masala shartida berilgan ikkita munosabat asosida ikkita tenglama tuziladi.
3. Tenglamalar sistemasi eng qulay usul bilan yechiladi.
4. Javobning masala shartiga mosligi tekshiriladi.

Misol. Ikki sonning yig'indisi 20, ayirmasi 6 ga teng. Bu sonlarni toping.
Yechilishi: x — katta son, y — kichik son.

$$\begin{cases} x + y = 20 \\ x - y = 6 \end{cases} \Rightarrow\ 2x = 26 \Rightarrow x = 13,\ y = 7.$$

Javob: 13 va 7.

$$Tekshirish: \begin{cases} 13 + 7 = 20 \\ 13 - 7 = 6 \end{cases}$$

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Amaliy qism

1–5. *Tenglashtirish usuli bilan yeching.*

1 $\begin{cases} y = 2x + 1 \\ y = 3x - 2 \end{cases}$

2 $\begin{cases} y = x + 5 \\ y = 2x + 1 \end{cases}$

3 $\begin{cases} y = 3x - 4 \\ y = x + 2 \end{cases}$

4 $\begin{cases} y = -x + 7 \\ y = 2x + 1 \end{cases}$

5 $\begin{cases} y = 4x - 3 \\ y = x + 6 \end{cases}$

6–20. *O'rniga qo'yish usuli bilan yeching.*

6 $\begin{cases} y = 2x \\ x + y = 6 \end{cases}$

7 $\begin{cases} y = 3x \\ x + y = 8 \end{cases}$

8 $\begin{cases} y = x - 3 \\ x + y = 7 \end{cases}$

9 $\begin{cases} y = 2x - 1 \\ x + y = 8 \end{cases}$

10 $\begin{cases} y = 4x \\ 3x + y = 14 \end{cases}$

11 $\begin{cases} x = 2y \\ x + y = 12 \end{cases}$

12 $\begin{cases} x = 3y \\ 2x - y = 10 \end{cases}$

13 $\begin{cases} y = -2x \\ x - y = 9 \end{cases}$

14 $\begin{cases} x - y = 5 \\ y = -4x \end{cases}$

15 $\begin{cases} y = x + 1 \\ 3x - y = 5 \end{cases}$

16 $\begin{cases} y = 2x - 3 \\ 4x - y = 11 \end{cases}$

17 $\begin{cases} x = y + 2 \\ 3x - 2y = 10 \end{cases}$

18 $\begin{cases} y = 1{,}5x \\ 5x - 2y = 8 \end{cases}$

19 $\begin{cases} y = 0{,}5x \\ 3x - 2y = 8 \end{cases}$

20 $\begin{cases} y = 2{,}5x \\ 4x - y = 6 \end{cases}$

21–30. *Avval nisbatni $y = kx$ ko'rinishiga keltiring, so'ngra o'rniga qo'yish usuli bilan yeching.*

21 $\begin{cases} \dfrac{y}{x} = 3 \\ 2x + y = 10 \end{cases}$

22 $\begin{cases} \dfrac{x}{y} = 2 \\ x + y = 12 \end{cases}$

23 $\begin{cases} \dfrac{y}{x} = 2 \\ x + y = 9 \end{cases}$

24 $\begin{cases} \dfrac{x}{y} = 3 \\ x - y = 8 \end{cases}$

25 $\begin{cases} \frac{y}{x} = 4 \\ x + y = 15 \end{cases}$

26 $\begin{cases} \frac{x}{y} = 5 \\ x - y = 12 \end{cases}$

27 $\begin{cases} \frac{y}{x} = 2 \\ 4x - y = 6 \end{cases}$

28 $\begin{cases} \frac{x}{y} = 4 \\ 3x - y = 22 \end{cases}$

29 $\begin{cases} \frac{y}{x} = 5 \\ x + y = 18 \end{cases}$

30 $\begin{cases} \frac{y}{x} = -3 \\ 2x + y = -3 \end{cases}$

31–40. *Qo'shish usuli bilan yeching.*

31 $\begin{cases} x - y = 5 \\ x + y = 13 \end{cases}$

32 $\begin{cases} 2x + y = 7 \\ -2x + 3y = 5 \end{cases}$

33 $\begin{cases} x - 2y = 7 \\ 3x + 2y = 13 \end{cases}$

34 $\begin{cases} x + y = 11 \\ x - y = 3 \end{cases}$

35 $\begin{cases} 3x + y = 14 \\ x - y = 2 \end{cases}$

36 $\begin{cases} 2x + y = 11 \\ 3x - y = 14 \end{cases}$

37 $\begin{cases} x + 3y = 12 \\ x - 3y = 0 \end{cases}$

38 $\begin{cases} 5x + 2y = 17 \\ x - 2y = 1 \end{cases}$

39 $\begin{cases} 2x - 3y = -1 \\ -2x + 5y = 7 \end{cases}$

40 $\begin{cases} 4x + y = 14 \\ -4x + 3y = 10 \end{cases}$

41–52. *Matnli masalalar. Tenglamalar sistemasini tuzib yeching.*

41 Ikki sonning yig'indisi 24, ayirmasi 8 ga teng. Bu sonlarni toping.

42 Bog'dagi olma va shaftoli daraxtlarining umumiy soni 50 ta. Olma daraxtlari shaftoli daraxtlaridan 10 taga ortiq bo'lsa, har bir turdagi daraxt nechtadan ekilgan?

43 Sinfdagi 28 ta o'quvchidan o'g'il bolalar qizlardan 4 taga ko'p bo'lsa, o'g'il bolalar sonini toping.

44 To'g'ri to'rtburchakning perimetri 36 sm, uzunligi enidan 4 sm ortiq. To'g'ri to'rtburchakning tomonlarini toping.

45 Ikki sonning yig'indisi 15 ga teng bo'lib, kattasini kichigiga bo'lganda 4 hosil bo'ladi. Bu sonlarni toping.

46 Avtoturargohda jami 20 ta avtomashina va motosikl bor. Ularning g'ildiraklari soni jami 56 ta bo'lsa, har biridan nechtadan bo'lgan?

47 Ikki sonning yig'indisi 40 ga teng va kattasi kichigidan 3 marta katta. Bu sonlarni toping.

48 Anvar va Bahodir birgalikda 1000 so'm to'pladi. Anvar Bahodirdan 200 so'm ko'p to'plagan bo'lsa, har biri qanchadan pul to'plagan?

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49 5 ta daftar va 3 ta ruchka 21 000 so'mga, 3 ta daftar va 5 ta ruchka esa 19 000 so'mga sotib olindi. Bitta daftar va bitta ruchkaning narxlari yig'indisini toping.

50 Otaning yoshi o'g'lining yoshidan 24 yosh katta. 4 yildan keyin otaning yoshi o'g'lining yoshidan 3 marta katta bo'ladi. Hozir ota necha yoshda?

51 Mehmonxonada bir o'rinli va ikki o'rinli xonalar soni birgalikda 30 ta. Bu xonalarga jami 45 ta mehmonni joylashtirish mumkin bo'lsa, bir o'rinli xonalar nechta?

52 Bekat oldida jami 18 ta 2 g'ildirakli va 3 g'ildirakli velosiped turibdi. Ularning g'ildiraklari soni jami 44 ta. 2 g'ildirakli velosipedlar soni nechta?

53–59. *Aralash misollar.*

53 $a+b=64,\ \dfrac{a}{b}=7$ tengliklar o'rinli bo'lsa, $a-b$ ning qiymatini toping.

54 Agar $\begin{cases}5x+2y=64\\ y=1{,}5x\end{cases}$ bo'lsa, xy ning qiymatini toping.

55 $\dfrac{7b-4a}{b}=2$ bo'lsa, $\dfrac{a}{b}$ ning qiymatini toping.

56 $\dfrac{x-3y+z}{x-y+z}=3$ bo'lsa, $\dfrac{x+5z}{x+2z}$ ning qiymatini toping.

57 $\dfrac{x+3y-z}{x+y-z}=3$ bo'lsa, $\dfrac{x+2z}{2x+z}$ ning qiymatini toping.

58 $x+y=90$ va $x-2y=0$ bo'lsa, $x-y$ ning qiymatini toping.

59 $x+y=60$ va $x=2y$ bo'lsa, x ning qiymatini toping.

Mavzu bo'yicha test savollari

1-variant

1. $\begin{cases} y = 2x + 1 \\ y = 3x - 1 \end{cases}$ sistemaning yechimini toping.
A) $(2;\ 5)$ B) $(1;\ 3)$ C) $(3;\ 7)$ D) $(2;\ 4)$

2. $\begin{cases} y = 3x \\ x + y = 12 \end{cases}$ sistemaning yechimini toping.
A) $(2;\ 6)$ B) $(4;\ 8)$ C) $(3;\ 9)$ D) $(1;\ 11)$

3. $\begin{cases} x + y = 10 \\ x - y = 4 \end{cases}$ sistemadan x ning qiymatini toping.
A) 4 B) 5 C) 6 D) 7

4. $\begin{cases} 2x + y = 8 \\ x - y = 1 \end{cases}$ sistemadan y ning qiymatini toping.
A) 2 B) 3 C) 4 D) 5

5. $\begin{cases} y = x + 2 \\ 2x + y = 11 \end{cases}$ sistemadan y ning qiymatini toping.
A) 3 B) 5 C) 7 D) 9

6. Ikki sonning yig'indisi 25, ayirmasi 7 ga teng. Kichik sonni toping.
A) 8 B) 9 C) 10 D) 11

7. $\begin{cases} \dfrac{x}{y} = 3 \\ x + y = 16 \end{cases}$ sistemadan x ning qiymatini toping.
A) 4 B) 8 C) 10 D) 12

8. $\begin{cases} x + y = 15 \\ \dfrac{x}{y} = 4 \end{cases}$ sistemadan y ning qiymatini toping.
A) 3 B) 5 C) 9 D) 12

9. $a + b = 50,\ \ a - b = 10$ bo'lsa, ab ning qiymatini toping.
A) 400 B) 500 C) 600 D) 800

10. $\begin{cases} 2x + y = 14 \\ y = 1{,}5x \end{cases}$ sistemadan x ning qiymatini toping.
A) 2 B) 3 C) 4 D) 5

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2-variant

1. $\begin{cases} y = x + 3 \\ y = 2x + 1 \end{cases}$ sistemaning yechimini toping.
A) $(1;\ 4)$ B) $(2;\ 5)$ C) $(3;\ 6)$ D) $(4;\ 7)$

2. $\begin{cases} y = 4x \\ x + y = 10 \end{cases}$ sistemaning yechimini toping.
A) $(2;\ 8)$ B) $(1;\ 9)$ C) $(5;\ 5)$ D) $(3;\ 7)$

3. $\begin{cases} x + y = 14 \\ x - y = 2 \end{cases}$ sistemadan y ning qiymatini toping.
A) 2 B) 4 C) 6 D) 8

4. $\begin{cases} 3x + y = 11 \\ x - y = 1 \end{cases}$ sistemadan x ning qiymatini toping.
A) 2 B) 3 C) 4 D) 5

5. $\begin{cases} y = 2x - 1 \\ x + y = 8 \end{cases}$ sistemadan x ning qiymatini toping.
A) 1 B) 2 C) 3 D) 4

6. Ikki sonning yig'indisi 30, ayirmasi 8 ga teng. Katta sonni toping.
A) 11 B) 13 C) 15 D) 19

7. $\begin{cases} \dfrac{y}{x} = 2 \\ x + y = 18 \end{cases}$ sistemadan y ning qiymatini toping.
A) 4 B) 6 C) 9 D) 12

8. $\begin{cases} x - y = 4 \\ x + y = 12 \end{cases}$ bo'lsa, xy ning qiymatini toping.
A) 20 B) 32 C) 40 D) 48

9. $a + b = 80$, $\dfrac{a}{b} = 3$ bo'lsa, $a - b$ ning qiymatini toping.
A) 20 B) 30 C) 40 D) 50

10. $\begin{cases} 5x + 2y = 24 \\ y = 1{,}5x \end{cases}$ sistemadan x ning qiymatini toping.
A) 1 B) 3 C) 5 D) 6

Javoblar

1. $(3;\ 7)$
2. $(4;\ 9)$
3. $(3;\ 5)$
4. $(2;\ 5)$
5. $(3;\ 9)$
6. $(2;\ 4)$
7. $(2;\ 6)$
8. $(5;\ 2)$
9. $(3;\ 5)$
10. $(2;\ 8)$
11. $(8;\ 4)$
12. $(6;\ 2)$
13. $(3;\ -6)$
14. $(1;\ -4)$
15. $(3;\ 4)$
16. $(4;\ 5)$
17. $(6;\ 4)$
18. $(4;\ 6)$
19. $(4;\ 2)$
20. $(4;\ 10)$
21. $(2;\ 6)$
22. $(8;\ 4)$
23. $(3;\ 6)$
24. $(12;\ 4)$
25. $(3;\ 12)$
26. $(15;\ 3)$
27. $(3;\ 6)$
28. $(8;\ 2)$
29. $(3;\ 15)$
30. $(3;\ -9)$
31. $(9;\ 4)$
32. $(2;\ 3)$
33. $(5;\ -1)$
34. $(7;\ 4)$
35. $(4;\ 2)$
36. $(5;\ 1)$
37. $(6;\ 2)$
38. $(3;\ 1)$
39. $(4;\ 3)$
40. $(2;\ 6)$
41. 16 va 8
42. Olma: 30 ta, shaftoli: 20 ta
43. 16 ta
44. Uzunligi: 11 sm, eni: 7 sm
45. 12 va 3
46. Avtomashina: 8 ta, motosikl: 12 ta
47. 30 va 10
48. Anvar: 600 so'm, Bahodir: 400 so'm
49. 5000 so'm
50. 32
51. 15
52. 10
53. 48
54. 96
55. $\frac{5}{4}$
56. 4
57. 1
58. 30
59. 40

Test javoblari

1-variant: 1) A 2) C 3) D 4) A 5) B 6) B 7) D 8) A 9) C 10) C
2-variant: 1) B 2) A 3) C 4) B 5) C 6) D 7) D 8) B 9) C 10) B

Mavzu bo'yicha interaktiv testlar
QR kodni skanerlab, bilimingizni sinab ko'ring

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15.

Matnli masalalar

Nazariy qism

✠ Asosiy qoida

Matnli masalani yechish bosqichlari:

1. Masala shartini diqqat bilan o'qing va uning mazmuni nima haqida ekanini yaxshilab tushuning.
2. **Noma'lum kattalikni** x harfi bilan belgilang (kerak bo'lsa boshqa kattaliklar ham x orqali ifodalanadi).
3. Masala shartiga ko'ra **tenglama** (yoki amallar zanjiri) tuzing.
4. Tenglamani yeching.
5. Topilgan natijani masala shartiga qo'yib **tekshiring** va javobning masala shartiga mosligini tekshiring.

Songa doir masalalar

✠ Asosiy qoida

Bunday masalalarda noma'lum sonni topish kerak bo'ladi. Asosiy g'oya: noma'lum sonni x deb belgilab, masala shartiga muvofiq tenglama tuziladi.

Misol: "Son 3 ga ko'paytirilib, hosil bo'lgan natijaga 5 qo'shilsa, 20 hosil bo'ladi."

$$3x + 5 = 20 \;\Rightarrow\; 3x = 15 \;\Rightarrow\; x = 5.$$

Foydali formulalar:

- Ikki sonning yig'indisi S, ayirmasi A bo'lsa:

$$\text{katta son} = \frac{S + A}{2}, \qquad \text{kichik son} = \frac{S - A}{2}.$$

- Ketma-ket natural sonlarning o'zaro bog'liqligi: $x,\ x+1,\ x+2,\ \ldots$
- Ketma-ket toq (yoki juft) natural sonlarning o'zaro bog'liqligi: $x,\ x+2,\ x+4,\ \ldots$

Harakatga doir masalalar

✠ Asosiy qoida

Masofa, tezlik va vaqt orasidagi bog'lanish:

$$\boxed{S = v \cdot t}$$

bu yerda S — masofa, v — tezlik, t — vaqt. Bu formuladan:

$$v = \frac{S}{t}, \qquad t = \frac{S}{v}.$$

Yoshga doir masalalar

✠ Asosiy qoida

Vaqt o'tishi bilan **hamma odamning yoshi bir xil miqdorga** oshadi. Demak, ikki odamning **yoshlari ayirmasi vaqt o'tishi bilan o'zgarmaydi**.

Agar hozir A ning yoshi — a, B ning yoshi — b bo'lsa:

- n yildan keyin: A ning yoshi $a + n$, B ning yoshi $b + n$ ga teng bo'ladi.
- n yil oldin: A ning yoshi $a - n$, B ning yoshi $b - n$ ga teng bo'ladi.

O'rta arifmetik qiymat

✎ Ta'rif

n ta sonning **o'rta arifmetik qiymati** deb shu sonlar yig'indisining ularning soniga nisbatiga aytiladi:

$$\overline{x} = \frac{x_1 + x_2 + x_3 + \ldots + x_n}{n}.$$

✠ Asosiy qoida

- Yig'indi = o'rta arifmetik × sonlar soni: $x_1 + x_2 + \ldots + x_n = \overline{x} \cdot n$.
- Ikki sonning o'rta arifmetigi: $\overline{x} = \dfrac{a+b}{2}$.
- $1, 2, 3, \ldots, n$ ketma-ket natural sonlarning o'rta arifmetigi: $\overline{x} = \dfrac{1+n}{2}$.

Misol: $5,\ 8,\ 11$ sonlarning o'rta arifmetigi: $\dfrac{5+8+11}{3} = \dfrac{24}{3} = 8$.

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Amaliy qism

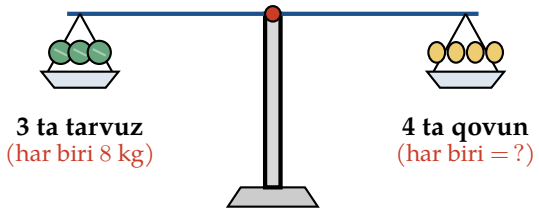
1–10. *Songa doir masalalar.*

1 Bir songa 5 qo'shib, 2 ga ko'paytirib, hosil bo'lgan natijadan 10 ayrilsa, 4 hosil bo'ladi. Berilgan sonni toping.

2 Ikki ketma-ket natural sonning yig'indisi 21 ga teng. Ularning kattasini toping.

3 Ikki sonning yig'indisi 18 ga, ayirmasi esa 12 ga teng. Ularning kattasini toping.

4 Tarozining bir pallasida 3 ta tarvuz, ikkinchisida 4 ta qovun bor va tarozi muvozanatda turibdi. Agar bitta tarvuzning vazni 8 kg bo'lsa, bitta qovunning vazni necha kg?



5 Bir sonning 3 baravariga 7 qo'shilsa, 25 hosil bo'ladi. Bu sonni toping.

6 Uchta ketma-ket natural sonning yig'indisi 36 ga teng. Ularning eng kattasini toping.

7 Ikki sonning yig'indisi 60 va biri ikkinchisidan 3 marta katta bo'lsa, sonlardan kichigini toping.

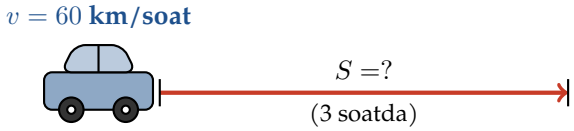
8 Bir son ikkinchisidan 14 ta ortiq va ularning yig'indisi 50 ga teng bo'lsa, kichik sonni toping.

9 Tarozining bir pallasida 5 ta olma va 2 ta nok, ikkinchisida 10 ta olma va 1 ta nok bo'lib, tarozi muvozanatda turibdi. Bitta olmaning massasi 100 g bo'lsa, bitta nokning massasi necha gramm?

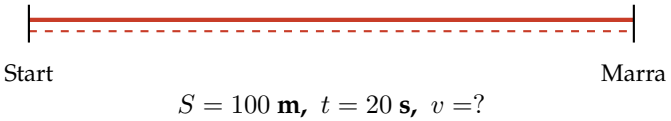
10 Bir son o'ylandi. Uni 4 ga ko'paytirib, natijadan 8 ayirildi, so'ng hosil bo'lgan son 3 ga bo'linsa, 8 hosil bo'ladi. O'ylangan sonni toping.

11–20. *Harakatga doir masalalar.*

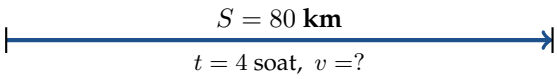
11 Avtomobil soatiga 60 km tezlik bilan 3 soat yo'l yursa, u qancha masofa bosib o'tadi?



12 Sportchi 100 metr masofani 20 sekundda yugurib o'tdi. Uning tezligi necha m/s?



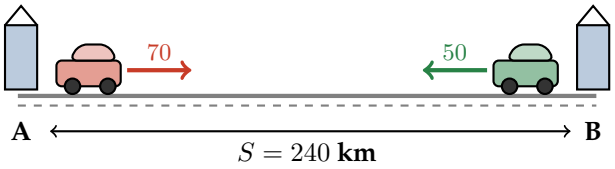
13 Velosipedchi 80 km masofani 4 soatda bosib o'tdi. Uning tezligini toping.



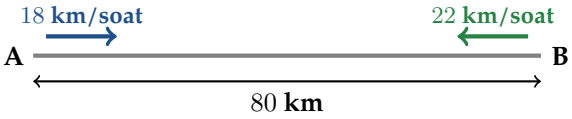
14 Poyezd soatiga 90 km tezlik bilan harakatlanmoqda. U 270 km masofani necha soatda bosib o'tadi?



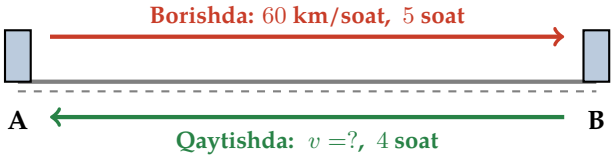
15 Ikki shahar orasidagi masofa 240 km. Bu shaharlardan bir vaqtning o'zida ikki avtomobil bir-biriga qarab yo'lga chiqdi. Birinchisining tezligi 70 km/soat, ikkinchisiniki 50 km/soat. Ular necha soatdan keyin uchrashadi?



16 Ikki shahar orasidagi masofa 80 km. Bir velosipedchi A shahridan, ikkinchisi B shahridan bir vaqtda bir-biriga qarab yo'lga chiqdi. Birinchisining tezligi 18 km/soat, ikkinchisiniki 22 km/soat bo'lsa, ular necha soatdan keyin uchrashadilar?

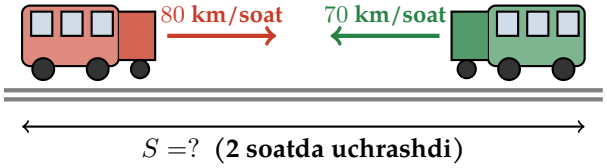


17 Avtomobil A shahridan B shahriga 60 km/soat tezlik bilan 5 soatda yetib bordi. Qaytishda esa shu yo'lni 4 soatda bosib o'tdi. Avtomobilning qaytishdagi tezligini toping.



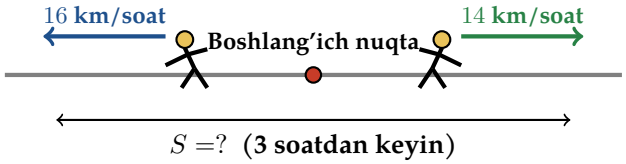
18 Sportchi 6 km masofani 30 daqiqada yugurib o'tdi. Uning tezligi necha km/soat?

19 Ikki poyezd ikki stansiyadan bir vaqtda bir-biriga qarab yo'lga chiqdi. Birinchisining tezligi 80 km/soat, ikkinchisiniki 70 km/soat. Ular 2 soatdan keyin uchrashishgan bo'lsa, stansiyalar orasidagi masofani toping.



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20 Ikki sportchi bitta nuqtadan bir vaqtda **teskari** (qarama-qarshi) yo'nalishlarda yo'lga chiqishdi. Birinchisining tezligi 16 km/soat, ikkinchisiniki 14 km/soat. 3 soatdan keyin ular orasidagi masofa qancha bo'ladi?

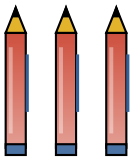


21–30. *Narxga doir masalalar.*

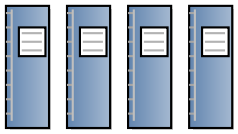
21 Bir daftarning narxi 3000 so'm bo'lsa, 5 ta daftar necha so'm turadi?

22 2 kg olma 16 000 so'm tursa, 5 kg olma necha so'm turadi?

23 Bitta ruchkaning narxi 2000 so'm, bitta daftarniki esa 4000 so'm. 3 ta ruchka va 4 ta daftar necha so'm turadi?



3 **ta ruchka**
(har biri 2000 so'm)



4 **ta daftar**
(har biri 4000 so'm)

24 6 ta tortning umumiy narxi 48 000 so'm. 1 ta tortning narxi qancha?

25 Bitta nonning narxi 2000 so'm bo'lsa, 12 000 so'mga nechta non olish mumkin?

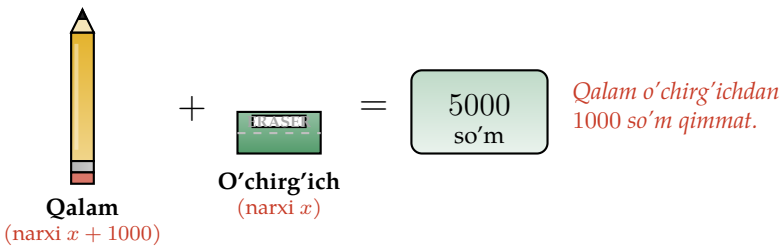
26 3 kg shakar 18 000 so'm tursa, 5 kg shakar necha so'm turadi?

27 Bir kishi 2 ta kitob va 3 ta daftar uchun 22 000 so'm to'ladi. Bitta kitob 5000 so'm bo'lsa, bir daftarning narxi qancha?

28 4 kg karam va 3 kg piyoz uchun 17 000 so'm to'landi. 1 kg karam 2000 so'm bo'lsa, 1 kg piyoz necha so'm turadi?

29 Do'konda 1 kg bananning narxi 20 000 so'm edi, chegirma e'lon qilingach 16 000 so'mga tushdi. 3 kg bananni chegirma bilan olganda qancha pul tejaladi?

30 O'quvchi do'konga 1 ta qalam va 1 ta o'chirg'ich uchun 5000 so'm to'ladi. Qalamning narxi o'chirg'ichnikidan 1000 so'm qimmat bo'lsa, har birining narxini toping.



31–40. *Yoshga doir masalalar.*

31 Anvar 12 yoshda, akasi esa undan 5 yosh katta. Anvarning akasi necha yoshda?

32 Otaning yoshi 35, o'g'lining yoshi 8 ga teng. O'g'il otasidan necha yosh kichik?

33 Ona bilan qizining yoshlari yig'indisi 45 ga teng. Ona qizidan 25 yosh katta bo'lsa, qizining yoshini toping.

34 Hozir aka 12 yoshda, uka 6 yoshda. 5 yildan keyin ularning yoshlari yig'indisi nechaga teng bo'ladi?

35 Aka 14 yoshda, ukasi 8 yoshda. Necha yil oldin akaning yoshi ukasining yoshidan 2 marta katta bo'lgan?

36 Otaning yoshi o'g'lining yoshidan 4 marta katta, ularning yoshlari ayirmasi 30 bo'lsa, otaning yoshini toping.

37 Karima 3 yil oldin 9 yoshda edi. 5 yildan keyin u necha yoshda bo'ladi?

38 Ikki opa-singilning yoshlari yig'indisi 30 yosh. Kattasi kichigidan 4 yosh katta bo'lsa, kichigining yoshini toping.

39 Otaning yoshi o'g'lining yoshidan 5 marta katta. 6 yildan keyin esa otaning yoshi o'g'lining yoshidan 3 marta katta bo'lsa, otaning hozirgi yoshini toping.

40 Hozir aka ukasidan 4 yosh katta. 5 yil oldin akaning yoshi ukasining yoshidan 3 marta katta bo'lgan bo'lsa, hozir akaning yoshi nechaga teng?

41–50. *O'rta arifmetik qiymat.*

41 12 va 8 sonlarning o'rta arifmetigi nechaga teng?

42 15, 21 va 30 sonlarning o'rta arifmetigi nechaga teng?

43 4, 8, 12, 16 va 20 sonlarning o'rta arifmetigi nechaga teng?

44 3, 11 va x sonlarning o'rta arifmetigi 8 ga teng bo'lsa, x ni toping.

45 To'rt sonning o'rta arifmetigi 15 ga teng. Ulardan uchtasi 10, 18 va 12 bo'lsa, to'rtinchi sonni toping.

46 Beshta sonning o'rta arifmetigi 8 ga teng. Bu sonlarga yana bitta son qo'shilgach, oltita sonning o'rta arifmetigi 10 ga teng bo'ladi. Qo'shilgan sonni toping.

47 Birdan boshlab ketma-ket olingan dastlabki nechta natural sonning o'rta arifmetigi 15 ga teng bo'ladi?

48 Birdan boshlab ketma-ket olingan dastlabki nechta natural sonning o'rta arifmetigi 10 ga teng bo'ladi?

49 Ikki sonning o'rta arifmetigi 17 ga teng. Bu sonlardan biri 14 bo'lsa, ikkinchisini toping.

50 Beshta sportchining og'irliklari: 65 kg, 68 kg, 72 kg, 70 kg va 75 kg. Ularning o'rtacha og'irligini toping.

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Mavzu bo'yicha test savollari

1-variant

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1. Bir songa 4 qo'shib, 3 ga ko'paytirilsa 27 hosil bo'ladi. Berilgan sonni toping.
   A) 4 B) 5 C) 6 D) 9

2. Ikki ketma-ket natural sonning yig'indisi 33 ga teng. Ularning kichigini toping.
   A) 15 B) 16 C) 17 D) 18

3. Avtomobil 70 km/soat tezlik bilan 4 soat yurdi. U qancha masofa bosib o'tgan?
   A) 260 km B) 270 km C) 280 km D) 300 km

4. Ikki shahar orasidagi masofa 300 km. Ulardan bir vaqtda ikki avtomobil bir-biriga qarab yo'lga chiqdi. Agar ularning tezliklari 60 km/soat va 90 km/soat bo'lsa, avtomobillar necha soatdan keyin uchrashishadi?
   A) 1 B) $1{,}5$ C) 2 D) $2{,}5$

5. 4 kg olma $24\,000$ so'm tursa, 7 kg olma necha so'm turadi?
   A) $36\,000$ B) $40\,000$ C) $42\,000$ D) $48\,000$

6. Botir 13 yoshda, akasi undan 7 yosh katta. Akaning yoshini toping.
   A) 6 B) 18 C) 20 D) 21

7. Ota bilan o'g'lining yoshlari yig'indisi 50. Ota o'g'lidan 30 yosh katta bo'lsa, otaning yoshini toping.
   A) 35 B) 40 C) 45 D) 30

8. 10 va 14 sonlarning o'rta arifmetigi nechaga teng?
   A) 11 B) 12 C) 13 D) 24

9. 5, 8 va x sonlarning o'rta arifmetigi 9 ga teng bo'lsa, x ni toping.
   A) 12 B) 13 C) 14 D) 15

10. Birdan boshlab ketma-ket olingan dastlabki nechta natural sonning o'rta arifmetigi 6 ga teng bo'ladi?
    A) 5 B) 6 C) 11 D) 12

2-variant

1. Bir songa 6 qo'shib, 2 ga ko'paytirilsa va natijadan 4 ayrilsa, 20 hosil bo'ladi. Noma'lum sonni toping.
   A) 4 B) 5 C) 6 D) 7
2. Ikki sonning yig'indisi 40, ayirmasi 10 ga teng bo'lsa, katta sonni toping.
   A) 15 B) 20 C) 25 D) 30
3. Velosipedchi 4 soatda 60 km yo'l bosib o'tgan bo'lsa, uning tezligini toping.
   A) 12 km B) 15 km C) 18 km D) 20 km
4. Poyezd 80 km/soat tezlik bilan 480 km masofani necha soatda bosib o'tadi?
   A) 4 B) 5 C) 6 D) 8
5. 3 ta ruchka 9000 so'm tursa, 7 ta ruchka necha so'm turadi?
   A) 18 000 B) 21 000 C) 24 000 D) 27 000
6. Karima 5 yil oldin 10 yoshda bo'lsa, hozir u necha yoshda?
   A) 12 B) 14 C) 15 D) 16
7. Aka 16 yoshda, uka 8 yoshda. Akaning yoshi ukasining yoshidan necha marta katta?
   A) 2 B) 3 C) 4 D) 8
8. 6, 12 va 18 sonlarning o'rta arifmetigi nechaga teng?
   A) 9 B) 10 C) 12 D) 15
9. To'rt sonning o'rta arifmetigi 9. Ulardan uchtasi 5, 8 va 12 ga teng bo'lsa, to'rtinchi sonni toping.
   A) 9 B) 10 C) 11 D) 13
10. Hozir o'g'ilning yoshi otasining yoshidan 4 marta kichik. Ularning yoshlari ayirmasi 27 bo'lsa, otaning yoshi nechaga teng?
    A) 32 B) 36 C) 40 D) 45

Javoblar

1. 2
2. 11
3. 15
4. 6 kg
5. 6
6. 13
7. 15
8. 18
9. 500 g
10. 8
11. 180 km
12. 5 m/s
13. 20 km/soat
14. 3 soat
15. 2 soat
16. 2 soat
17. 75 km/soat
18. 12 km/soat
19. 300 km
20. 90 km
21. 15 000 so'm
22. 40 000 so'm
23. 22 000 so'm
24. 8000 so'm
25. 6 ta
26. 30 000 so'm
27. 4000 so'm
28. 3000 so'm
29. 12 000 so'm
30. Qalam: 3000 so'm, o'chirg'ich: 2000 so'm
31. 17 yosh
32. 27 yosh
33. 10

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34. 28 yosh
35. 2 yil oldin
36. 40 yosh
37. 17 yosh
38. 13
39. 30
40. 11
41. 10
42. 22
43. 12
44. 10
45. 20
46. 20
47. 29
48. 19
49. 20
50. 70 kg

Test javoblari

1-variant: 1) B 2) B 3) C 4) C 5) C 6) C 7) B 8) B 9) C 10) C
2-variant: 1) C 2) C 3) B 4) C 5) B 6) C 7) A 8) C 9) C 10) B

Mavzu bo'yicha interaktiv testlar
QR kodni skanerlab, bilimingizni sinab ko'ring

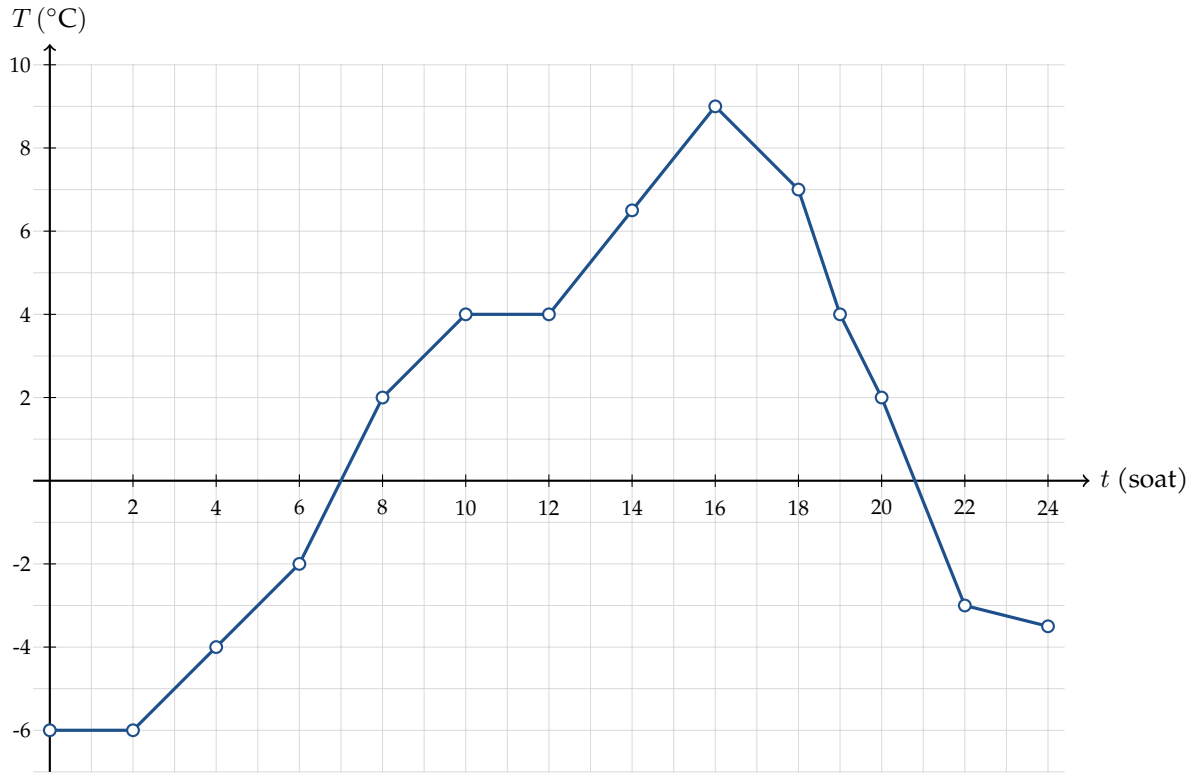
16.

Jadvallar va grafiklar

Misollar va masalalar

1–10. *1-misol. Sutka davomida havo harorati o'zgarishining grafigi.*

Quyidagi grafikda kunning 24 soati davomida havo haroratining (°C da) o'zgarishi ko'rsatilgan. Grafikka qarab savollarga javob bering.



1. Sutkalik eng past harorat necha gradus bo'lgan?
2. Sutkalik eng baland harorat necha gradus bo'lgan?
3. Eng baland harorat soat nechada kuzatilgan?
4. Soat 14:00 da harorat necha gradus bo'lgan?
5. Soat 8:00 da harorat necha gradus bo'lgan?
6. Kunning qaysi vaqtlarida havo harorati 2°C bo'lgan?

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7 Kunning qaysi vaqtlarida harorat $4°C$ bo'lgan?

8 Sutkadagi eng baland va eng past harorat necha gradusga farq qiladi?

9 Soat $4{:}00$ dan $14{:}00$ gacha harorat necha gradusga o'zgargan?

10 Sutka davomida harorat taxminan necha soat manfiy $(T < 0)$ bo'lgan?

11–18. *2-misol. Sinflararo futbol musobaqasi jadvali.*

5-sinflar o'rtasida o'tkazilgan futbol musobaqasi natijalari quyidagi jadvalda berilgan. Har g'alaba uchun 3 ochko, durrang uchun 1 ochko, mag'lubiyat uchun 0 ochko beriladi. Katakdagi $a{:}b$ yozuvi — mos satrdagi sinfning shu ustundagi sinf bilan o'yini natijasi (a — satrdagi sinf kiritgan gollar soni, b — ustundagi sinf kiritgan gollar soni).

Sinf	**5-A**	**5-B**	**5-C**	**5-D**	**5-E**	**5-F**
5-A	—	2:3	1:2	0:0	1:2	2:1
5-B	3:2	—	5:0	0:1	4:2	2:0
5-C	2:1	0:5	—	1:1	2:2	1:0
5-D	0:0	1:0	1:1	—	0:2	2:2
5-E	2:1	2:4	2:2	2:0	—	1:1
5-F	1:2	0:2	0:1	2:2	1:1	—

11 5-A sinfning 5-B sinfi bilan bo'lgan o'yini natijasi qanday yakunlangan?

12 5-B sinf nechta o'yinda g'alaba qozongan?

13 5-A sinf jami necha ochko to'plagan?

14 Musobaqada qaysi sinf eng ko'p ochko to'plab chempion bo'lgan?

15 5-E sinf nechta o'yinda mag'lub bo'lgan?

16 Musobaqada nechta o'yin durrang bilan yakunlangan?

17 5-C sinf butun musobaqada jami nechta gol kiritgan?

18 5-D sinfning ochkosi 5-F sinfning ochkosidan necha marta ko'p?

19–25. *3-misol. Pochta xizmati tariflari jadvali.*

Dilshod shahar pochtasi orqali jo'natma yubormoqchi. Pochta ikki xil tarif bo'yicha xizmat taklif qiladi. Tariflar quyidagi jadvalda berilgan:

Vazn (g)	1-usul (so'm)	2-usul (so'm)
$1-70$	10 000	14 000
$71-200$	11 000	14 000
$201-300$	12 000	15 000
$301-450$	14 000	15 000
$451-600$	16 000	16 000
$601-750$	18 000	17 000
$751-900$	20 000	18 000
$901-1000$	25 000	20 000

19 50 grammlik jo'natmani 1-usul bilan yuborish necha so'm turadi?

20 350 grammlik jo'natmani 2-usul bilan yuborish necha so'm turadi?

21 830 grammlik jo'natmani 1-usul bilan yuborish narxi qancha bo'ladi?

22 830 grammlik jo'natmani 2-usul bilan yuborish narxi qancha bo'ladi?

23 830 grammlik jo'natma uchun qaysi usul arzonroq va qancha so'mga arzon?

24 1-usulda 600 grammlik jo'natma 200 grammlik jo'natmadan necha so'mga qimmat?

25 Qaysi vazn oralig'ida ikkala usul ham bir xil narxda?

26–33. *4-misol. 4-sinf o'quvchilarining 3-chorakdagi davomati.*

Jadvalda 4-A, 4-B va 4-C sinflarining yanvar, fevral, mart oylaridagi o'rtacha kunlik davomati (o'quvchilar soni) ko'rsatilgan.

Sinf	Yanvar	Fevral	Mart
4-A	27	25	26
4-B	24	24	23
4-C	25	23	24

26 4-A sinfda yanvar oyida o'rtacha nechta o'quvchi darsga qatnashgan?

27 Mart oyida 4-B sinfda o'rtacha nechta o'quvchi darsga qatnashgan?

28 Fevralda eng ko'p davomat qaysi sinfda kuzatilgan?

29 4-C sinfning yanvar, fevral, mart oylaridagi o'rtacha davomatini toping.

30 Uch oy davomida 4-A sinfning o'rtacha davomatini toping.

31 Mart oyida 4-B va 4-C sinflar davomati farqi nechtaga teng?

32 Uch oy yig'indisi bo'yicha qaysi sinfda davomat eng yuqori bo'lgan?

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33 4-C sinfda fevral oyida yanvarga nisbatan davomat nechta o'quvchiga kamaygan?

34–41. *5-misol. "Top Food" gamburger savdosi jadvali.*

"Top Food" oshxonasida gamburger to'plamlari quyidagi narxlarda sotiladi.

Gamburgerlar soni	1	2	3	4	5
Narxi (so'm)	31 000	55 000	85 000	110 000	144 000

34 Bitta gamburger qancha turadi?

35 4 ta gamburger qancha turadi?

36 5 talik to'plamda bitta gamburger o'rtacha qancha turadi?

37 Bitta gamburger eng arzonga tushadigan to'plam(lar)ni toping.

38 Nozim 12 ta gamburgerni eng kam pul sarflab necha so'mga sotib oladi?

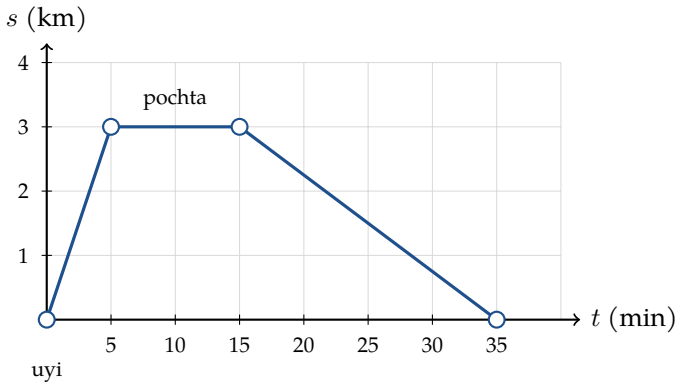
39 Agar Nozim faqat bittalab gamburger sotib olsa, 12 ta uchun qancha to'laydi va eng tejamli usuldan necha so'mga ko'p pul to'laydi?

40 10 ta gamburgerni eng kam pul sarflab necha so'mga sotib olish mumkin?

41 Nozim 12 ta gamburgerni eng tejamli yo'l bilan olib, kassirga 500 ming so'm pul uzatdi. Nozim qancha qaytim oladi?

42–51. *6-misol. Velosipedchining harakat grafigi.*

Velosipedchi uydan pochtaga borib, u yerda biroz turdi va keyin uyiga qaytdi. Grafikda undan **uyigacha bo'lgan masofa** (km) **vaqtga** (minutda) bog'liq holda berilgan.



42 Velosipedchining uydan pochtagacha bo'lgan masofasi necha km?

43 Velosipedchi pochtaga borishga necha minut sarflagan?

44 Velosipedchi pochtada necha minut turgan?

45 Velosipedchi pochtadan uyga qaytishga necha minut sarflagan?

46 Velosipedchining uydan pochtagacha bo'lgan o'rtacha tezligi necha km/soat?

47 Velosipedchining pochtadan uyga qaytishdagi o'rtacha tezligi necha km/soat?

48 Velosipedchi jami necha km yo'l bosib o'tgan?

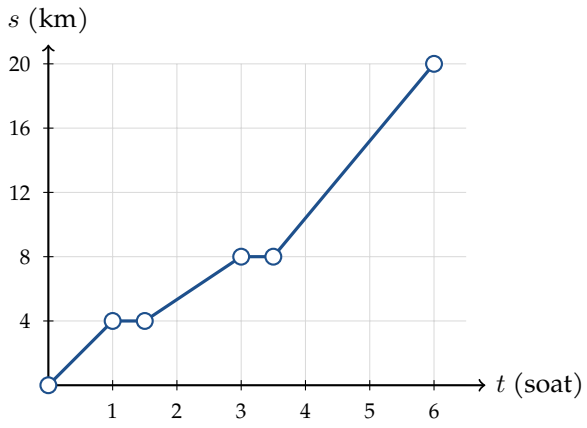
49 Velosipedchining butun safardagi (uydan chiqqanidan to qaytib kelguniga qadar) o'rtacha tezligi necha km/soat?

50 Velosipedchi qaysi yo'lda (borishda yoki qaytishda) tezroq harakatlangan?

51 Velosipedchi uyidan jo'naganidan to qaytib kelguniga qadar jami necha minut sarflagan?

52–62. *7-misol. Turistlarning harakat grafigi.*

Turistlar yig'ilish joyiga qadar yo'l bosib bordilar. Grafikda bosib o'tilgan **masofa** (km) **vaqtga** (soatda) bog'liq holda berilgan.



52 Turistlar yig'ilish joyigacha jami necha km yo'l yurishgan?

53 Butun yo'lga turistlar necha soat sarflashgan?

54 Birinchi 1 soat ichida turistlar necha km yo'l yurishgan?

55 Turistlar yo'lga chiqqanidan keyin 8 km masofaga necha soatdan so'ng yetib kelishgan?

56 Turistlar yo'lda jami necha marta dam olishgan?

57 Turistlar dam olish uchun jami necha soat sarflashgan?

58 Ikkinchi dam olishdan keyin manzilga yetib kelguncha necha soat o'tgan?

59 Birinchi 1 soatdagi o'rtacha tezlik qanday bo'lgan?

60 Yo'lning oxirgi qismida (3,5 soatdan keyin) o'rtacha tezlik qancha bo'lgan?

61 Butun yo'l davomidagi (dam olish vaqti bilan birga) o'rtacha tezlik qancha?

62 Yo'lning qaysi qismida turistlar eng tez harakatlanganlar?

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63–70. *8-misol. Avtomobillarning yoqilg'i sarfi jadvali.*

Jadvalda A, B, C, D, E avtomobillari bosib o'tgan masofa va shu masofaga sarflangan benzin miqdori keltirilgan.

Avtomobil	A	B	C	D	E
Sayohat uzunligi (km)	144	95	345	70	78
Benzin sarfi (litr)	16	10	23	7	9

63 C avtomobili 345 km yo'lda qancha litr benzin sarflaydi?

64 A avtomobilining 1 km uchun benzin sarfi qancha litr (*kasr ko'rinishida*)?

65 B avtomobilining 1 km uchun benzin sarfi qancha litr?

66 Qaysi avtomobil 1 km masofaga eng kam benzin sarflaydi?

67 Qaysi avtomobil 1 km masofaga eng ko'p benzin sarflaydi?

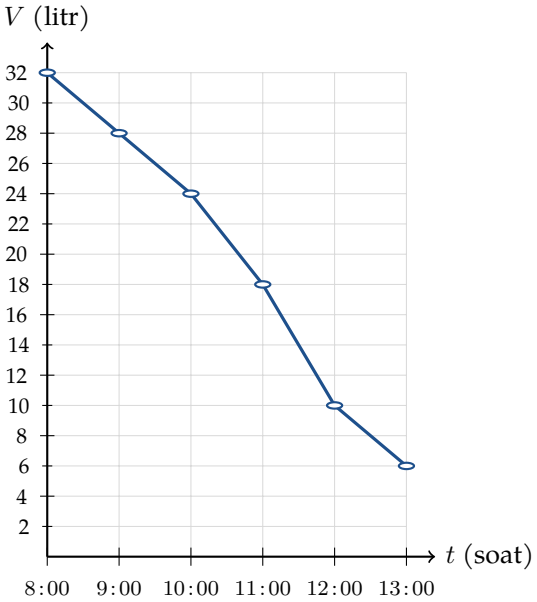
68 D avtomobili 350 km yo'lga necha litr benzin sarflashi mumkin?

69 Agar A va C avtomobillari jadvalda ko'rsatilgan mos masofalarni yursa, jami benzin sarfi qancha bo'ladi?

70 E avtomobili 156 km masofaga qancha litr benzin sarflaydi?

71–79. *9-misol. Avtomobil bakidagi yoqilg'ining sarflanish grafigi.*

Avtomobil soat 8:00 dan 13:00 gacha yo'lda bo'ldi. Quyidagi grafikda uning bakidagi yoqilg'i miqdori (litrda) vaqt davomida qanday kamayganligi ko'rsatilgan.



71 Soat 8:00 da bakda necha litr yoqilg'i bor edi?

72 Soat 13:00 da bakda necha litr yoqilg'i qoldi?

73 Soat 8:00 dan 13:00 gacha jami qancha yoqilg'i sarflandi?

74 Soat 10:00 dan 13:00 gacha qancha yoqilg'i sarflandi?

75 10:00–13:00 oralig'ida sarflangan yoqilg'i boshlang'ich yoqilg'ining qancha qismini tashkil etadi?

76 Soat 10:00 da bakda necha litr yoqilg'i bor edi?

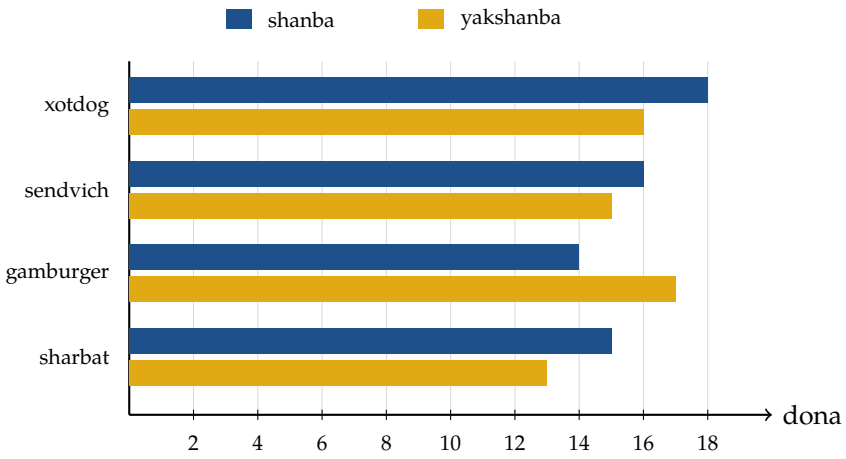
77 Qaysi 1 soatlik oraliqda eng ko'p yoqilg'i sarflandi va qancha?

78 Agar avtomobil 1 km uchun 0,2 litr sarflasa, 8:00–13:00 oralig'ida necha km yurgan?

79 5 soat davomida o'rtacha (1 soatda) qancha yoqilg'i sarflangan?

80–87. *10-misol. Shanba va yakshanba kunlari savdosi diagrammasi.*

Diagrammada shanba va yakshanba kunlari xotdog, sendvich, gamburger va sharbatlardan qancha miqdorda sotilganligi ko'rsatilgan.



80 Shanba kuni nechta xotdog sotilgan?

81 Yakshanba kuni nechta sendvich sotilgan?

82 Ikki kunda jami nechta gamburger sotilgan?

83 Ikki kunda jami nechta sharbat sotilgan?

84 Shanba kuni jami nechta mahsulot sotilgan?

85 Yakshanba kuni jami nechta mahsulot sotilgan?

86 Ikki kunda eng ko'p sotilgan mahsulot qaysi va u jami nechta sotilgan?

87 Shanba va yakshanba kunlari to'rt xil mahsulotdan jami necha dona sotilgan?

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88–95. *11-misol. Tezkor savol-javob o'yini natijalari jadvali.*

Sinfda har kuni 10 ta savoldan iborat tezkor savol-javob o'yini o'tkaziladi. Har bir to'g'ri javob uchun 5 ball beriladi, har bir xato javob uchun esa 2 ball ayiriladi. 3 kun mobaynida o'quvchilar nechta savolga to'g'ri javob berganligi quyidagi jadvalda ko'rsatilgan.

O'quvchining ismi	Dushanba	Seshanba	Chorshanba
Diyora	7	8	7
Sherzod	8	9	9
Jahongir	7	10	8
Muslima	9	8	7

88 Diyora dushanba kuni nechta to'g'ri javob bergan?

89 Jahongir seshanba kuni nechta xato javob bergan?

90 Chorshanba kuni eng ko'p to'g'ri javobni kim bergan?

91 Muslima 3 kun davomida jami nechta to'g'ri javob bergan?

92 Sherzod 3 kun davomida jami nechta xato javob bergan?

93 Diyora dushanba kuni umumiy necha ball to'plagan?

94 Jahongir 3 kun davomida jami necha ball to'plagan?

95 3 kunlik umumiy ball bo'yicha eng yuqori va eng past ball orasidagi farq nechiga teng?

96–100. *12-misol. Bozordagi mahsulotlar narxi jadvali.*

Quyidagi jadvalda bozordagi mahsulotlar narxi va xarid qilinadigan miqdori berilgan.

Nomi	1 kg narxi (so'm)	Xarid miqdori (kg)
Pomidor	9 000	3
Baqlajon	13 000	4
Bodring	8 000	5

96 3 kg pomidorning narxi necha so'm bo'ladi?

97 4 kg baqlajonning narxi necha so'm bo'ladi?

98 5 kg bodringning narxi necha so'm bo'ladi?

99 Jami necha kilogramm mahsulot xarid qilingan?

100 Jami xarid qilingan mahsulot uchun jami qancha so'm to'lash kerak?

Mavzu bo'yicha test savollari

1-variant

1. Yoz oyida do'kon 4 hafta davomida quyidagicha savdo qildi: 1-hafta 250 ta, 2-hafta 320 ta, 3-hafta 280 ta, 4-hafta 350 ta mahsulot. 4 hafta davomida jami nechta mahsulot sotilgan?
   A) 1100 B) 1150 C) 1200 D) 1250

2. Sutka davomida soat $6{:}00$ da havo harorati $-3°\text{C}$, soat $16{:}00$ da $+9°\text{C}$ bo'lgan. Bu vaqtlar orasida harorat necha gradusga ko'tarilgan?
   A) $6°\text{C}$ B) $9°\text{C}$ C) $12°\text{C}$ D) $15°\text{C}$

3. Bozorda 1 kg olma $8\,000$ so'm, 1 kg nok $12\,000$ so'm. Baxtiyor 3 kg olma va 2 kg nok sotib olsa, u qancha pul to'laydi?
   A) $40\,000$ B) $44\,000$ C) $48\,000$ D) $52\,000$

4. Bir yilda Anvar 25 ta, Bekzod 18 ta, Doniyor 32 ta va Sherzod 27 ta kitob o'qigan bo'lsa, birgalikda nechta kitob o'qilgan?
   A) 92 B) 102 C) 112 D) 122

5. Avtomobil bakidagi yoqilg'i miqdori soat $8{:}00$ da 40 litr, soat $14{:}00$ da 16 litr bo'lgan. 6 soatda qancha yoqilg'i sarflangan?
   A) 16 litr B) 20 litr C) 24 litr D) 30 litr

6. Sinflararo musobaqada 8-A sinf 4 ta g'alaba, 1 ta durrang va 2 ta mag'lubiyat qozondi (3-1-0 tizimda). 8-A sinf jami necha ochko to'plagan?
   A) 11 B) 12 C) 13 D) 14

7. Yopilmagan jo'mrakdan 1 daqiqada 60 ta tomchi suv tushadi. 1 soatda nechta tomchi tushadi?
   A) 600 B) $1\,200$ C) $1\,800$ D) $3\,600$

8. Asal narxi: 1 kg $80\,000$ so'm. Mijoz 250 g asal sotib oldi. U qancha to'laydi?
   A) $15\,000$ B) $20\,000$ C) $25\,000$ D) $30\,000$

9. Turistlar 3 soatda 12 km, keyin 1 soat dam, so'ng yana 2 soatda 8 km yo'l yurdi. Jami vaqt bo'yicha o'rtacha tezligni toping.
   A) $2{,}5$ km/soat B) 3 km/soat C) $\dfrac{10}{3}$ km/soat D) 4 km/soat

10. 4 ta o'quvchining har biri 5 tadan savolga javob berdi. Ularning bergan to'g'ri javoblari mos ravishda 4, 3, 5, 3 ta. To'g'ri javoblar umumiy savollarning necha foizini tashkil etadi?
    A) 60% B) 65% C) 70% D) 75%

2-variant

1. Quyida dastlabki 4 oydagi savdo ko'rsatkichlari berilgan: yanvar 240, fevral 280, mart 310, aprel 270. Eng ko'p savdo qaysi oyda bo'lgan?
   A) yanvar B) fevral C) mart D) aprel

2. Sutka davomida soat $4{:}00$ da havo harorati $-5°\text{C}$, soat $14{:}00$ da $+7°\text{C}$ bo'lgan. Bu vaqtlar orasida harorat qanchaga o'zgargan?
   A) $8°\text{C}$ B) $10°\text{C}$ C) $12°\text{C}$ D) $14°\text{C}$

3. Bir ishchi 8 soatda 96 ta detal tayyorlasa, u soatiga o'rtacha nechta detal tayyorlaydi?
   A) 10 B) 11 C) 12 D) 14

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4. Quyida 5 ta o'quvchining olgan bahosi keltirilgan: 4, 5, 4, 3, 5. O'quvchilarning o'rtacha bahosi nechaga teng?
A) 4 B) 4,2 C) 4,5 D) 5

5. Statistikaga ko'ra A maktabda 120 ta, B — 150 ta, C — 90 ta, D — 140 ta o'quvchi o'qiydi. Jami nechta o'quvchi bor?
A) 480 B) 500 C) 520 D) 540

6. Avtomobil 4 soatda 280 km bosib o'tdi. Avtomobilning o'rtacha tezligi qancha?
A) 60 km/soat B) 65 km/soat C) 70 km/soat D) 75 km/soat

7. Soatiga 4 dona mahsulot ishlab chiqaradigan stanok 24 soatda nechta mahsulot ishlab chiqaradi?
A) 72 B) 84 C) 96 D) 108

8. Sport musobaqasida 4 sportchining natijalari: 11,2 s; 11,5 s; 10,9 s va 11,7 s. Eng tez yugurgan sportchining natijasi qancha?
A) 11,7 s B) 11,5 s C) 11,2 s D) 10,9 s

9. Telefon tarifiga ko'ra birinchi 5 daqiqa bepul, qolgan har bir daqiqa 200 so'm. 12 daqiqalik suhbat necha so'm turadi?
A) 1 200 B) 1 400 C) 1 500 D) 2 400

10. Quyida 3 sinfning o'quvchilari soni keltirilgan: 5-A — 28, 5-B — 24, 5-C — 22 ta. Eng ko'p o'quvchili sinf qolgan ikki sinfning o'rtacha o'quvchilar sonidan nechtaga ko'p?
A) 3 B) 4 C) 5 D) 6

Javoblar

1. $-6°$C

2. 9°C

3. soat 16:00 da

4. 6,5°C

5. 2°C

6. soat 8:00 va 20:00 da

7. soat 10:00 dan 12:00 gacha va soat 19:00 da

8. 15°C

9. 10,5°C ga oshgan

10. taxminan 10 soat

11. 2:3 (mag'lub)

12. 4 ta

13. 4 ochko

14. 5-B sinf (12 ochko)

15. 1 ta

16. 5 ta

17. 6 ta gol

18. 3 marta

19. 10 000 so'm

20. 15 000 so'm

21. 20 000 so'm

22. 18 000 so'm

23. 2-usul; 2 000 so'mga arzon

24. 5 000 so'mga

25. $451 - 600$ g oralig'ida

26. 27 ta

27. 23 ta

28. 4-A sinfda (25 ta)

29. 24 ta

30. 26 ta

31. 1 ta (4-C ko'p)

32. 4-A (78 jami)

33. 2 o'quvchiga

34. 31 000 so'm

35. 110 000 so'm

36. 28 800 so'm

37. 2 va 4 talik to'plamlar

38. 330 000 so'm

39. 372 000 so'm; 42 000 so'mga ko'p

40. 275 000 so'm

41. 170 000 so'm

42. 3 km

43. 5 minut

44. 10 minut

45. 20 minut

46. 36 km/soat

47. 9 km/soat

48. 6 km

49. $\approx 10{,}29$ km/soat

50. borishda

51. 35 minut

52. 20 km

53. 6 soat

54. 4 km

55. 3 soat

56. 2 marta

57. 1 soat

58. 2,5 soat

59. 4 km/soat

60. 4,8 km/soat

61. $\frac{10}{3} \approx 3{,}33$ km/soat

62. oxirgi qism (3,5–6 soat)

63. 23 litr

64. $\frac{1}{9}$ litr

65. $\frac{2}{19}$ litr

66. C avtomobil

67. E avtomobil

68. 35 litr

69. 39 litr

70. 18 litr

71. 32 litr

72. 6 litr

73. 26 litr

74. 18 litr

75. $\frac{9}{16}$ qism

76. 24 litr

77. 11:00–12:00; 8 litr

78. 130 km

79. 5,2 litr/soat

80. 18 ta

81. 15 ta

82. 31 ta

83. 28 ta

84. 63 ta

85. 61 ta

86. xotdog; 34 ta

87. 124 ta

88. 7 ta

89. 0 ta

90. Sherzod

91. 24 ta

92. 4 ta

93. 29 ball

94. 115 ball

95. 28 ball

96. 27 000 so'm

97. 52 000 so'm

98. 40 000 so'm

99. 12 kg

100. 119 000 so'm

Test javoblari

1-variant: 1) C 2) C 3) C 4) B 5) C 6) C 7) D 8) B 9) C 10) D

2-variant: 1) C 2) C 3) C 4) B 5) B 6) C 7) C 8) D 9) B 10) C

Mavzu bo'yicha interaktiv testlar

QR kodni skanerlab, bilimingizni sinab ko'ring

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17.

Burchaklar va masofalar

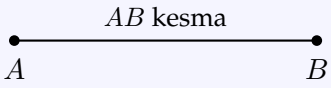
Nazariy qism

Kesma va uning uzunligi

✎ Ta’rif

Kesma deb to’g’ri chiziqning ikki nuqta bilan chegaralangan qismiga aytiladi. Bu ikki nuqta kesmaning **uchlari** deyiladi.

Belgilanishi: AB — uchlari A va B nuqtalar bo’lgan kesma. Uning uzunligi ham qisqacha AB bilan belgilanadi (masalan, $AB = 12$ sm).

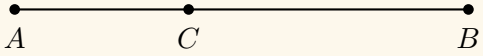


Kesmani nuqta orqali bo’lish

✠ Asosiy qoida

Agar C nuqta AB kesmaga tegishli bo’lsa, u holda

$$\boxed{AB = AC + CB}$$



Bundan kelib chiqadi:

$$AC = AB - CB, \qquad CB = AB - AC.$$

Agar to’g’ri chiziqda $A,\ B,\ C,\ D$ nuqtalar shu tartibda joylashgan bo’lsa:

$$AD = AB + BC + CD, \qquad AC = AB + BC, \qquad BD = BC + CD.$$

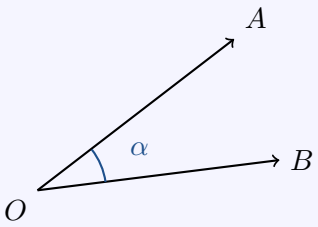
Burchak tushunchasi

✎ Ta'rif

Bir nuqtadan chiquvchi ikki nurdan iborat shakl **burchak** deyiladi.

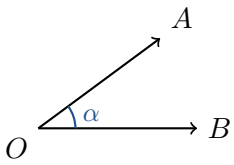
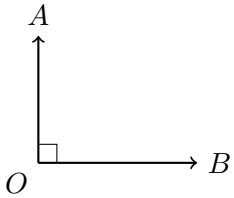
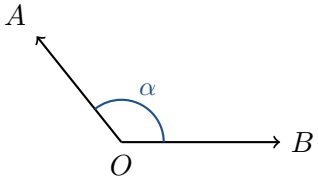
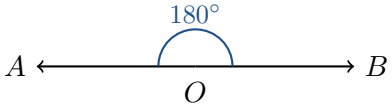
- Nurlarning umumiy nuqtasi — burchakning **uchi** deyiladi;
- Nurlar — burchakning **tomonlari** deyiladi.
- Burchak **gradus** ($^\circ$) larda o'lchanadi;

Belgilanishi: $\angle AOB$, $\angle O$, $\angle 1$, α, β, γ, ...



$\angle AOB = \alpha$ — uchi O, tomonlari OA, OB.

Burchak turlari

1) O'tkir burchak. $0^\circ < \alpha < 90^\circ$. To'g'ri burchakdan kichik. Misol: 30°, 45°, 60°, 89°.	
2) To'g'ri burchak. $\alpha = 90^\circ$. Tomonlari o'zaro **perpendikulyar** ($OA \perp OB$).	
3) O'tmas burchak. $90^\circ < \alpha < 180^\circ$. To'g'ri burchakdan katta, lekin yoyiqdan kichik. Misol: 100°, 135°, 179°.	
4) Yoyiq burchak. $\alpha = 180^\circ$. Tomonlari bir to'g'ri chiziqda yotadi.	
5) $\alpha = 360^\circ$. Nur o'zining boshlang'ich nuqtasi atrofida bir marta to'la aylanib, dastlabki holatiga qaytadi.	

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Qo'shni burchaklar

✎ Ta'rif

Bittadan tomoni ustma-ust tushib, qolgan tomonlari bir-birini to'ldiruvchi nurlardan iborat bo'lgan ikkita burchak **qo'shni burchaklar** deyiladi.

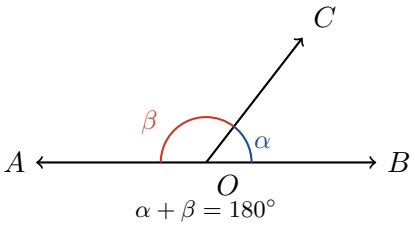
✠ Asosiy qoida

Qo'shni burchaklar yig'indisi $180°$ ga teng:

$$\angle AOC + \angle COB = 180°.$$

Demak, agar bir burchakni bilsak, ikkinchisini quyidagicha topamiz:

$$\angle COB = 180° - \angle AOC.$$



$\alpha + \beta = 180°$

Vertikal burchaklar

✠ Asosiy qoida

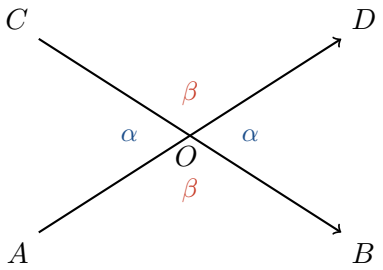
Vertikal burchaklar o'zaro teng:

$$\angle AOC = \angle BOD, \qquad \angle AOB = \angle COD.$$

Bundan tashqari, qo'shni bo'lgan har qaysi juftlik $180°$ ni tashkil etadi:

$$\angle AOC + \angle AOB = 180°.$$

Demak, ikki kesishuvchi to'g'ri chiziqdan hosil bo'lgan to'rtta burchakdan **faqat bittasi ma'lum bo'lsa, qolgan uchtasini topish mumkin**.



$\alpha + \beta = 180°,\ \angle AOC = \angle BOD = \alpha.$

Burchak bissektrisasi

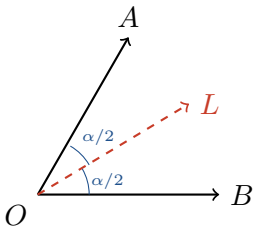
✎ Ta'rif

Burchakning uchidan chiqib, uni teng ikki burchakka ajratuvchi nur **burchak bissektrisasi** deyiladi.

✠ Asosiy qoida

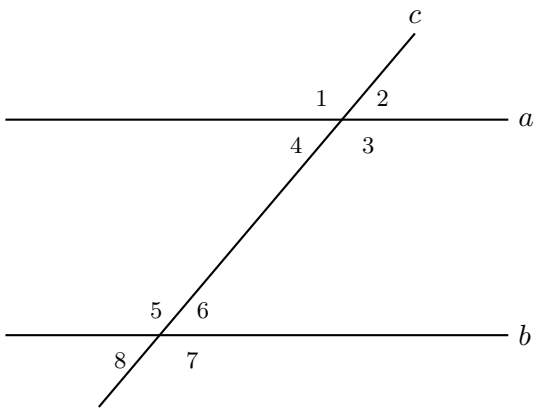
Agar OL — $\angle AOB$ ning bissektrisasi bo'lsa, u holda

$$\angle AOL = \angle LOB = \frac{\angle AOB}{2}.$$



Parallel to'g'ri chiziqlar va kesuvchi chiziq

$a \parallel b$ to'g'ri chiziqlar uchinchi c chiziq (**kesuvchi**) bilan kesishganda **8 ta burchak** hosil bo'ladi. Ulardan bir nechta muhim juftliklarini bilish zarur:



✠ Asosiy qoida

$a \parallel b$ bo'lsa, quyidagilar o'rinli:

1) Mos burchaklar teng:

$$\angle 1 = \angle 5, \quad \angle 2 = \angle 6, \quad \angle 3 = \angle 7, \quad \angle 4 = \angle 8.$$

2) Almashinuvchi ichki burchaklar teng:

$$\angle 4 = \angle 6, \qquad \angle 3 = \angle 5.$$

3) Almashinuvchi tashqi burchaklar teng:

$$\angle 1 = \angle 7, \qquad \angle 2 = \angle 8.$$

4) Bir tomonli ichki burchaklar yig'indisi 180°:

$$\angle 4 + \angle 5 = 180^\circ, \qquad \angle 3 + \angle 6 = 180^\circ.$$

5) Vertikal burchaklar teng:

$$\angle 1 = \angle 3, \quad \angle 2 = \angle 4, \quad \angle 5 = \angle 7, \quad \angle 6 = \angle 8.$$

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Uchburchakning ichki burchaklari yig'indisi

✠ Asosiy qoida

Har qanday uchburchakning **ichki burchaklari yig'indisi 180°** ga teng:

$$\angle A + \angle B + \angle C = 180^\circ.$$

Xususiy hollar:

- **To'g'ri burchakli** uchburchakda ikkita o'tkir burchak yig'indisi 90°;
- **Teng tomonli** uchburchakning har bir burchagi 60°;
- **Teng yonli** uchburchakda asosdagi burchaklar teng.

Teng yonli uchburchak xossasi

✎ Ta'rif

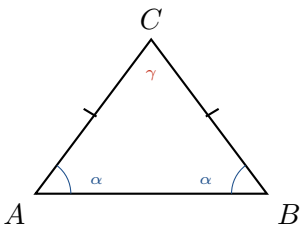
Ikki tomoni o'zaro teng bo'lgan uchburchak **teng yonli uchburchak** deyiladi. Teng tomonlari — **yon tomonlar**, uchinchi tomoni — **asos** deb ataladi. Yon tomonlar uchrashadigan uch — uchburchakning **uchi** deyiladi.

✠ Asosiy qoida

Asosiy xossa. Teng yonli uchburchakda **asosdagi ikki burchak o'zaro teng**:

$$\boxed{\angle A = \angle B.}$$

Aksincha: agar uchburchakning ikki burchagi teng bo'lsa, u teng yonli uchburchak.



To'rtburchakning ichki burchaklari yig'indisi

✠ Asosiy qoida

Har qanday **to'rtburchak**ning ichki burchaklari yig'indisi **360°** ga teng:

$$\angle A + \angle B + \angle C + \angle D = 360^\circ.$$

✓ Eslatma

- **To'g'ri to'rtburchak** va **kvadratning** har bir burchagi 90°.

Soat millari va burchaklar

✠ Asosiy qoida

Minut mili (uzun mil) bir to'la aylanani **60 minutda** bosib o'tadi:

$$\frac{360^\circ}{60} = 6^\circ \quad \text{har bir minutda.}$$

Soat mili (qisqa mil) bir to'la aylanani **12 soatda** bosib o'tadi:

$$\frac{360^\circ}{12} = 30^\circ \quad \text{har bir soatda.}$$

Bir minutda soat mili qanchaga buriladi?

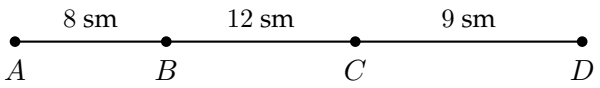
$$\frac{30^\circ}{60} = 0{,}5^\circ \quad \text{har bir minutda.}$$

Vaqt	Minut mili buriladi	Soat mili buriladi
1 minut	6°	$0{,}5^\circ$
1 soat (60 min)	360° (to'liq aylana)	30°

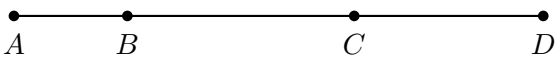
Amaliy qism

1–10. *Kesma va masofa. Kesmalar yig'indisi.*

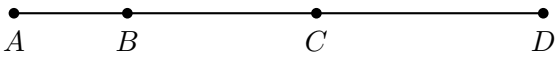
1 Chizmaga ko'ra AD kesma uzunligini toping.



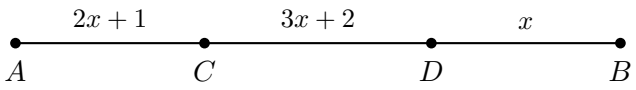
2 A, B, C, D nuqtalar berilgan. $AC = 20$ sm, $BD = 22$ sm, $AD = 32$ sm. BC kesma uzunligini toping.



3 A, B, C, D nuqtalar bir to'g'ri chiziqda tartib bilan joylashgan. $AB = 7$ sm, $AC = 18$ sm, $BD = 24$ sm. AD ni toping.

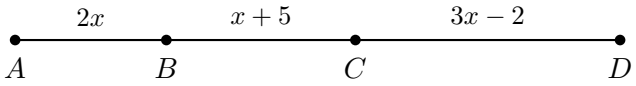


4 Chizmaga ko'ra $AD = 33$ bo'lsa, AB kesma uzunligini toping.

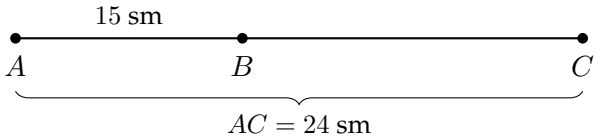


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5 $A,\ B,\ C,\ D$ nuqtalar bir to'g'ri chiziqda shu tartibda joylashgan. $AB = 2x$, $BC = x + 5$, $CD = 3x - 2$, $AD = 39$. BD kesma uzunligini toping.



6 Chizmaga ko'ra BC kesma uzunligini toping.



7 C nuqta AB kesmaga tegishli. $AC = 11$ sm, $AB = 27$ sm bo'lsa, CB kesma uzunligini toping.

8 B nuqta A va C nuqtalar orasida joylashgan. $AB = 14$ sm, $BC = 9$ sm bo'lsa, AC kesma uzunligini toping.

9 $A,\ B,\ C,\ D$ nuqtalar bir to'g'ri chiziqda shu tartibda joylashgan. $AB = 6$ sm, $BC = 8$ sm, $CD = 11$ sm bo'lsa, AD kesma uzunligini toping.

10 B nuqta A va C nuqtalar orasida. $AC = 36$ sm, AB esa BC dan 8 sm uzun bo'lsa, AB va BC kesmalarni toping.

11–60. *Burchak, uning turlari, qo'shni va vertikal burchaklar, bissektrisa.*

11 Quyidagi burchaklarning turini aniqlang: $a)\ 35°;\ b)\ 90°;\ c)\ 117°;\ d)\ 180°;\ e)\ 89°;\ f)\ 360°$.

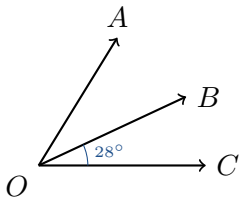
12 Yoyiq burchakning $\frac{1}{4}$ qismi necha gradusga teng?

13 Yoyiq burchakning $\frac{3}{5}$ qismi necha gradusga teng?

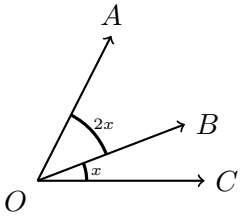
14 To'g'ri burchakning $\frac{2}{3}$ qismi necha gradusga teng?

15 Yoyiq burchakning $\frac{5}{6}$ qismi necha gradusga teng?

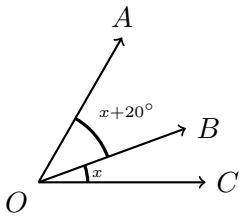
16 $\angle AOC = 47°$, $\angle BOC = 28°$ va OB nur $\angle AOC$ ning ichida yotadi. $\angle AOB$ ni toping.



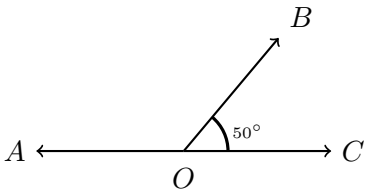
17 $\angle AOC = 63°$. Rasmda ko'rsatilgandek $\angle AOB$ ning kattaligi $2x$, $\angle BOC$ niki x. x ni toping.



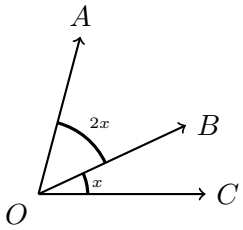
18 $\angle AOC = 60°$. OB nur shu burchakni ikkiga ajratadi va bunda $\angle AOB = x + 20°$, $\angle BOC = x$. x ni toping.



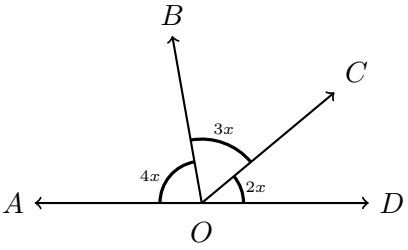
19 OB nur yoyiq $\angle AOC = 180°$ ning ichidan o'tadi. $\angle BOC = 50°$. $\angle AOB$ ni toping.



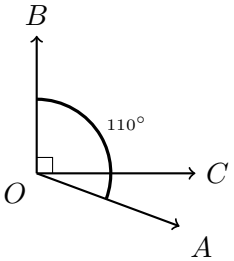
20 Bir nuqtadan uchta nur (OA, OB, OC) chiqqan va ular ketma-ket joylashgan. $\angle AOB = 2x$, $\angle BOC = x$. Agar $\angle AOC = 75°$ bo'lsa, $\angle AOB$ ni toping.



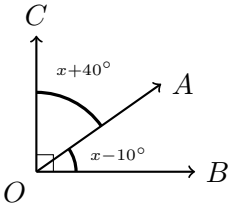
21 Bir nuqtadan to'rtta nur ketma-ket chiqqan: $\angle AOB = 4x$, $\angle BOC = 3x$, $\angle COD = 2x$ va $\angle AOD$ yoyiq. $\angle BOC$ ni toping.



22 $OB \perp OC$ va $\angle AOB = 110°$. $\angle AOC$ ni toping .

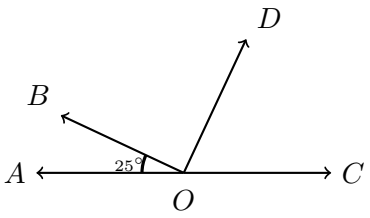


23 $CO \perp OB$. A nuqta $\angle COB$ ning ichida joylashgan; $\angle AOB = x - 10°$, $\angle COA = x + 40°$. x ni toping.

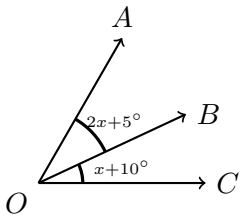


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24 $BO \perp OD$, $\angle BOA = 25°$. $\angle COD$ ni toping .



25 $\angle AOC = 60°$ va OB uning ichidan o'tadi. Agar $\angle AOB = 2x + 5°$, $\angle BOC = x + 10°$ bo'lsa, x ni toping.



26 Ikki to'g'ri chiziq O nuqtada kesishadi. Hosil bo'lgan to'rtta burchakdan biri $35°$ bo'lsa, qolgan uchtasini toping.

27 Ikki to'g'ri chiziqning kesishishidan hosil bo'lgan qo'shni burchaklardan biri ikkinchisidan 4 marta katta. Burchaklarning kichigini toping.

28 Ikki to'g'ri chiziqning kesishishidan hosil bo'lgan qo'shni burchaklardan biri ikkinchisidan 3 marta katta bo'lsa, ularni toping.

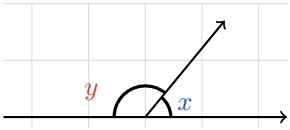
29 Agar qo'shni burchaklardan biri ikkinchisidan $40°$ ortiq bo'lsa, ularni toping.

30 Qo'shni burchaklarning ayirmasi $52°$ ga teng. Bu burchaklarning kattasini toping.

31 Qo'shni burchaklarning ayirmasi $30°$ ga teng bo'lsa, bu burchaklarning kichigini toping.

32 Agar qo'shni burchaklarning gradus o'lchovlari $4{:}5$ kabi nisbatda bo'lsa, ulardan kichigini toping.

33 Chizmadan kichik burchakni toping.



$y - x = 48°$

34 Ikki to'g'ri chiziq kesishishidan hosil bo'lgan qo'shni burchaklarning gradus o'lchovlari $7{:}2$ nisbatda bo'lsa, ulardan kichigini toping.

35 Ikki to'g'ri chiziq kesishishidan hosil bo'lgan to'rtta burchakdan uchtasining yig'indisi $290°$ ga teng. Eng kichik burchakni toping.

36 AB va CD to'g'ri chiziqlar O nuqtada kesishadi. $\angle AOD$ va $\angle COB$ burchaklarning yig'indisi $140°$ ga teng. $\angle AOC$ burchakni toping.

37 Ikki to'g'ri chiziqning kesishishidan hosil bo'lgan qo'shni burchaklardan biri ikkinchisidan 4 marta kichik. Bu to'g'ri chiziqlar orasidagi katta burchakni toping.

38 Ikki to'g'ri chiziqning kesishishidan hosil bo'lgan kichik burchak $54°$. Bu to'g'ri chiziqlar orasidagi katta burchakni toping.

39 Ikki to'g'ri chiziq kesishishidan hosil bo'lgan to'rtta burchakdan biri ikkinchisidan 5 marta kichik. Shu burchaklarning kattasini toping.

40 Ikki to'g'ri chiziq kesishadi. Hosil bo'lgan burchaklarning kattaligi nisbati $7{:}3$ ga teng bo'lsa, bu burchaklarning kichigini toping.

41 Ikki to'g'ri chiziq kesishishidan hosil bo'lgan burchaklarning kichigi $30°$ ga teng. Bu burchakka qo'shni bo'lmagan burchakni toping.

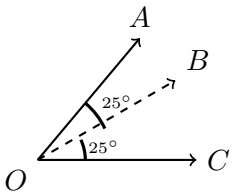
42 $58°$ ga teng burchakning bissektrisasi bilan tomonlaridan biri orasidagi burchak nimaga teng?

43 Burchakning bissektrisasi uning tomoni bilan $35°$ li burchak tashkil qilsa, burchakning o'zini toping.

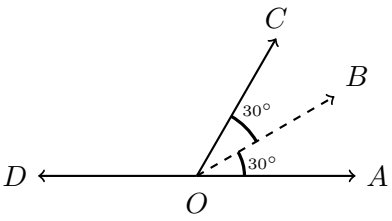
44 Burchakning bissektrisasi uning tomonlaridan biri bilan $90°$ li burchak tashkil qilsa, burchakning o'zini toping.

45 Bissektrisa burchakning bir tomoni bilan $52°$ li burchak tashkil etadi. Bu burchakning ikkinchi tomoni bilan bissektrissa orasidagi burchakni toping.

46 OB — $\angle AOC$ ning bissektrisasi va $\angle AOB = 25°$. $\angle AOC$ ni toping.

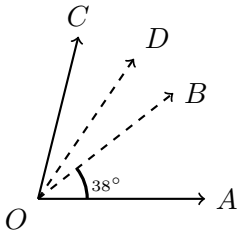


47 OB — $\angle AOC$ ning bissektrisasi va $\angle AOB = 30°$. Agar D nuqta A ning qarama-qarshi nurida (ya'ni A, O, D bir chiziqda) bo'lsa, $\angle DOC$ ni toping.



48 Qo'shni burchaklar bissektrisalari orasidagi burchakni toping.

49 $\angle AOC$ ning bissektrisasi OB va $\angle AOB = 38°$. Agar OD nur $\angle BOC$ ning bissektrisasi bo'lsa, $\angle AOD$ ni toping.



50 Yoyiq burchak $4{:}5{:}3$ nisbatda uchta burchakka bo'lingan. Hosil bo'lgan burchaklarning eng kichigini toping.

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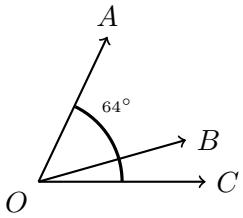
51 Yoyiq burchak $7{:}5{:}6$ nisbatda uchta burchakka bo'lingan. Ulardan eng kattasini toping.

52 Yoyiq burchak uchta bo'lakka bo'lingan bo'lib, ular butunning $\frac{1}{6}$, $\frac{1}{3}$ va $\frac{1}{2}$ qismini tashkil etadi. Eng katta bo'lakning burchak o'lchovini toping.

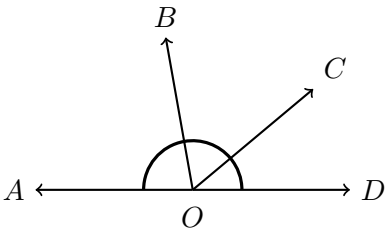
53 Qo'shni burchaklardan birining $\frac{2}{5}$ qismi ikkinchisining 10% ini tashkil etsa, bu burchaklarni toping.

54 Ikki kesishuvchi to'g'ri chiziqdan hosil bo'lgan to'rtta burchakdan biri qolgan uchtasining yig'indisidan 40° ga kichik. Shu burchakni toping.

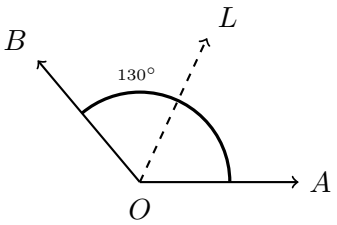
55 $\angle AOC$ ichidan OB nur o'tadi. $\angle AOB$ burchak $\angle BOC$ burchakdan 3 marta katta. Agar $\angle AOC = 64^\circ$ bo'lsa, $\angle BOC$ ni toping.



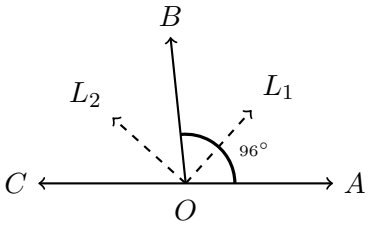
56 O nuqtadan to'rtta nur (OA, OB, OC, OD) chiqqan va ular ketma-ket joylashgan. A, O, D nuqtalar bir to'g'ri chiziqda yotadi (ya'ni $\angle AOD = 180^\circ$). $\angle AOB = 4x + 20^\circ$, $\angle BOC = 3x + 10^\circ$, $\angle COD = 2x + 6^\circ$ bo'lsa, $\angle BOC$ ni toping.



57 $\angle AOB = 130^\circ$. OL — uning bissektrisasi. $\angle AOL$ ni toping.



58 $\angle AOB = 96^\circ$. OL_1 — $\angle AOB$ ning, OL_2 esa $\angle AOB$ ga qo'shni burchakning bissektrisasi. $\angle L_1OL_2$ ni toping.

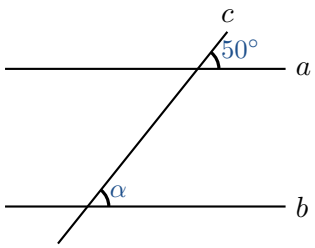


59 Ikki perpendikulyar to'g'ri chiziq kesishishidan hosil bo'lgan burchaklardan birining bissektrisasi ikkinchi to'g'ri chiziq bilan qanday burchak hosil qiladi?

60 Burchakning bissektrisasi tomonlaridan biri bilan 42° tashkil etsa, bu burchakka qo'shni bo'lgan burchakning yarmini toping.

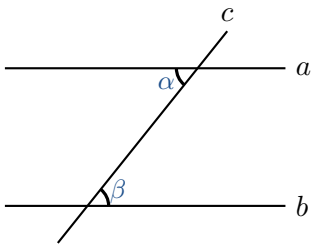
61–80. *Parallel to'g'ri chiziqlar va kesuvchi.*

61 $a \parallel b$. Kesuvchi chiziq a bilan $50°$ li burchak tashkil etadi. b chiziq bilan kesuvchi orasidagi mos burchakni toping.

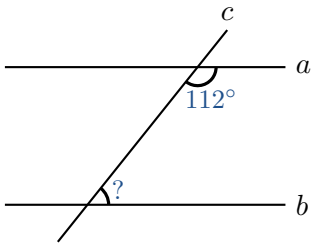


62 $a \parallel b$. Kesuvchi a bilan $130°$ li o'tmas burchak hosil qiladi. b chiziqdagi mos burchakni toping.

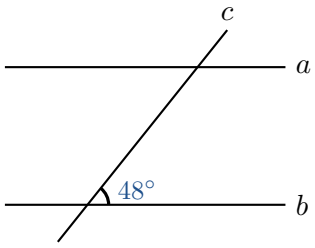
63 $a \parallel b$ va $\alpha = 65°$. Bunda α va β ning **almashinuvchi ichki** burchaklar ekanligi ma'lum. β ni toping.



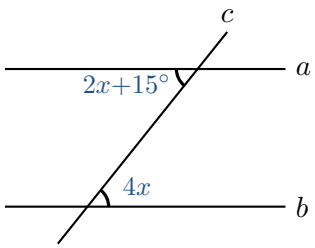
64 $a \parallel b$. **Bir tomonli ichki** burchaklardan biri $112°$ ga teng. Ikkinchisini toping.



65 $a \parallel b$. Kesuvchi b chiziq bilan $48°$ li burchak tashkil etadi. a chiziq tomonida hosil bo'lgan to'rtta burchakdan eng kattasini toping.

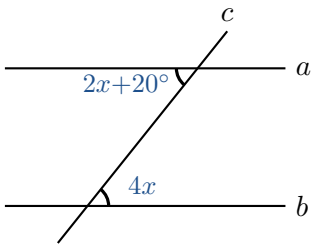


66 $a \parallel b$. Rasmga ko'ra $\alpha = 4x$ ni toping.

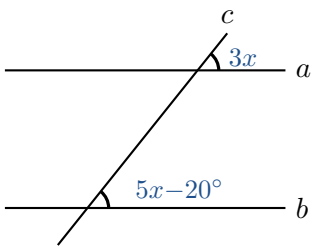


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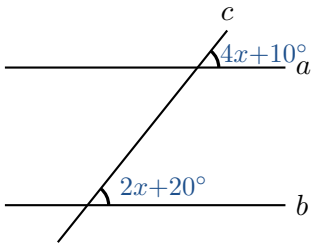
67 $a \parallel b$. $\beta = 4x$ ni toping.



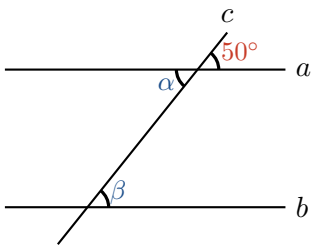
68 $a \parallel b$. a tomonidagi mos burchak $3x$, b tomonidagi mos burchak $5x - 20°$. $\alpha = 3x$ ni toping.



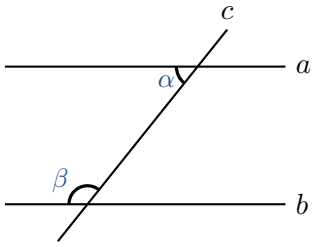
69 $a \parallel b$. a chiziq bilan kesuvchi orasidagi burchak $4x + 10°$, b chiziq bilan kesuvchi orasidagi mos burchak $2x + 20°$ ga teng. x ni toping.



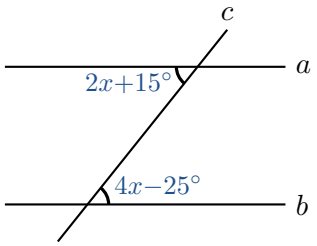
70 $a \parallel b$. Berilgan chizmaga ko'ra α va β ni toping.



71 $a \parallel b$. Rasmga ko'ra α va β — **bir tomonli ichki** burchaklar; $\alpha = 3x$, $\beta = 6x - 18°$. α ni toping.



72 $a \parallel b$. Almashinuvchi ichki burchaklardan biri $2x + 15°$, ikkinchisi $4x - 25°$ ga teng. x ni toping.



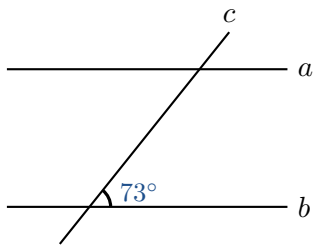
73 $a \parallel b$. Bir tomonli ichki burchaklarning yig'indisi $9x$ bo'lsa, x ni toping.

74 $a \parallel b$ va kesuvchidan hosil bo'lgan to'rtta o'tmas burchakning yig'indisi 440° bo'lsa, shu burchaklarni toping.

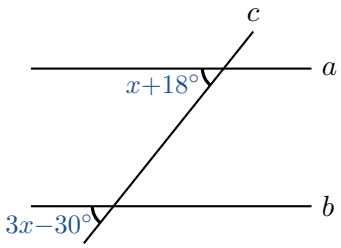
75 $a \parallel b$. Kesuvchi chiziq a va b to'g'ri chiziqlar bilan 8 ta burchak hosil qildi. Hosil bo'lgan burchaklardan o'tkir bo'lganlarining yig'indisi 244° bo'lsa, ularni toping.

76 Tekislikdagi ikki to'g'ri chiziq o'zaro kesishmaydigan bo'lsa, ular qanday joylashgan bo'ladi?

77 $a \parallel b$. Kesuvchi chiziq b bilan 73° li burchak hosil qiladi. a chiziq tomonida hosil bo'lgan barcha o'tkir burchaklar yig'indisini toping.

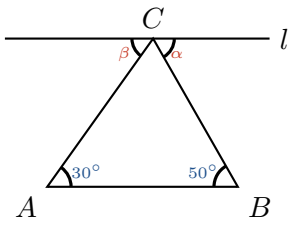


78 $a \parallel b$. x ni toping.



79 $a \parallel b$. Mos burchaklardan biri yoyiq burchakning $\frac{2}{5}$ qismiga teng. Ikkinchi mos burchakni toping.

80 l to'g'ri chiziq ABC uchburchakning C uchidan o'tadi va AB tomoniga parallel. $\angle CAB = 30^\circ$, $\angle CBA = 50^\circ$. $\alpha - \beta$ ni toping .



81–95. *Uchburchakning ichki va tashqi burchaklari.*

81 ABC uchburchakda $\angle BAC = 70^\circ$, $\angle BCA = 60^\circ$ bo'lsa, $\angle ABC$ ni toping.

82 Uchburchakning ichki burchaklaridan ikkitasi 50° va 80° ga teng bo'lsa, uning 3-ichki burchagini toping.

83 Uchburchakning ichki burchaklaridan ikkitasi 70° va 80° ga teng bo'lsa, uning 3-ichki burchagini toping.

84 Agar uchburchakning ichki burchaklari $2{:}3{:}4$ nisbatda bo'lsa, uning eng katta burchagini toping.

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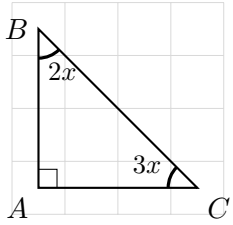
85 Agar uchburchakning ichki burchaklari $1{:}2{:}3$ nisbatda bo'lsa, uchburchakning turini aniqlang.

86 Uchburchak burchaklari $5{:}6{:}7$ nisbatda. Uchburchakning eng katta burchagini toping.

87 $\triangle ABC$ da $\angle BAC = x + 10°$, $\angle ABC = x + 20°$, $\angle BCA = x$. x ni toping.

88 $\triangle ABC$ da $\angle BAC = 3x$, $\angle ABC = 7x$, $\angle BCA = 2x$. $\angle ABC$ ni toping.

89 $\triangle ABC$ da $\angle A = 90°$. Rasmdan $\angle ABC = 2x$, $\angle ACB = 3x$ bo'lsa, $\angle ACB$ ni toping.



90 Teng yonli uchburchakning asosdagi burchaklaridan biri $40°$ bo'lsa, uning uchidagi burchakni toping.

91 Teng yonli uchburchakning uchidagi burchagi $80°$ ga teng bo'lsa, uning asosidagi burchaklarini toping.

92 Uchburchakning ichki ikkita burchagi $70°$ va $60°$ ga teng bo'lsa, uchburchakning bu burchaklarga qo'shni bo'lmagan tashqi burchagini toping.

93 Uchburchakning ichki burchaklari $3{:}2{:}1$ nisbatda. Uning eng katta tashqi burchagini toping.

94 Uchburchakning bir burchagi $50°$, qolgan ikkitasi nisbati $2{:}3$ kabi. Bu burchaklarni toping.

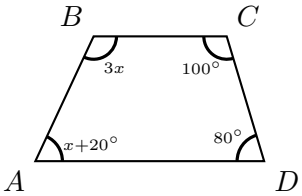
95 Uchburchakning burchaklari ketma-ket arifmetik progressiyani tashkil etadi (ya'ni o'rtadagisi qolganlarining yarim yig'indisiga teng). Uchburchak o'rta burchagini aniqlang.

96–100. *To'rtburchakning ichki burchaklari.*

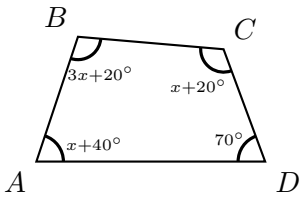
96 To'rtburchakning uchta burchagi $80°$, $100°$ va $90°$ ga teng. To'rtinchi burchagini toping.

97 To'rtburchakning to'rt burchagidan ikkitasi $60°$ va $130°$, qolganlaridan biri to'g'ri burchak bo'lsa, to'rtinchi burchakni toping.

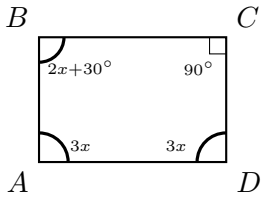
98 $ABCD$ to'rtburchakda $\angle BAD = x + 20°$, $\angle ADC = 80°$, $\angle DCB = 100°$, $\angle CBA = 3x$. x ni toping.



99 $ABCD$ to'rtburchakda $\angle BAD = x+40°$, $\angle ADC = 70°$, $\angle DCB = x+20°$, $\angle CBA = 3x+20°$. x ni toping.



100 $ABCD$ to'rtburchakning burchaklari mos ravishda $3x$, $2x+30°$, $90°$ va $3x$ ga teng bo'lsa, x ni toping.



101–110. *Soat millari va burchaklar.*

101 Soat 12:00 bo'lganda minut va soat millari orasidagi burchakni toping.

102 Soatning minut mili 8 minutda necha gradusga buriladi?

103 Soatning minut mili 42 minutda necha gradusga buriladi?

104 Soatning minut mili 30 minutda necha gradusga buriladi va bu qanday burchak hosil qiladi?

105 Soatning minut mili 45 minutda necha gradusga buriladi?

106 Soatning soat mili 1 soatda necha gradusga buriladi?

107 Soatning soat mili 4 soatda necha gradusga buriladi va bu qaysi burchak turi?

108 Soatning soat mili 20 minutda necha gradusga buriladi?

109 Soat 3:00 bo'lganda minut va soat millari orasidagi burchakni toping.

110 Soat 2:00 bo'lganda minut va soat millari orasidagi burchak qancha?

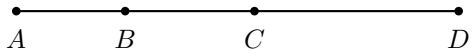
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Mavzu bo'yicha test savollari

1-variant

1. C nuqta A va B orasida. $AC = 13$ sm, $CB = 9$ sm. AB kesma uzunligini toping.
   A) 4 sm B) 13 sm C) 22 sm D) 26 sm

2. $A,\ B,\ C,\ D$ nuqtalar berilgan. $AC = 25$ sm, $BD = 27$ sm, $AD = 45$ sm. BC kesma uzunligini toping.



   A) 11 B) 8 C) 10 D) 7

3. Burchakning bissektrisasi uning tomoni bilan $60°$ li burchak tashkil qilsa, burchakning o'zini toping.
   A) $70°$ B) $130°$ C) $120°$ D) $30°$

4. Ikki to'g'ri chiziqning kesishishidan hosil bo'lgan burchaklardan biri ikkinchisidan 4 marta kichik. Bu to'g'ri chiziqlar orasidagi katta burchakni toping.
   A) $144°$ B) $75°$ C) $36°$ D) $115°$

5. Ikki to'g'ri chiziqning kesishishidan hosil bo'lgan burchaklardan kichigi $25°$ ga teng. Bu chiziqlar orasidagi katta burchakni toping.
   A) $110°$ B) $155°$ C) $75°$ D) $115°$

6. AB va CD to'g'ri chiziqlar O nuqtada kesishadi. $\angle AOD$ va $\angle COB$ ning yig'indisi $230°$. $\angle AOC$ ni toping.
   A) $70°$ B) $120°$ C) $65°$ D) $115°$

7. Uchburchakning burchaklari $1{:}2{:}3$ nisbatda. Uchburchakning turini aniqlang.
   A) o'tkir burchakli B) to'g'ri burchakli C) o'tmas burchakli D) teng tomonli

8. $a \parallel b$. a va b larni kesuvchi to'g'ri chiziq a bilan $70°$ li burchak hosil qiladi. b bilan hosil qiladigan o'tmas burchakni toping.
   A) $70°$ B) $90°$ C) $110°$ D) $130°$

9. Soatning minut mili 25 minutda qancha gradusga buriladi?
   A) $90°$ B) $120°$ C) $150°$ D) $180°$

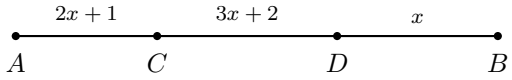
10. To'rtburchakning uchta burchagi $80°$, $90°$ va $100°$ ga teng. To'rtinchi burchakni toping.
    A) $90°$ B) $100°$ C) $80°$ D) $110°$

2-variant

---

1. D nuqta AB kesma ichida. $AD = 12$, $DB = 7$. AB kesma uzunligini toping.
   A) 5 B) 12 C) 19 D) 24

2. Chizmadagi $AB = 63$ bo'lsa, CB kesma uzunligini toping.



   A) 10 B) 32 C) 42 D) 53

3. Burchakning bissektrisasi uning tomoni bilan 40° li burchak tashkil qilsa, burchakning o'zini toping.
   A) 40° B) 80° C) 100° D) 20°

4. Ikki to'g'ri chiziq kesishadi. Hosil bo'lgan burchaklarning kattaligi nisbati $5{:}4$ ga teng bo'lsa, kichigini toping.
   A) 80° B) 100° C) 110° D) 105°

5. Qo'shni burchaklardan biri ikkinchisining $\frac{1}{5}$ qismi. Kattasini toping.
   A) 30° B) 36° C) 144° D) 150°

6. Yoyiq burchakning $\frac{2}{3}$ qismi necha gradusga teng?
   A) 90° B) 108° C) 120° D) 135°

7. Uchburchakning burchaklari $2{:}3{:}4$ nisbatda bo'lsa, eng katta burchagini toping.
   A) 40° B) 60° C) 80° D) 90°

8. $a \parallel b$. Bir tomonli ichki burchaklardan biri 115° ga teng. Ikkinchisini toping.
   A) 55° B) 65° C) 75° D) 85°

9. Soatning soat mili 3 soatda necha gradusga buriladi va bu qaysi burchak turi?
   A) 60°, o'tkir B) 90°, to'g'ri C) 120°, o'tmas D) 180°, yoyiq

10. $ABCD$ to'rtburchakda $\angle A = 70^\circ$, $\angle B = 110^\circ$, $\angle C = 80^\circ$. $\angle D$ ni toping.
    A) 80° B) 90° C) 100° D) 110°

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Javoblar

1. 29 sm

2. 10 sm

3. 31 sm

4. 39

5. 27

6. 9 sm

7. 16 sm

8. 23 sm

9. 25 sm

10. $AB = 22$ sm, $BC = 14$ sm

11. a) o'tkir; b) to'g'ri;
c) o'tmas; d) yoyiq;
e) o'tkir; f) to'la

12. $45°$

13. $108°$

14. $60°$

15. $150°$

16. $19°$

17. $21°$

18. $20°$

19. $130°$

20. $50°$

21. $60°$

22. $20°$

23. $30°$

24. $65°$

25. $15°$

26. $145°$, $35°$, $145°$

27. $36°$

28. $45°$ va $135°$

29. $70°$ va $110°$

30. $116°$

31. $75°$

32. $80°$

33. $66°$

34. $40°$

35. $70°$

36. $110°$

37. $144°$

38. $126°$

39. $150°$

40. $54°$

41. $30°$

42. $29°$

43. $70°$

44. $180°$

45. $52°$

46. $50°$

47. $120°$

48. $90°$

49. $57°$

50. $45°$

51. $70°$

52. $90°$

53. $36°$ va $144°$

54. $160°$

55. $16°$

56. $58°$

57. $65°$

58. $90°$

59. $45°$

60. $48°$

61. $50°$

62. $130°$

63. $65°$

64. $68°$

65. $132°$

66. $30°$

67. $40°$

68. $30°$

69. $5°$

70. $50°$, $50°$

71. $66°$

72. $20°$

73. $20°$

74. Har biri $110°$

75. Har biri $61°$

76. parallel

77. $146°$

78. $24°$

79. $72°$

80. $\alpha - \beta = 50° - 30° = 20°$

81. $50°$

82. $50°$

83. $30°$

84. $80°$

85. to'g'ri burchakli

86. $70°$

87. $50°$

88. $105°$

89. $54°$

90. $100°$

91. $50°$, $50°$

92. $130°$

93. $150°$

94. $52°$ va $78°$

95. $60°$

96. $90°$

97. $80°$

98. $40°$

99. $42°$

100. $30°$

101. $0°$ (millar ustma-ust tushadi)

102. $48°$

103. $252°$

104. $180°$, yoyiq burchak

105. $270°$

106. $30°$

107. $120°$, o'tmas burchak

108. $10°$

109. $90°$

110. $60°$

Test javoblari

1-variant: 1) C 2) D 3) C 4) A 5) B 6) C 7) B 8) C 9) C 10) A

2-variant: 1) C 2) C 3) B 4) A 5) D 6) C 7) C 8) B 9) B 10) C

Mavzu bo'yicha interaktiv testlar

QR kodni skanerlab, bilimingizni sinab ko'ring

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18.

Kombinatorika

Nazariy qism

✎ Ta’rif

Kombinatorika — matematikaning biror to’plam elementlarini qandaydir shartlar asosida tanlash va joylashtirish haqidagi bo’limidir.

“Ko’paytirish” va “Qo’shish” qoidalari

✠ Asosiy qoida

Ko’paytirish qoidasi
Agar A element n ta usul bilan, B element m ta usul bilan tanlanishi mumkin bo’lsa, u holda A va B juftlik

$$n \cdot m$$

ta usul bilan tanlanishi mumkin.
Kalit so’z: “**VA**” (“HAM”).

✠ Asosiy qoida

Qo’shish qoidasi
Agar A element n ta usul bilan, B element esa m ta usul bilan tanlanishi mumkin bo’lsa, u holda A yoki B elementni

$$n + m$$

ta usul bilan tanlash mumkin.
Kalit so’z: “**YOKI**”.

✓ Eslatma

Qaysi qoidani qachon ishlatamiz?
- Agar har **ikkala** tanlovni(birga) amalga oshirish kerak bo’lsa — **KO’PAYTIRAMIZ**.
- Agar **ikkisidan birini** tanlash kerak bo’lsa — **QO’SHAMIZ**.

Tushunish uchun misollar

✓ **Eslatma**

1-misol. Anvarda 3 ta ruchka (qizil, ko'k, qora) va 2 ta yozuv materiali (daftar, bloknot) bor. U konspekt yozish uchun **1 ta ruchka VA 1 ta yozuv materiali** tanlamoqchi. Necha xil tanlash mumkin?

Yechim. Anvar **ham** ruchka, **ham** yozuv vositasi tanlaydi — **VA qoidasi**.

Quyidagi jadvalda barcha juftliklar keltirilgan:

	Daftar	**Bloknot**
Qizil ruchka	qizil + daftar	qizil + bloknot
Ko'k ruchka	ko'k + daftar	ko'k + bloknot
Qora ruchka	qora + daftar	qora + bloknot

Javob: $3 \cdot 2 = 6$ xil.

✓ **Eslatma**

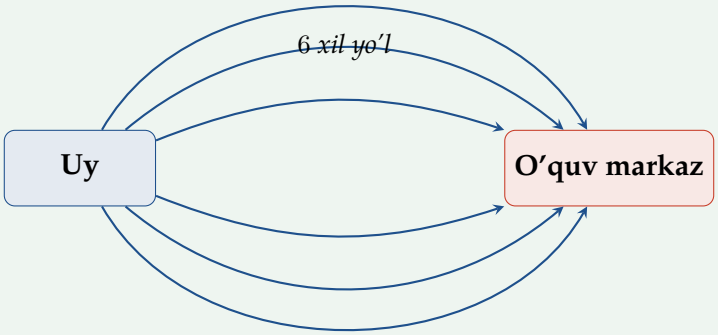
2-misol. Anvarda yana o'sha 3 ta ruchka va 2 ta yozuv vositasi bor. U **faqat bittasini** (yoki bitta ruchkani, yoki bitta yozuv vositasini) sovg'a qilmoqchi. Anvar necha xil tanlov qilishi mumkin?

Yechim. Bu yerda **yoki ruchka, yoki yozuv vositasi** — demak, **qo'shish qoidasi**.

$$3 + 2 = 5 \quad \text{xil.}$$

✓ **Eslatma**

3-misol. Anvarning uyidan o'quv markazigacha 6 ta har xil yo'l mavjud. Anvar darsga **borib-kelishi** necha xil bo'lishi mumkin?
(Borgan yo'lidan qaytib kelish mumkin emas.)



Yechim. Anvar avval boradi (yo'l-1), keyin qaytadi (yo'l-2). Ikkala harakat ham bajariladi — demak, **ko'paytirish qoidasi**.

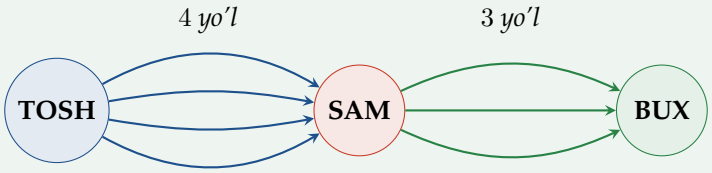
Borish 6 xil, qaytish esa, borgan yo'lidan qaytmagani uchun — 5 xil.

$$6 \cdot 5 = 30 \quad \text{xil.}$$

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✓ Eslatma

4-misol. Toshkentdan Samarqandga 4 xil yo'l, Samarqanddan Buxoroga 3 xil yo'l mavjud. Toshkentdan Buxoroga (Samarqand orqali) necha xil yo'l bilan borish mumkin?



Yechim. "Toshkent–Samarqand" **VA** "Samarqand–Buxoro".

$$4 \cdot 3 = 12 \quad \text{xil.}$$

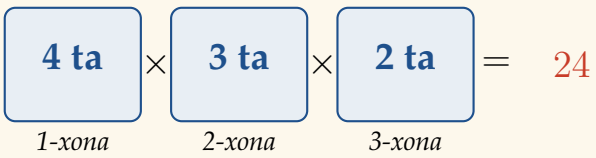
Raqamlardan son tuzish

✠ Asosiy qoida

Asosiy g'oya. n xonali sonni tuzish uchun — har bir xona uchun raqam tanlash kerak. Bunda"ko'paytirish" qoidasi ishlatiladi.

Diqqat: Sonning birinchi (eng chap) raqami **0 bo'la olmaydi**!

Misol. $1, 2, 3, 4$ raqamlaridan foydalanib (raqamlarni takrorlamasdan) nechta 3 xonali son tuzish mumkin?



$$= \quad 24$$

1-xonaga 4 ta raqamning istalganini, 2-xonaga qolgan 3 tasi, 3-xonaga qolgan 2 tasini tanlaymiz: $4 \cdot 3 \cdot 2 = 24$ ta.

✓ Eslatma

Takrorlanish bormi yoki yo'qmi?

• **Takrorlanmasa**: har xonada qolgan raqamlar soni yoziladi(**oldingidan bitta kamayib boradi**): $n \cdot (n-1) \cdot (n-2) \cdot \ldots$

• **Takrorlanish mumkin bo'lsa**: har bir xonada n tadan holat bo'ladi: $n \cdot n \cdot n \cdot \ldots$

Lavozimga tanlash — “boshliq va yordamchi masalasi”

✓ **Eslatma**

Misol. 5 ta nomzoddan boshliq va uning yordamchisini necha xil usulda tanlash mumkin?
Boshliqni tanlash: 5 xil yordamchini tanlash esa, (qolgan 4 tadan biri) 4 xil, demak:

$$5 \cdot 4 = 20 \quad \text{xil.}$$

Diqqat: “Aziz boshliq, Botir yordamchi” va “Botir boshliq, Aziz yordamchi” — bu **ikki har xil** variant!

Amaliy qism

I qism. Ko’paytirish va qo’shish qoidalariga doir masalalar

1–20.

1. Anvarda 4 xil ko’ylak va 3 xil shim bor. Bir juft (1 ta ko’ylak va 1 ta shim) kiyimni necha xil usulda tanlash mumkin?

2. Stol ustida 5 xil olma va 4 xil nok bor. Bitta mevani (yoki olma, yoki nok) necha xil tanlash mumkin?

3. Mansurda 3 xil daftar va 2 xil ruchka bor. U konspekt uchun 1 ta daftar va 1 ta ruchkani necha xil usulda tanlashi mumkin?

4. Anvarning uyidan o’quv markazigacha 6 ta har xil yo’l bor. Borgan yo’lidan qaytmasdan, Anvar necha xil usulda darsga borib-kelishi mumkin?

5. Sport do’konida 5 xil futbolka va 3 xil shim bor. Bir juft sport kompletni (1 ta futbolka va 1 ta shim) necha xil tanlash mumkin?

6. Restoran menyusida 4 xil sho’rva, 6 xil ikkinchi taom va 3 xil ichimlik bor. Tushlikka buyurtmani(har biridan bittadan) necha xil usulda qilish mumkin?

7. Toshkentdan Samarqandga 4 xil yo’l, Samarqanddan Buxoroga 5 xil yo’l bor. Toshkentdan Buxoroga (Samarqand orqali) necha xil usulda borish mumkin?

8. Sinfda 8 ta o’g’il va 5 ta qiz bor. Bittasini (1 o’g’il yoki 1 qiz) navbatchi qilib necha xil tanlash mumkin?

9. Sport zalida 8 ta o’zbekistonlik va 10 ta qozog’istonlik sportchi bor. Musobaqa uchun har bir mamlakatdan bittadan sportchini necha xil usulda tanlash mumkin?

10. Kafeda 5 xil pitsa va 6 xil ichimlik bor. 1 ta pitsa va 1 ta ichimlikni necha xil tanlash mumkin?

11. Garderobda 4 xil ko’ylak, 5 xil shim va 3 xil poyabzal bor. To’liq kiyim jamlanmasi necha xil usulda tanlanadi?

12. Og’abekda 7 xil qalam va 6 xil qog’oz bor. U rasm chizish uchun qalam va qog’ozni necha xil usulda tanlashi mumkin?

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13 Agar 2 ta ko'cha, har bir ko'chada 3 ta mehmonxona va har bir mehmonxonada 4 xil xona bo'lsa, bitta xonani necha xil usulda band qilish mumkin?

14 Marketda 4 xil non, 3 xil sut va 5 xil pishloq bor. Har bir mahsulotdan bittadan necha xil usulda xarid qilish mumkin?

15 Sinfda 12 ta o'quvchi bor. 1 ta o'quvchini navbatchilikka necha xil tanlash mumkin?

16 Bog'da 6 ta olma va 4 ta nok daraxti bor. 1 ta olma va 1 ta nok daraxtini necha xil usulda tanlash mumkin?

17 Kompyuter o'yinida 4 xil qahramon va har birining 3 xil quroli bor. Qahramon va uning qurolini necha xil usulda tanlash mumkin?

18 Anvarda 3 xil nonushta varianti, 4 xil tushlik varianti va 2 xil kechki ovqat varianti bor. U bir kunlik ratsionni necha xil tuzishi mumkin?

19 7 xil olma va 5 xil nokdan bittasini necha xil usulda xarid qilish mumkin?

20 Telefon do'konida 5 xil telefon, 3 xil g'ilof va 2 xil himoya oynasi sotiladi. To'liq jamlanmani necha xil usulda tanlash mumkin?

21–30.

21 Sayohat yo'nalishi: Toshkent→Samarqand — 5 xil yo'l, Samarqand→Buxoro — 4 xil yo'l va Buxoro→Xiva — 3 xil yo'l. Toshkentdan Xivaga necha xil usulda borish mumkin?

22 Anvarning uyidan o'quv markaziga 7 xil yo'l bilan borish mumkin. Borgan yo'lidan qaytmaslik sharti bilan, Anvar necha xil usulda darsga borib-kelishi mumkin?

23 Kafe menyusida 4 xil birinchi taom, 5 xil ikkinchi taom, 6 xil ichimlik va 3 xil desert bor. To'liq buyurtmani (har bir turdan bittadan) necha xil usulda berish mumkin?

24 Maktabda 7 ta ayol o'qituvchi va 4 ta erkak o'qituvchi bor. Direktorlikka nomzodni necha xil usulda tanlash mumkin?

25 Aviakassada Toshkent–Dubay yo'nalishi uchun 5 ta reys, Dubay–London yo'nalishi uchun 4 ta reys bor. Toshkentdan Londonga (Dubay orqali) necha xil reys bilan borish mumkin?

26 Sumkada 5 ta qizil, 4 ta yashil, 3 ta ko'k qalam bor. Bitta qalamni necha xil usulda tanlab olish mumkin?

27 10 nomzoddan direktor va uning o'rinbosarini necha xil usulda tanlash mumkin?

28 6 o'quvchidan 3 tasini 3 ta stulga necha xil usulda joylashtirish mumkin?

29 Toshkent shahrida "Qorako'l ziyo" o'quv markazining 4 ta filiali bo'lib, har bir filialida 5 tadan matematika ustozi ishlaydi va har bir ustozda 6 tadan guruhi bor. Mashg'ulotlarga yozilishning necha xil usuli mavjud?

30 Restoranda 3 ta ofitsiant bo'lib, har birining 4 ta xizmat ko'rsatadigan stoli bor. Mijozning necha xil stol tanlash imkoniyati bor?

31–40.

31 0, 1, 2, 3, 4, 5 raqamlaridan foydalanib (raqamlar takrorlanmasin) nechta 3 xonali son tuzish mumkin?

32 Vazada 3 ta oq, 4 ta qizil, 5 ta sariq va 6 ta binafsharang lola bor. 4 ta har xil rangli guldan iborat buket necha xil usulda yasaladi?

33 Qulf kodi 4 ta belgidan iborat va har bir belgi 0 dan 9 gacha raqamlardan biri bo'lsa, necha xil 4 ta belgili kod mavjud?

34 5 xonali kod: birinchi raqam 0 bo'lmasligi kerak, qolganlari 0 dan 9 gacha raqamlar (raqamlar takrorlanishi mumkin). Necha xil kod mavjud?

35 5 ta o'quvchini 5 ta stulga necha xil usulda joylashtirish mumkin?

36 3 ta kitobni 3 ta javonga (har javonda 1 tadan kitob) necha xil usulda terish mumkin?

37 Hamyonda 3 ta dollar, 4 ta yevro va 5 ta funt sterling qog'oz pullari bor. Har xil pul birligidan bittadan olishning necha xil usuli mavjud?

38 Telefon paroli 4 ta raqamdan iborat, 0 va 1 raqamlari ishlatilmaydi (qolgan raqamlar takrorlanishi mumkin). Necha xil parol terish mumkin?

39 3 ta savoldan iborat testning har savolida 4 ta javob varianti (A, B, C, D) bor. To'liq javob varianti necha xil bo'lishi mumkin?

40 4 ta o'rindiqqa 4 kishi necha xil joylasha oladi?

II qism.

41–45.

41 5 nomzoddan boshliq va uning yordamchisini necha xil tanlash mumkin?

42 2, 3, 4 raqamlaridan foydalanib (raqamlar takrorlanmasin) nechta 2 xonali son tuzish mumkin?

43 Vazada 3 ta oq va 4 ta qizil lolalar bor. 1 ta oq va 1 ta qizil loladan iborat guldasta necha xil usulda yasaladi?

44 2, 3, 4, 5 raqamlaridan foydalanib (raqamlar takrorlanishi mumkin) nechta 2 xonali son tuzish mumkin?

45 Sinfdagi 8 ta o'quvchidan sinf sardorini necha xil tanlash mumkin?

46–60.

46 6 ta nomzoddan boshliq va yordamchini necha xil tanlash mumkin?

47 8 ta nomzoddan 1 ta prezident, 1 ta vitse-prezident va 1 ta kotibni necha xil tanlash mumkin?

48 10 ta o'quvchidan iborat sinfda sinf sardori va uning yordamchisini necha xil usulda tanlash mumkin?

49 2, 3, 4, 5, 6, 7 raqamlaridan foydalanib (raqamlar takrorlanishi mumkin) nechta 2 xonali son tuzish mumkin?

50 2, 3, 4, 5, 6, 7 raqamlaridan foydalanib (raqamlar takrorlanmasin) nechta 2 xonali son tuzish mumkin?

51 2, 3, 4, 5, 6, 7 raqamlaridan foydalanib (raqamlar takrorlanishi mumkin) nechta 3 xonali son tuzish mumkin?

52 2, 3, 4, 5, 6, 7 raqamlaridan foydalanib (raqamlar takrorlanmasin) nechta 3 xonali son tuzish mumkin?

53 2, 3, 4, 5, 6, 7 raqamlaridan foydalanib (raqamlar takrorlanmasin) nechta 4 xonali son tuzish mumkin?

54 Vazada 3 ta oq, 4 ta qizil va 5 ta sariq lolalar bor. 3 xil rangli guldasta necha xil yasaladi?

55 6 ta nomzoddan boshliq, yordamchi va mas'ul kotibni necha xil usulda tanlash mumkin?

56 12 ta aholidan rais va yordamchini necha xil tanlash mumkin?

57 2, 4, 6, 8 raqamlaridan foydalanib (raqamlar takrorlanmasin) nechta 3 xonali son tuzish mumkin?

58 2, 4, 6, 8 raqamlaridan foydalanib (raqamlar takrorlanishi mumkin) nechta 3 xonali son tuzish mumkin?

59 1, 2, 3 raqamlaridan foydalanib (raqamlar takrorlanmasin) tuzilgan barcha 3 xonali sonlar yig'indisini toping.

60 5, 6, 7, 9 raqamlaridan foydalanib (raqamlar takrorlanishi mumkin) nechta 2 xonali toq son tuzish mumkin?

61–65.

61 0, 2, 4, 6, 8 raqamlaridan foydalanib (raqamlar takrorlanmasin) nechta 3 xonali son tuzish mumkin?

62 2, 3, 4, 5, 6, 7 raqamlaridan foydalanib (raqamlar takrorlanishi mumkin) nechta 3 xonali toq son tuzish mumkin?

63 2, 3, 4, 5, 6, 7 raqamlaridan foydalanib (raqamlar takrorlanmasin) nechta 3 xonali juft son tuzish mumkin?

64 12 do'stdan iborat kompaniya sayohatga otlanmoqda. Ular orasidan 4 tasini quyidagi rollarga tanlashadi: haydovchi, navigator, DJ va oshpaz. Buni necha xil usulda amalga oshirish mumkin?

65 Akmalning shkafida 4 ta ko'ylak, 5 ta shim, 6 juft poyabzal va 3 ta kurtka bor. U har biridan bittadan tanlab, necha xil kiyim-bosh to'plamini hosil qila oladi?

Mavzu bo'yicha test savollari

1-variant

1. Restoranda 4 xil pitsa va 5 xil ichimlik bor. Bittadan tanlab, necha xil buyurtma berish mumkin?
   A) 9 B) 16 C) 20 D) 25

2. Eshik kodi 2 xonali bo'lib, har xonaga $2,\ 3,\ 4,\ 5$ raqamlaridan birini qo'yish mumkin. Nechta turli kod tuzish mumkin?
   A) 12 B) 16 C) 20 D) 24

3. 6 ta nomzoddan tadbir uchun bosh boshlovchi va yordamchi boshlovchini necha xil tanlash mumkin?
   A) 12 B) 30 C) 36 D) 60

4. Sevara uyidan ishxonasigacha 5 xil yo'l bilan boradi. Borib-kelishda har xil yo'ldan yursa, necha xil bo'ladi?
   A) 10 B) 20 C) 25 D) 30

5. Burger tayyorlashda 4 xil non, 5 xil go'sht va 6 xil sous bor. Har biridan bittadan tanlab, necha xil burger yasaladi?
   A) 15 B) 60 C) 90 D) 120

6. 12 ta xodimdan oyning eng yaxshi xodimini necha xil tanlash mumkin?
   A) 1 B) 6 C) 11 D) 12

7. Do'konda 5 xil telefon brendi, har birining 4 xil rangi va 3 xil xotira hajmi mavjud. Telefonga necha xil buyurtma berish mumkin?
   A) 12 B) 35 C) 60 D) 120

8. 3 ta do'st kinoteatrda qatorga necha xil tartibda o'tira oladi?
   A) 3 B) 6 C) 9 D) 27

9. Bozorda 7 xil olma va 5 xil nok sotilmoqda. Bittasini sotib olishning necha xil yo'li bor?
   A) 7 B) 5 C) 12 D) 35

10. Kompaniyada 8 xodim ishlaydi. Ulardan bosh direktor, moliyaviy direktor va kotib tanlashning necha xil yo'li bor?
    A) 24 B) 56 C) 336 D) 512

2-variant

1. Kinoteatrda 6 xil film va 4 xil popkorn ta'mi mavjud. Bittadan tanlab necha xil tarzda dam olish mumkin?
   A) 10 B) 18 C) 24 D) 30

2. Bayram uchun 7 xil video o'yin va 5 xil stol o'yini bor. Faqat 1 ta o'yin o'ynashning necha xil yo'li bor?
   A) 12 B) 35 C) 30 D) 7

3. "X-Faktor" finalida 4 ishtirokchi qoldi. Birinchi, ikkinchi va uchinchi o'rin sohiblari necha xil tartibda aniqlanishi mumkin?
   A) 12 B) 24 C) 36 D) 64

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4. Telekanal 10 ta jurnalistidan bosh muxbir va uning o'rinbosarini tanlamoqchi. Buni necha xil usulda amalga oshirish mumkin?
   A) 20 B) 45 C) 90 D) 100

5. Wi-Fi paroli 3 raqamli bo'lib, har raqam 1, 3, 5, 7 dan biri bo'lishi mumkin (raqamlar takrorlanishi mumkin). Nechta turli parol tuzish mumkin?
   A) 12 B) 24 C) 64 D) 81

6. Sushi to'plamida 3 xil guruch, 5 xil baliq va 4 xil sous mavjud. Har biridan bittadan tanlab necha xil sushi tayyorlash mumkin?
   A) 12 B) 15 C) 60 D) 90

7. Anvar uyidan buvisinikiga 3 xil yo'l bilan, buvisinikidan tog'asinikiga 5 xil yo'l bilan borishi mumkin. U uyidan tog'asinikiga necha xil yo'l bilan borishi mumkin?
   A) 8 B) 15 C) 25 D) 30

8. Sayohat agentligida 5 xil shahar va 3 xil mehmonxona darajasi taklif qilinmoqda. Bittadan tanlab necha xil sayohat paketi tanlash mumkin?
   A) 8 B) 12 C) 15 D) 25

9. Konsert dasturida 5 ta artist sahnaga ketma-ket chiqishi kerak. Ularni necha xil tartibda joylashtirish mumkin?
   A) 60 B) 120 C) 240 D) 600

10. Sovg'a do'konida 5 xil gul buketi va 4 xil o'rama qog'oz mavjud. Bittadan tanlab necha xil sovg'a tayyorlash mumkin?
    A) 9 B) 18 C) 20 D) 25

Javoblar

1. 12
2. 9
3. 6
4. 30
5. 15
6. 72
7. 20
8. 13
9. 80
10. 30
11. 60
12. 42
13. 24
14. 60
15. 12
16. 24
17. 12
18. 24
19. 12
20. 30
21. 60
22. 42
23. 360
24. 11
25. 20
26. 12
27. 90
28. 120
29. 120
30. 12
31. 100
32. 360
33. 10 000
34. 90 000
35. 120
36. 6
37. 60
38. 4096
39. 64
40. 24
41. 20
42. 6
43. 12
44. 16
45. 8

46. 30
47. 336
48. 90
49. 36
50. 30
51. 216
52. 120
53. 360
54. 60
55. 120
56. 132
57. 24
58. 64
59. 1332
60. 12
61. 48
62. 108
63. 60
64. 11 880
65. 360

Test javoblari

1-variant: 1) C 2) B 3) B 4) B 5) D 6) D 7) C 8) B 9) C 10) C
2-variant: 1) C 2) A 3) B 4) C 5) C 6) C 7) B 8) C 9) B 10) C

Mavzu bo'yicha interaktiv testlar
QR kodni skanerlab, bilimingizni sinab ko'ring

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Qaydlar uchun

Qaydlar uchun

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Variantlar

1-variant

1. 1 dan 50 gacha sonlar ichida 3 raqami necha marta ishtirok etadi?

A) 14 B) 15 C) 13 D) 16

2. Hisoblang: $(-11)^2 + 12 \cdot (-10)$

A) -1 B) 21 C) 2 D) 1

3. Hisoblang: $\frac{1}{3} + 0{,}2$

A) $\frac{7}{15}$ B) $\frac{2}{3}$ C) $\frac{8}{15}$ D) $\frac{11}{15}$

4. Hisoblang: $\sqrt{225}$

A) 15 B) 13 C) 14 D) 25

5. Ikki sonning yig'indisi 180 ga teng, nisbati esa $4:5$ kabi. Bu sonlarning musbat ayirmasini toping.

A) 10 B) 36 C) 20 D) 18

6. $144 : x = 36$ tenglamani yeching.

A) 6 B) 8 C) 12 D) 4

7. a sonining 30% i 90 ga teng. a ni toping.

A) 300 B) 270 C) 150 D) 360

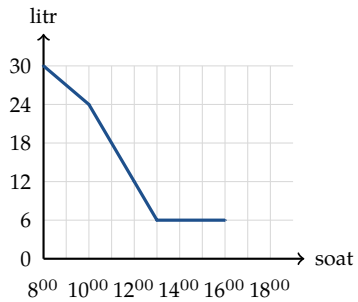
8. Gul do'konida qizil atirguldan 3 xil, oq atirguldan 2 xil gul bor. Bitta atirgulni tanlashning necha usuli bor?

A) 6 B) 5 C) 9 D) 12

9. Avtomobil yoqilg'isining vaqt bo'yicha o'zgarishi grafikda berilgan. 10^{00} dan 13^{00} gacha qancha litr yoqilg'i sarflangan?

A) 24 B) 12

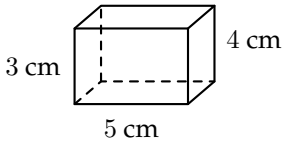
C) 18 D) 6



10. To'g'ri burchakli parallelepipedning qirralari 3, 4 va 5 cm. Eng katta yoq yuzini toping.

A) 20 cm^2 B) 15 cm^2

C) 12 cm^2 D) 60 cm^2



2-variant

---

1. 6 ga bo'linadigan ikki xonali sonlar nechta?

A) 14 B) 16 C) 13 D) 15

2. Hisoblang: $-\big((-130-70)-(130+(-30))\big)$

A) 300 B) 200 C) -300 D) 100

3. Hisoblang: $2\frac{1}{2} : 2\frac{1}{4}$

A) $1\frac{1}{9}$ B) $\frac{9}{10}$ C) $\frac{5}{6}$ D) $\frac{6}{5}$

4. Aka ukasidan 10 yosh katta, hozir ularning yoshlari nisbati $3:1$. Necha yildan keyin yoshlar nisbat $2:1$ bo'ladi?

A) 4 B) 8 C) 5 D) 10

5. $\frac{x+7}{4-x} = \frac{2}{9}$ tenglamani yeching.

A) 3 B) -5 C) 6 D) -4

6. Mahsulotning dastlabki narxi $55\,000$ so'm edi va $44\,000$ so'mga tushirildi. Mahsulot narxiga necha foiz chegirma qilingan?

A) 11% B) 25% C) 40% D) 20%

7. Ko'paytuvchilarga ajrating: $6ab^2c - 18abc$

A) $6ab(c-3)$ B) $6abc(b-3)$ C) $6abc(b-2)$ D) $abc(b-18)$

8. 3 ta tovuq va 4 ta o'rdakdan tovuq va o'rdak juftligini necha xil usulda tanlash mumkin?

A) 7 B) 24 C) 12 D) 14

9. n-burchakli prizmaning uchlari 16 ta bo'lsa, prizma qirralari sonini toping.

A) 16 B) 18 C) 32 D) 24

10. 3 km 4 dm $= \boxed{?}$ cm

A) 300400 B) 3000040 C) 300040 D) 300004

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3-variant

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1. Raqamlari ko'paytmasi 12 ga teng bo'lgan ikki xonali sonlar nechta?

A) 3 B) 5 C) 4 D) 6

2. Hisoblang: $-\big((-12)-88\big)+(52+49)$

A) 200 B) 199 C) 1 D) 201

3. Hisoblang: $\dfrac{111}{333}+\dfrac{222}{666}-\dfrac{333}{999}$

A) $\dfrac{1}{3}$ B) $\dfrac{1}{9}$ C) 1 D) $\dfrac{2}{3}$

4. Bog'dagi olma va nok daraxtlarining soni $5:3$ nisbatda. Jami 96 ta daraxt bo'lsa, olma daraxti nechta?

A) 36 B) 48 C) 60 D) 40

5. $10\cdot(x-10)=6x$ tenglamani yeching.

A) 20 B) 15 C) 35 D) 25

6. Bankka $5\,000\,000$ so'm qo'yildi.Agar qo'yilgan pul 1 yilda 25% ga oshsa, qo'yilgan pulning 1 yildan keyingi miqdorini toping.

A) $6\,200\,000$ B) $6\,000\,000$ C) $5\,250\,000$ D) $6\,250\,000$

7. $(a^4)^2\cdot a^3$ ifodani soddalashtiring.

A) a^9 B) a^{11} C) a^8 D) a^{24}

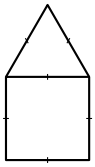
8. 10 nafar do'st bir-biri bilan qo'l berib ko'rishdi. Jami nechta qo'l berish bo'lgan?

A) 90 B) 45 C) 20 D) 100

9. Rasmda muntazam uchburchak va kvadrat berilgan. Kvadratning yuzi 36 ga teng bo'lsa, uchburchakning perimetrini toping.

A) 16 B) 18

C) 24 D) 12



10. $2\ \text{m}^3\ 5\ \text{dm}^3 = \boxed{?}\ \text{dm}^3$

A) 2005 B) 2050 C) 2500 D) 205

4-variant

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1. Ikki xonali natural sonlar ichida 7 raqami necha marta ishtirok etadi?

A) 18 B) 17 C) 19 D) 20

2. Hisoblang: $|-10|+|-20|-|-10|\cdot 3$

A) 60 B) 30 C) 45 D) 0

3. Hisoblang: $\frac{5}{6}:\frac{15}{24}+\frac{1}{2}:\frac{1}{4}$

A) $3\frac{1}{3}$ B) $\frac{8}{3}$ C) $\frac{7}{2}$ D) $\frac{11}{3}$

4. $P=5a-2b$ va $Q=5b-2a$ bo'lsa, $P+Q$ ni toping.

A) $3(a+b)$ B) $3(a-b)$ C) $7(a-b)$ D) $3(a-c)$

5. $2019-2x=2021$ tenglamani yeching.

A) 1 B) -1 C) 4 D) 11

6. 14 gektar dalaning 80% iga paxta ekilgan. Dalaning qancha yeriga paxta ekilgan?

A) $13{,}2$ ga B) $15{,}2$ ga C) $11{,}2$ ga D) $10{,}3$ ga

7. Soddalashtiring: $(10-4a):a-2$

A) $a-1$ B) $\frac{10}{a}-6$ C) 1 D) $\frac{2a}{3}-1$

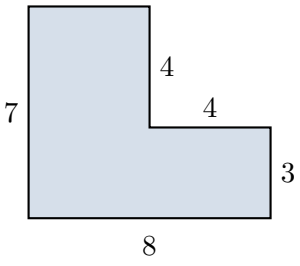
8. $0,1,2,3$ raqamlaridan foydalanib nechta 2 xonali son tuzish mumkin(raqamlar takrorlanishi mumkin)?

A) 20 B) 9 C) 12 D) 6

9. Berilgan shaklning perimetrini toping (o'lchamlar cm da).

A) 50 B) 60

C) 30 D) 35



10. $40\,700$ m^2 ni gektar va arda ifodalang.

A) 40 ga 7 ar B) 4 ga 7 ar C) 407 ar D) 4 ga 700 ar

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5-variant

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1. Barcha raqamlari juft bo'lgan ikki xonali sonlar nechta?

A) 20 B) 25 C) 30 D) 28

2. Koordinata o'qida $M(-4)$ nuqtani necha birlik va qaysi tomonga siljitsak $N(-9)$ ga boradi?

A) 5 birlik o'ngga B) 4 birlik chapga C) 4 birlik o'ngga D) 5 birlik chapga

3. Hisoblang: $2{,}3(6) + 1{,}(6)$

A) 4 B) $4{,}(6)$ C) $\frac{121}{30}$ D) $4{,}(5)$

4. Hisoblang: $\sqrt{\frac{144}{25}}$

A) $\frac{12}{5}$ B) $\frac{144}{25}$ C) $\frac{12}{25}$ D) $\frac{6}{5}$

5. Ikki sonning yig'indisi 220ga, nisbati esa 9ga teng bo'lsa, shu sonlarning kattasini toping.

A) 200 B) 198 C) 196 D) 194

6. $\begin{cases} x + y = 90 \\ x - 2y = 0 \end{cases}$ dan x ning qiymatini toping.

A) 30 B) 45 C) 50 D) 60

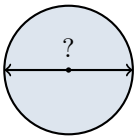
7. Korxonadagi matoning 16% idan ko'ylak tikildi. Bunga 320 metr mato ketgan bo'lsa, korxonada jami necha metr mato bo'lgan?

A) 1900 B) 2000 C) 1600 D) 3000

8. Savatda 3 xil 3 ta meva bor. Ularni tokchaga necha xil tartibda terish mumkin?

A) 5 B) 4 C) 6 D) 8

9. Doira yuzi 9π ga teng bo'lsa, doira diametrini toping.



A) 3 B) 12

C) 6 D) 9

10. 30 m/s necha km/s ga teng?

A) 0,3 B) 0,03 C) 0,003 D) 0,0003

6-variant

1. 50 ning bo'luvchisi bo'lgan ikki xonali sonlar nechta?

A) 2 B) 4 C) 5 D) 3

2. Hisoblang: $(-12)^2 - 11 \cdot 12$

A) 12 B) -12 C) 310 D) -276

3. Hisoblang: $\dfrac{17}{22} - \dfrac{3}{5}$

A) $\dfrac{19}{110}$ B) $\dfrac{119}{10}$ C) $\dfrac{2}{11}$ D) $\dfrac{3}{22}$

4. Ko'paytuvchilarga ajrating: $51a - 57abc$

A) $3a(17 - 16bc)$ B) $3a(17 - 19bc)$ C) $3a(17 + 19bc)$ D) $3a(19 - 17bc)$

5. $\dfrac{3x+7}{x-7} = \dfrac{7}{2}$ tenglamani yeching.

A) 34 B) -34 C) -63 D) 63

6. Tadbirkor nonni $13\,860$ so'mdan sotib 10% foyda qildi. Mahsulotning tan narxini toping.

A) $12\,500$ B) $12\,600$ C) $13\,600$ D) $15\,246$

7. 4-burchakli piramidaning qirralari sonini toping.

A) 5 B) 6 C) 8 D) 10

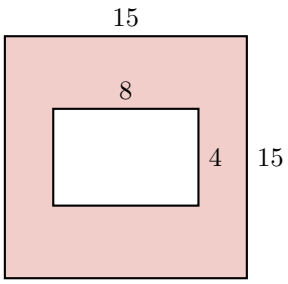
8. Hisoblang: $|a| - |b| + |-c|$, bunda $a = 12$, $b = -5$, $c = 8$.

A) 15 B) -15 C) 25 D) 7

9. Tomoni 15 ga teng bo'lgan kvadrat ichida tomonlari 8 va 4 ga teng to'g'ri to'rtburchak joylashtirilgan. Bo'yalgan soha yuzini toping.

A) 161 B) 193

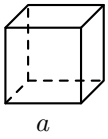
C) 200 D) 32



10. Kubning to'la sirti 54 cm^2 bo'lsa, kubning hajmini toping.

A) 18 cm^3 B) 36 cm^3

C) 54 cm^3 D) 27 cm^3



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7-variant

1. 90 va 60 sonlarining eng katta umumiy bo'luvchisini toping.

A) 10 B) 30 C) 15 D) 180

2. Hisoblang: $-2\frac{1}{3} \cdot 6$

A) -12 B) -10 C) -8 D) -14

3. Hisoblang: $\frac{1}{3} - 0{,}1$

A) $\frac{7}{15}$ B) $\frac{7}{30}$ C) $\frac{11}{15}$ D) $\frac{13}{30}$

4. Ikki sonning yig'indisi 320ga va nisbati 7ga teng bo'lsa, ularning kichigini toping.

A) 40 B) 60 C) 270 D) 280

5. $5(2x - 4) = 9x + 6$ tenglama ildizining yarmini toping.

A) 13 B) 26 C) 30 D) 15

6. Sutdan 12% qaymoq olinadi. 15 kg sutdan qancha qaymoq olinadi?

A) 1,8 g B) 18 kg C) 1,8 kg D) 18 g

7. Soddalashtiring: $(x + 2)(x^2 - 2x + 4) - 4$

A) $x^3 - 4$ B) $x^3 + 8$ C) $x^3 + 4$ D) $x^3 - 8$

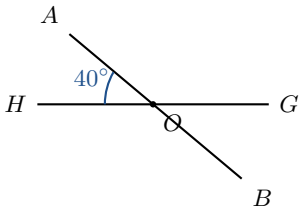
8. 3 ta turli mevani muzlatkichga necha usulda terish mumkin?

A) 3 B) 6 C) 5 D) 9

9. $\angle AOH = 40°$ bo'lsa, chizmadan foydalanib $\angle BOG + \angle AOG$ ni toping.

A) 140° B) 100°

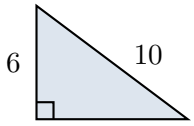
C) 40° D) 180°



10. To'g'ri burchakli uchburchakning bir kateti 6, gipotenuzasi 10 ga teng bo'lsa, uning yuzini toping.

A) 30 B) 60

C) 25 D) 24



8-variant

1. 48 sonining natural bo'luvchilari nechta?

A) 8 B) 12 C) 10 D) 16

2. Hisoblang: $27 + (-5) \cdot 4 - 3$

A) 12 B) 4 C) -8 D) 0

3. Hisoblang: $3\frac{1}{2} \cdot 2\frac{2}{5}$

A) $6\frac{1}{5}$ B) $5\frac{3}{7}$ C) $8\frac{2}{5}$ D) $6\frac{3}{5}$

4. $a + b = 320$, $\frac{a}{b} = 39$ bo'lsa, $a - b$ ning qiymatini toping.

A) 299 B) 302 C) 307 D) 304

5. $\frac{10 + x}{4 - x} = 1$ tenglamani yeching.

A) -2 B) -3 C) -1 D) 0

6. Devorning 15% i bo'yalgan bo'lsa, uning qancha qismi bo'yalmagan?

A) 0,8 B) 0,9 C) $\frac{17}{20}$ D) 0,75

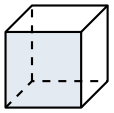
7. Birhadning koeffitsiyentini toping: $-45x^2 \cdot \left(-\frac{1}{2}xy^2\right) \cdot \left(-\frac{2}{9}x\right)$

A) 5 B) -10 C) $-\frac{1}{2}$ D) -5

8. 3 ta xatni 3 ta turli konvertga necha xil usulda joylash mumkin?

A) 5 B) 6 C) 1 D) 9

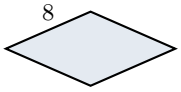
9. Rasmda berilgan kubning qirrasi 3 cm. Uning to'la sirti yuzini toping.



A) 27 cm^2 B) 36 cm^2

C) 9 cm^2 D) 54 cm^2

10. Rombning tomoni 8 ga teng. Rombning perimetrini toping.



A) 32 B) 24

C) 36 D) 40

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9-variant

1. 1 dan 100 gacha natural sonlar ichida 6 raqami necha marta ishtirok etadi?

A) 19 B) 18 C) 17 D) 20

2. Hisoblang: $-2{,}4 : (-0{,}8) + 7$

A) 40 B) 10 C) 4 D) 16

3. Hisoblang: $\frac{1}{3} + 0{,}25$

A) $\frac{7}{11}$ B) $\frac{1}{12}$ C) $\frac{7}{12}$ D) $\frac{4}{7}$

4. Hisoblang: $\sqrt{49} + \sqrt{9} - \sqrt{4}$

A) 8 B) 10 C) 6 D) 12

5. $3x - 4 = 2x + 6$ tenglamaning ildizi x ga teng bo'lsa, $x + 2$ ni toping.

A) 12 B) 11 C) 14 D) 10

6. 600 gektar yerning 150 gektari umumiy maydonning necha foizini tashkil etadi?

A) 50% B) 25% C) 15% D) 30%

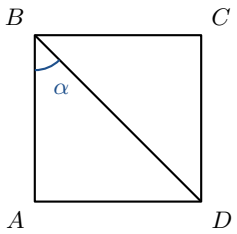
7. Soddalashtiring: $a(b - c) - b(c - a) - c(a - b)$

A) $2ab + 2ac$ B) $2b - 2ac$ C) $2ab - 2ac$ D) $4ab - 2ac$

8. 3, 5, 7 va 9 raqamlaridan (takrorlanishsiz) nechta uch xonali son tuzish mumkin?

A) 18 B) 24 C) 16 D) 20

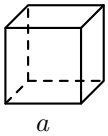
9. $ABCD$ kvadrat bo'lsa, diagonal hosil qilgan α burchakni toping.



A) 45° B) 40°

C) 90° D) 180°

10. Kubning asosining yuzi 49 ga teng bo'lsa, kubning to'la sirtini toping.



A) 156 B) 256

C) 194 D) 294

10-variant

1. 18 ning butun bo'luvchilari soni nechta?

A) 12 B) 6 C) 9 D) 24

2. Hisoblang: $36 \cdot (11 - 13) + 64 \cdot (17 - 19)$

A) 200 B) 100 C) -100 D) -200

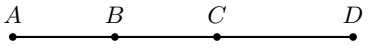
3. Hisoblang: $\frac{1}{3} - 0{,}2$

A) $\frac{7}{15}$ B) $\frac{2}{15}$ C) $\frac{1}{15}$ D) $\frac{4}{15}$

4. A, B, C, D nuqtalar berilgan: $AC = 25$ cm, $BD = 27$ cm, $AD = 45$ cm. BC kesma uzunligini toping.

A) 7 cm B) 11 cm

C) 8 cm D) 10 cm



5. $2 \cdot (3 - x) = 2 \cdot (3 - x) - 5$ tenglama nechta yechimga ega?

A) cheksiz ko'p B) 1 ta C) yechimga ega emas D) 2 ta

6. Do'konda 1900 ta non bor edi. Agar tushlikkacha nonning 75% i sotilgan bo'lsa, do'konda qancha non qolgan?

A) 1425 ta B) 475 ta C) 500 ta D) 750 ta

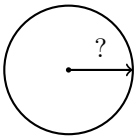
7. Hisoblang: $2{,}4 \cdot 6000000 \cdot 10^{-7}$

A) 14,4 B) 144 C) 1,44 D) 0,144

8. Aylananing uzunligi 10π ga teng bo'lsa, aylana radiusini toping.

A) 4 B) 10

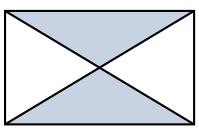
C) 8 D) 5



9. Rasmdagi to'g'ri to'rtburchakning necha foizi bo'yalgan?

A) 65% B) 75%

C) 25% D) 50%



10. 5400 kg ni tonna va sentnerda ifodalang.

A) 5 t 400 sr B) 5 t 40 sr C) 5 t 4 sr D) 5 t 4000 sr

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11-variant

1. $(2a^3b)^3$ ifodani soddalashtiring.

A) $8a^9b^3$ B) $6a^9b^3$ C) $8a^6b^3$ D) $2a^9b^3$

2. Hisoblang: $-28-3-(-31)$

A) -2 B) 0 C) 2 D) -62

3. Hisoblang: $4\frac{2}{3}:3\frac{2}{3}$

A) $\frac{13}{11}$ B) 1 C) $\frac{14}{11}$ D) $\frac{12}{11}$

4. $a+b=64$, $\frac{a}{b}=7$ bo'lsa, $a-b$ ning qiymatini toping.

A) 48 B) 32 C) 24 D) 36

5. $\frac{2+x}{4-x}=\frac{2}{9}$ tenglamani yeching.

A) $\frac{10}{11}$ B) $\frac{11}{10}$ C) $-\frac{11}{10}$ D) $-\frac{10}{11}$

6. 400 soni 30% ga oshirildi. Hosil bo'lgan sonning 30% ini toping.

A) 520 B) 156 C) 150 D) 400

7. Ko'paytuvchilarga ajrating: $8abc^2-16abc$

A) $8ab(c-3)$ B) $8abc(c-3)$ C) $abc(b-18)$ D) $8abc(c-2)$

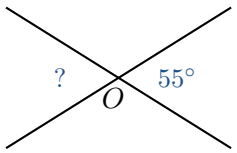
8. Savatda 4 ta olma va 3 ta nok bor. Savatdan 1 ta mevani necha xil usulda olish mumkin?

A) 12 B) 1 C) 7 D) 24

9. Rasmda berilgan vertikal burchakni toping.

A) $125°$ B) $90°$

C) $35°$ D) $55°$



10. 200 litr necha dm^3 ga teng?

A) 200 B) 20 C) 2 D) 2000

12-variant

1. 8 ga bo'linadigan ikki xonali sonlar nechta?

A) 10 B) 12 C) 13 D) 11

2. Hisoblang: $22 - (-7) - 38$

A) 9 B) -9 C) -29 D) 29

3. Hisoblang: $\frac{11}{12} - \frac{4}{3} + \frac{1}{6}$

A) $-0{,}25$ B) $0{,}25$ C) $0{,}35$ D) $-0{,}35$

4. Nisbati $3 : 5$ bo'lgan ikki sonning yig'indisi 160. Ularning musbat ayirmasini toping.

A) 20 B) 40 C) 60 D) 80

5. $\frac{9x + 13}{19x + 20} = \frac{1}{2}$ tenglamani yeching.

A) -6 B) 2 C) -2 D) 6

6. Mahsulot narxi avval 800 ming edi va biroz muddatdan keyin 600 ming bo'ldi. Mahsulot narxi necha foizga kamaygan?

A) 15% B) 20% C) 35% D) 25%

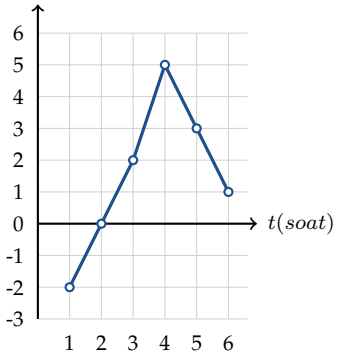
7. Hisoblang: $\sqrt{16} \cdot \sqrt{81}$

A) 13 B) 97 C) 36 D) 1296

8. 5 ta bo'sh stulga 2 o'quvchini necha xil usulda o'tqazish mumkin?

A) 10 B) 20 C) 25 D) 7

9. Grafikda sutkalik harorat $^\circ C$ da berilgan. Eng past va eng baland harorat orasidagi farqni toping.



A) 5° B) 9°

C) 7° D) 12°

10. 58 km 4 dm necha santimetrga teng?

A) 5800040 B) 5800400 C) 580004 D) 58040

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13-variant

1. Raqamlari ko'paytmasi 6 ga teng bo'lgan ikki xonali sonlar nechta?

A) 3 B) 4 C) 2 D) 1

2. Hisoblang: $-23 + (-10) - (-33)$

A) 0 B) -66 C) 66 D) -46

3. Hisoblang: $1\frac{1}{2} \cdot \frac{1}{3}$

A) 3 B) 1 C) $\frac{1}{2}$ D) $\frac{1}{3}$

4. $a^8 : a^3 \cdot a^2$ ifodani soddalashtiring.

A) a^3 B) a^{10} C) a^{13} D) a^7

5. $\frac{4x+7}{3} = \frac{x-7}{2}$ tenglamani yeching.

A) -5 B) -6 C) -3 D) -7

6. Firma ishlab chiqargan 1200 dona mahsulotning 92% i a'lo navli. A'lo navli mahsulotlar sonini toping.

A) 1114 B) 1000 C) 1004 D) 1104

7. $(2a^2 - 6a^3) \cdot 2a$ ifodani soddalashtiring.

A) $4a^2 + 12a^4$ B) $4a^4 - 12a^3$ C) $4a^3 - 12a^4$ D) $12a^4 - 4a^3$

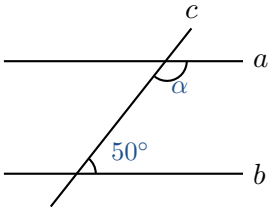
8. Fermada 3 ta tovuq va 4 ta o'rdak bor. Tovuq va o'rdak juftligini necha xil tanlash mumkin?

A) 12 B) 7 C) 24 D) 14

9. $a \parallel b$. Kesuvchi b chiziq bilan $50°$ li burchak hosil qiladi. Rasmdagi α burchakni toping.

A) $70°$ B) $130°$

C) $90°$ D) $40°$



10. 4 km 3 dm necha santimetrga teng?

A) 400030 B) 40030 C) 400300 D) 40300

14-variant

1. 4 ga bo'linadigan ikki xonali sonlar nechta?

A) 20 B) 22 C) 23 D) 24

2. Hisoblang: $(-13)^2 + 14 \cdot (-12)$

A) -1 B) 2 C) 3 D) 1

3. $\frac{14}{15}$ ni o'nli kasr shaklida yozing.

A) $0{,}933$ B) $0{,}9(3)$ C) $0{,}3(9)$ D) $0{,}399$

4. Otaning 10 qadami uzunligi 8 metr, o'g'lining 5 qadami uzunligi 3 metr. Otaning bir qadami o'g'lining bir qadamidan necha cm uzun?

A) 30 B) 40 C) 20 D) 10

5. Soddalashtiring: $4(3x - 2) - 3(2x - 5)$

A)$18x + 7$ B) $6x - 23$ C) $6x + 7$ D) $6x - 7$

6. Aka ukasidan 10 yosh katta, hozir nisbat $3 : 1$. Necha yildan keyin $2 : 1$ bo'ladi?

A) 5 B) 4 C) 8 D) 10

7. Soddalashtiring : $57a^2bc : (19abc)$

A) $4ab$ B) a C) $5bc$ D) $3a$

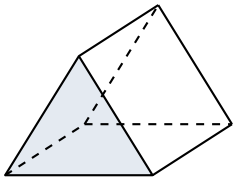
8. Sport zalda 8 o'zbek, 10 qozoq sportchi bor. 1 o'zbek va 1 qozoqni necha xil tanlash mumkin?

A) 40 B) 36 C) 72 D) 80

9. Rasmda berilgan uchburchakli prizmaning yon yoqlari sonini toping.

A) 3 B) 2

C) 4 D) 5



10. $7{,}2$ tonna yukning chorak (to'rtdan bir) qismi sifatsiz chiqti. Qancha yuk sifatsiz?

A) $0{,}9$ kg B) $1{,}8$ t C) 90 kg D) 900 kg

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15-variant

1. 1 dan 80 gacha bo'lgan natural sonlar ichida 4 raqami necha marta ishtirok etadi?

A) 17 B) 19 C) 18 D) 20

2. Hisoblang: $-\big((-100-60)-(100+(-40))\big)$

A) 200 B) -220 C) 220 D) 100

3. Hisoblang: $\frac{5}{6}:\frac{15}{24}+\frac{1}{2}:\frac{1}{4}$

A) $\frac{8}{3}$ B) $\frac{7}{2}$ C) $3\frac{1}{3}$ D) $\frac{11}{3}$

4. Ikki sonning yig'indisi 300, nisbati esa 4 ga teng bo'lsa, ularning musbat ayirmasini toping.

A) 160 B) 180 C) 200 D) 150

5. $\frac{x+7}{4-x}=\frac{2}{9}$ tenglamani yeching.

A) -5 B) 3 C) 6 D) -4

6. Bug'doy tegirmonda yanchilganda 81% i un bo'ladi. 24,3 tonna un olish uchun necha tonna bug'doy kerak?

A) 32 B) 40 C) 28 D) 30

7. Soddalashtiring: $(2a^2+6a):2a$

A) $a+3$ B) $a+2$ C) $3-a$ D) $a+4$

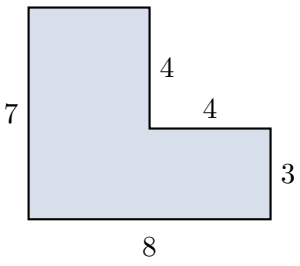
8. 5 xil sabzavotdan 2 tasini olib necha xil salat tayyorlash mumkin?

A) 4 B) 8 C) 12 D) 10

9. Berilgan shaklning perimetrini toping (o'lchamlar cm da).

A) 36 B) 30

C) 20 D) 24



10. 50 800 m^2 ni gektar va arda ifodalang.

A) 5 ga 8 ar B) 50 ga 8 ar C) 508 ar D) 5 ga 800 ar

16-variant

1. 72 sonining natural bo'luvchilari nechta?

A) 10 B) 11 C) 9 D) 12

2. Hisoblang: $-1{,}2 : (-0{,}2) + 5$

A) 11 B) 10 C) 12 D) 13

3. Hisoblang: $3\frac{1}{5} \cdot 1\frac{7}{8}$

A) 5 B) 6 C) 7 D) 8

4. $P = 6a - 3b$ va $Q = 6b - 3a$ bo'lsa, $P + Q$ ni toping.

A) $3(a - b)$ B) $9(a - b)$ C) $3(b - a)$ D) $3(a + b)$

5. $\frac{2x-1}{3} = \frac{3x+2}{3} + 2$ tenglamani yeching.

A) -3 B) 9 C) 3 D) -9

6. a sonining 36% i, 108 ning 24% iga teng. a ni toping.

A) 72 B) 36 C) 108 D) 54

7. Hisoblang: $\sqrt{36} \cdot \sqrt{4} + \sqrt{25}$

A) 12 B) 14 C) 17 D) 21

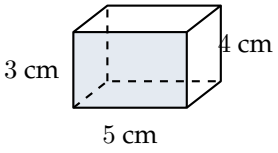
8. 4 xil kitobdan 2 tasini tanlab, javonga yonma-yon necha xil tartibda qo'yish mumkin?

A) 6 B) 12 C) 8 D) 16

9. Rasmda o'lchamlari 5 cm, 4 cm va 3 cm bo'lgan parallelepiped berilgan. Uning hajmini toping.

A) 60 cm^3 B) 12 cm^3

C) 47 cm^3 D) 120 cm^3



10. 3 t 5 sr ni kilogrammlarda ifodalang.

A) 3050 kg B) 3005 kg C) 3500 kg D) 3550 kg

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17-variant

1. 12 va 18 sonlarining eng kichik umumiy karralisini toping.

A) 36 B) 6 C) 72 D) 216

2. Hisoblang: $|a| - |b| + |-c|$, bunda $a = 15$, $b = -8$, $c = 9$.

A) 2 B) 32 C) -12 D) 16

3. $\frac{13}{15}$ ni o'nli kasr shaklida yozing.

A) 0,42 B) $0,4(6)$ C) $0,8(6)$ D) $0,2(6)$

4. Umumiy uzunligi 36 cm bo'lgan arqon $5:4$ nisbatda qirqildi. Kichik arqon kattasidan necha cm qisqa?

A) 5 B) 4 C) 12 D) 9

5. $\frac{2x-1}{3} = \frac{3x+2}{3} + 2$ tenglamani yeching.

A) -3 B) 9 C) -9 D) 3

6. Korxonadagi matoning 16% idan ko'ylak tikildi va bunga 320 metr mato ishlatilgan bo'lsa, korxonada jami necha metr mato bo'lgan?

A) 2020 cm B) 20000 m C) 200 m D) 2000 m

7. Birhadning koeffitsiyentini toping: $36x \cdot \left(-\frac{1}{2}xy\right) \cdot \frac{3}{9}x^2$

A) 6 B) 4 C) -6 D) 18

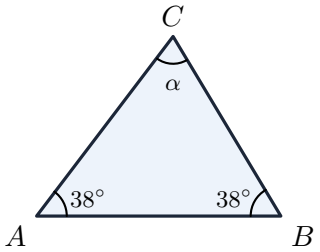
8. 4 ta turli mevani muzlatkichga necha usulda terish mumkin?

A) 30 B) 24 C) 25 D) 12

9. $\angle CAB = \angle CBA$ va rasmdagi ma'lumotlardan foydalanib α ni toping.

A) 104° B) 37°

C) 74° D) 80°



10. 4300 kgni tonna va sentnerda ifodalang.

A) 4 t 30 sr B) 4 t 3 sr C) 4 t 300 sr D) 4 t 3000 sr

18-variant

1. 1 dan 40 gacha sonlar ichida 2 raqami necha marta ishtirok etadi?

A) 13 B) 15 C) 12 D) 14

2. Hisoblang: $(-12)^2 - 11 \cdot 12$

A) 12 B) -12 C) 310 D) -276

3. Hisoblang: $2{,}3(6) + 1{,}(6)$

A) 4 B) $4{,}(6)$ C) $\frac{121}{30}$ D) $4{,}(5)$

4. Uch sonning o'rta arifmetigi 20 ga teng. Agar ulardan ikkitasi 15 va 22 bo'lsa, uchinchi sonni toping.

A) 21 B) 25 C) 23 D) 27

5. $2 \cdot (x - 3) + 7 = 2x + 1$ tenglama nechta yechimga ega?

A) yechimga ega emas B) 1 ta C) 2 ta D) cheksiz ko'p

6. Mansur bankka 4 500 000 so'm qo'ydi va 1 yildan so'ng puliga 25% ustama qo'shildi. U qancha so'm foyda ko'rgan?

A) 6 500 000 B) 562 000 C) 1 125 000 D) 6 000 000

7. Soddalashtiring: $a \cdot \left(\frac{b}{c} - \frac{d}{c}\right)$

A) $a \cdot \frac{b-d}{2c}$ B) $a \cdot \frac{b-d}{c}$ C) $a \cdot \frac{d-b}{c}$ D) $a \cdot \frac{b-d}{bc}$

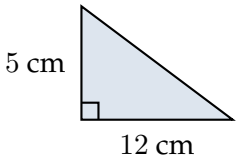
8. 10 nafar do'st bir-biri bilan qo'l berib ko'rishdi. Jami nechta qo'l berish bo'lgan?

A) 90 B) 45 C) 20 D) 100

9. To'g'ri burchakli uchburchakning katetlari 5 va 12 cm. Uchburchak yuzini toping.

A) 30 B) 60

C) 17 D) 120



10. 8600 kg ni tonna va sentnerda ifodalang.

A) 5 t 400 sr B) 8 t 6 sr C) 5 t 40 sr D) 5 t 4000 sr

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19-variant

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1. 9 ga bo'linadigan ikki xonali sonlar nechta?

A) 9 B) 11 C) 12 D) 10

2. Hisoblang: $-2{,}4 : (-0{,}8) + 7$

A) 40 B) 4 C) 16 D) 10

3. $\frac{5}{6}$ ni o'nli kasr shaklida yozing.

A) $0{,}42$ B) $0{,}4(6)$ C) $0{,}8(3)$ D) $0{,}2(6)$

4. Sardor va 2 ta ukasining umumiy vazni 162 kg. Sardorning vazni 2 ukasi vaznlari yig'indisiga teng bo'lsa, Sardorning vaznini toping.

A) 85 B) 75 C) 70 D) 81

5. $8(x-1) = 6x$ tenglamani yeching.

A) 3 B) 5 C) 4 D) 2

6. 28 cm li kesma $6:1$ nisbatda bo'lindi. Kichik bo'lakning uzunligini toping.

A) 24 B) 4

C) 14 D) 21

7. Soddalashtiring: $-3a - (a+4) + 4(a+1)$

A) 1 B) 0 C) $4a$ D) $2a$

8. 3 ta turli mevani muzlatkichga necha usulda terish mumkin?

A) 3 B) 5 C) 6 D) 9

9. Uchburchakning asosi 7 m, balandligi 4 m. Uchburchak yuzini toping.

A) 14 B) 15 C) 16 D) 28

10. 2 m^3 5 dm^3 ni santimetr kubda ifodalang.

A) $2\,005\,000$ B) $200\,500$ C) $205\,000$ D) $2\,050\,000$

20-variant

1. 96 va 64 sonlarining eng katta umumiy bo'luvchisini toping.

A) 8 B) 16 C) 32 D) 192

2. Hisoblang: $|-10|+|-20|-|-10|\cdot 3$

A) 0 B) 60 C) 30 D) 45

3. Hisoblang: $\frac{1}{3}+0{,}1$

A) $\frac{13}{15}$ B) $\frac{2}{5}$ C) $\frac{17}{30}$ D) $\frac{13}{30}$

4. $\frac{x-3y+z}{x-y+z}=3$ bo'lsa, $\frac{x+5z}{x+2z}$ ni toping.

A) 1 B) 4 C) 2 D) 3

5. $\frac{x}{4}-\frac{x}{3}=\frac{1}{6}$ tenglamani yeching.

A) 4 B) -3 C) -2 D) 6

6. 300 metr mato olingan bo'lsa va u jami matoning 12% ini tashkil etsa, fabrikada jami qancha metr mato bor?

A) 1600 B) 2500 C) 2000 D) 1800

7. Hisoblang: $\sqrt{100}-\sqrt{36}+\sqrt{49}$

A) 9 B) 13 C) 23 D) 11

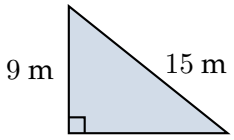
8. 8 ta turli nomzoddan direktor va uning o'rinbosarini necha xil usulda tanlash mumkin?

A) 56 B) 64 C) 42 D) 28

9. Kateti 9 m, gipotenuzasi esa 15 m bo'lgan to'g'ri burchakli uchburchakning yuzini toping.

A) 54 m^2 B) 60 m^2

C) 108 m^2 D) 67,5 m^2



10. 1 sutka 4 soat necha minutga teng?

A) 1680 B) 48000 C) 2400 D) 100800

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21-variant

1. 1 dan 70 gacha bo'lgan natural sonlar ichida 1 raqami necha marta ishtirok etadi?

A) 16 B) 18 C) 17 D) 15

2. Hisoblang: $27 + (-5) \cdot 4 - 3$

A) 12 B) -8 C) 0 D) 4

3. Hisoblang: $\frac{11}{12} - \frac{4}{3} + \frac{1}{6}$

A) 0,25 B) $-0{,}25$ C) 0,35 D) $-0{,}35$

4. Aka ukasidan 12 yosh katta, hozir nisbat $4:1$. Akaning hozirgi yoshini toping.

A) 16 B) 4 C) 12 D) 20

5. $\frac{3x+2}{2x-1} = 2$ tenglamani yeching.

A) 4 B) -4 C) 1 D) -2

6. 900 gramm eritma tarkibida 18 gramm tuz bor. Eritmaning necha foizini tuz tashkil qiladi?

A) 9% B) 2% C) 20% D) 10%

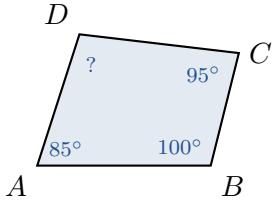
7. Hisoblang: $\sqrt{49} \cdot \sqrt{4} + \sqrt{9}$

A) 14 B) 17 C) 11 D) 21

8. 10 ta bo'sh stulga 2 o'quvchini necha xil usulda joylashtirish mumkin?

A) 60 B) 45 C) 80 D) 90

9. To'rtburchakning uch burchagi rasmda berilgan. To'rtinchi burchakni toping.



A) 70° B) 90°

C) 100° D) 80°

10. 3 km 2 dm $= \boxed{?}$ cm

A) 300200 B) 3000020 C) 300020 D) 300002

22-variant

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1. Raqamlari ko'paytmasi 8 ga teng bo'lgan ikki xonali sonlar nechta?

A) 3 B) 5 C) 4 D) 2

2. Hisoblang: $-\big((-15)-85\big)+(45+50)$

A) 195 B) 200 C) 190 D) 5

3. Hisoblang: $13\frac{1}{5} : 0{,}8$

A) 33 B) 15 C) 16 D) $\frac{33}{2}$

4. $0{,}875$ ni oddiy kasr ko'rinishida yozing.

A) $\frac{4}{5}$ B) $\frac{3}{4}$ C) $\frac{5}{6}$ D) $\frac{7}{8}$

5. $144 : x = 36$ tenglamani yeching.

A) 10 B) 4 C) 8 D) 16

6. Uzum quritilganda 36% vaznini yo'qotadi. 4 tonna mayiz olish uchun necha kg uzum kerak?

A) 6250 B) 14500 C) 12500 D) 16500

7. Ko'paytuvchilarga ajrating: $3a - 51abc$

A) $3a(17-bc)$ B) $3a(1+17bc)$ C) $3a(1-17bc)$ D) $3a(1+19bc)$

8. Gul do'konida qizil atirguldan 4 xil, oq atirguldan 2 xil gul bor. Bitta gul tanlashning necha usuli mavjud?

A) 6 B) 8 C) 60 D) 12

9. $x+y=6$ va $xy=5$ bo'lsa, $x^2+2xy+y^2$ ifodaning qiymatini toping.

A) 25 B) 36 C) 30 D) 41

10. $2{,}4$ soatning $\frac{1}{6}$ qismi necha daqiqa bo'ladi?

A) 240 B) 12 C) 25 D) 24

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23-variant

1. 1 dan 90 gacha bo'lgan natural sonlar ichida 8 raqami necha marta ishtirok etadi?

A) 19 B) 18 C) 20 D) 17

2. Hisoblang: $-28-3-(-31)$

A) -2 B) 2 C) 0 D) -62

3. Hisoblang: $\frac{1}{3}+0{,}25$

A) $\frac{7}{11}$ B) $\frac{1}{12}$ C) $\frac{4}{7}$ D) $\frac{7}{12}$

4. Otaning 10 qadami 8 metr, o'g'lining 5 qadami 3 metr. Otaning qadami o'g'lining qadamidan necha cm uzun?

A) 30 B) 20 C) 40 D) 10

5. $2019-2x=2021$ tenglamani yeching.

A) 1 B) -1 C) 4 D) 11

6. Do'konda 1900 ta non bor. Tushlikkacha 75% i sotildi. Qancha non qoldi?

A) 475 ta B) 1425 ta C) 500 ta D) 750 ta

7. Soddalashtiring: $(4+8a):2-3a$

A) $a+3$ B) $a-2$ C) $2+a$ D) 4

8. Fermada 4 ta tovuq va 5 ta o'rdak bor. Tovuq va o'rdak juftligini necha xil tanlash mumkin?

A) 20 B) 9 C) 40 D) 14

9. Sakkiz burchakli prizmaning yoqlari sonini toping.

A) 8 B) 16 C) 10 D) 12

10. 3500 metrga to'g'ri keladigan javobni toping.

A) 3 km 500 cm B) 3 km 50000 dm C) 30 km 500 m D) 3 km 500 m

24-variant

1. 6 ga bo'linadigan ikki xonali sonlar nechta?

A) 14 B) 15 C) 16 D) 13

2. Hisoblang: $|-10|+|-20|-|-10|\cdot 3$

A) 60 B) 30 C) 0 D) 45

3. Hisoblang: $2\frac{1}{6}+1\frac{1}{3}$

A) $\frac{1}{2}$ B) $3\frac{2}{3}$ C) $4\frac{2}{3}$ D) $3\frac{1}{2}$

4. 2 ta sonning yig'indisi 180, nisbati esa 5 ga teng. Sonlarning kattasini toping.

A) 120 B) 164 C) 150 D) 144

5. $\frac{10+x}{4-x}=1$ tenglamani yeching.

A) -3 B) -2 C) -1 D) 0

6. Bug'doy tegirmonda yanchilganda 19% kepak chiqsa, 24,3 tonna un olish uchun necha tonna bug'doy kerak?

A) 32 B) 45 C) 25 D) 30

7. $a+b=5$ va $ab=4$ bo'lsa, a^2b+ab^2 ifodaning qiymatini toping.

A) 16 B) 20 C) 24 D) 9

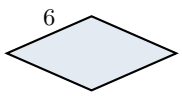
8. 3 ta xatni 3 ta turli konvertga necha xil usulda joylash mumkin?

A) 6 B) 5 C) 1 D) 9

9. Rombning tomoni 6 ga teng. Rombning perimetrini toping.

A) 18 B) 21

C) 27 D) 24



10. 10 m/s necha km/s ga teng?

A) 0,1 B) 0,001 C) 0,01 D) 0,0001

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25-variant

1. 60 sonining natural bo'luvchilari nechta?

A) 12 B) 10 C) 11 D) 9

2. Hisoblang: $(-13)^2 + 14 \cdot (-12)$

A) 1 B) -1 C) 2 D) 3

3. Hisoblang: $\frac{111}{333} - \frac{222}{666} + \frac{333}{999}$

A) $\frac{2}{3}$ B) $\frac{1}{3}$ C) 1 D) $\frac{4}{3}$

4. $a + b = 320$, $\frac{a}{b} = 39$ bo'lsa, $a - b$ ni toping.

A) 299 B) 302 C) 304 D) 307

5. $\frac{x+7}{4-x} = \frac{2}{9}$ tenglamani yeching.

A) 3 B) 6 C) -4 D) -5

6. Sutdan 12% qaymoq olinadi. 15 kg sutdan qancha qaymoq olinadi?

A) 1,8 g B) 18 kg C) 18 g D) 1,8 kg

7. Hisoblang: $\sqrt{\frac{81}{16}}$

A) $\frac{81}{16}$ B) $\frac{9}{4}$ C) $\frac{3}{4}$ D) $\frac{9}{8}$

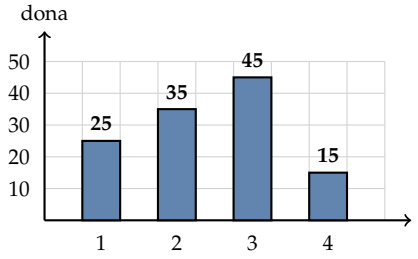
8. 5 xil sabzavotdan 2 tasini olib necha xil salat tayyorlash mumkin?

A) 4 B) 8 C) 12 D) 10

9. Ustun diagrammada 4 kunlik savdo natijalari berilgan. Berilganlar asosida sotishning kunlik o'rtacha ko'rsatkichini toping.

A) 120 B) 30

C) 40 D) 45



10. 512 500 grammni tonnada ifodalang.

A) 5,125 t B) 0,0521 t C) 0,5125 t D) 0,00521 t

26-variant

1. Raqamlari ko'paytmasi 4 ga teng bo'lgan ikki xonali sonlar nechta?

A) 3 B) 4 C) 2 D) 5

2. Hisoblang: $-\big((-12)-88\big)+(52+49)$

A) 200 B) 199 C) 201 D) 1

3. Hisoblang: $3\frac{2}{3}+2\frac{5}{10}$

A) $\frac{37}{6}$ B) $\frac{11}{3}$ C) $\frac{37}{8}$ D) $\frac{45}{8}$

4. Aka ukasidan 10 yosh katta va yoshlarining hozirgi nisbati $3:1$. Necha yildan keyin ular yoshlarining nisbati $2:1$ kabi bo'ladi?

A) 4 B) 8 C) 10 D) 5

5. $\frac{9x+13}{19x+20}=\frac{1}{2}$ tenglamani yeching.

A) -6 B) 2 C) 6 D) -2

6. 120 soni 480 ning necha foizini tashkil etadi?

A) 20% B) 25% C) 30% D) 40%

7. $(57a^2bc):(19abc)$ ni soddalashtiring.

A) $4ab$ B) $3a$ C) a D) $5bc$

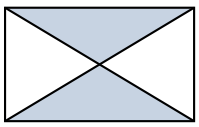
8. 20 nafar do'st qo'l berib ko'rishdi. Jami nechta qo'l berish bo'lgan?

A) 190 B) 145 C) 200 D) 100

9. Rasmdagi to'g'ri to'rtburchakning necha foizi bo'yalgan?

A) 65% B) 75%

C) 25% D) 50%



10. 4 km 3 dm $=\boxed{?}$ cm

A) 40030 B) 400300 C) 40300 D) 400030

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27-variant

1. 1 dan 55 gacha bo'lgan natural sonlar ichida 5 raqami necha marta ishtirok etadi?

A) 12 B) 11 C) 13 D) 10

2. Hisoblang: $(-12)^2 - 11 \cdot 12$

A) -12 B) 12 C) 310 D) -276

3. Hisoblang: $\frac{17}{22} - \frac{3}{5}$

A) $\frac{19}{110}$ B) $\frac{119}{10}$ C) $\frac{2}{11}$ D) $\frac{3}{22}$

4. $(a^3)^4 : a^5$ ifodani soddalashtiring.

A) a^2 B) a^9 C) a^{12} D) a^7

5. $10 \cdot (x - 10) = 6x$ tenglamani yeching.

A) 20 B) 15 C) 25 D) 35

6. a sonining 36% i, 108 ning 24% iga teng. a ni toping.

A) 36 B) 72 C) 108 D) 54

7. $(2x^3y^2)^3$ ifodani soddalashtiring.

A) $6x^9y^5$ B) $8x^9y^3$ C) $8x^6y^6$ D) $8x^9y^6$

8. Sport zalda 8 o'zbek, 10 qozoq sportchi. 1 o'zbek va 1 qozoqni necha xil tanlash mumkin?

A) 40 B) 36 C) 72 D) 80

9. Kubning to'la sirtining yuzi 96 cm^2 ga teng. Kubning hajmini toping.

A) 64 cm^3 B) 96 cm^3 C) 16 cm^3 D) 512 cm^3

10. 3 km 40 m $= \boxed{?}$ cm

A) 300400 B) 3000040 C) 304000 D) 300004

28-variant

1. 7 ga bo'linadigan ikki xonali sonlar nechta?

A) 12 B) 13 C) 14 D) 11

2. Hisoblang: $22 - (-7) - 38$

A) -9 B) 9 C) -29 D) 29

3. Hisoblang: $6{,}42 + \frac{9}{8} - 1{,}045$

A) 6,5 B) 6,4 C) 6 D) 1,25

4. 2 ta sonning yig'indisi 200, nisbati esa $4 : 6$ kabi. Sonlarning kattasini toping.

A) 160 B) 80 C) 100 D) 120

5. $5(2x - 4) = 9x + 6$ tenglama ildizining yarmini toping.

A) 26 B) 30 C) 13 D) 15

6. Bankka 5 000 000 so'm qo'yildi.Agar qo'yilgan pul 1 yilda 25% ga oshsa, 1 yildan keyingi pul miqdorini toping.

A) 6 200 000 B) 6 000 000 C) 6 250 000 D) 5 250 000

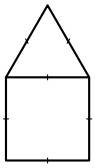
7. Soddalashtiring: $-3a - (a - 4) + 4(a - 1)$

A) 6 B) $4a$ C) 8 D) 0

8. Qulflash kodi 3 ta belgidan iborat va har bir belgi 0–9 raqamlaridan biri bo'la olsa, nechta kod tuzish mumkin (raqamlar takrorlanishi mumkin)?

A) 810 B) 720 C) 1000 D) 900

9. Rasmda muntazam uchburchak va kvadrat berilgan. Agar uchburchakning perimetri 24 ga teng bo'lsa, kvadratning yuzini toping.



A) 36 B) 100

C) 16 D) 64

10. 5400 kg ni tonna va sentnerda ifodalang.

A) 5 t 400 sr B) 5 t 4 sr C) 5 t 40 sr D) 5 t 4000 sr

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29-variant

1. $\overline{3*6}$ soni 9 ga bo'linsa, yulduzcha o'rnida quyidagi raqamlardan qaysi biri bo'lishi kerak?

A) 5 B) 3 C) 9 D) 6

2. Hisoblang: $-2{,}4 : (-0{,}8) + 7$

A) 40 B) 4 C) 16 D) 10

3. Hisoblang: $2\frac{1}{2} : 2\frac{1}{4}$

A) $\frac{9}{10}$ B) $1\frac{1}{9}$ C) $\frac{5}{6}$ D) $\frac{6}{5}$

4. $P = 5a - 2b$ va $Q = 5b - 2a$ bo'lsa, $P + Q$ ni toping.

A) $3(a-b)$ B) $7(a-b)$ C) $3(a+b)$ D) $3(a-c)$

5. $\frac{x+7}{4-x} = \frac{2}{9}$ tenglamani yeching.

A) -5 B) 3 C) 6 D) -4

6. Devorning 25% i bo'yalgan bo'lsa, uning qancha qismi bo'yalmagan?

A) $0{,}8$ B) $0{,}9$ C) $\frac{17}{20}$ D)$0{,}75$

7. Hisoblang: $\sqrt{64+36}$

A) 14 B) 8 C) 50 D) 10

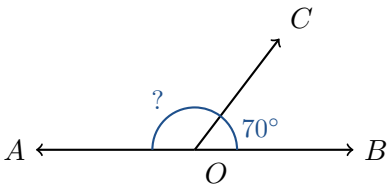
8. 5 ta turli rangdan 2 tasini necha xil usulda tanlash mumkin?

A) 10 B) 7 C) 20 D) 25

9. Rasmda qo'shni burchaklar berilgan. Noma'lum burchakni toping.

A) $70°$ B) $110°$

C) $80°$ D) $120°$



10. $90\,200$ m^2 ni gektar va arda ifodalang.

A) 9 ga 2 ar B) 90 ga 2 ar C) 90 ga 20 ar D) 9 ga 200 ar

30-variant

1. 96 va 72 sonlarining eng katta umumiy bo'luvchisini toping.

A) 24 B) 8 C) 12 D) 48

2. Hisoblang: $36 \cdot (11 - 13) + 64 \cdot (17 - 19)$

A) 200 B) 100 C) -100 D) -200

3. Hisoblang: $\frac{1}{3} - 0{,}1$

A) $\frac{7}{15}$ B) $\frac{7}{30}$ C) $\frac{11}{15}$ D) $\frac{13}{30}$

4. Bola tug'ilganda otasi 24 yosh edi. Hozir bolaning yoshi otasidan 3 marta kichik bo'lsa, bolaning yoshini toping.

A) 11 B) 15 C) 16 D) 12

5. $2 \cdot (3 - x) = 2 \cdot (3 - x) - 5$ tenglama nechta yechimga ega?

A) cheksiz ko'p B) yechimga ega emas C) 1 ta D) 2 ta

6. Tadbirkor mahsulotni 11 880 so'mdan sotib 10% ziyon qildi. Mahsulotning tan narxini toping.

A) 12 500 B) 13 600 C) 13 200 D) 14 600

7. Soddalashtiring: $a \cdot \left(\frac{b}{c} - \frac{d}{c}\right)$

A) $a \cdot \frac{b-d}{2c}$ B) $a \cdot \frac{d-b}{c}$ C) $a \cdot \frac{b-d}{c}$ D) $a \cdot \frac{b-d}{bc}$

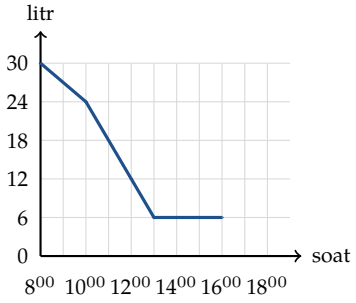
8. 7 nomzoddan 2 kishilik jamoani necha xil usulda tuzish mumkin?

A) 24 B) 8 C) 12 D) 21

9. Avtomobil yoqilg'isining vaqt bo'yicha o'zgarishi grafikda berilgan. 8^{00} dan 14^{00} gacha qancha litr yoqilg'i sarflangan?

A) 6 B) 18

C) 24 D) 12



10. 2 m^3 5 dm^3 ni santimetr kubda ifodalang.

A) 2 005 000 B) 200 500 C) 205 000 D) 2 050 000

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Javoblar

1-variant

1) B 2) D 3) C 4) A 5) C 6) D 7) A 8) B 9) C 10) A

2-variant

1) D 2) A 3) A 4) C 5) B 6) D 7) B 8) C 9) D 10) C

3-variant

1) C 2) D 3) A 4) C 5) D 6) D 7) B 8) B 9) B 10) A

4-variant

1) C 2) D 3) A 4) A 5) B 6) C 7) B 8) C 9) C 10) B

5-variant

1) A 2) D 3) C 4) A 5) B 6) D 7) B 8) C 9) C 10) B

6-variant

1) D 2) A 3) A 4) B 5) D 6) B 7) C 8) A 9) B 10) D

7-variant

1) B 2) D 3) B 4) A 5) A 6) C 7) C 8) B 9) D 10) D

8-variant

1) C 2) B 3) C 4) D 5) B 6) C 7) D 8) B 9) D 10) A

9-variant

1) D 2) B 3) C 4) A 5) A 6) B 7) C 8) B 9) A 10) D

10-variant

1) A 2) D 3) B 4) A 5) C 6) B 7) C 8) D 9) D 10) C

11-variant

1) A 2) B 3) C 4) A 5) D 6) B 7) D 8) C 9) D 10) A

12-variant

1) D 2) B 3) A 4) B 5) D 6) D 7) C 8) B 9) C 10) A

13-variant

1) B 2) A 3) C 4) D 5) D 6) D 7) C 8) A 9) B 10) A

14-variant

1) B 2) D 3) B 4) C 5) C 6) A 7) D 8) D 9) A 10) B

15-variant

1) C 2) C 3) C 4) B 5) A 6) D 7) A 8) D 9) B 10) A

16-variant

1) D 2) A 3) B 4) D 5) D 6) A 7) C 8) B 9) A 10) C

17-variant

1) A 2) D 3) C 4) B 5) C 6) D 7) C 8) B 9) A 10) B

18-variant

1) D 2) A 3) C 4) C 5) D 6) C 7) B 8) B 9) A 10) B

19-variant

1) D 2) D 3) C 4) D 5) C 6) B 7) B 8) C 9) A 10) A

20-variant

1) C 2) A 3) D 4) B 5) C 6) B 7) D 8) A 9) A 10) A

21-variant

1) C 2) D 3) B 4) A 5) A 6) B 7) B 8) D 9) D 10) C

22-variant

1) C 2) A 3) D 4) D 5) B 6) A 7) C 8) A 9) B 10) D

23-variant

1) A 2) C 3) D 4) B 5) B 6) A 7) C 8) A 9) C 10) D

24-variant

1) B 2) C 3) D 4) C 5) A 6) D 7) B 8) A 9) D 10) C

25-variant

1) A 2) A 3) B 4) C 5) D 6) D 7) B 8) D 9) B 10) C

26-variant

1) A 2) C 3) A 4) D 5) C 6) B 7) B 8) A 9) D 10) D

27-variant

1) A 2) B 3) A 4) D 5) C 6) B 7) D 8) D 9) A 10) C

28-variant

1) B 2) A 3) A 4) D 5) C 6) C 7) D 8) C 9) D 10) B

29-variant

1) C 2) D 3) B 4) C 5) A 6) D 7) D 8) A 9) B 10) A

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30-variant

1) A 2) D 3) B 4) D 5) B 6) C 7) C 8) D 9) C 10) A

Qaydlar uchun

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Foydalanilgan adabiyotlar:

1. Xaydarov B. Q. *Matematika 5-sinf*: umumiy o'rta ta'lim maktablarining 5-sinfi uchun darslik, 1-qism. – Toshkent, 2020. – 144 b.
2. Xaydarov B. Q. *Matematika 5-sinf*: umumiy o'rta ta'lim maktablarining 5-sinfi uchun darslik, 2-qism. – Toshkent, 2020. – 144 b.
3. Ismailov Sh. [va boshq.] *Matematika 6-sinf*: darslik. – Toshkent: Respublika ta'lim markazi, 2022. – 240 b.
4. Akmalov A. [va boshq.] *Algebra 7-sinf*: darslik. – Toshkent: Respublika ta'lim markazi, 2022. – 192 b.
5. Xaydarov B., Tashtemirova N., Asrorov I. *Geometriya 7-sinf*: darslik. – Toshkent: Respublika ta'lim markazi, 2022. – 192 b.

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MAJBURIY MATEMATIKA